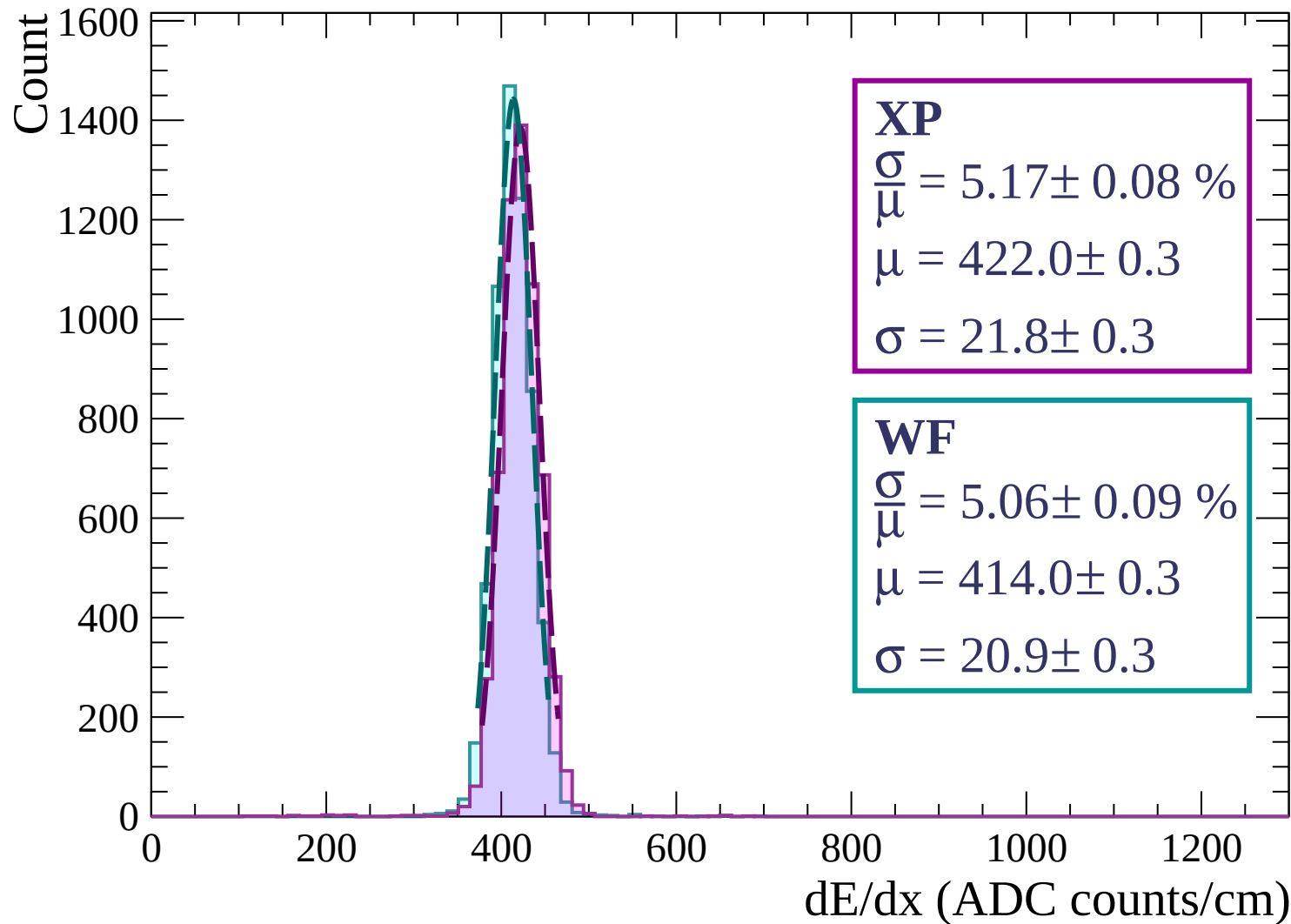
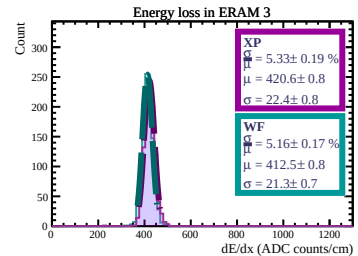
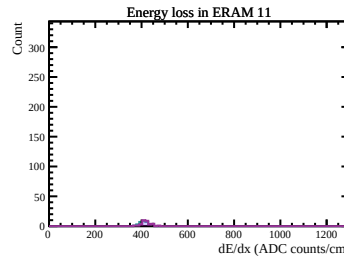
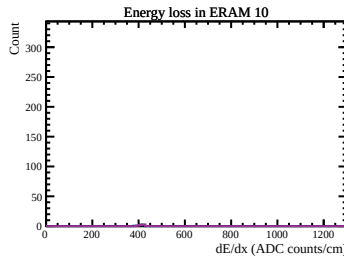
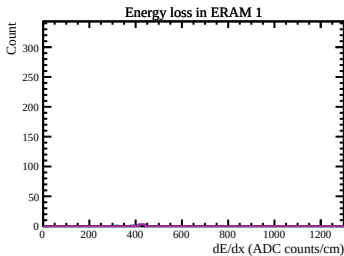
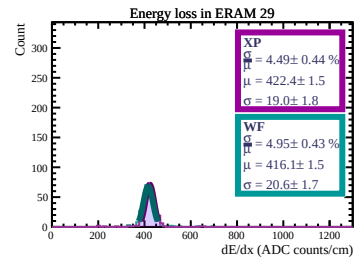
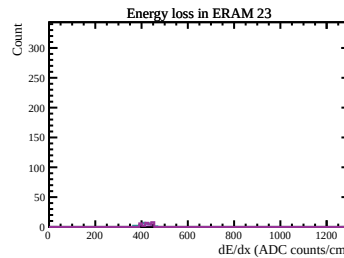
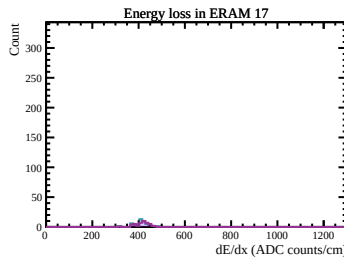
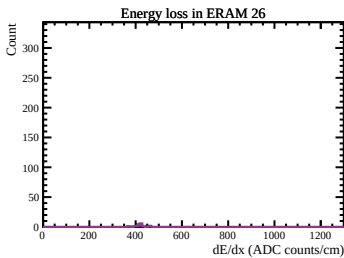
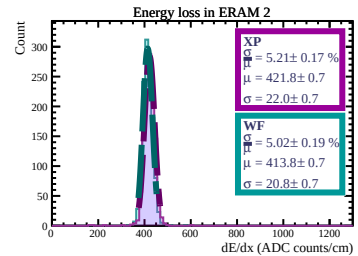
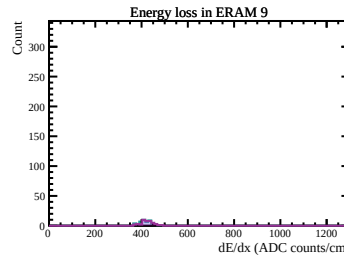
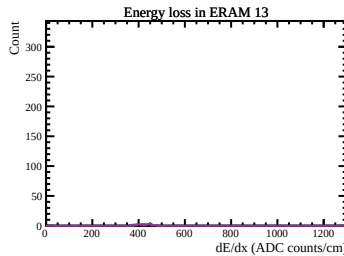
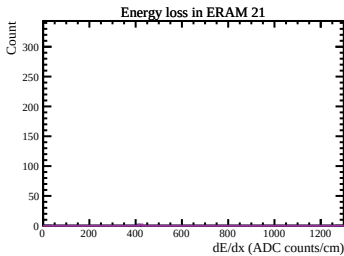
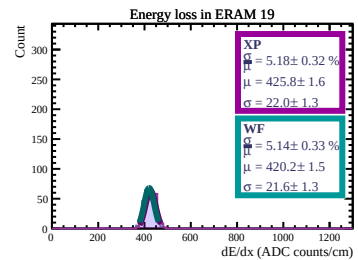
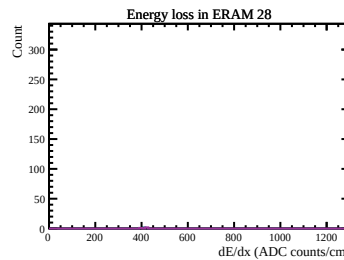
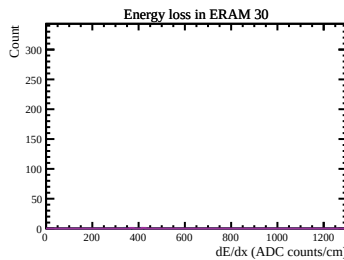
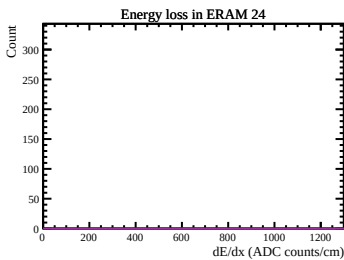
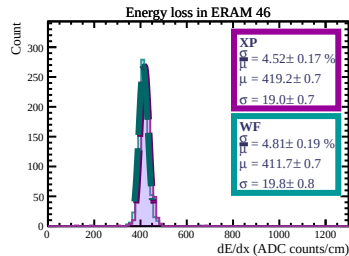
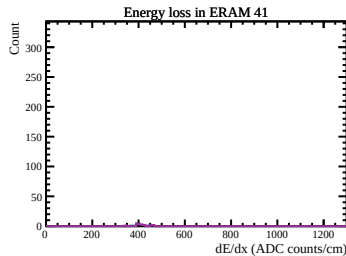
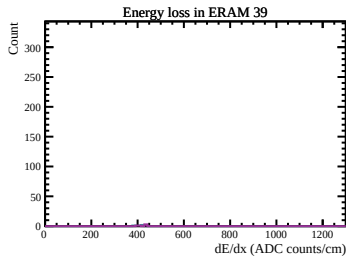
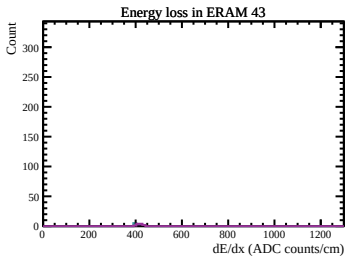
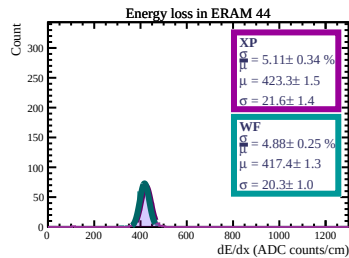
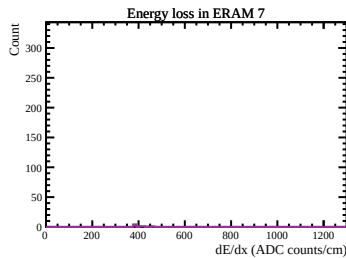
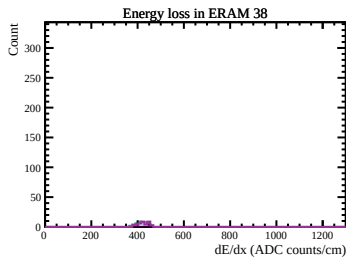
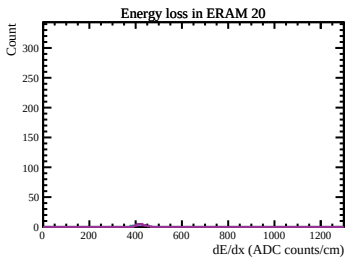
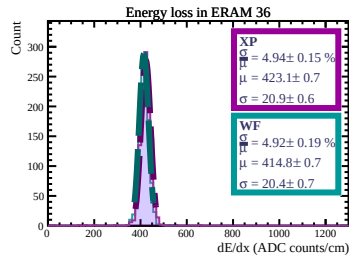
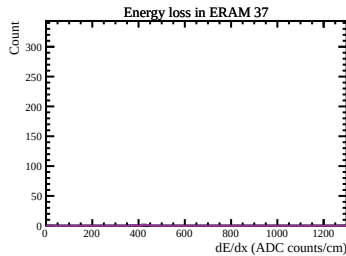
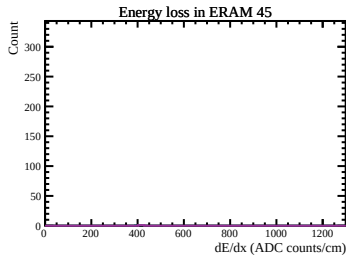
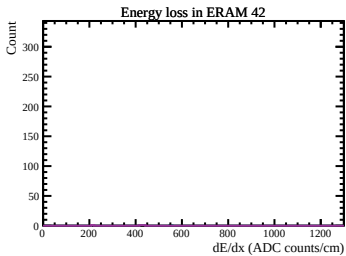
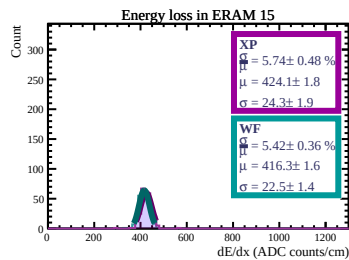
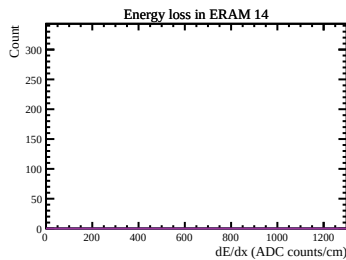
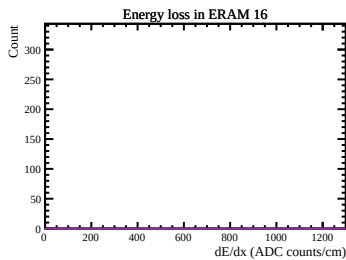
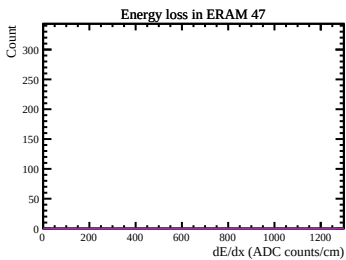
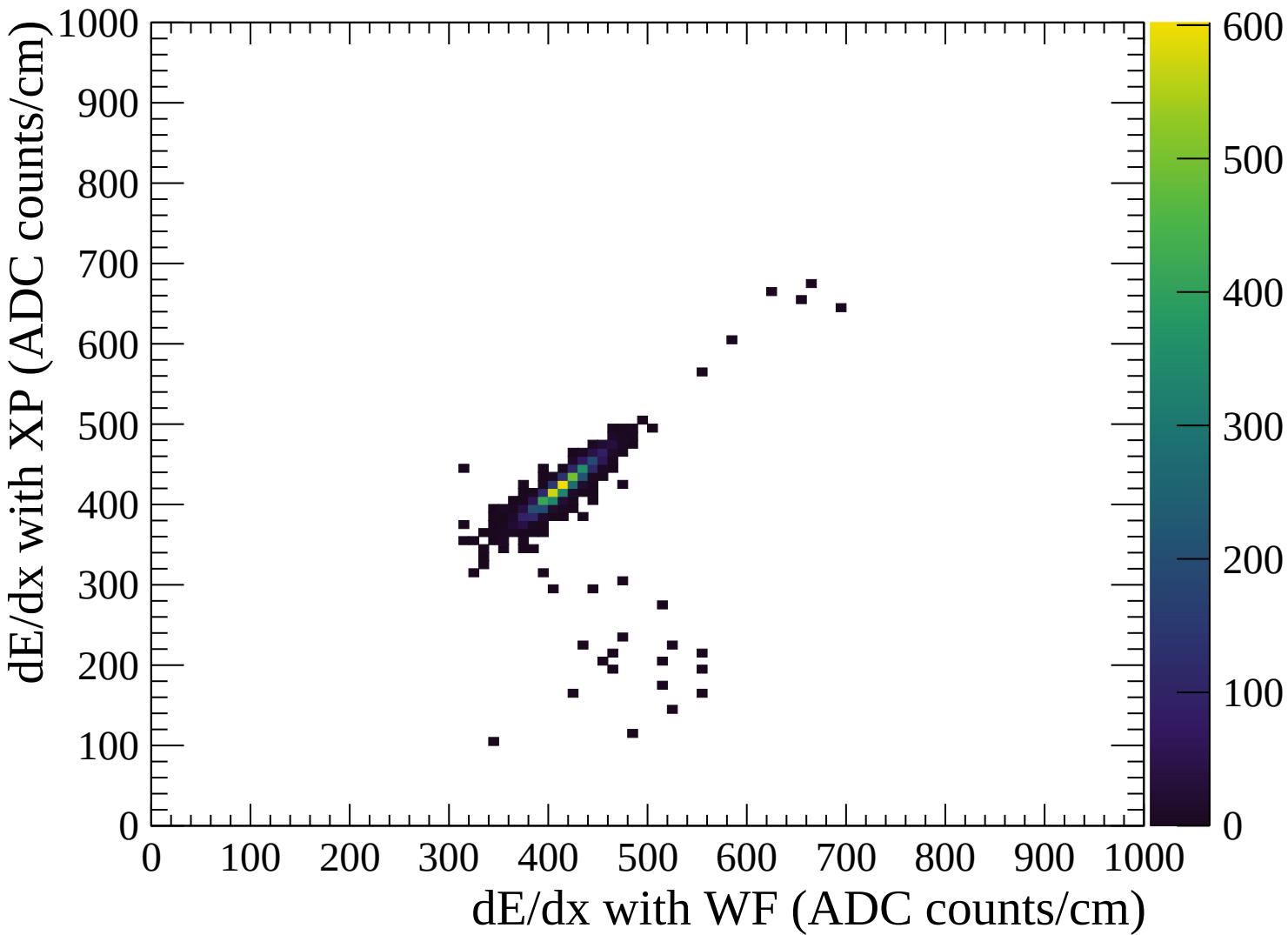
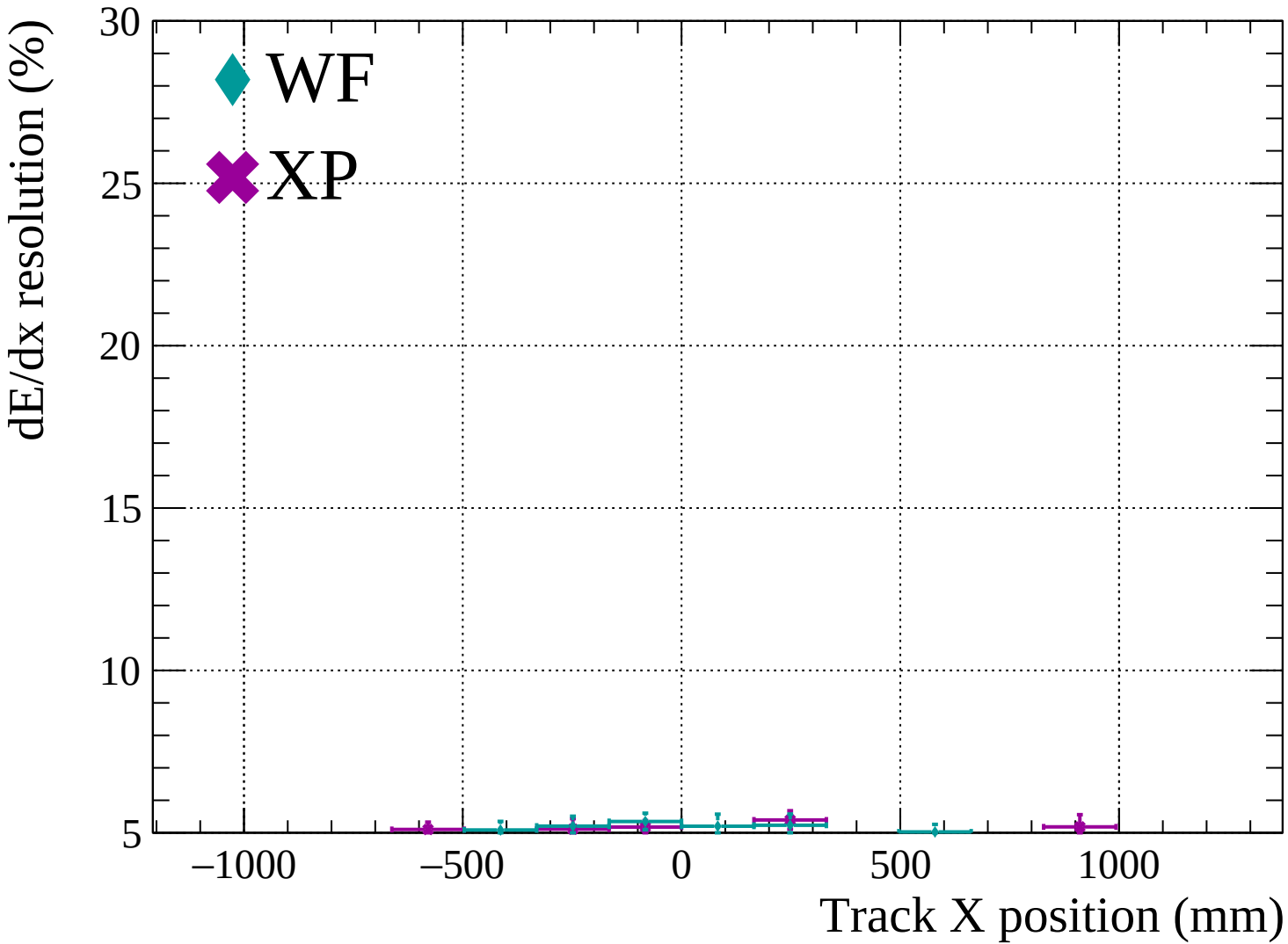


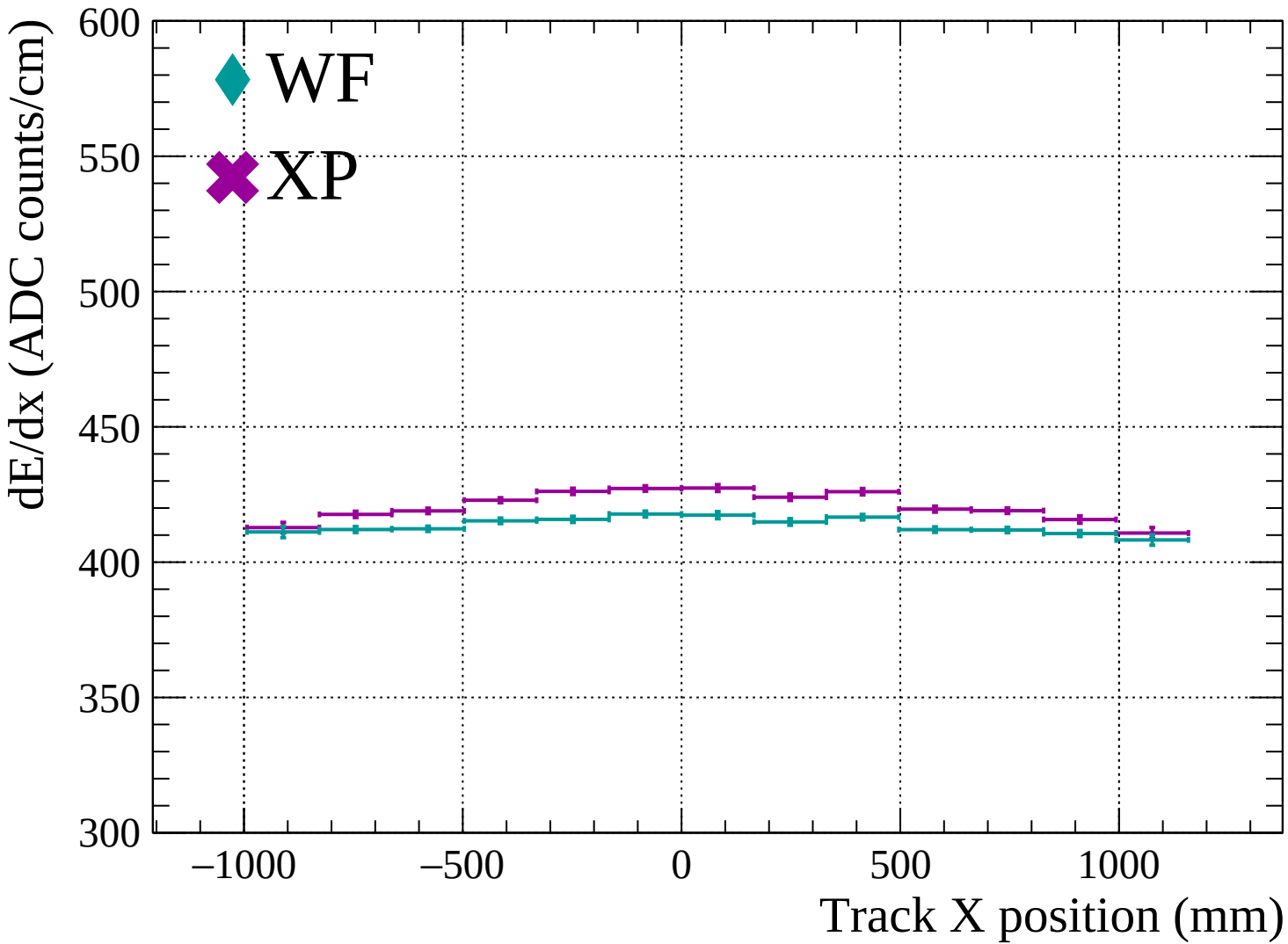


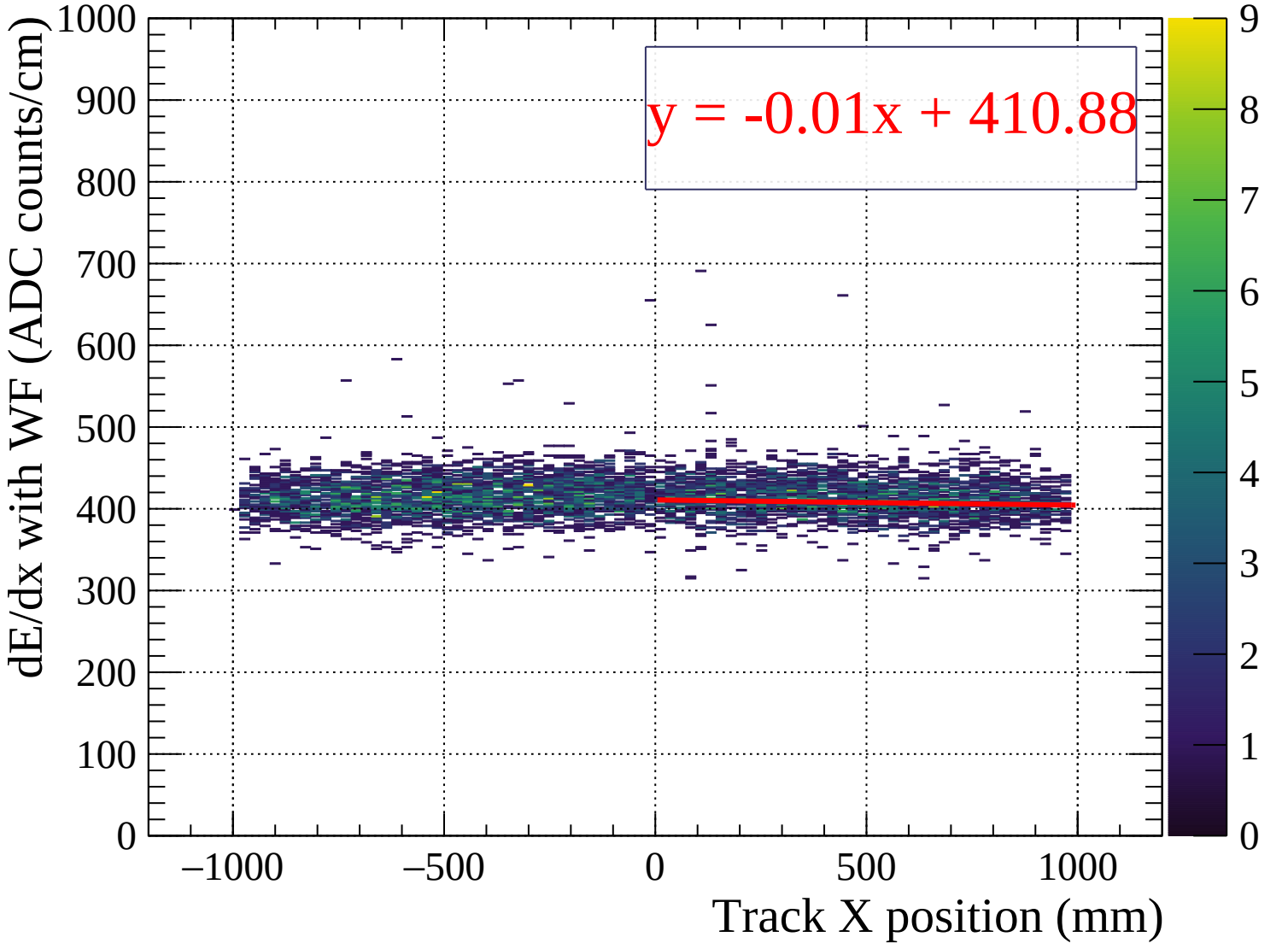


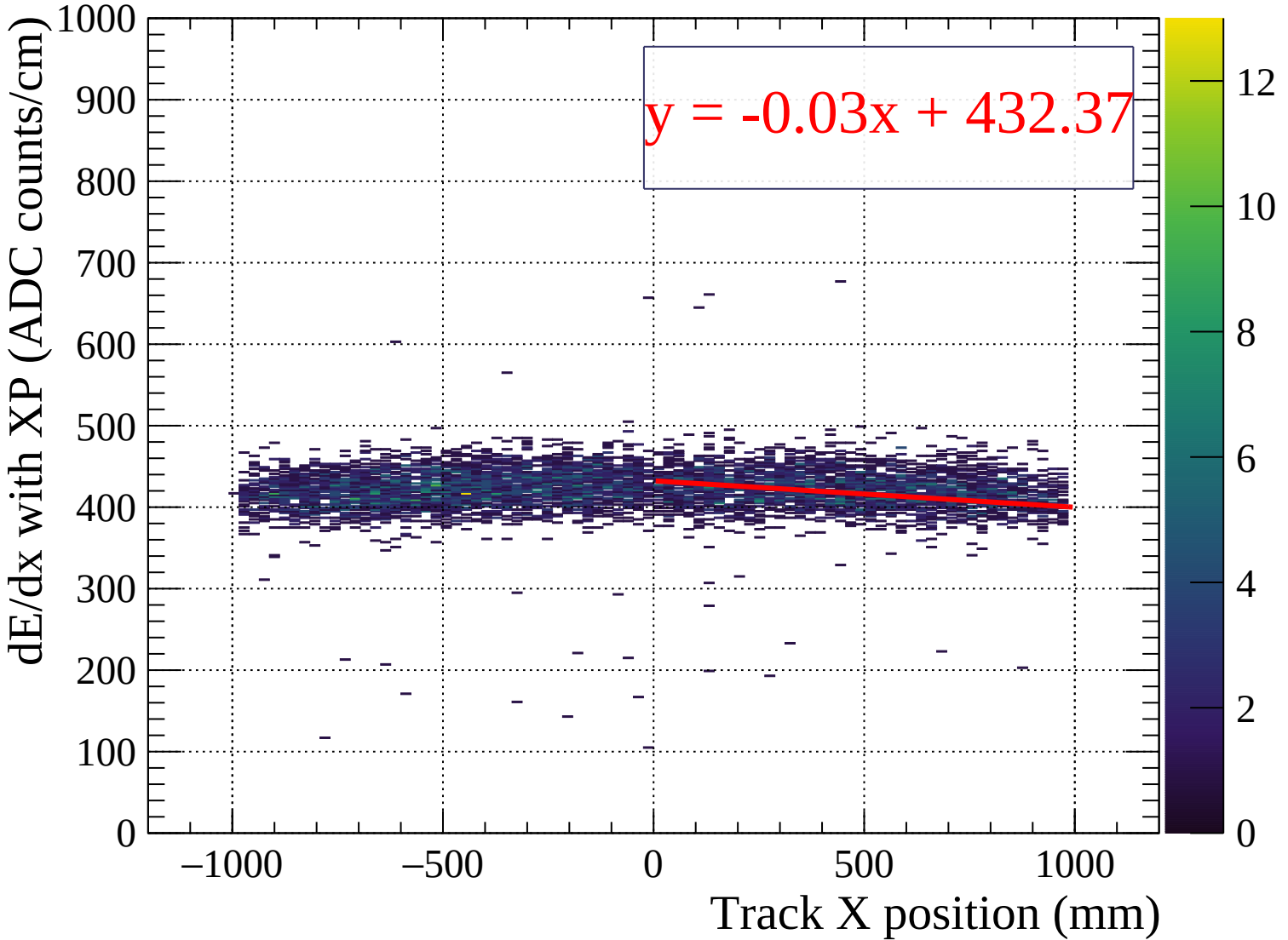


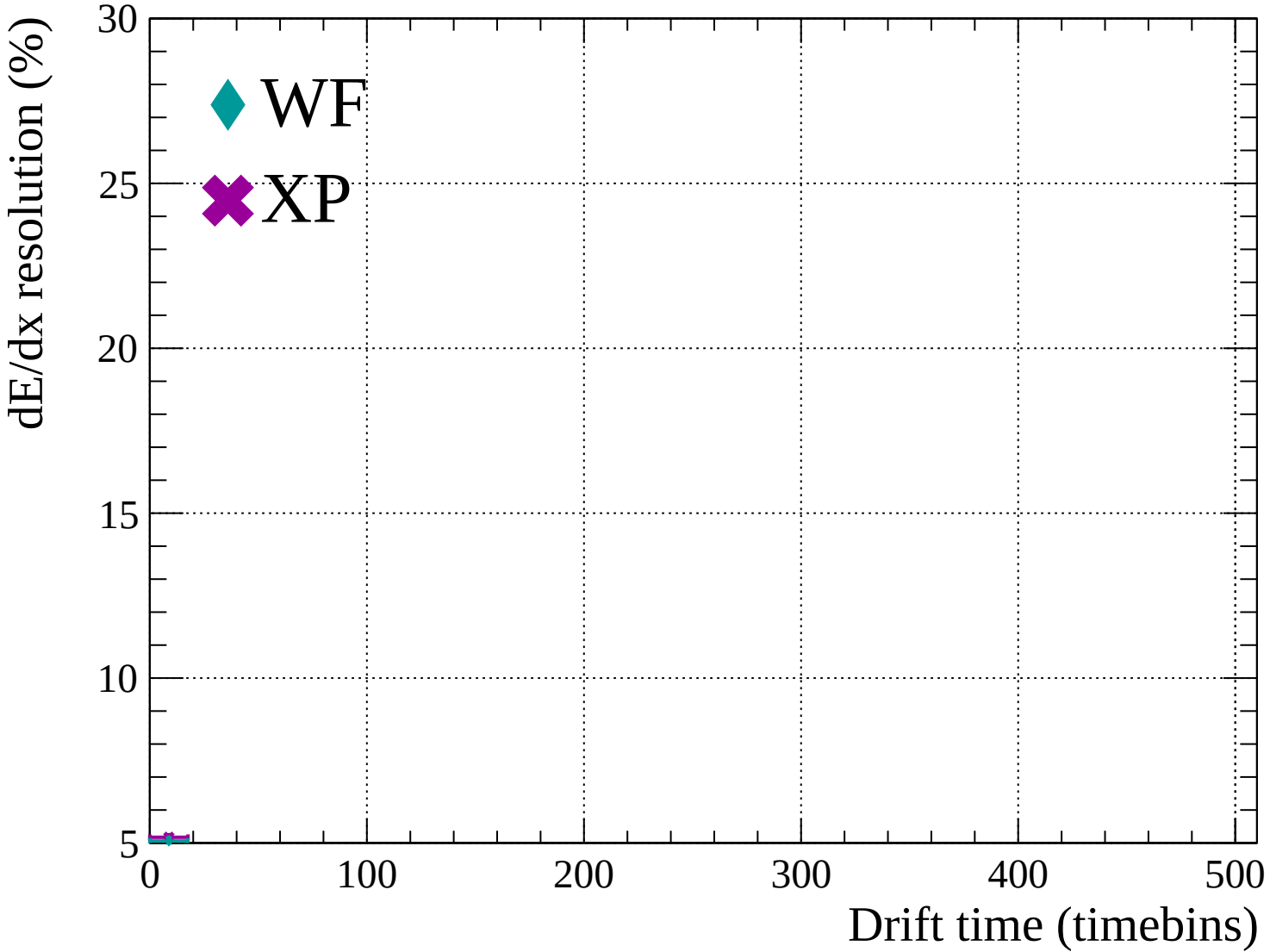


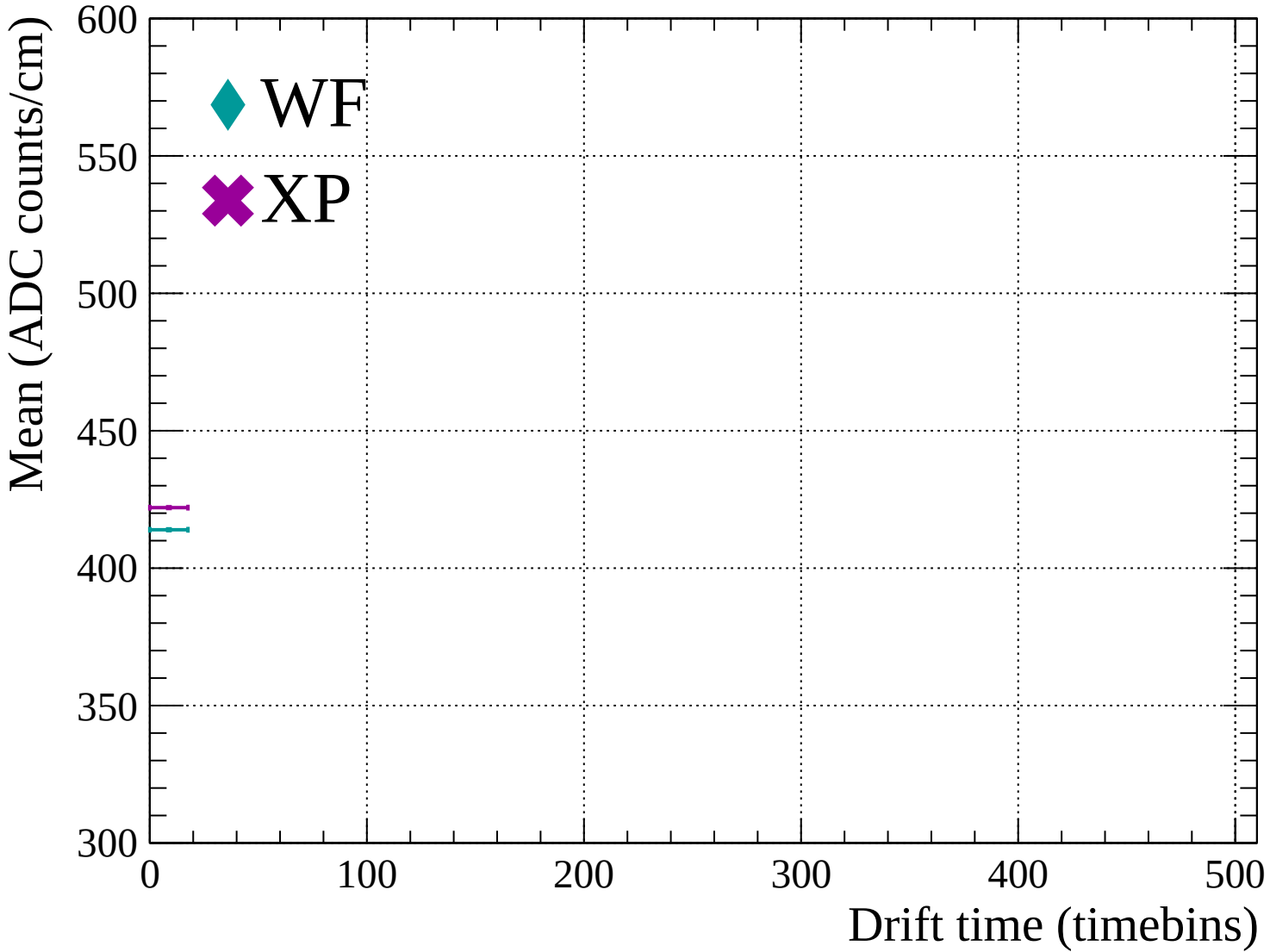


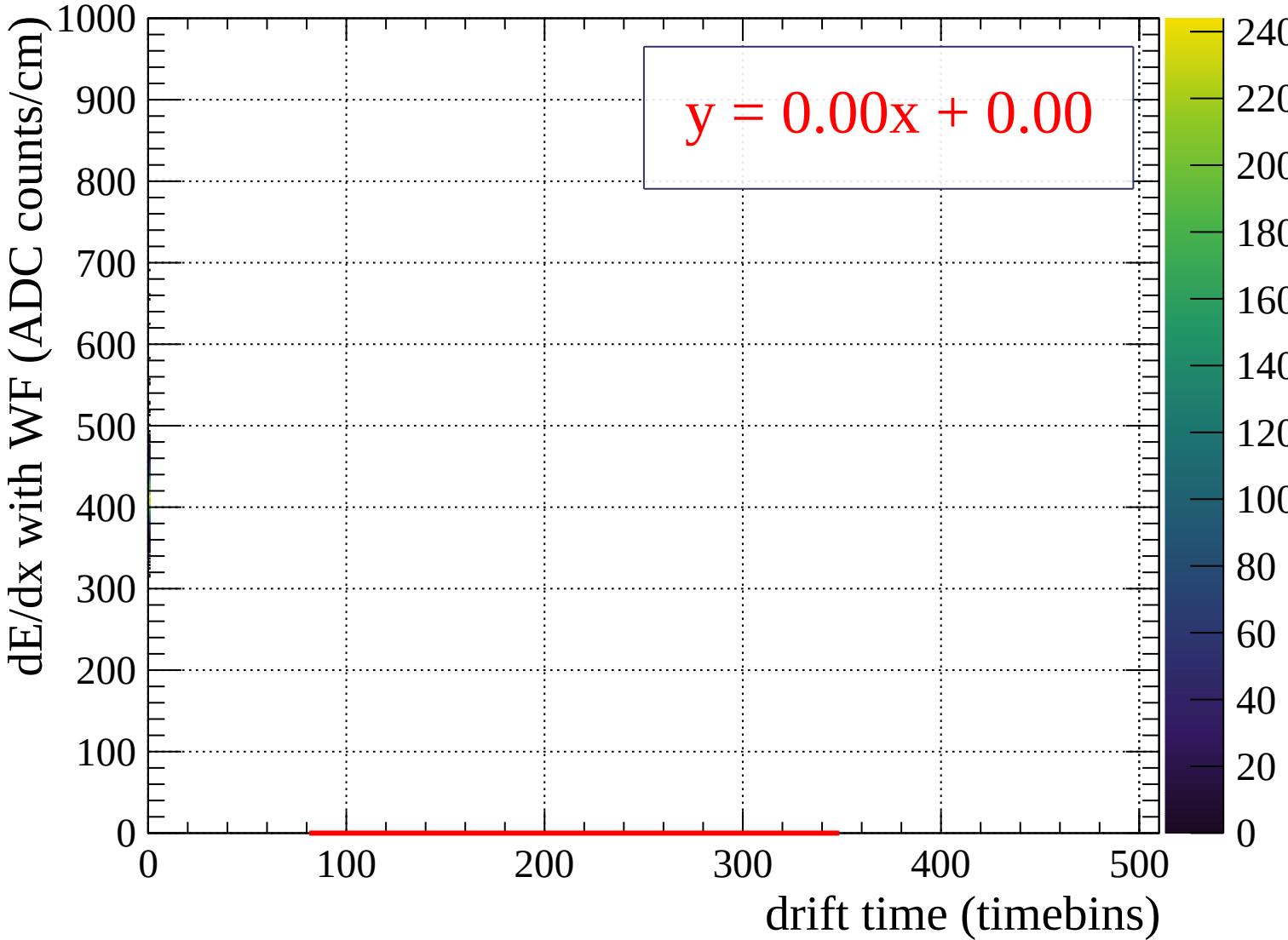


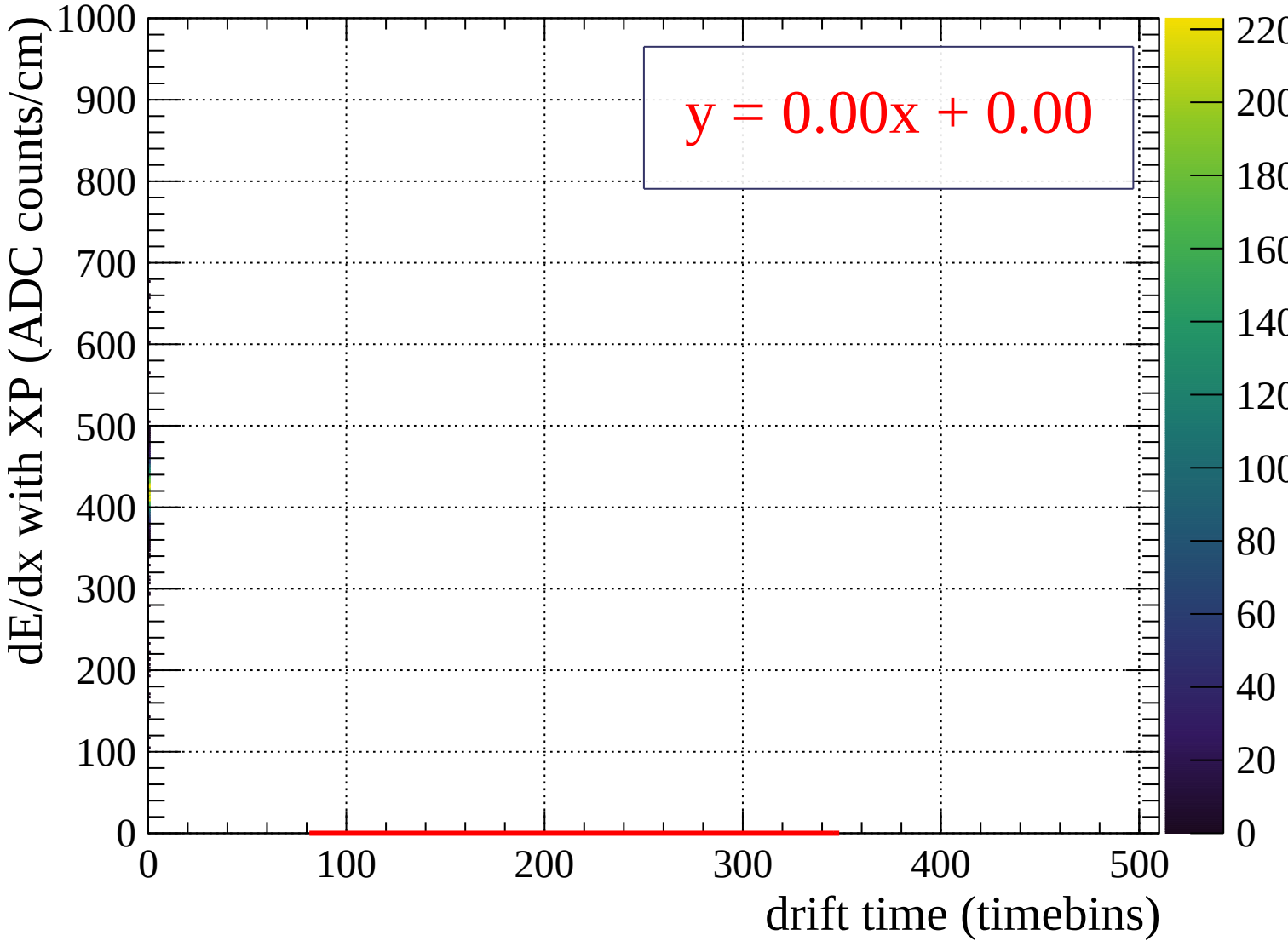


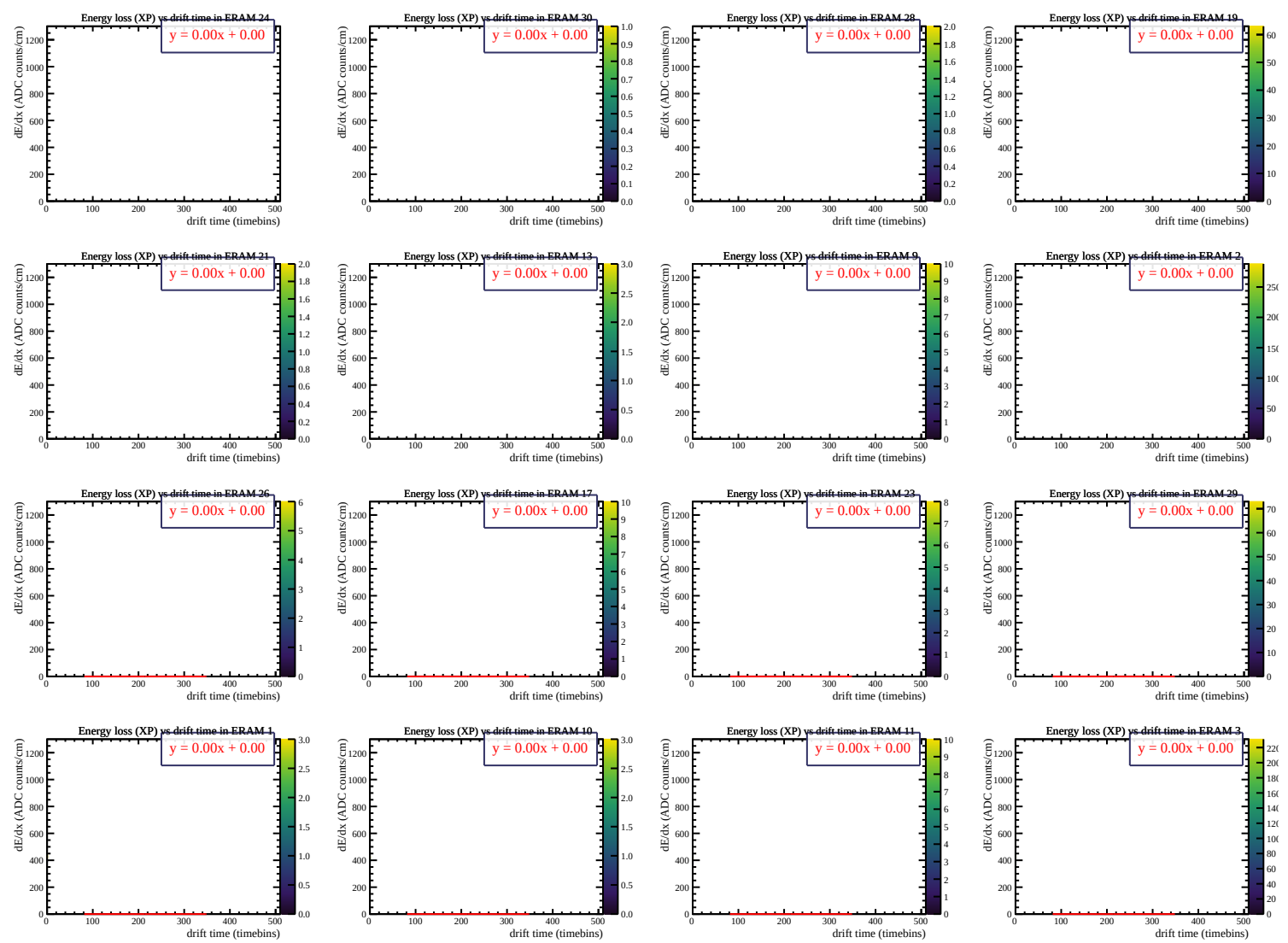


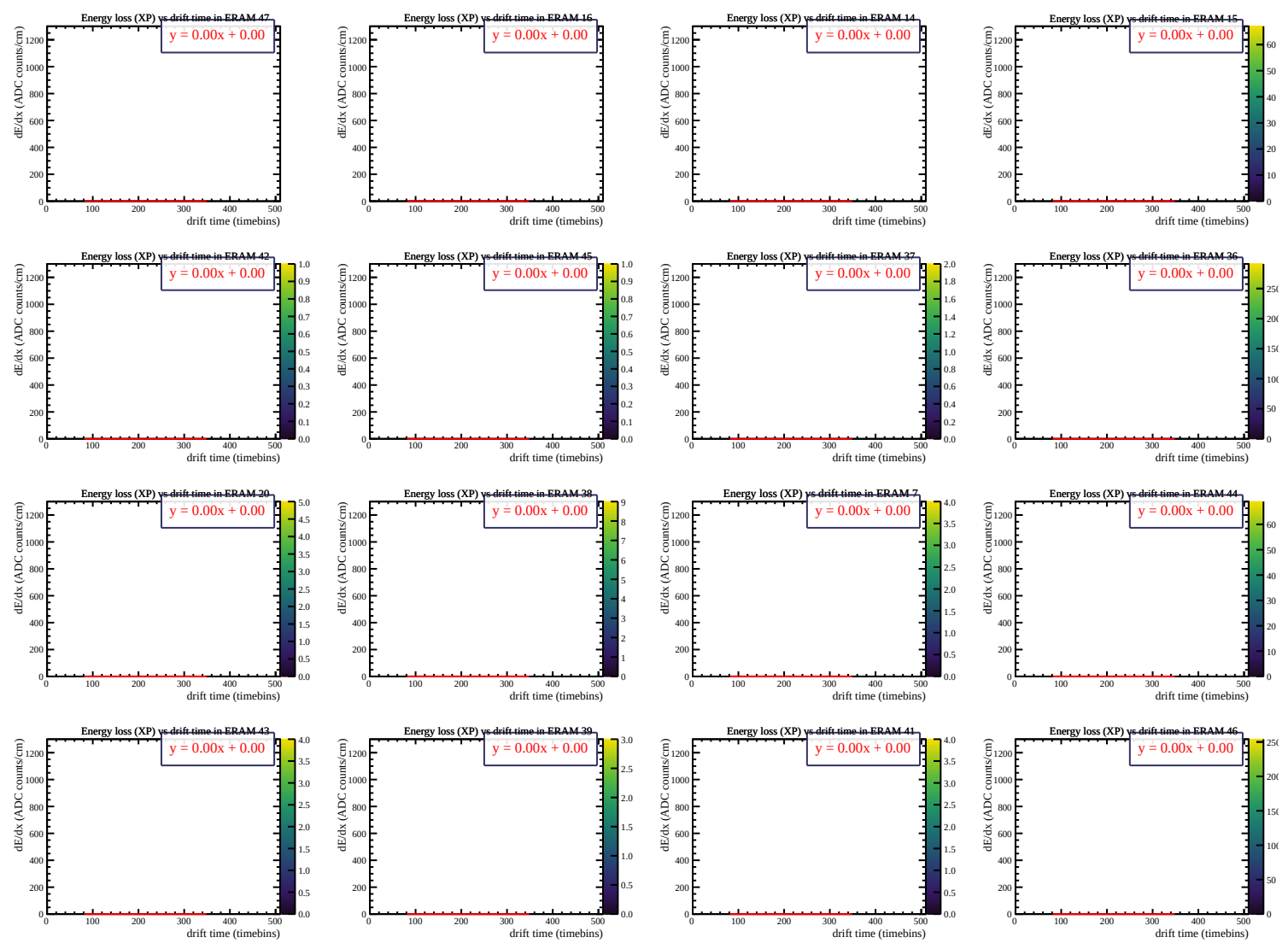


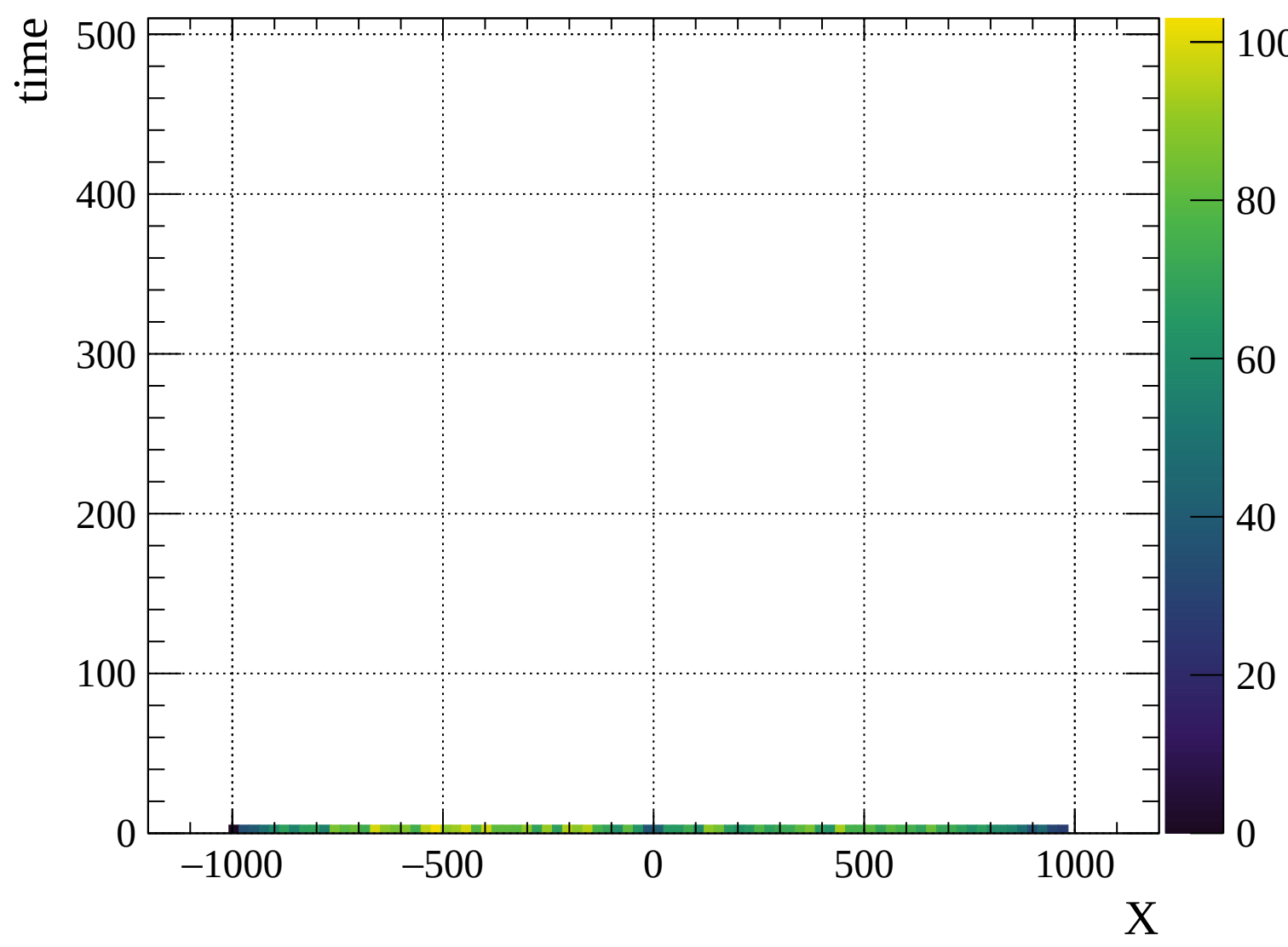


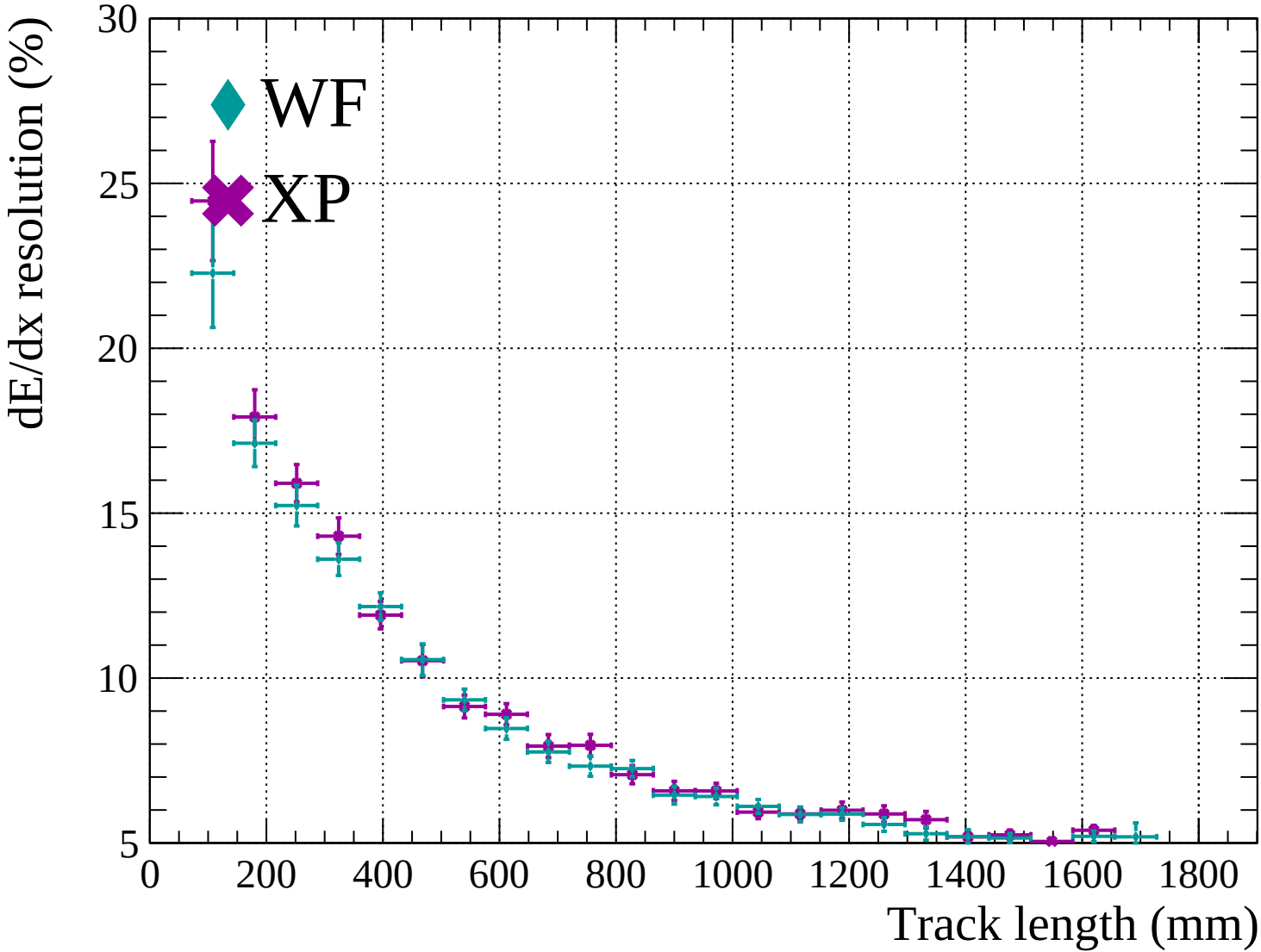


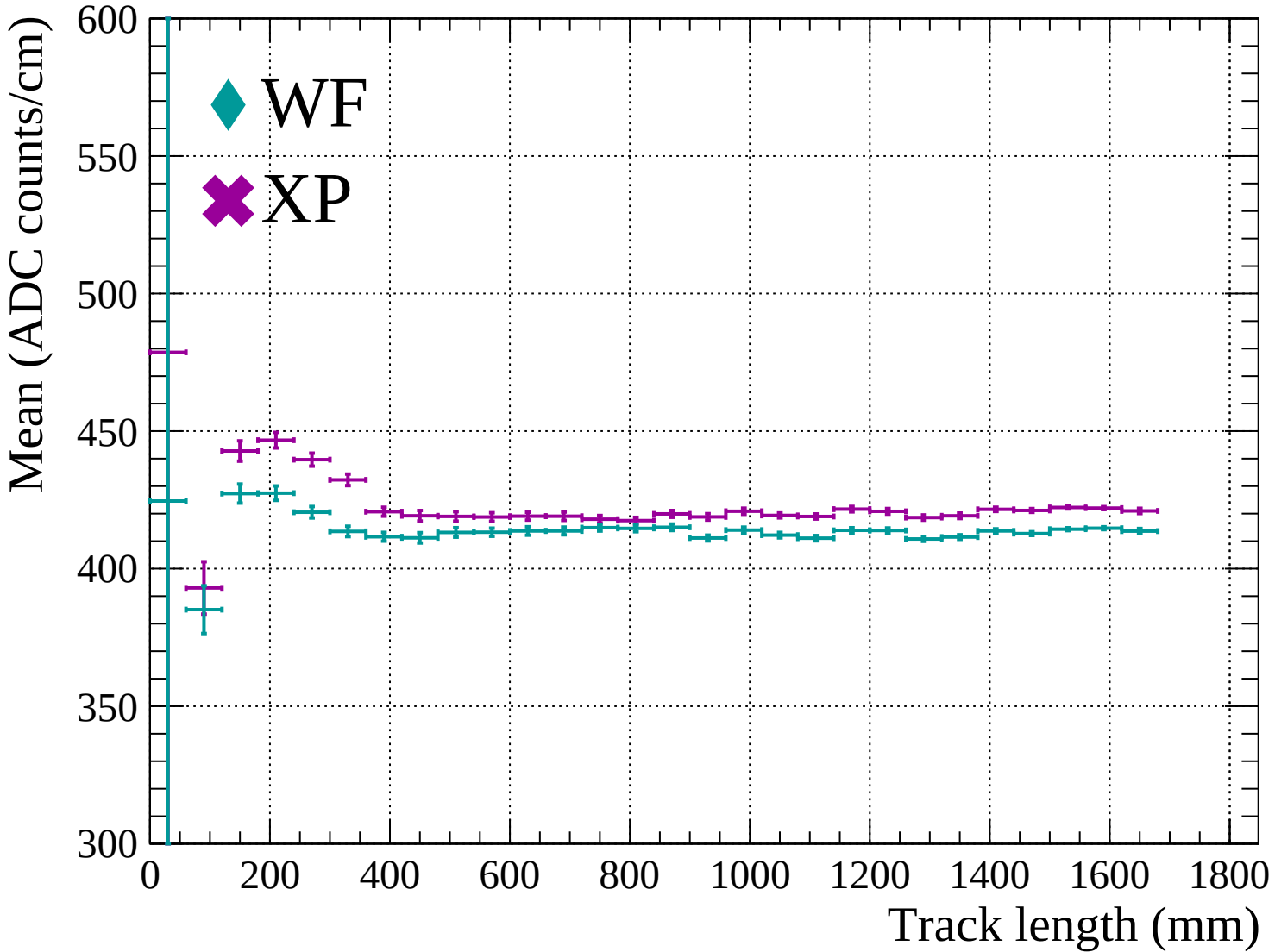


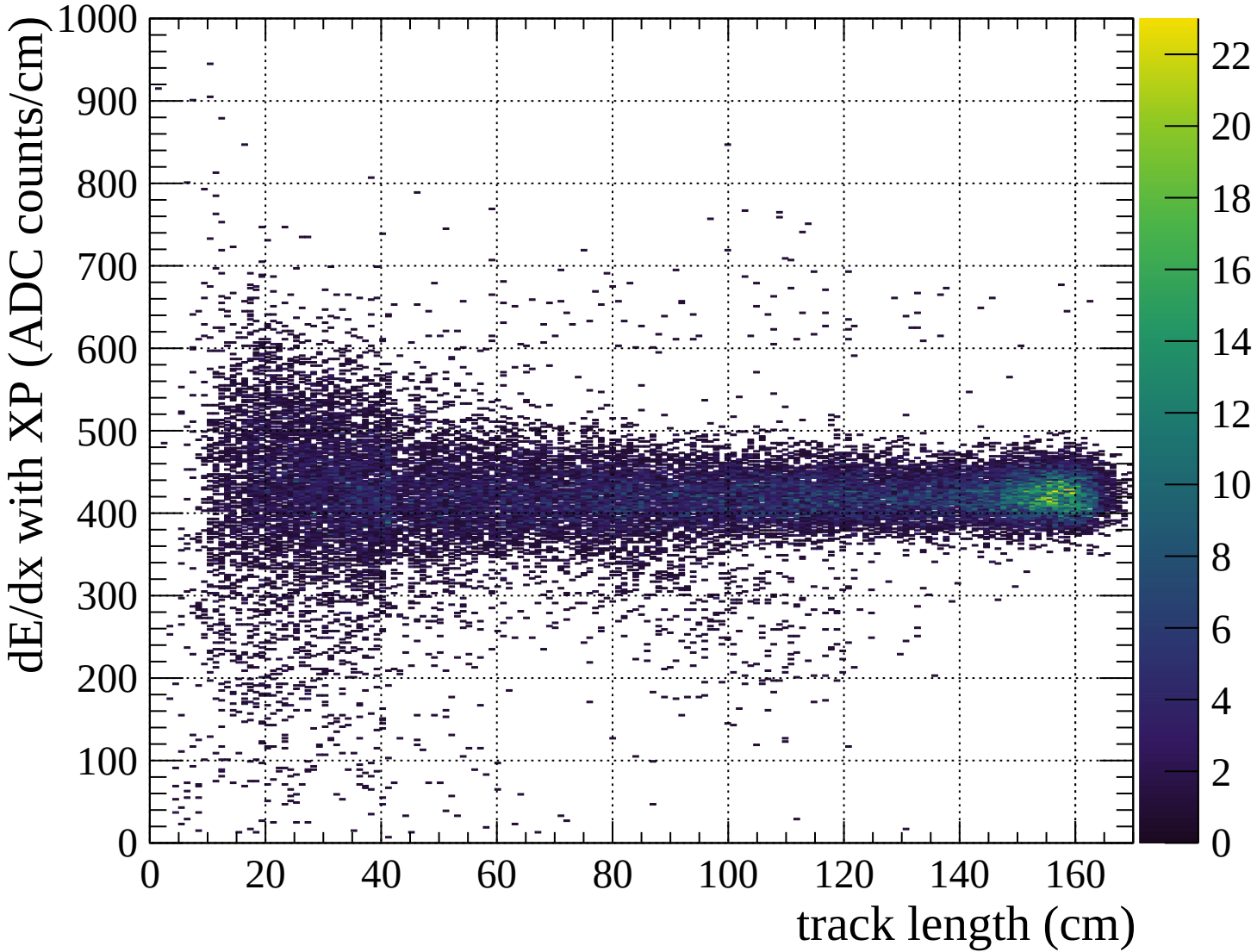


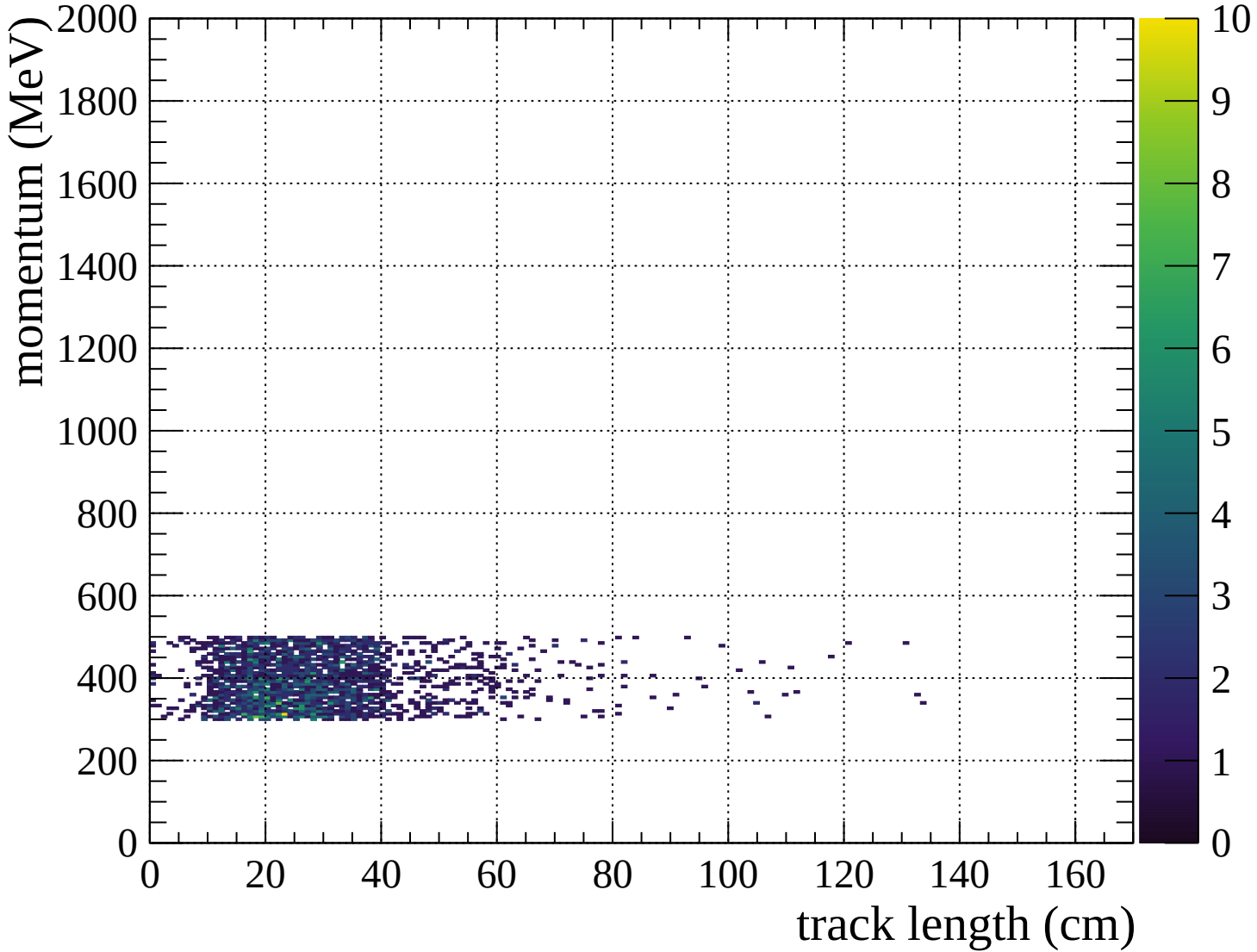


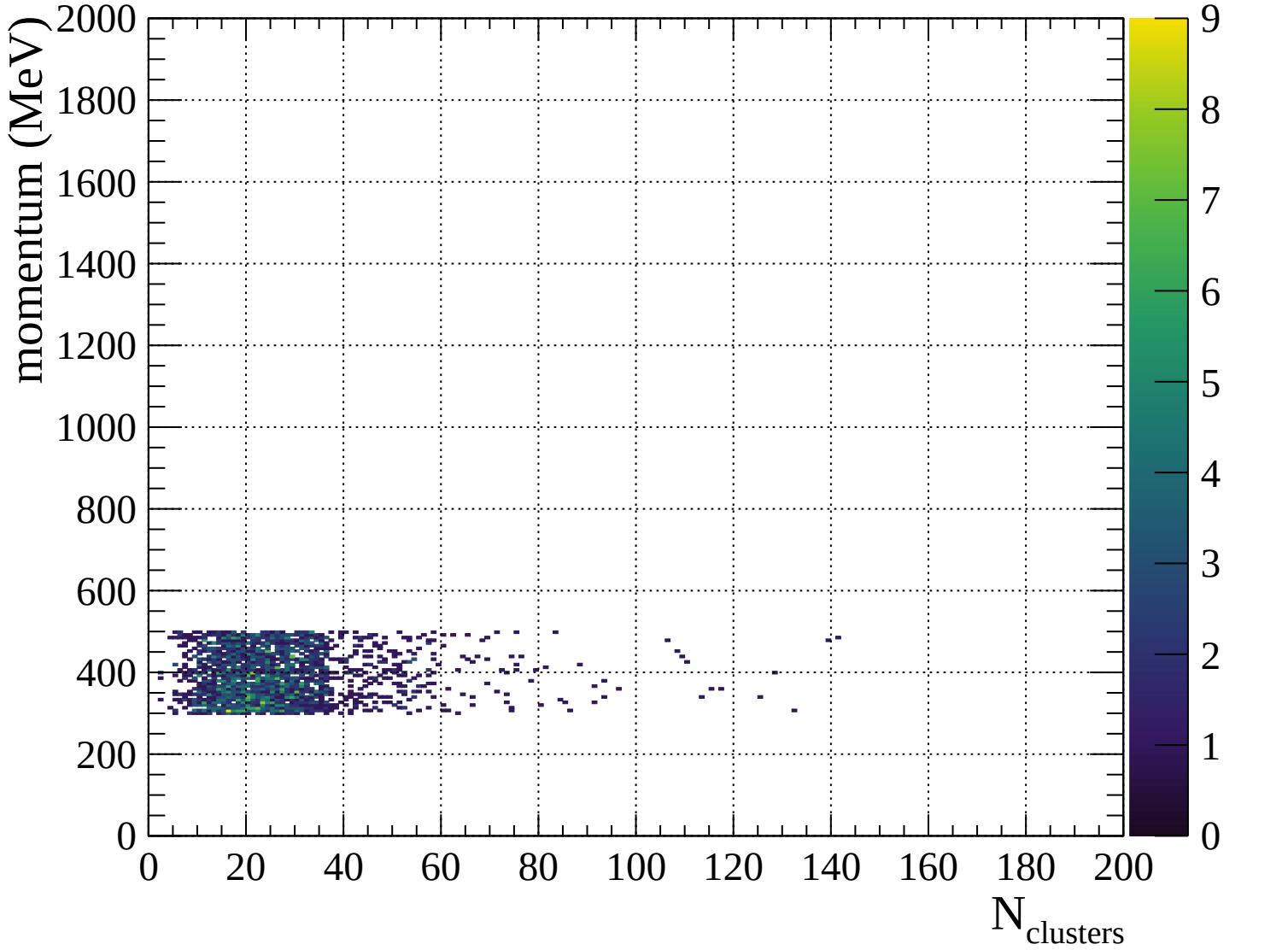


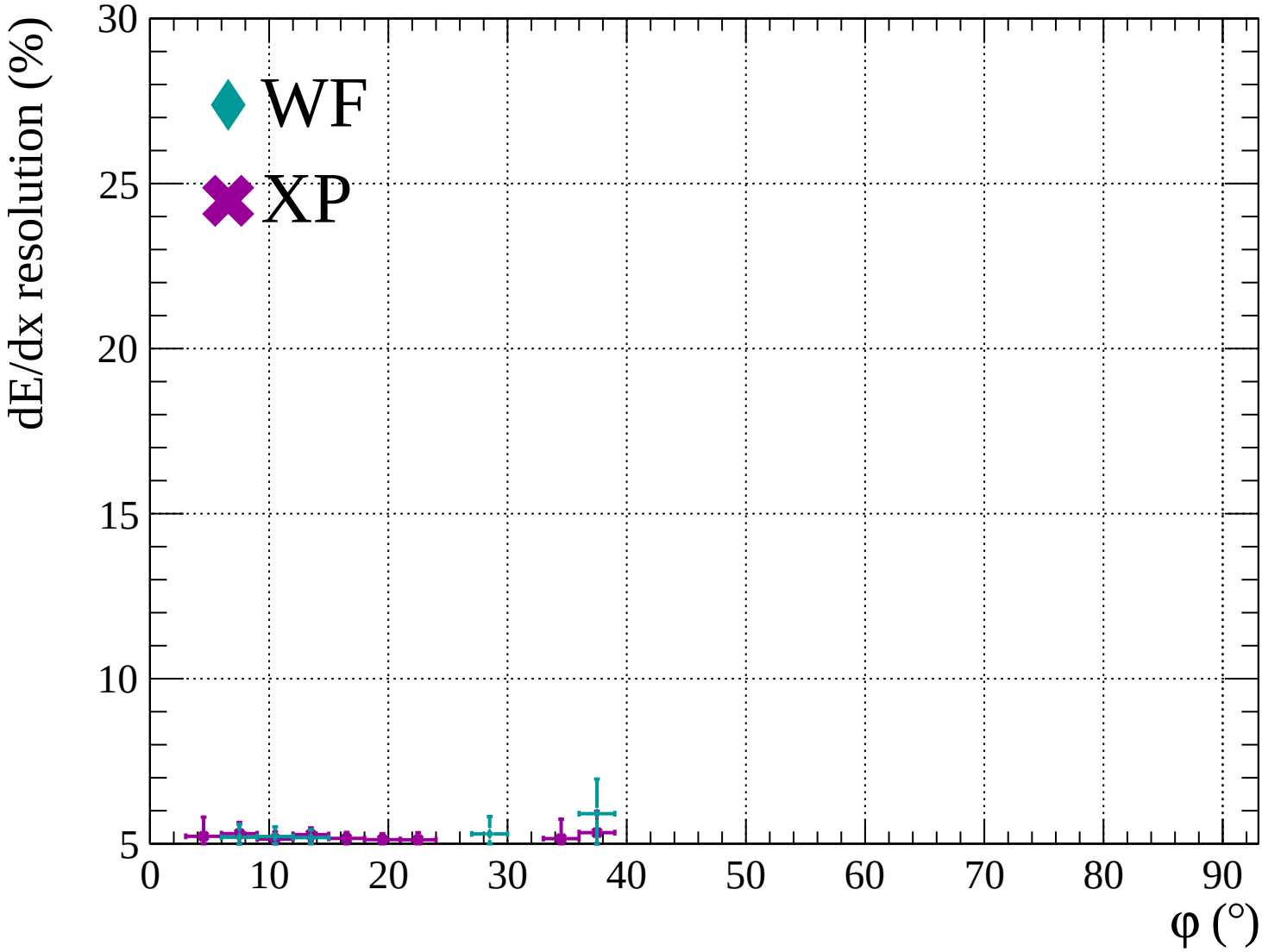


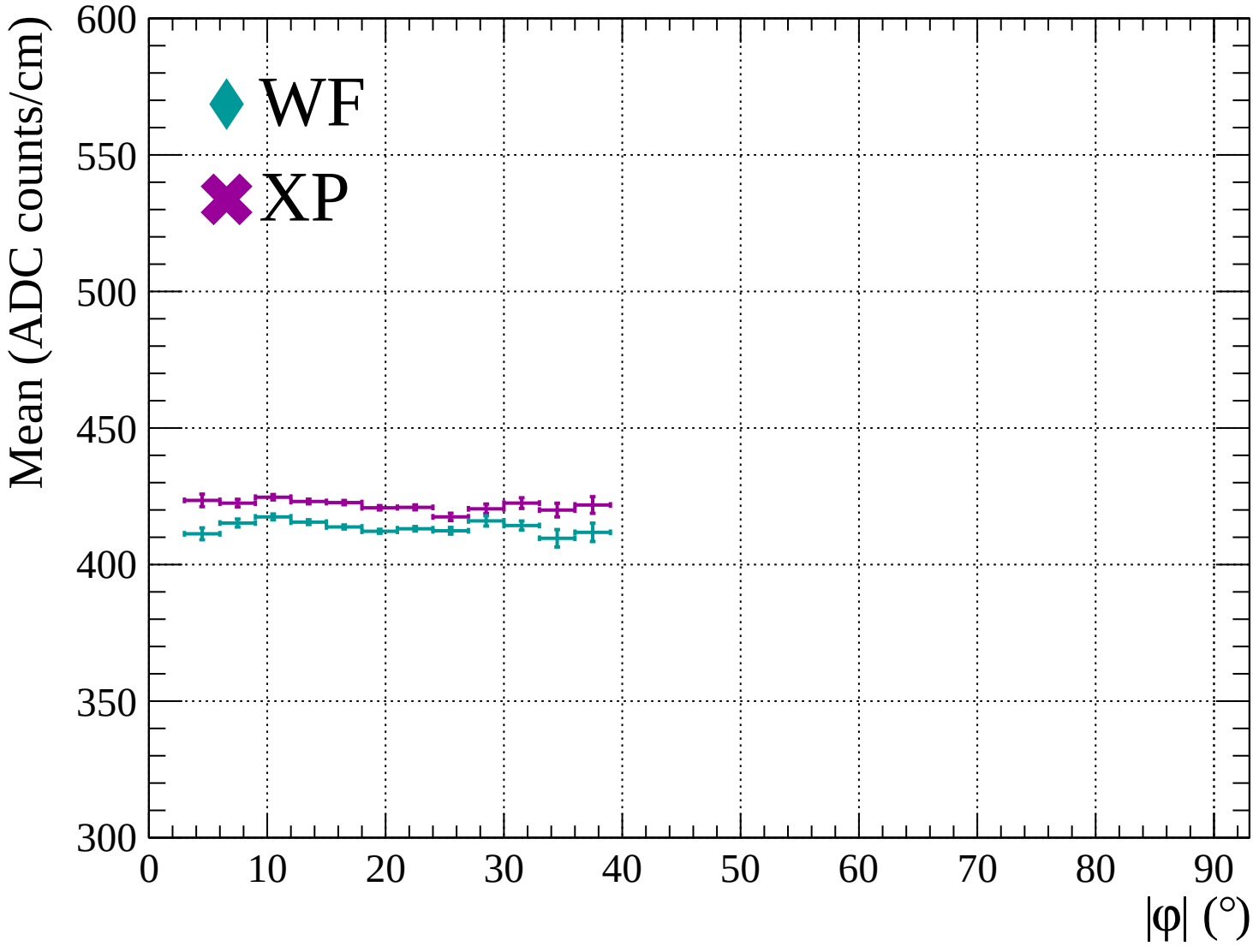


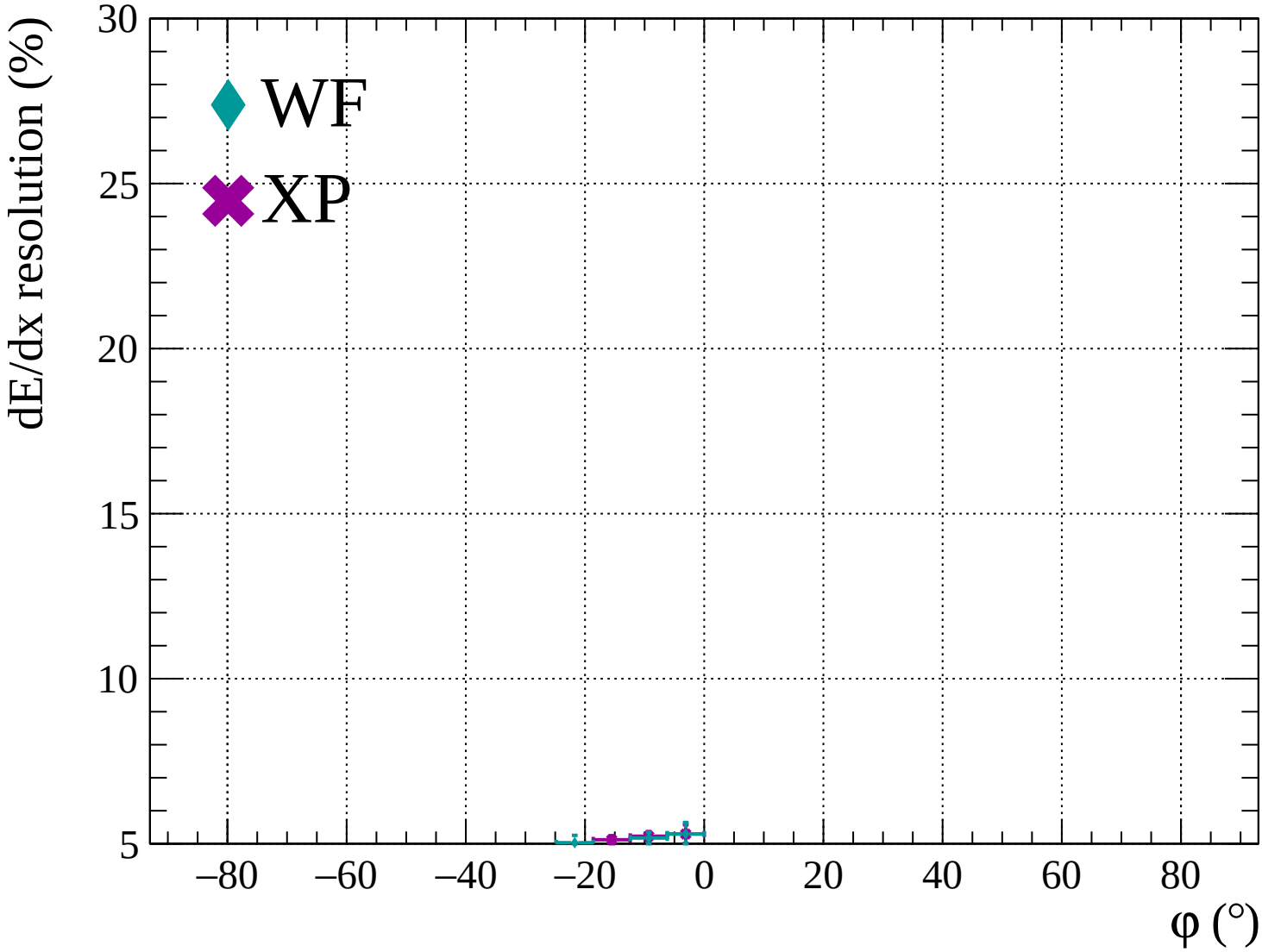


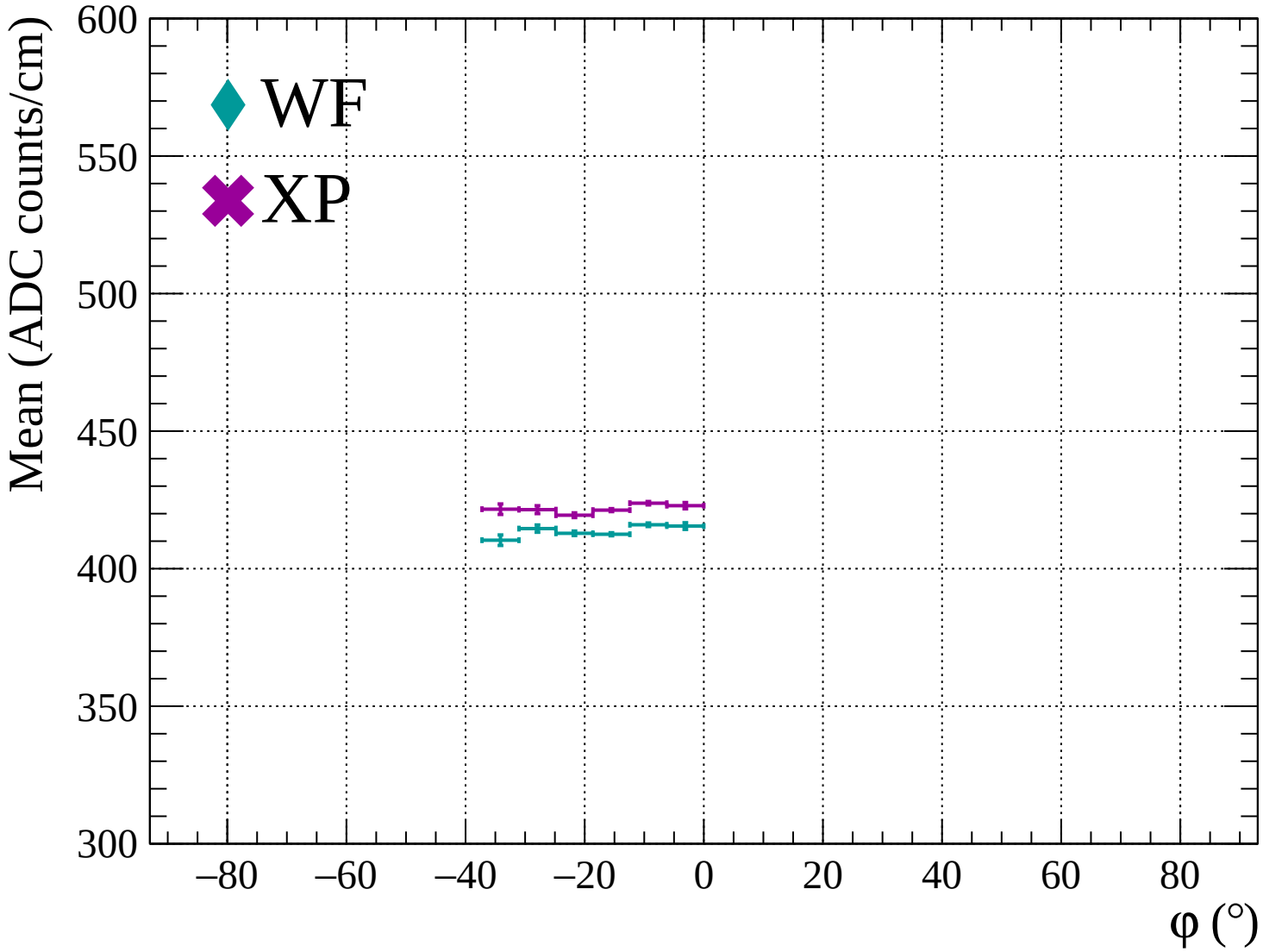


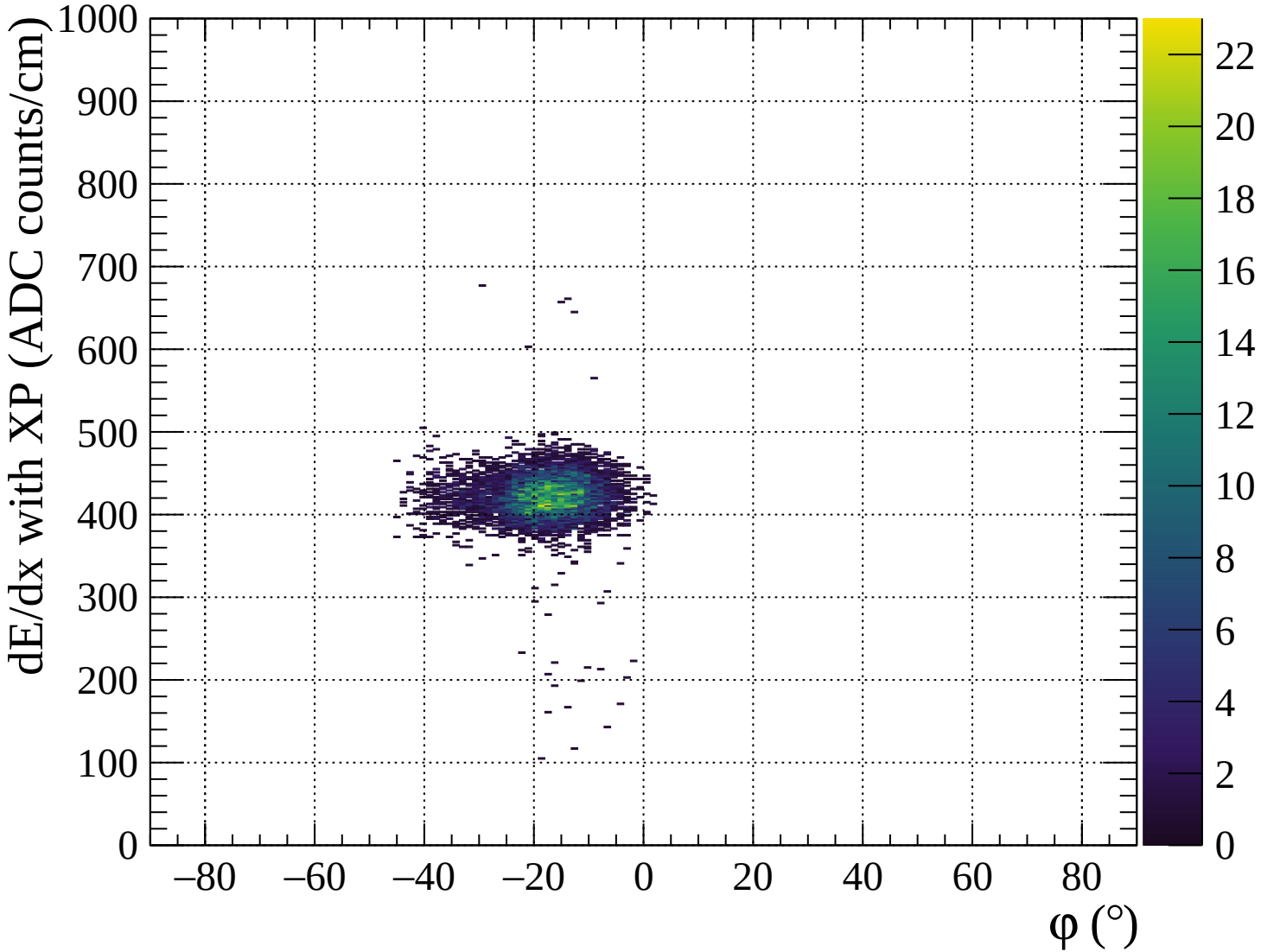


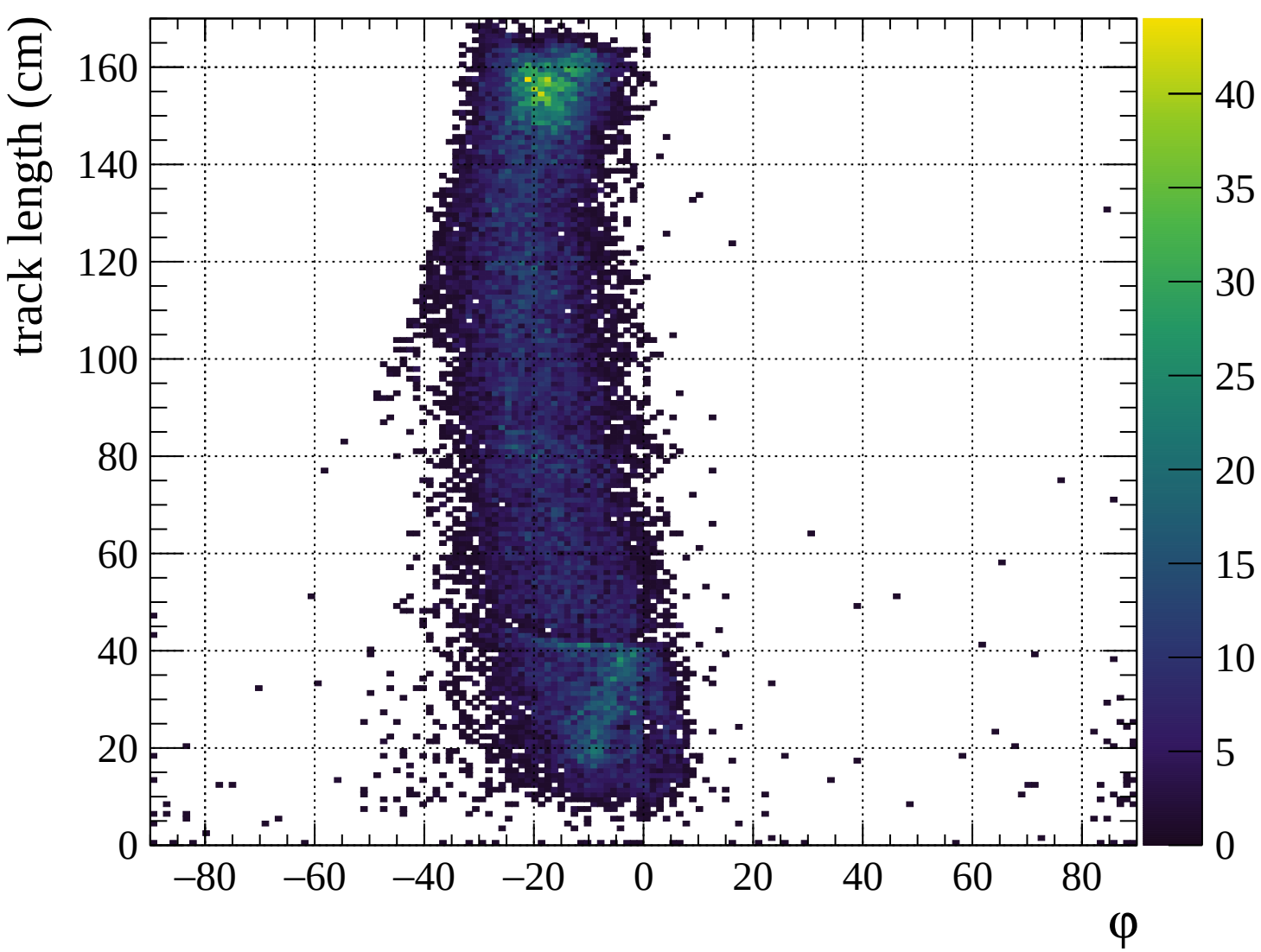


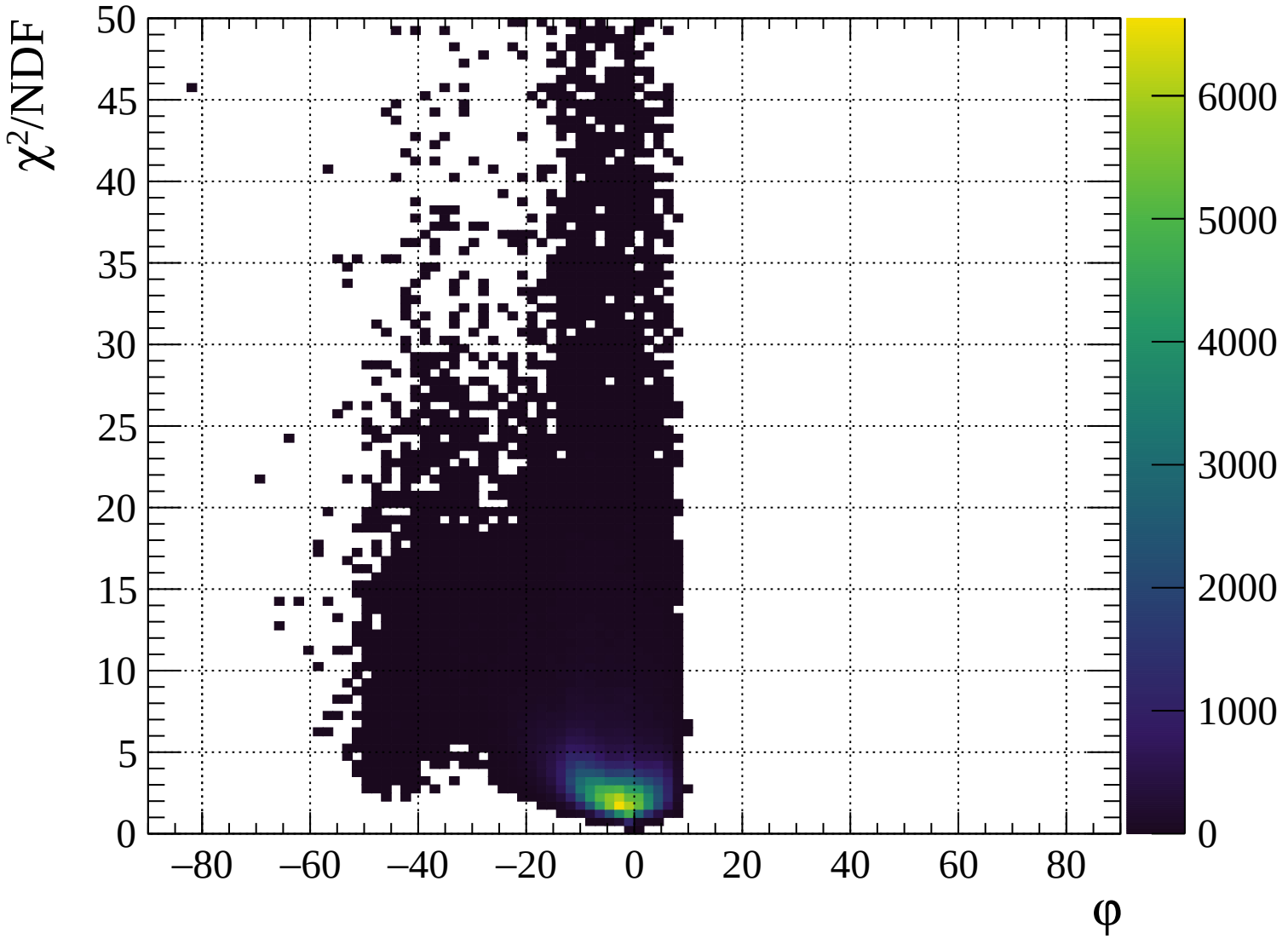


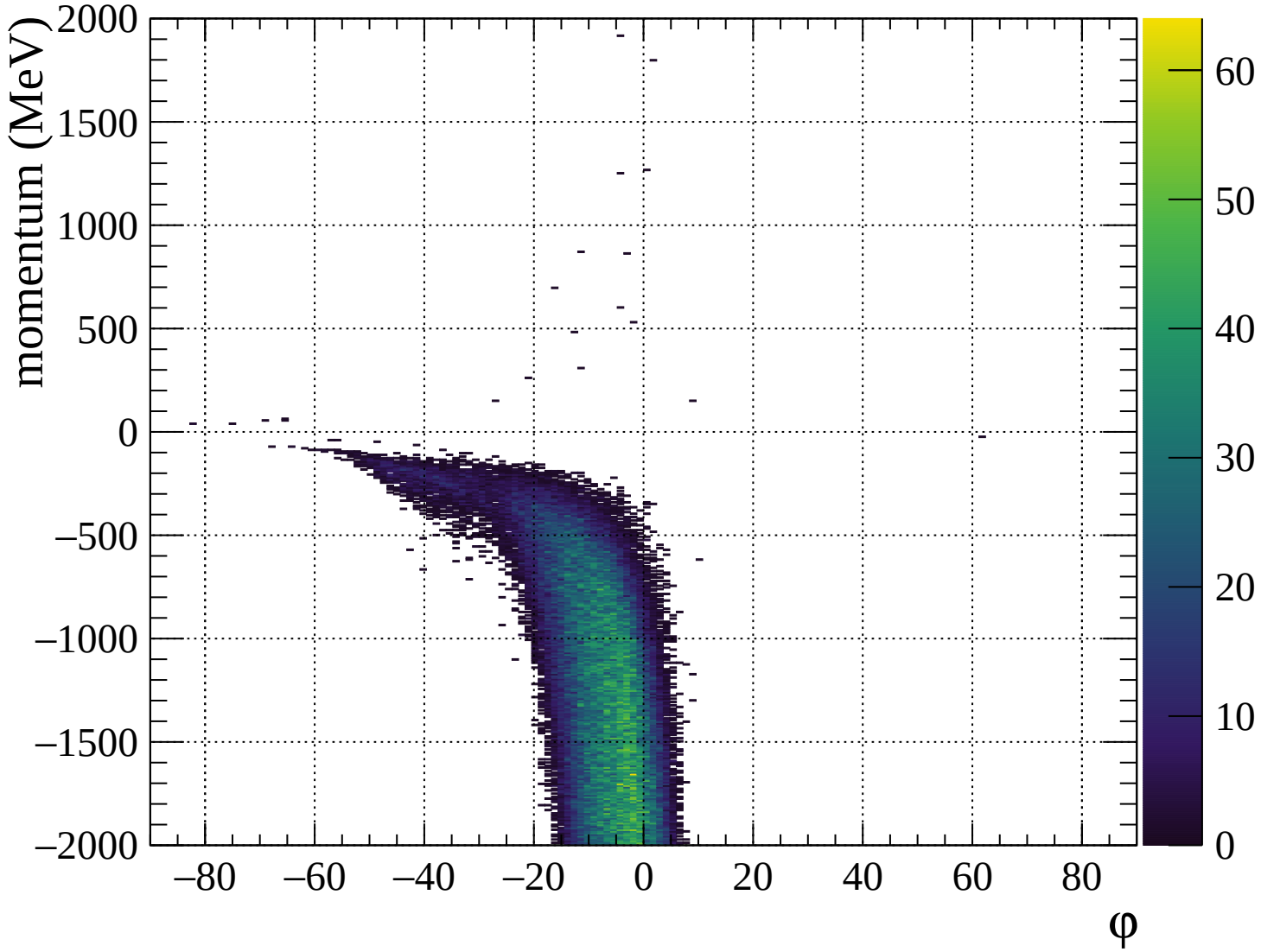


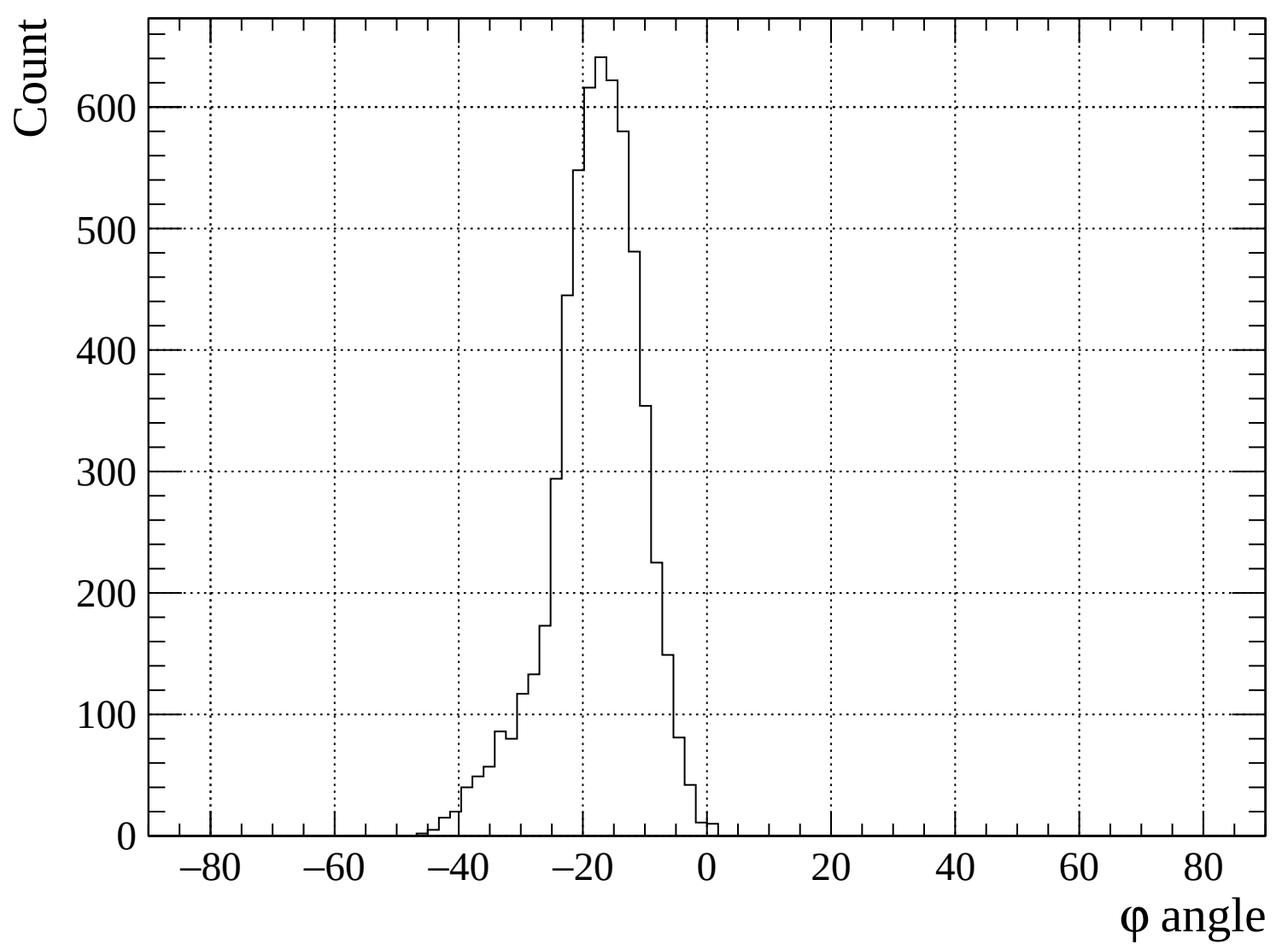


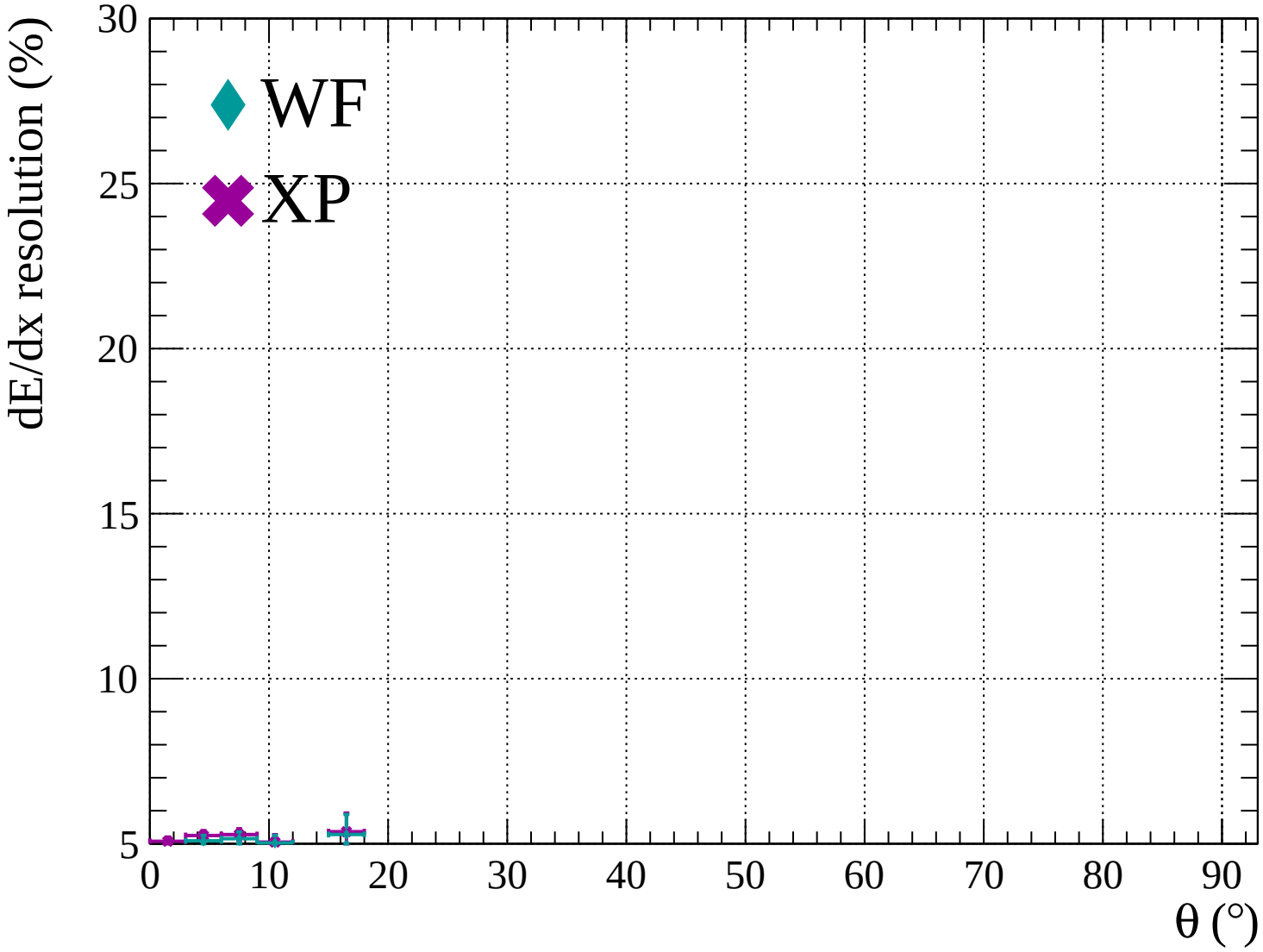


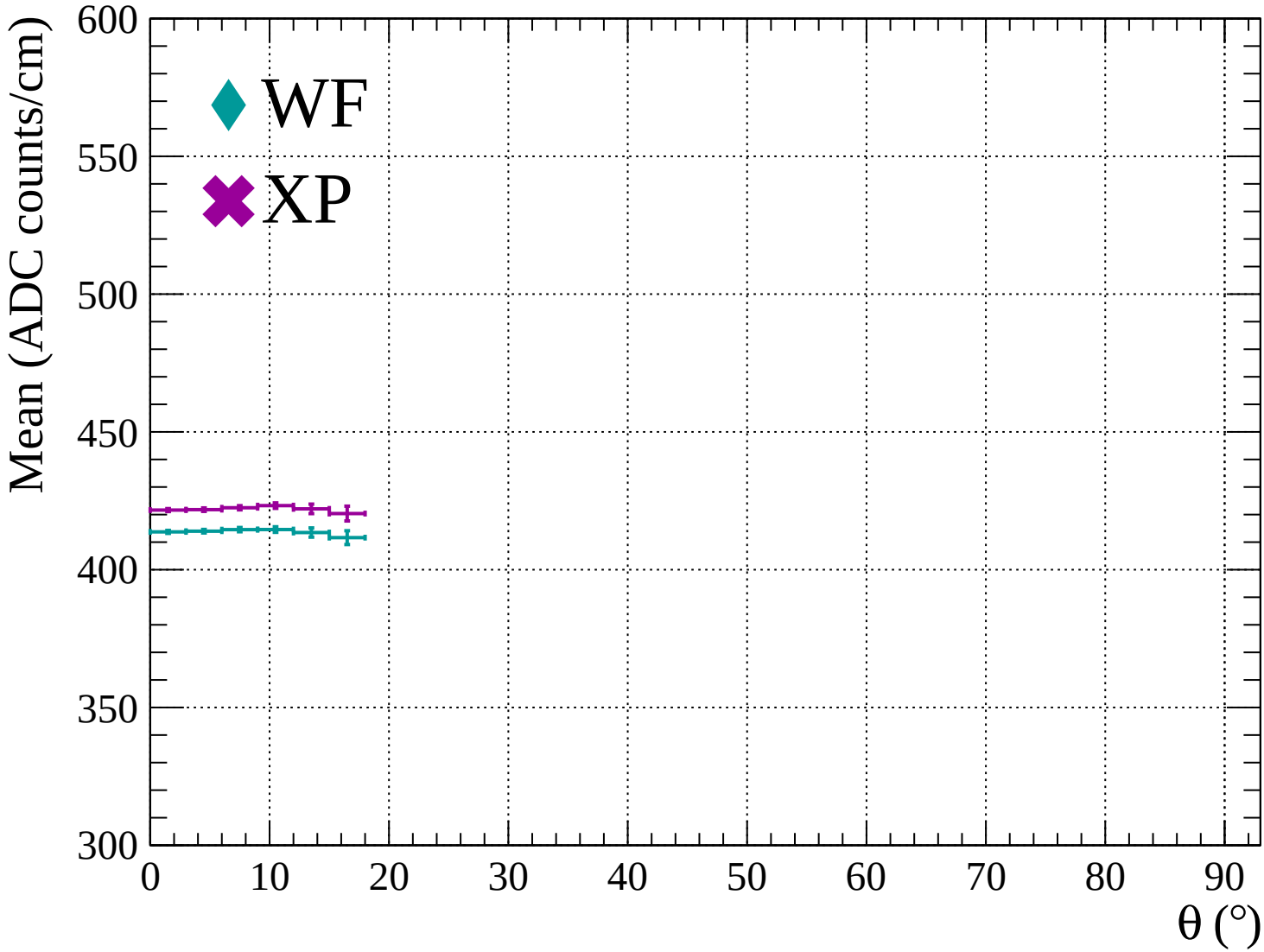


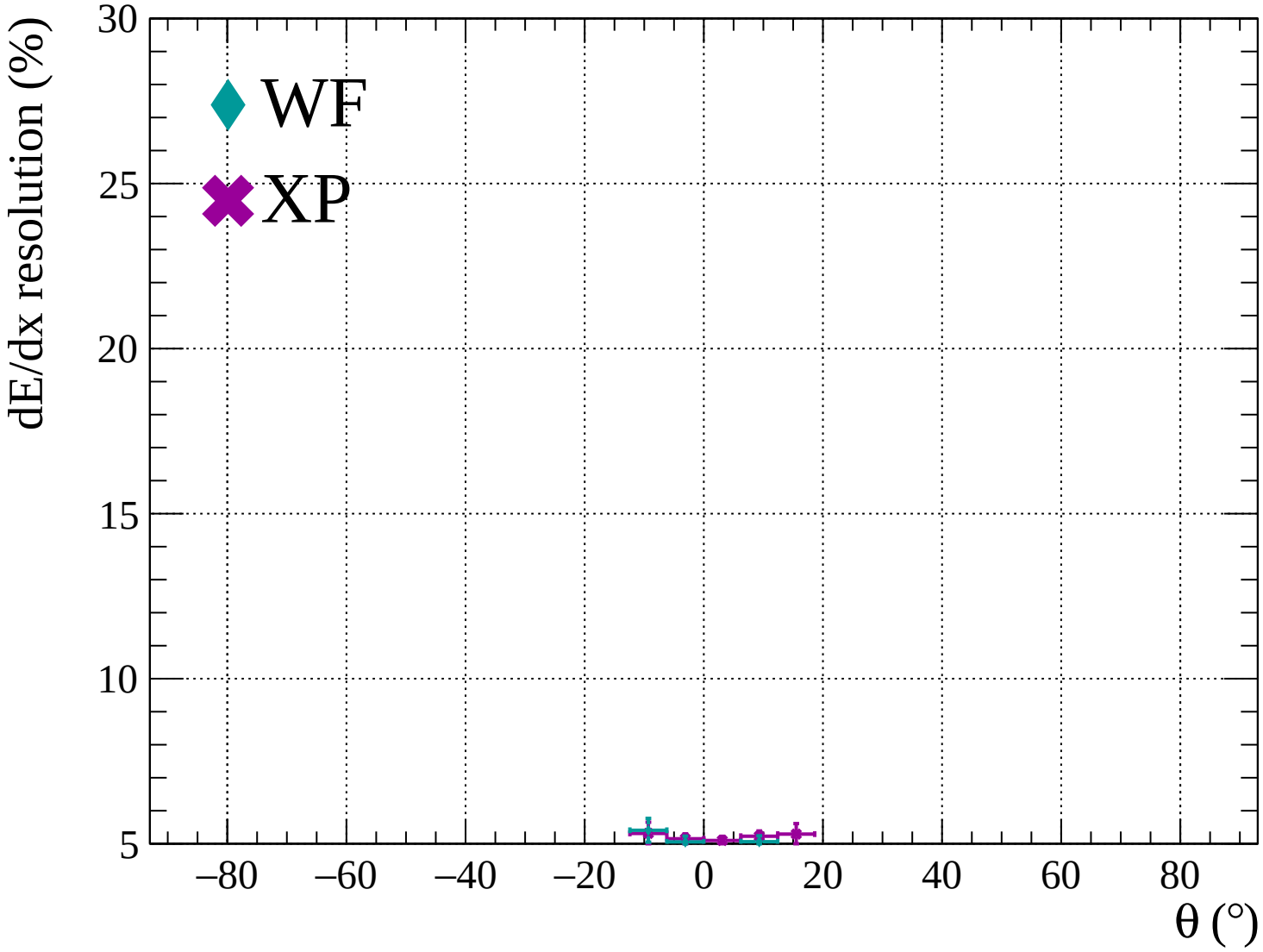


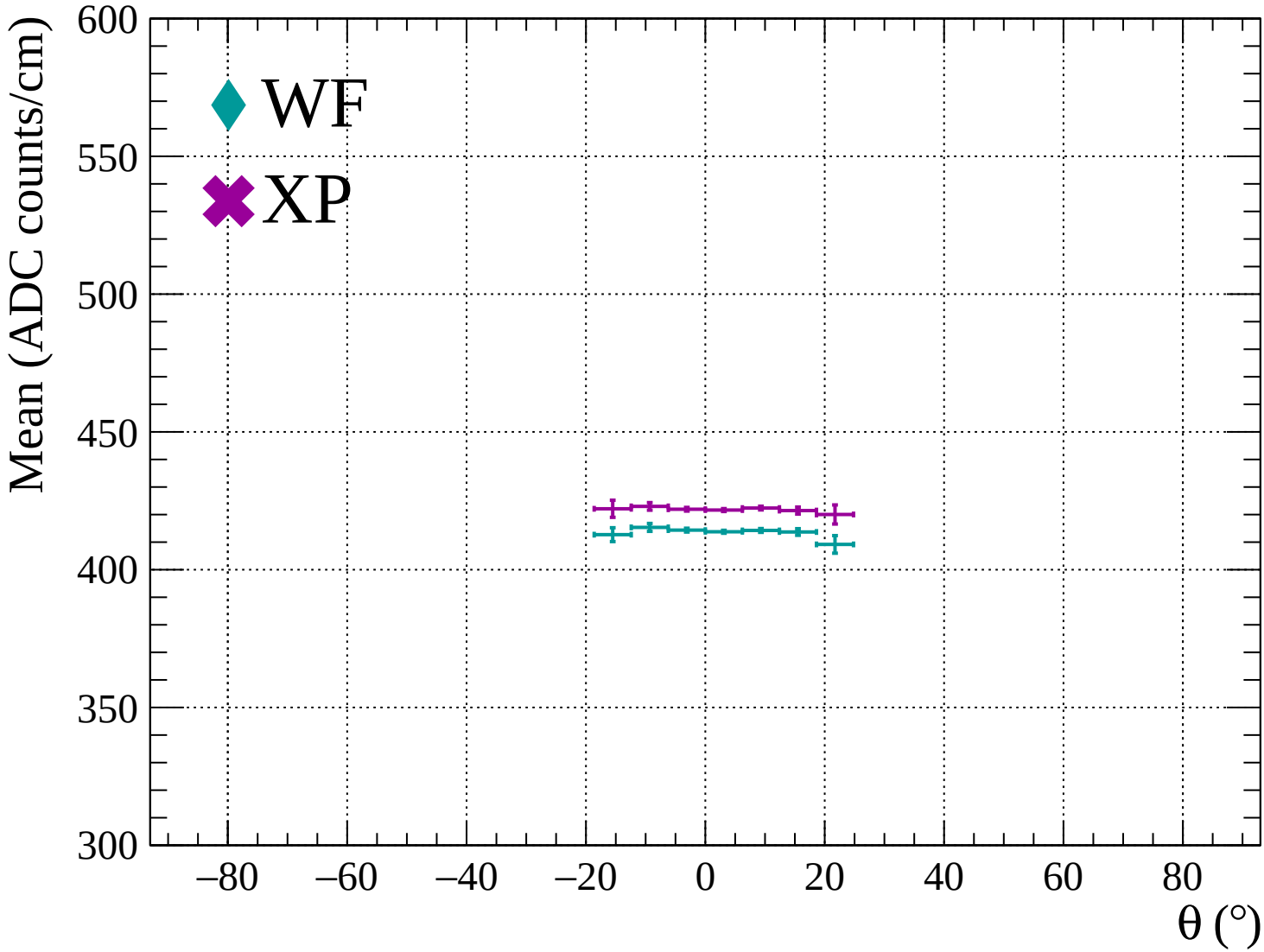


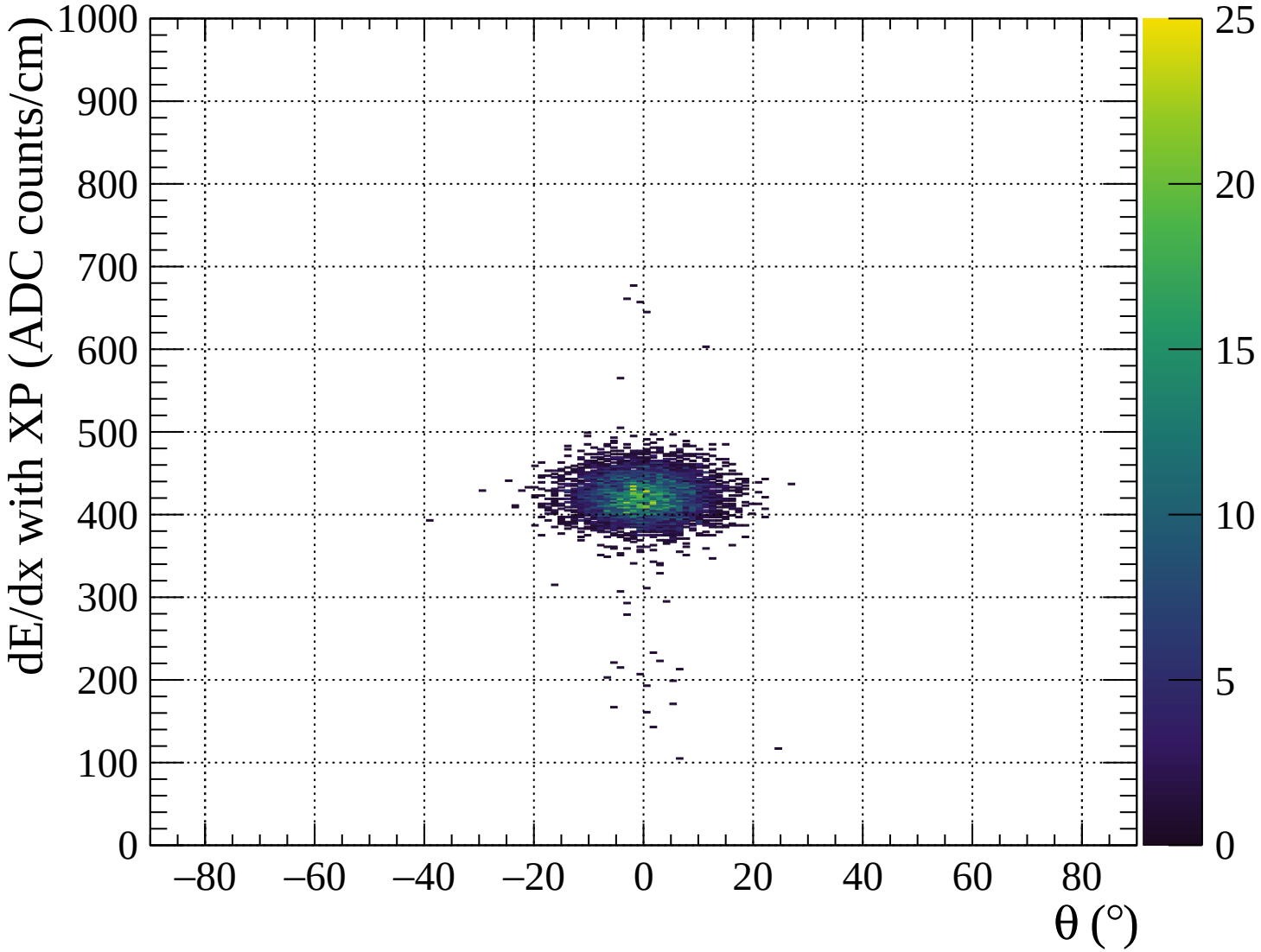


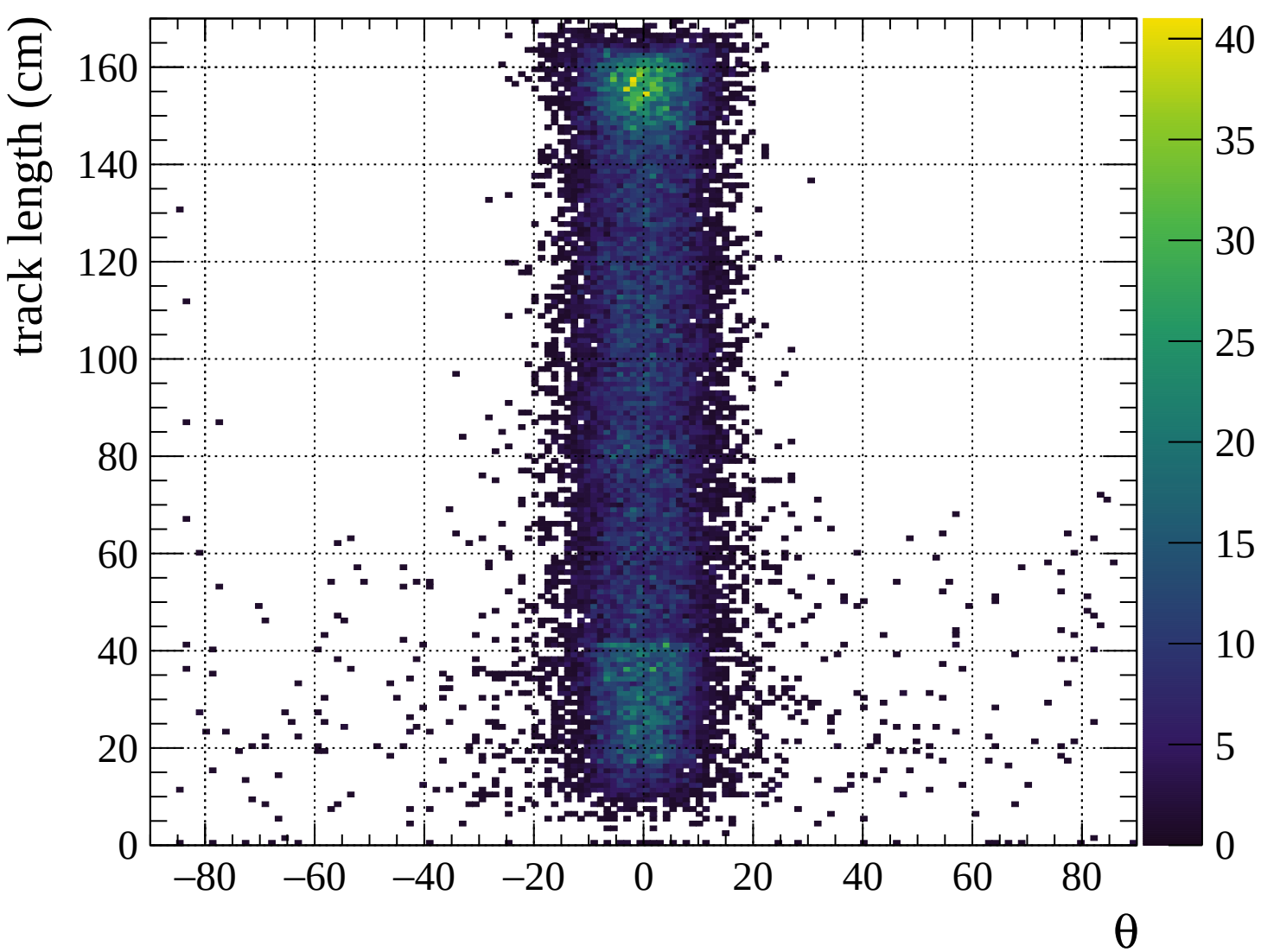


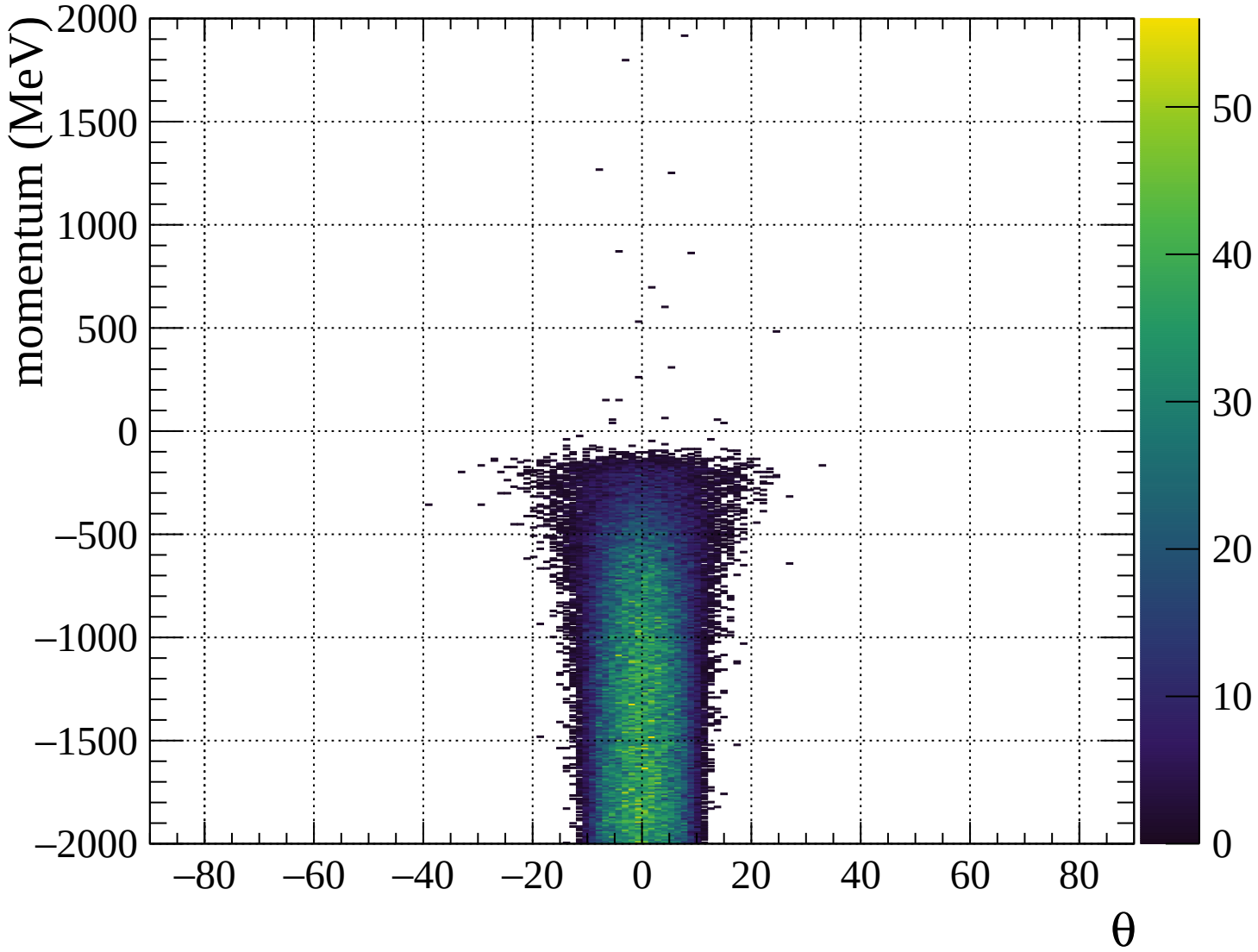


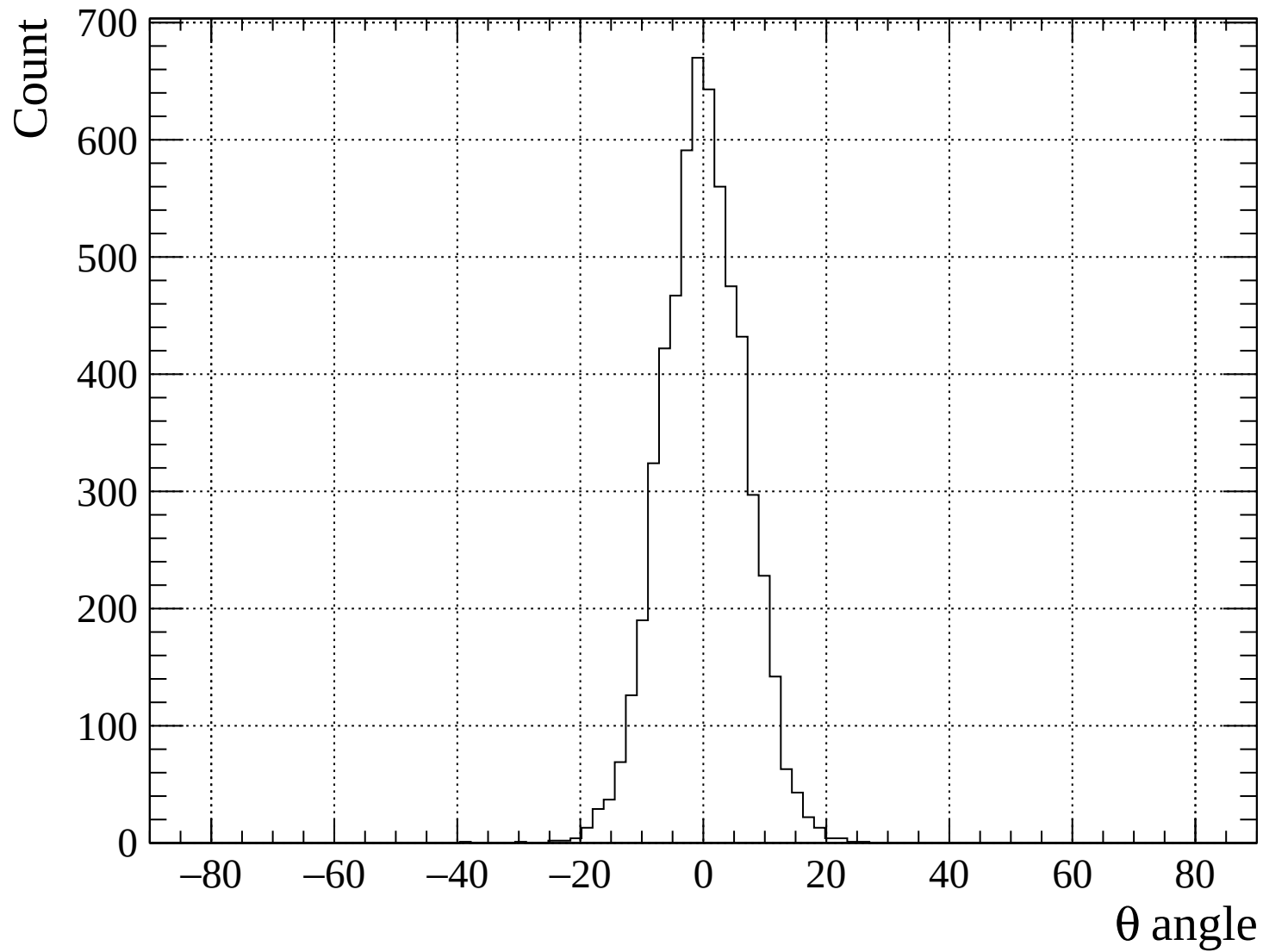


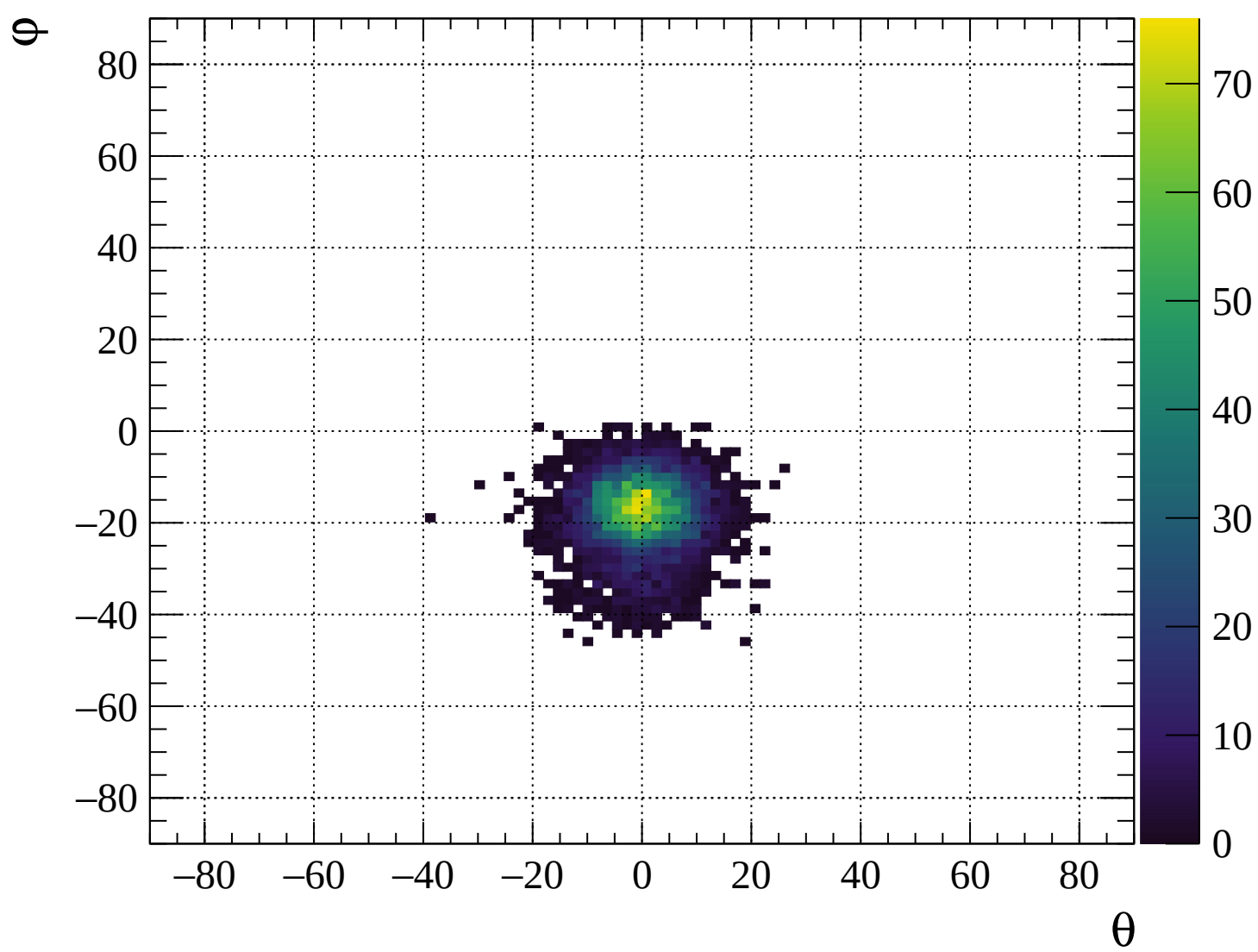


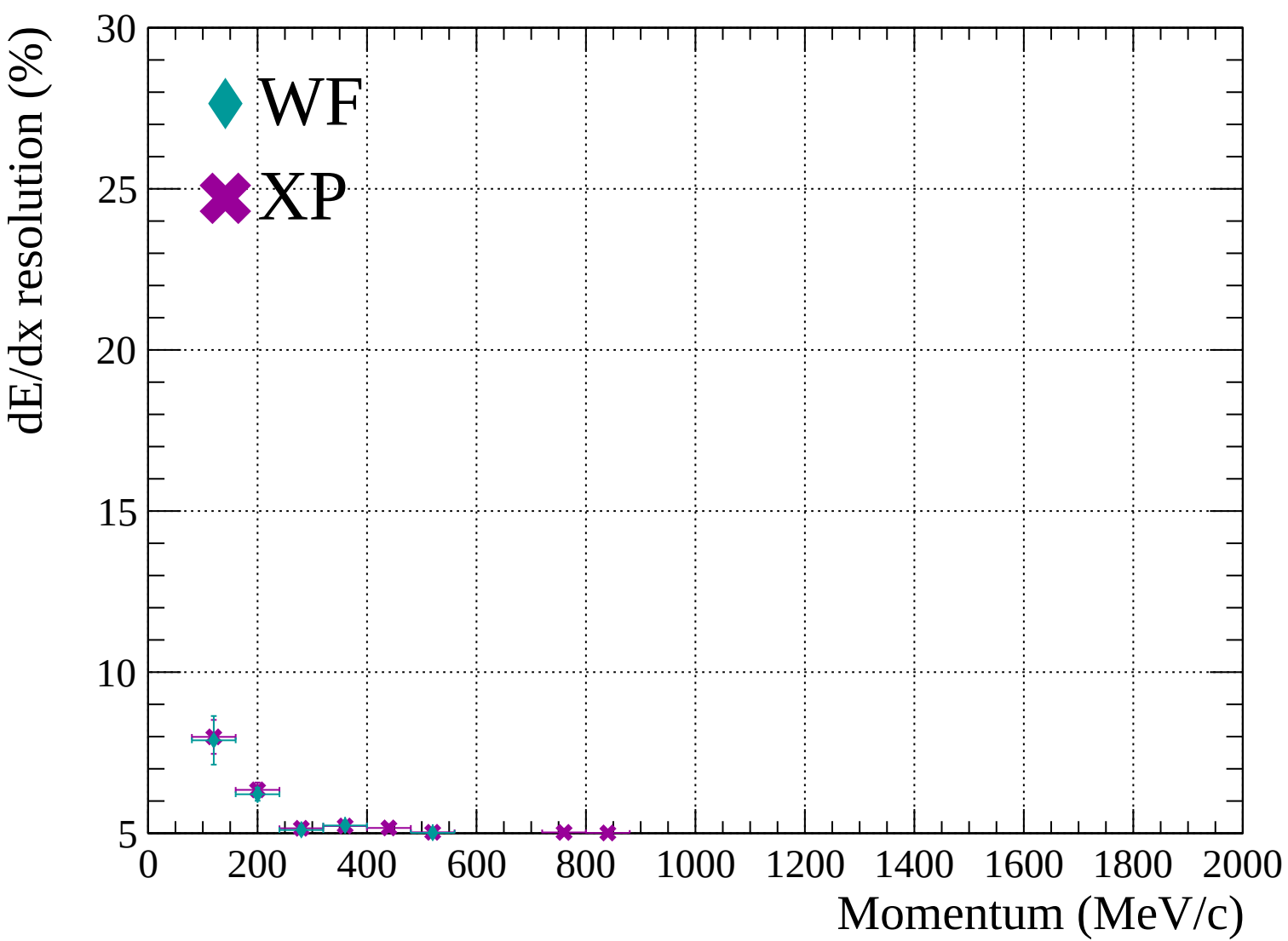


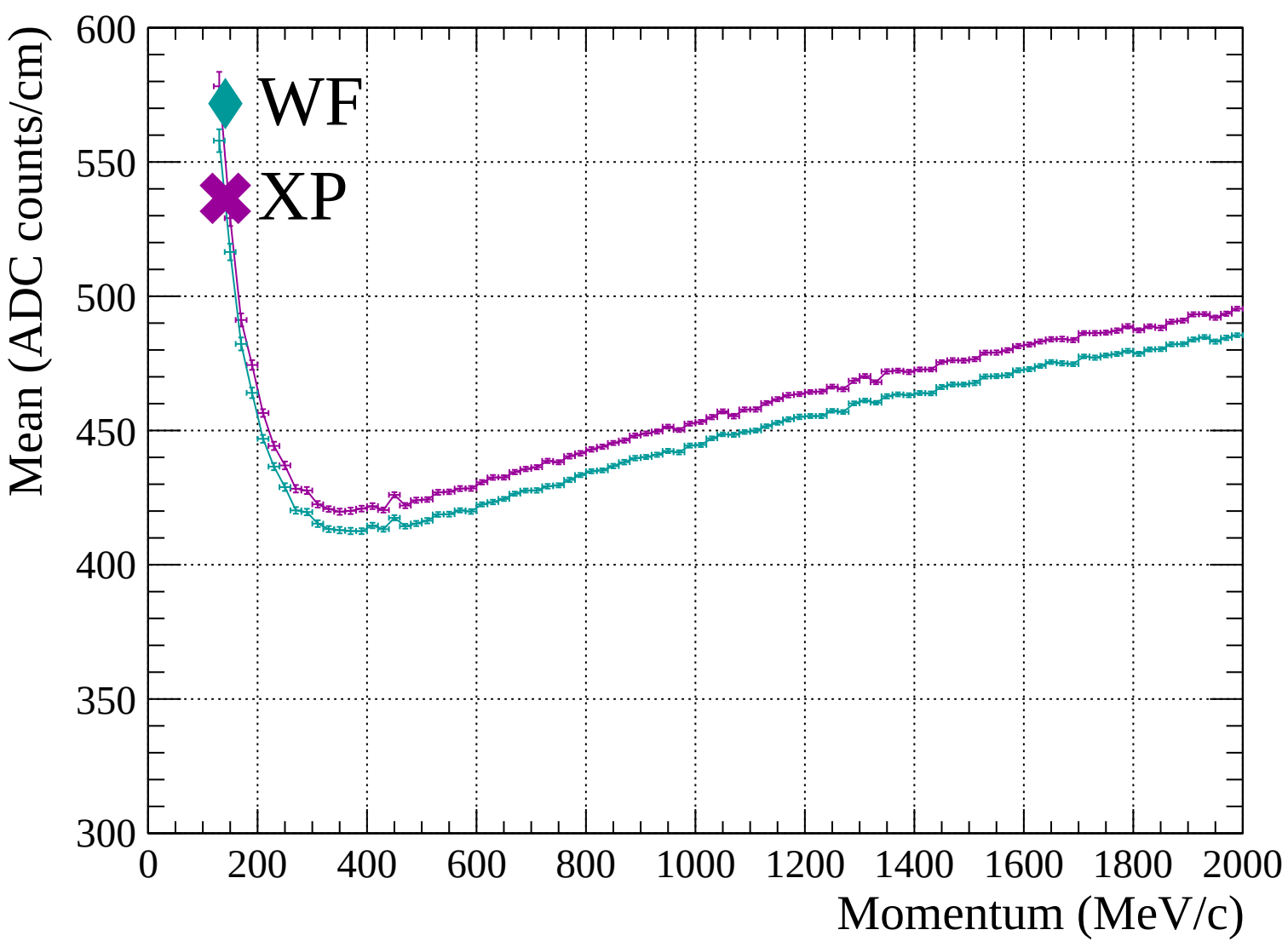


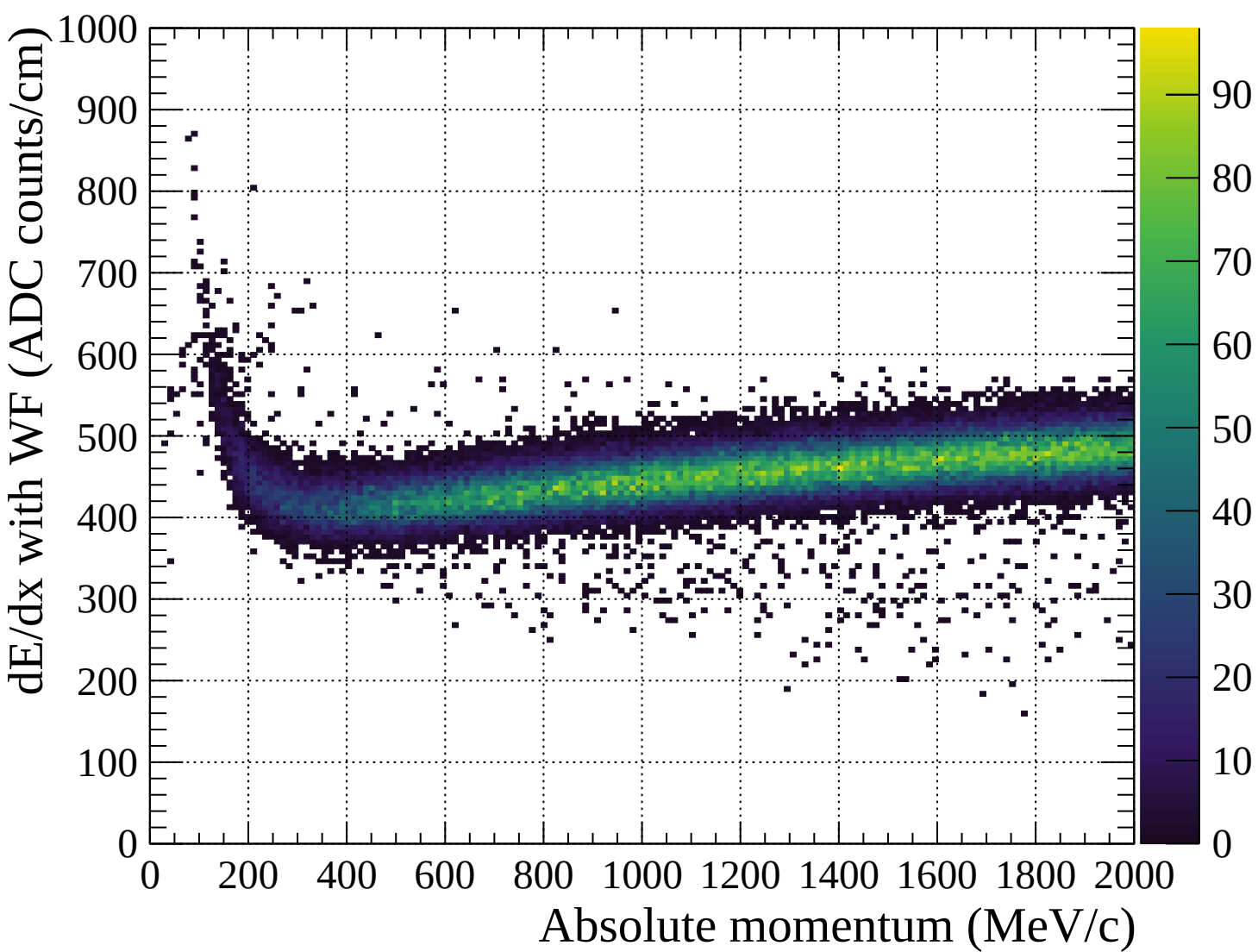


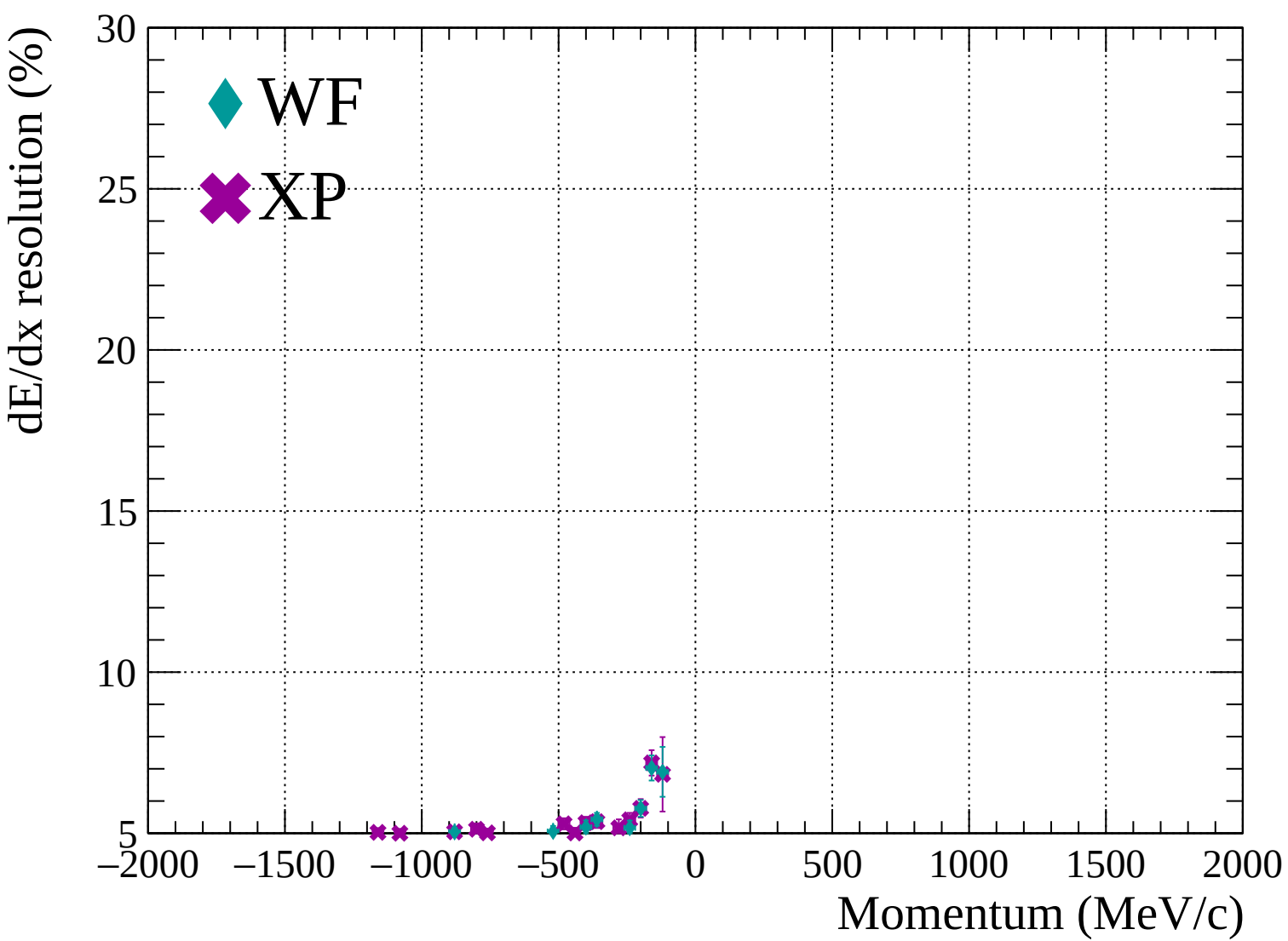


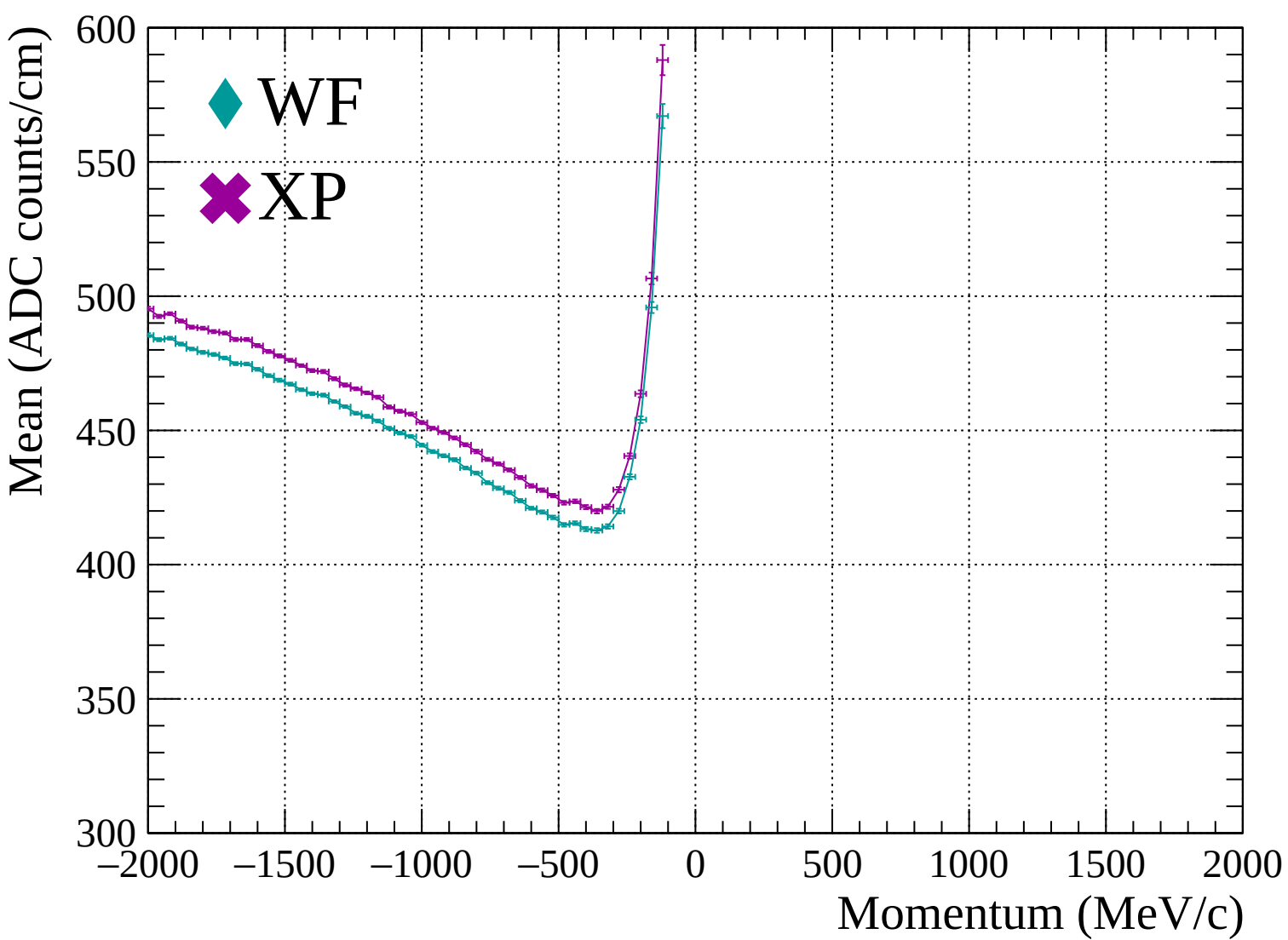


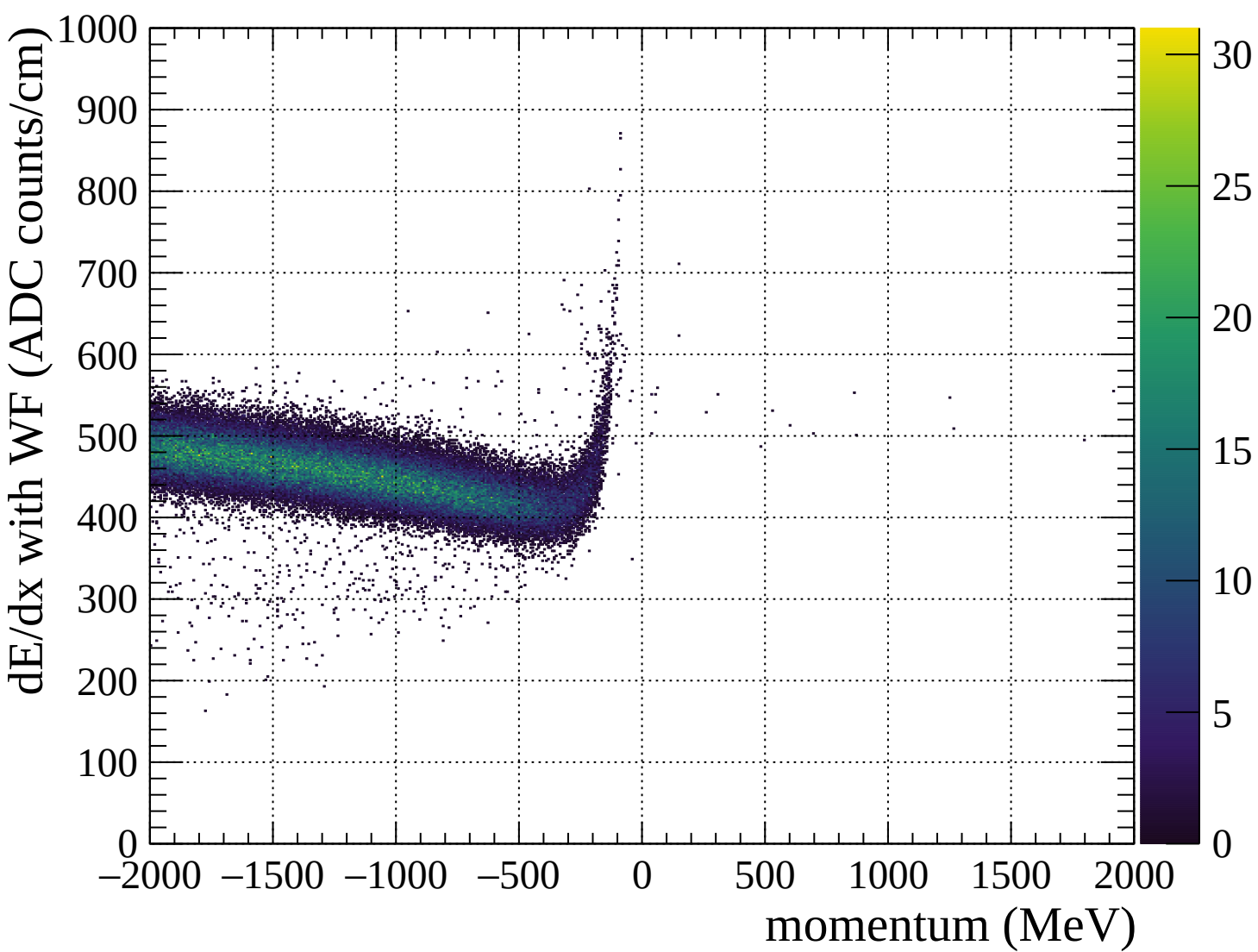


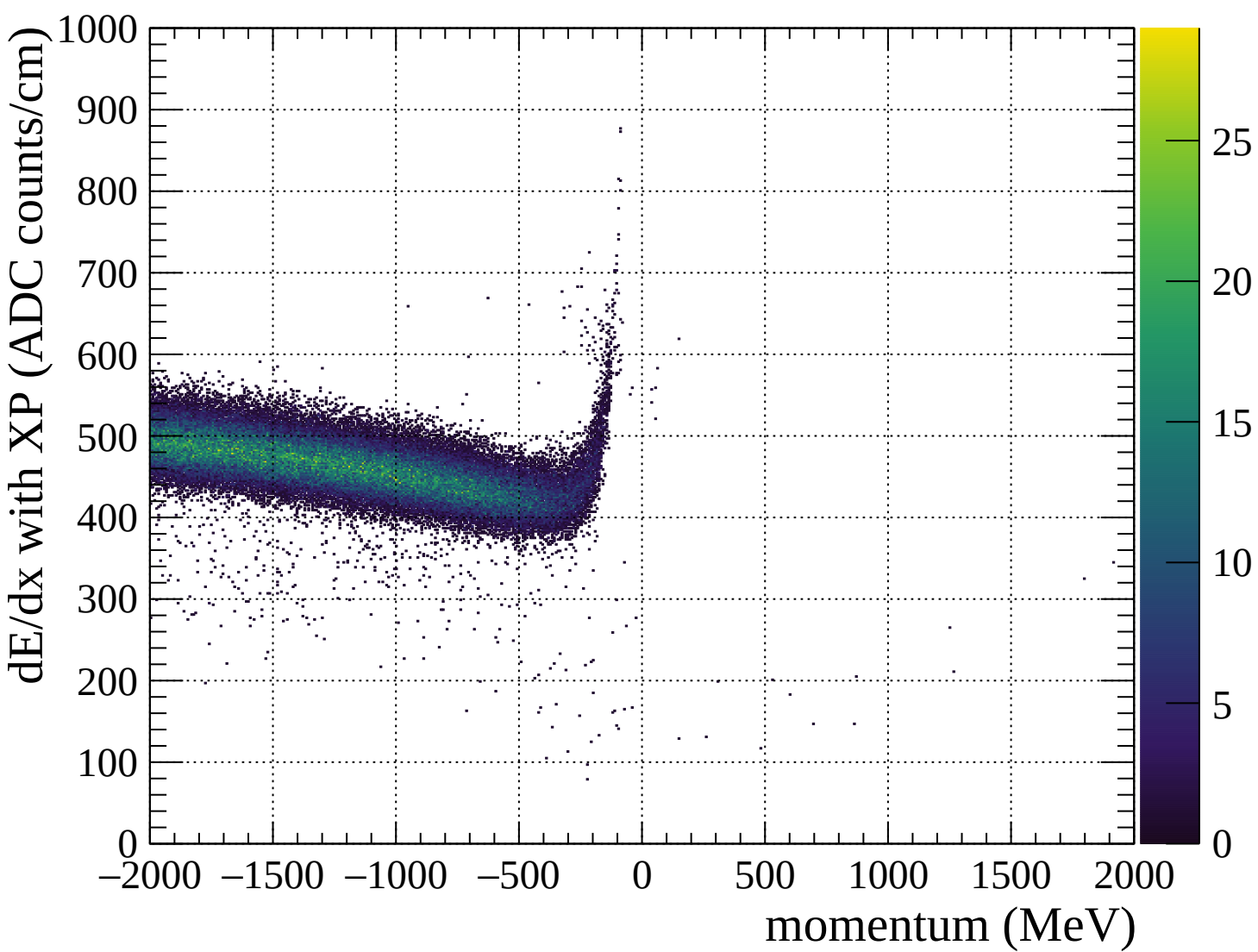


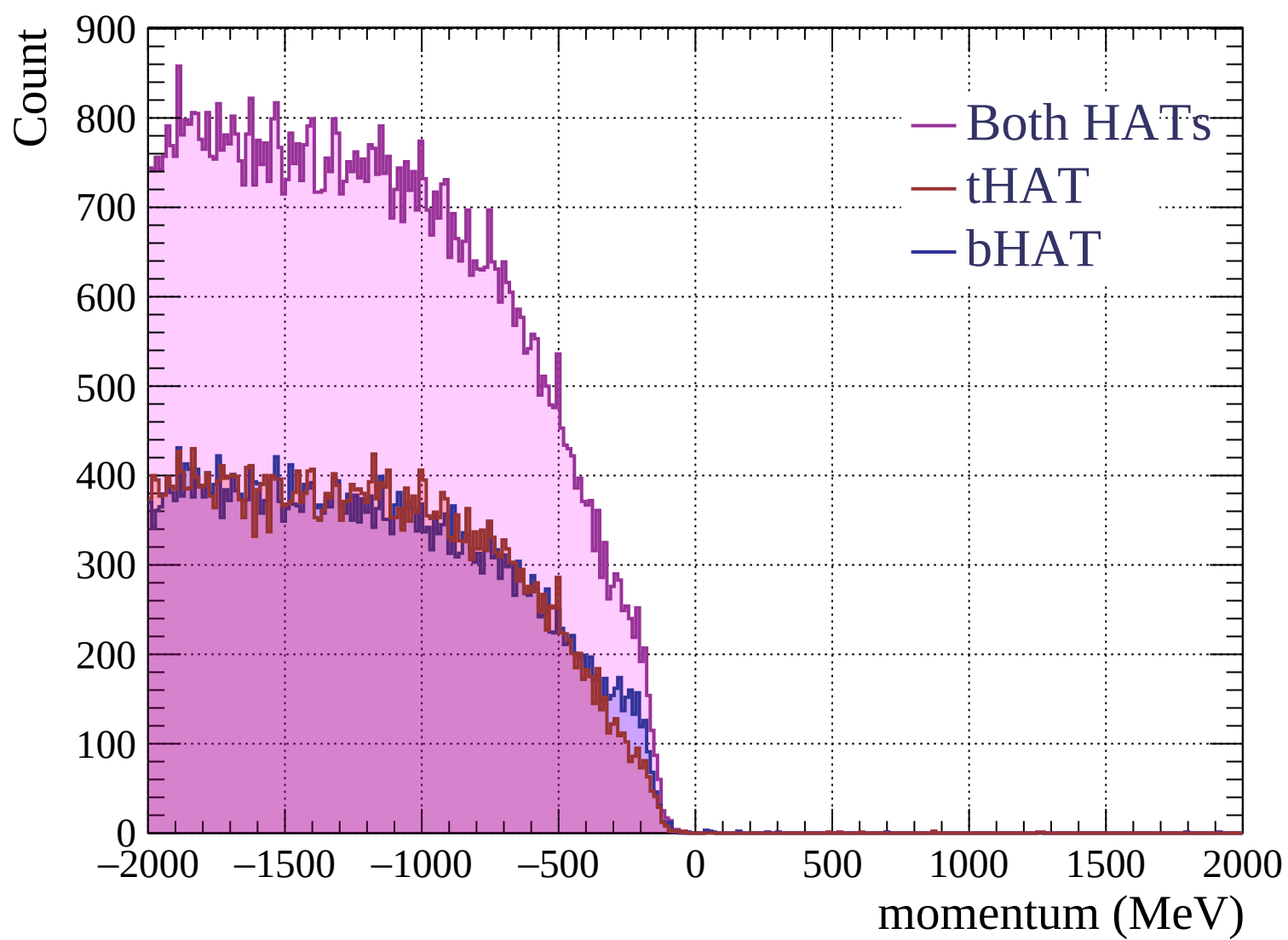


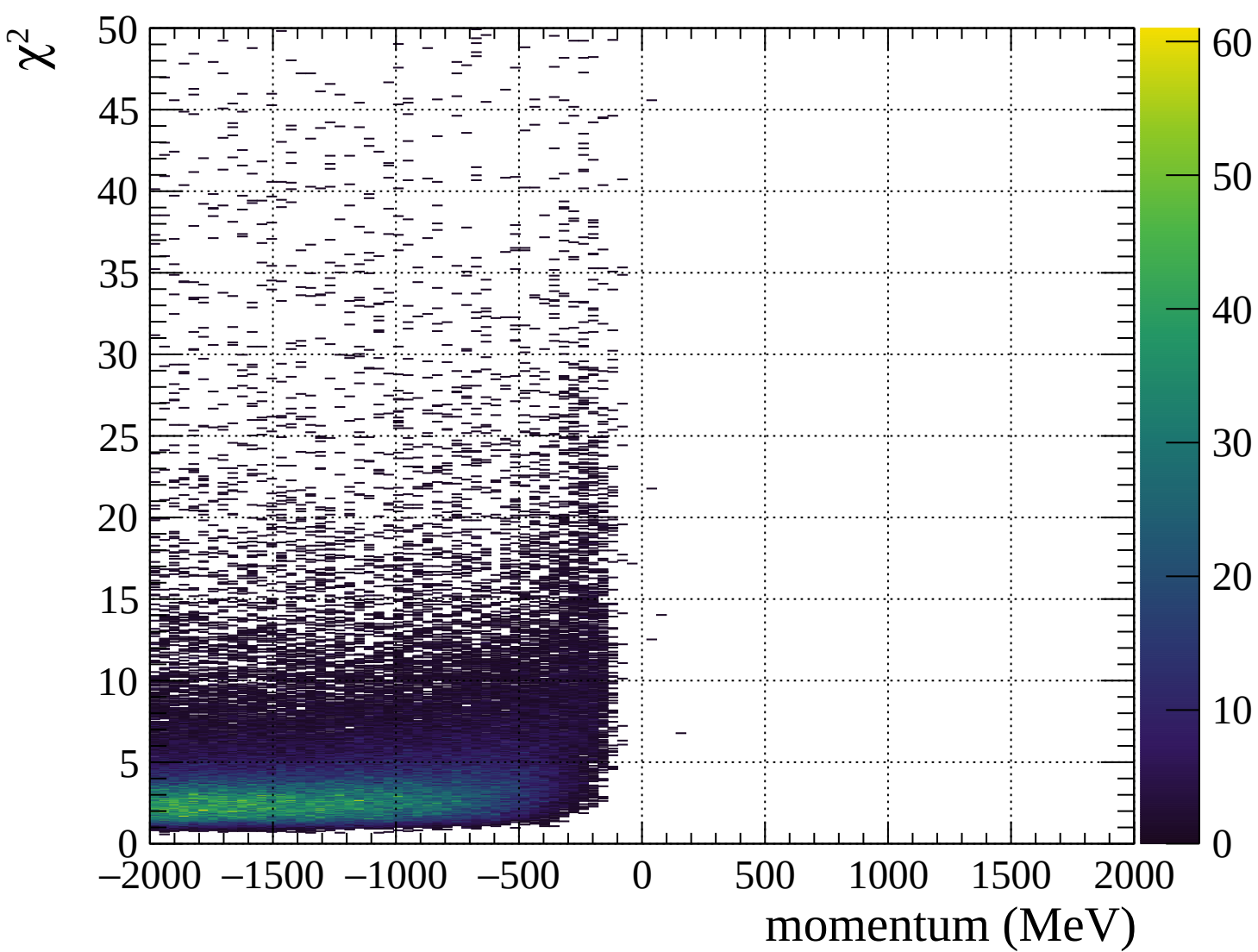


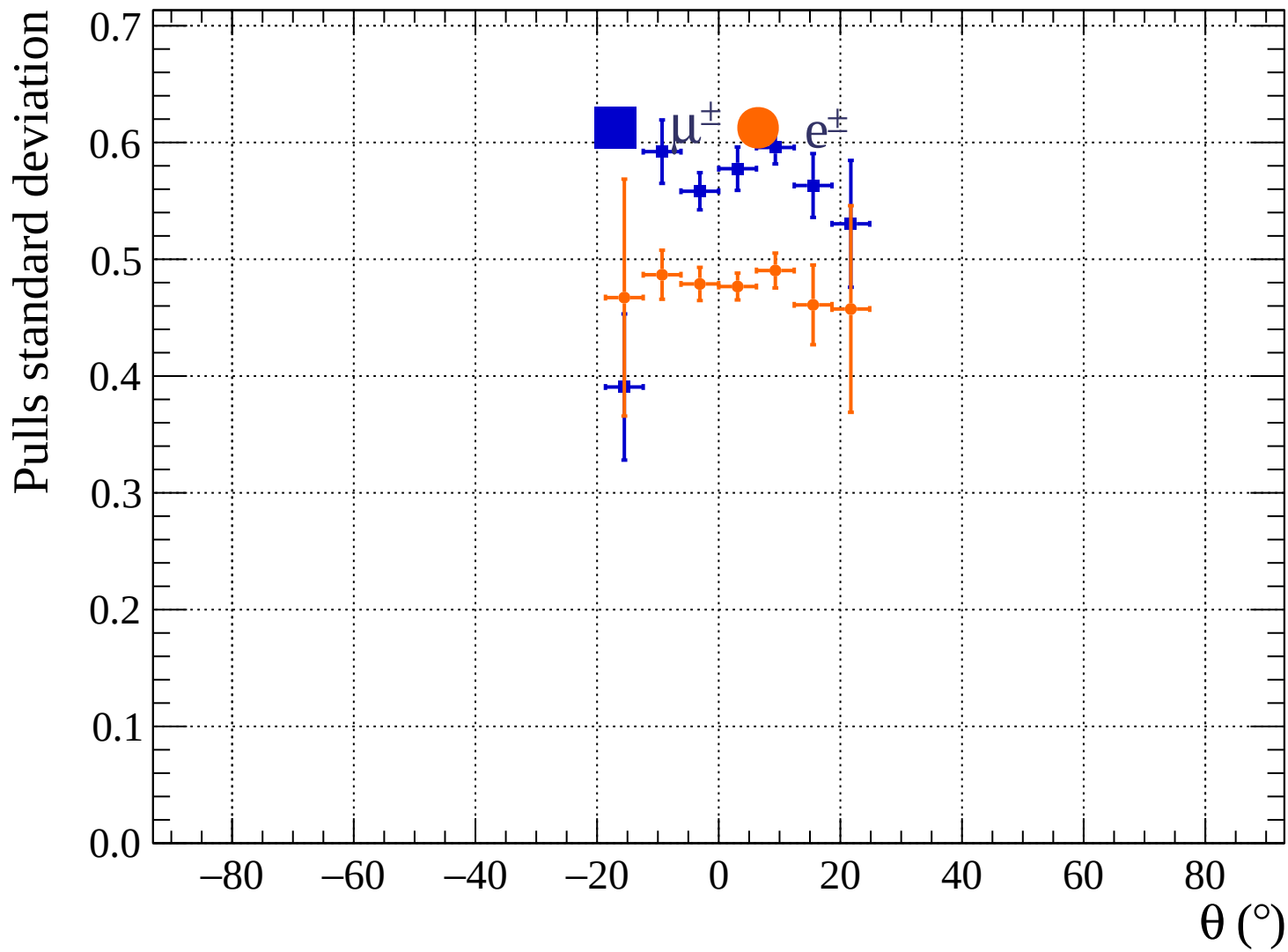


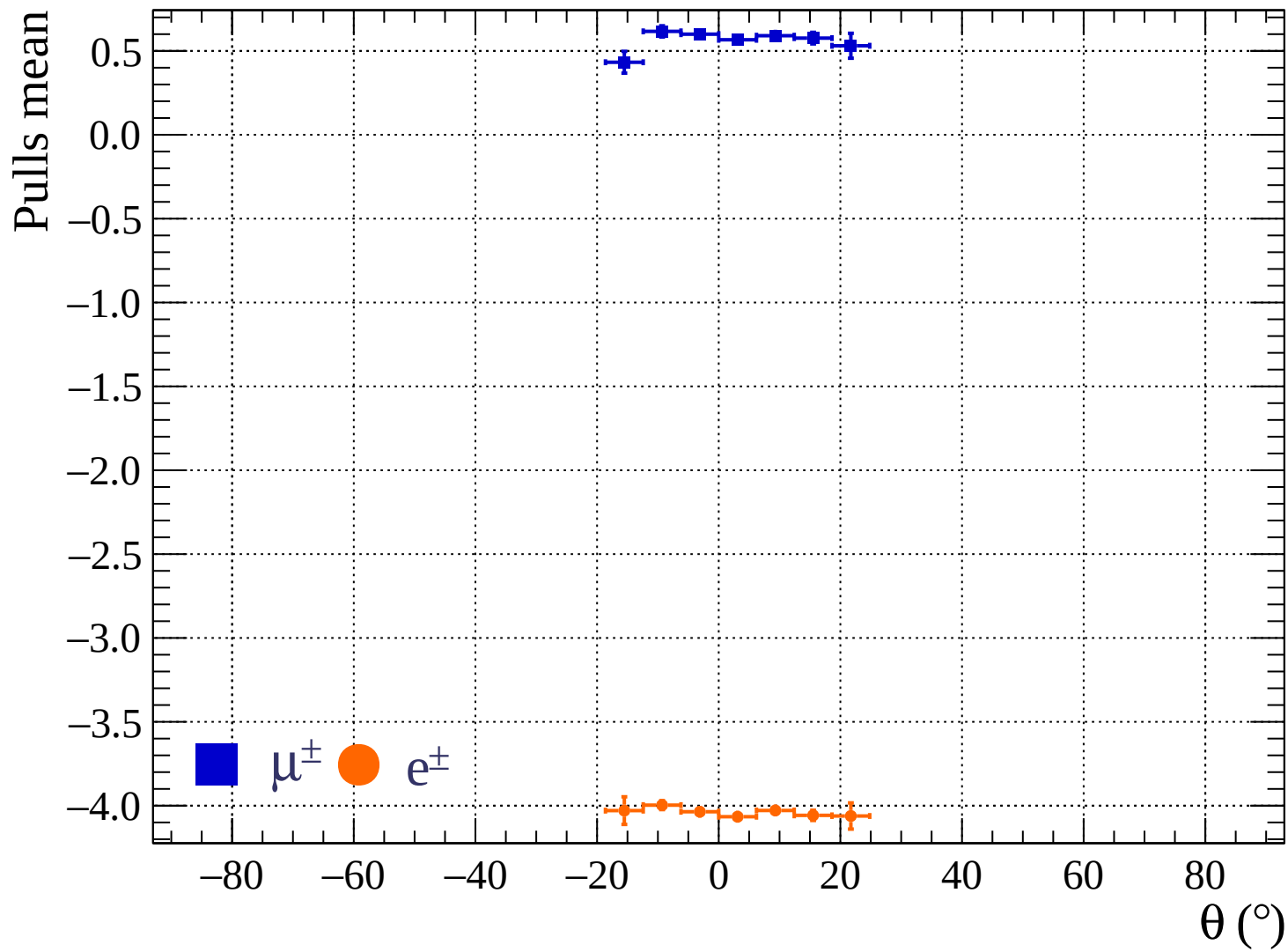




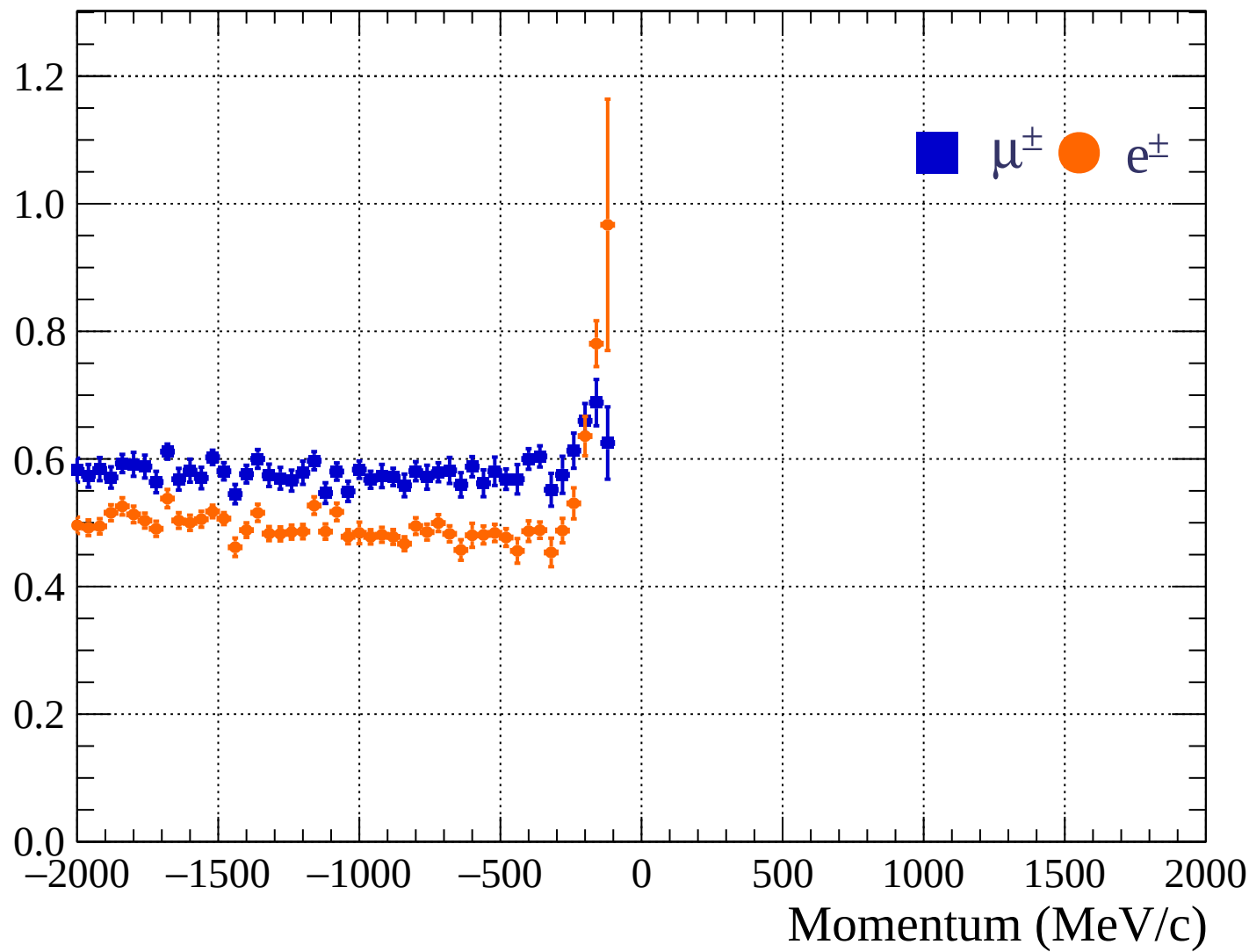




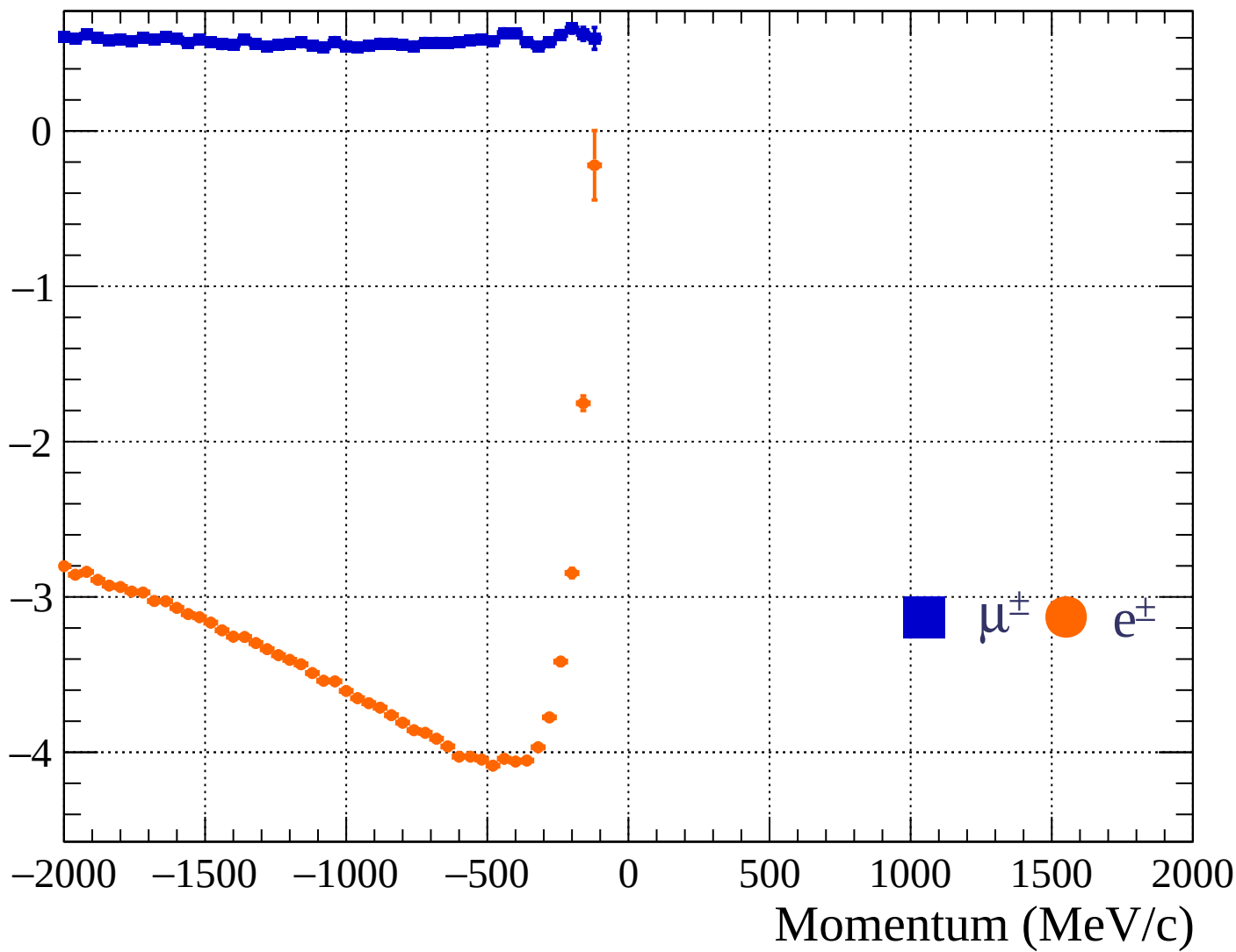


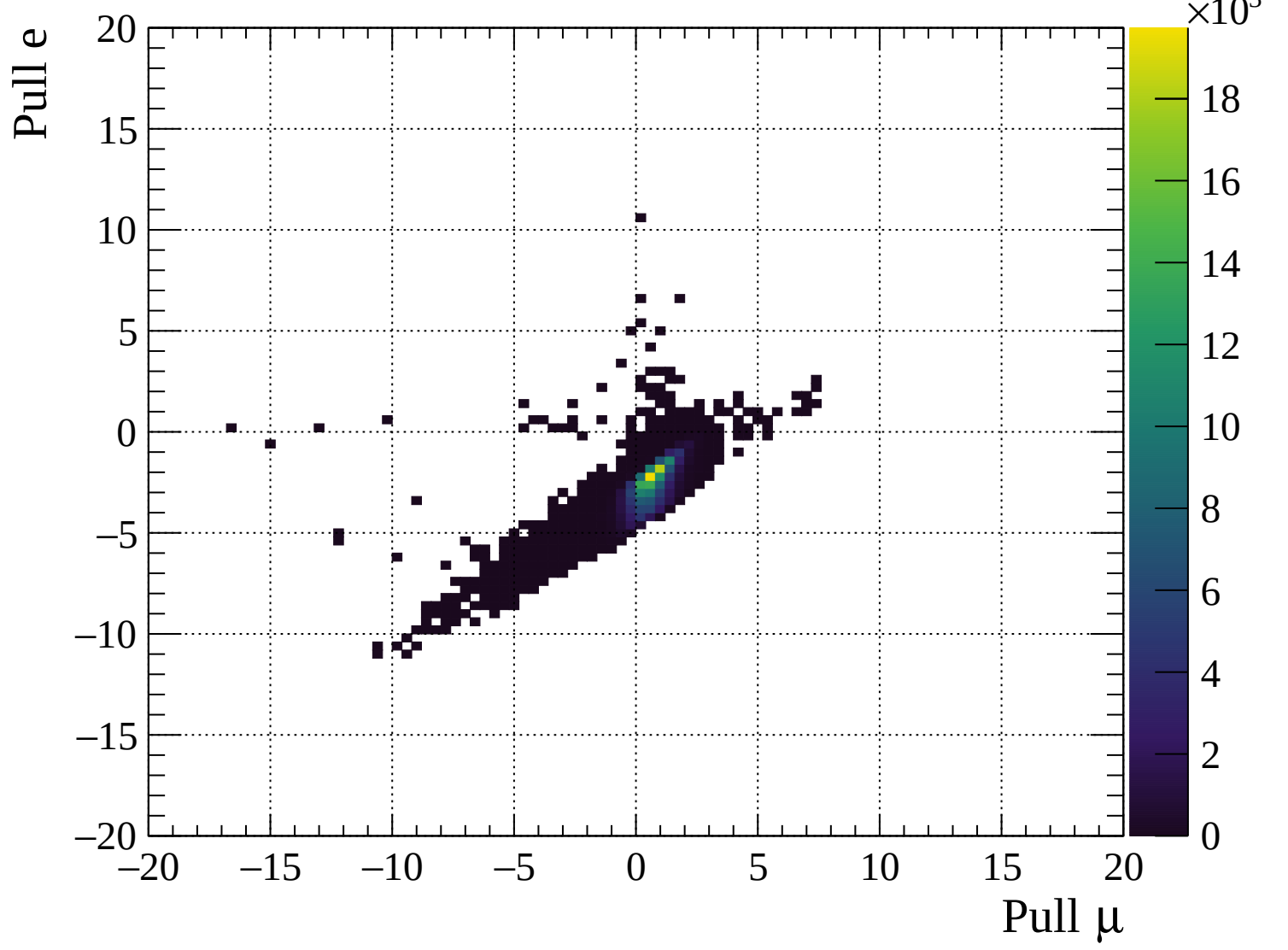


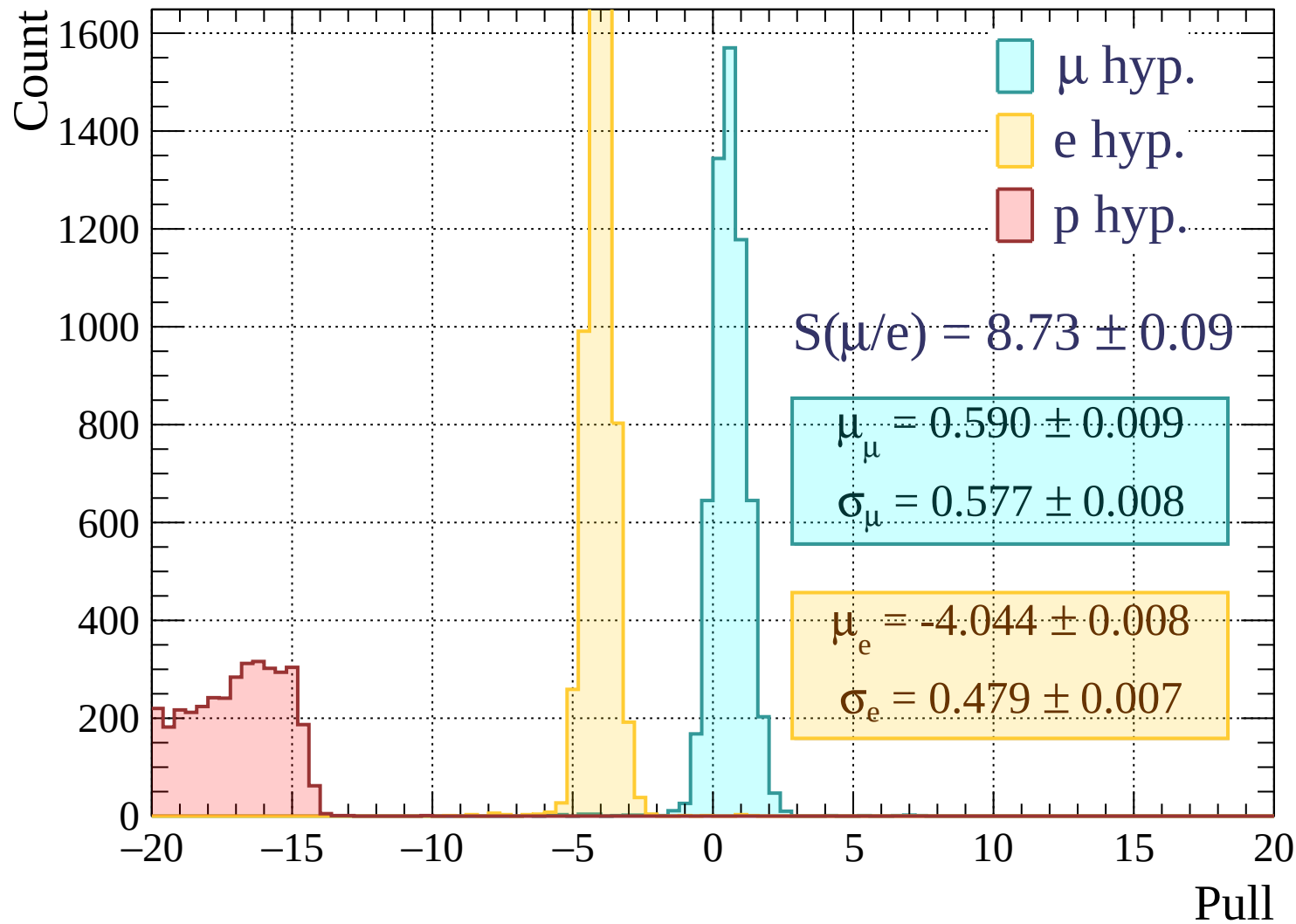
Pulls standard deviation

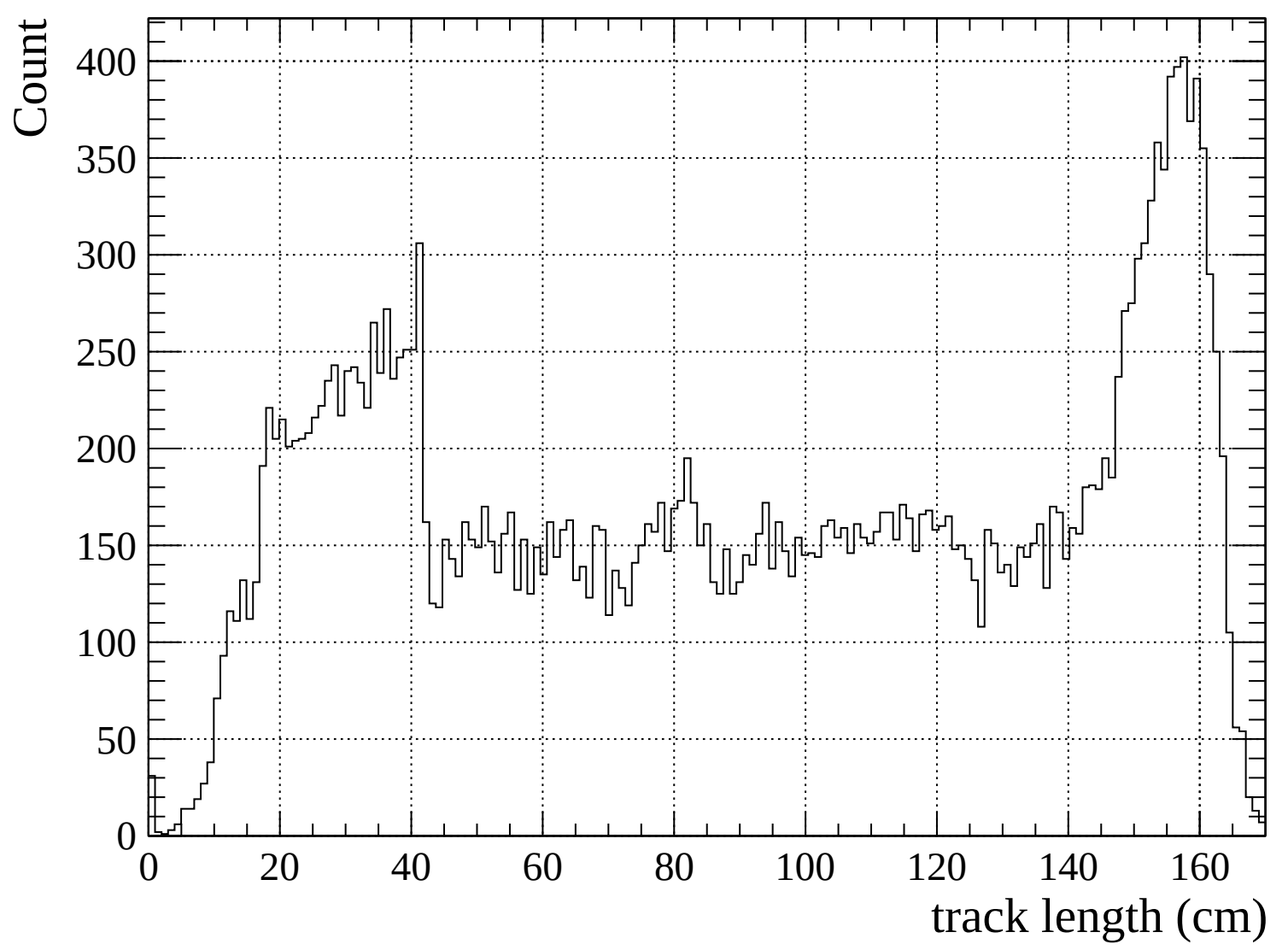


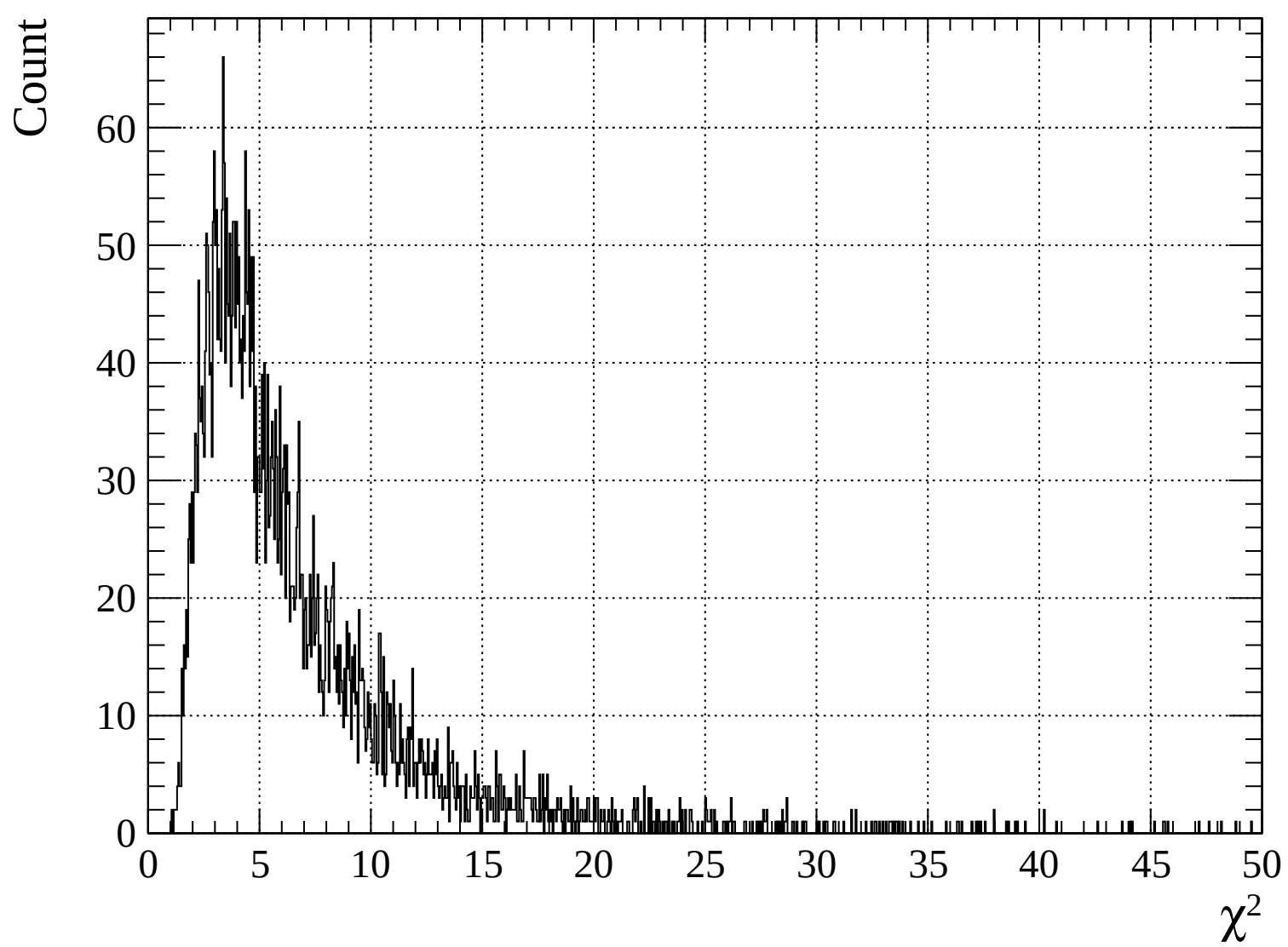
Pulls mean

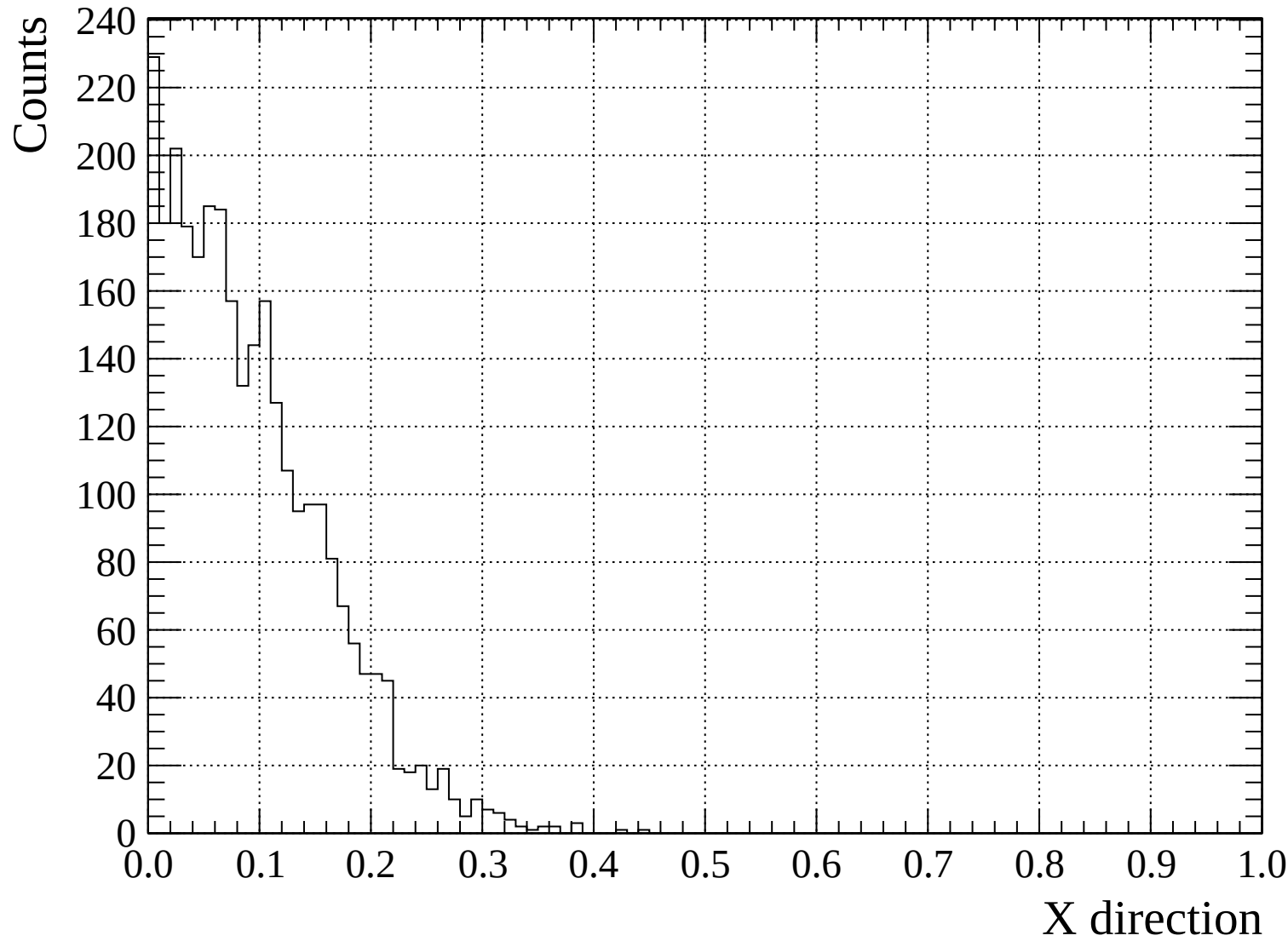


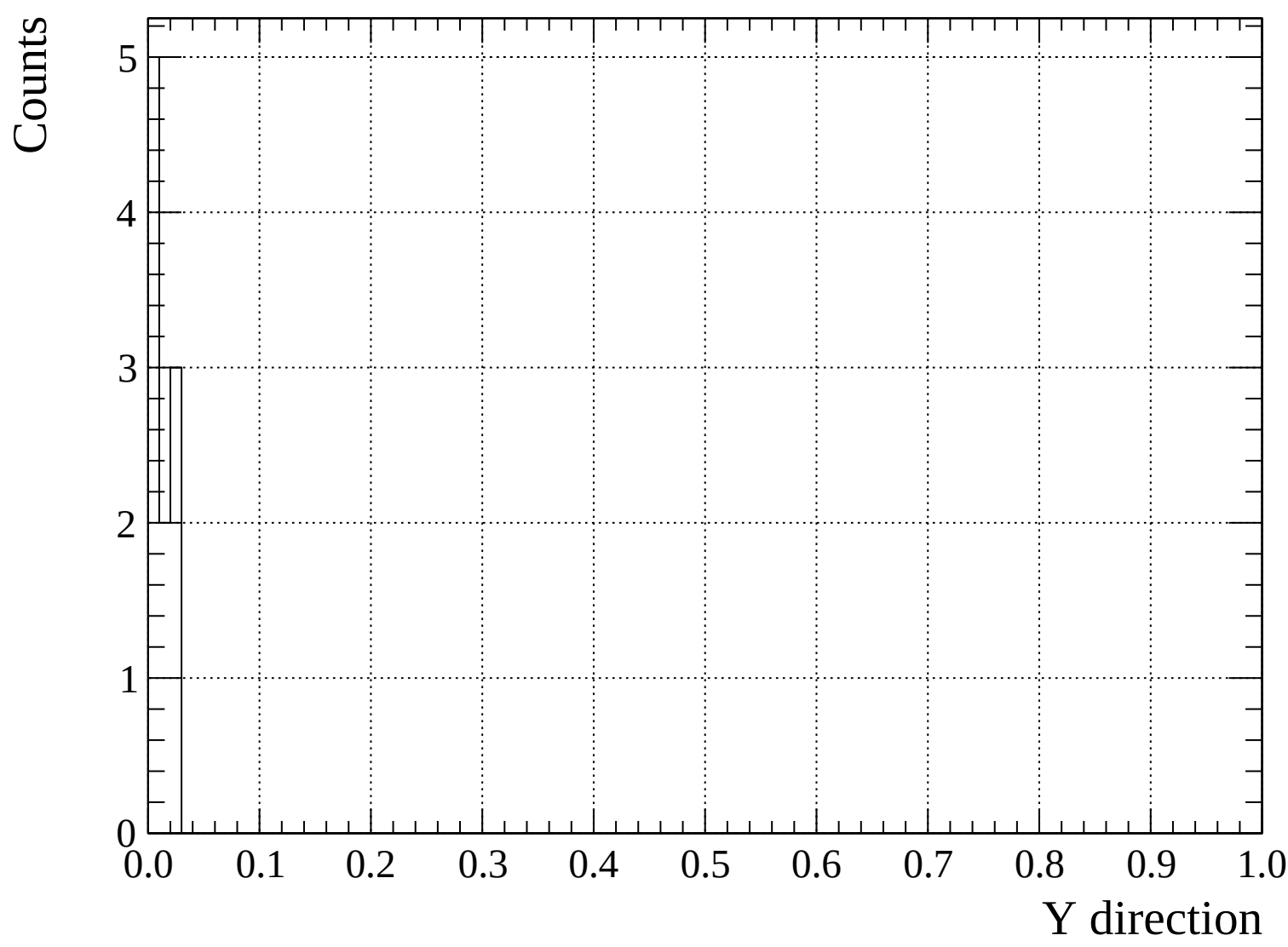


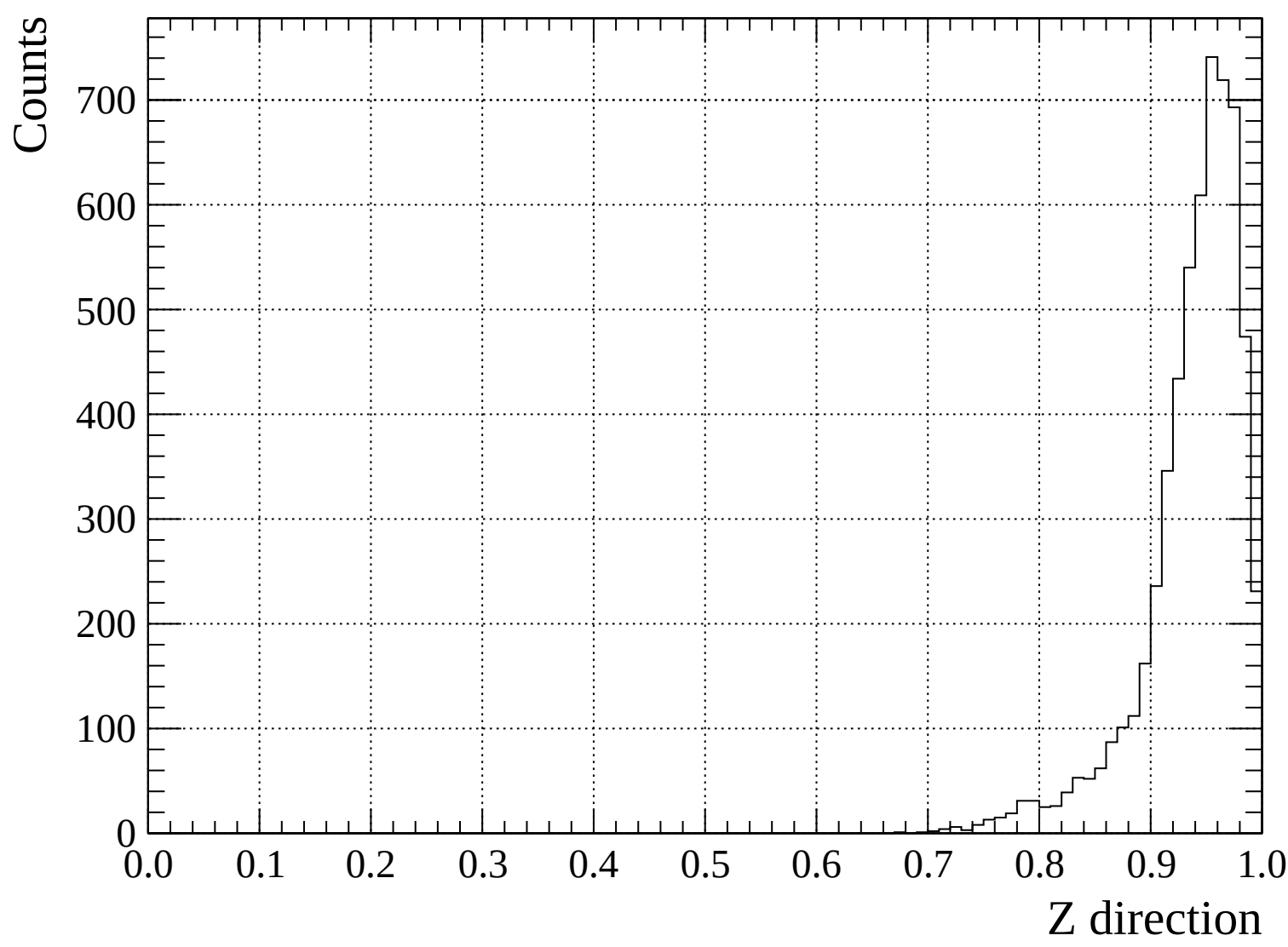


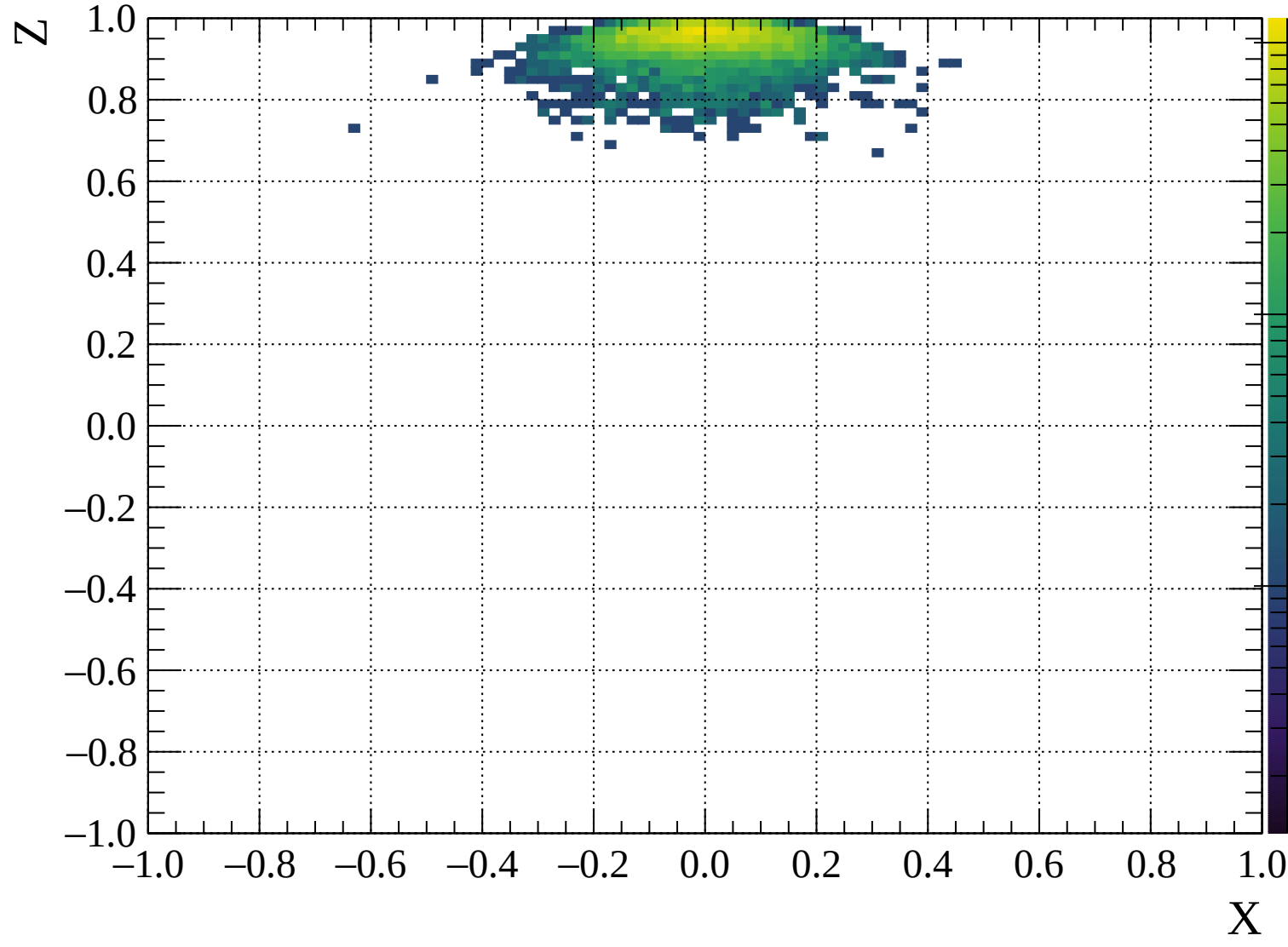


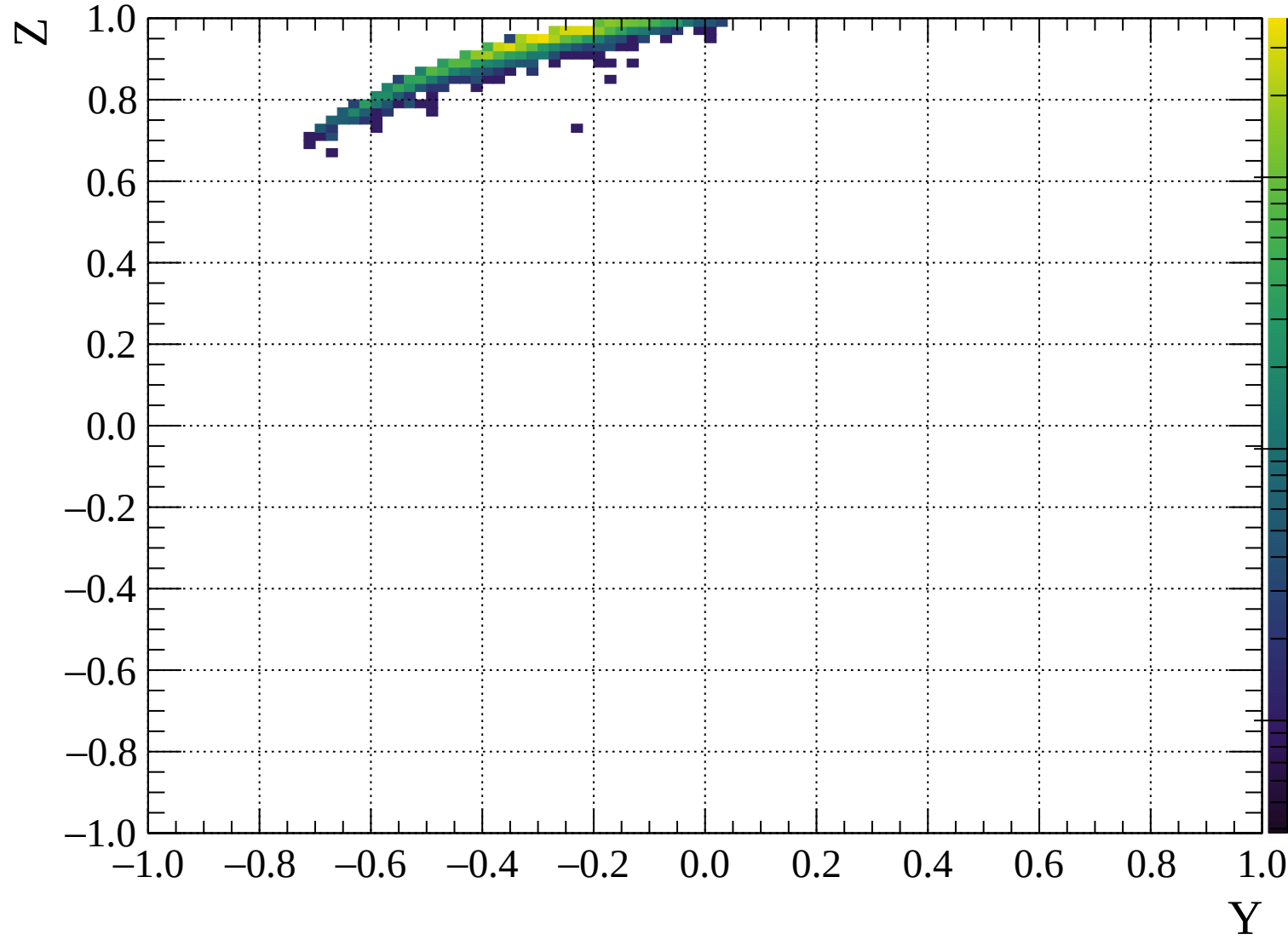


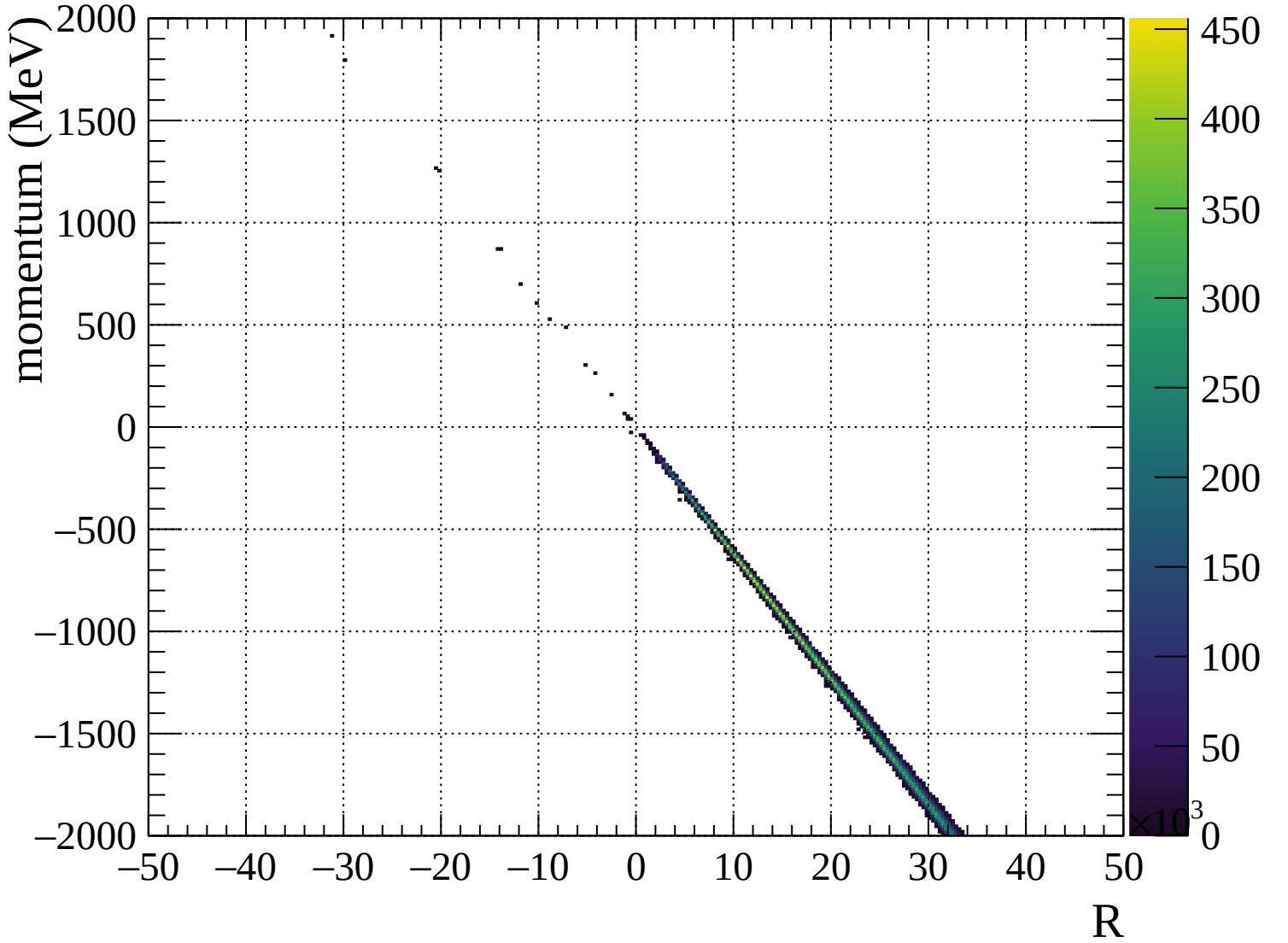


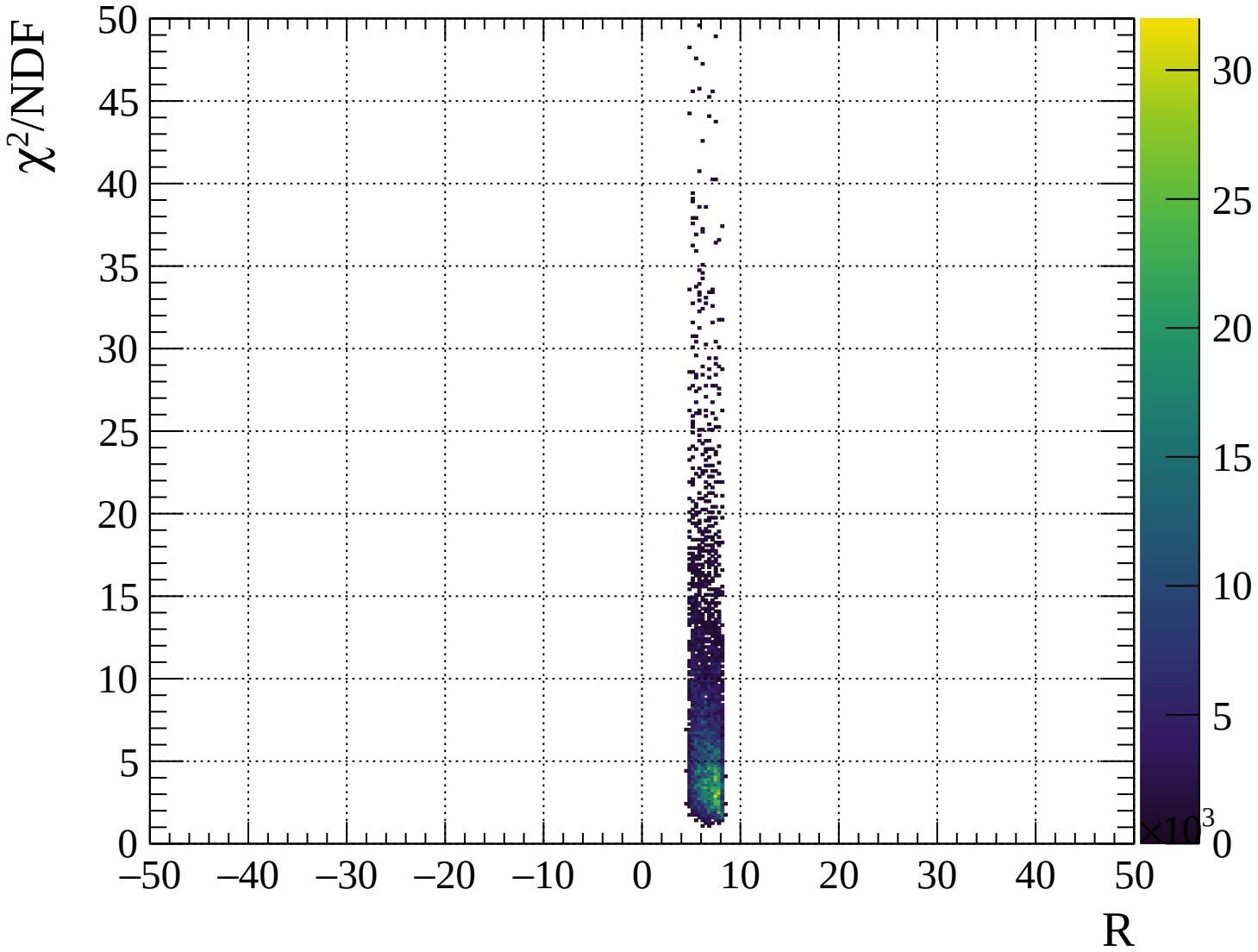






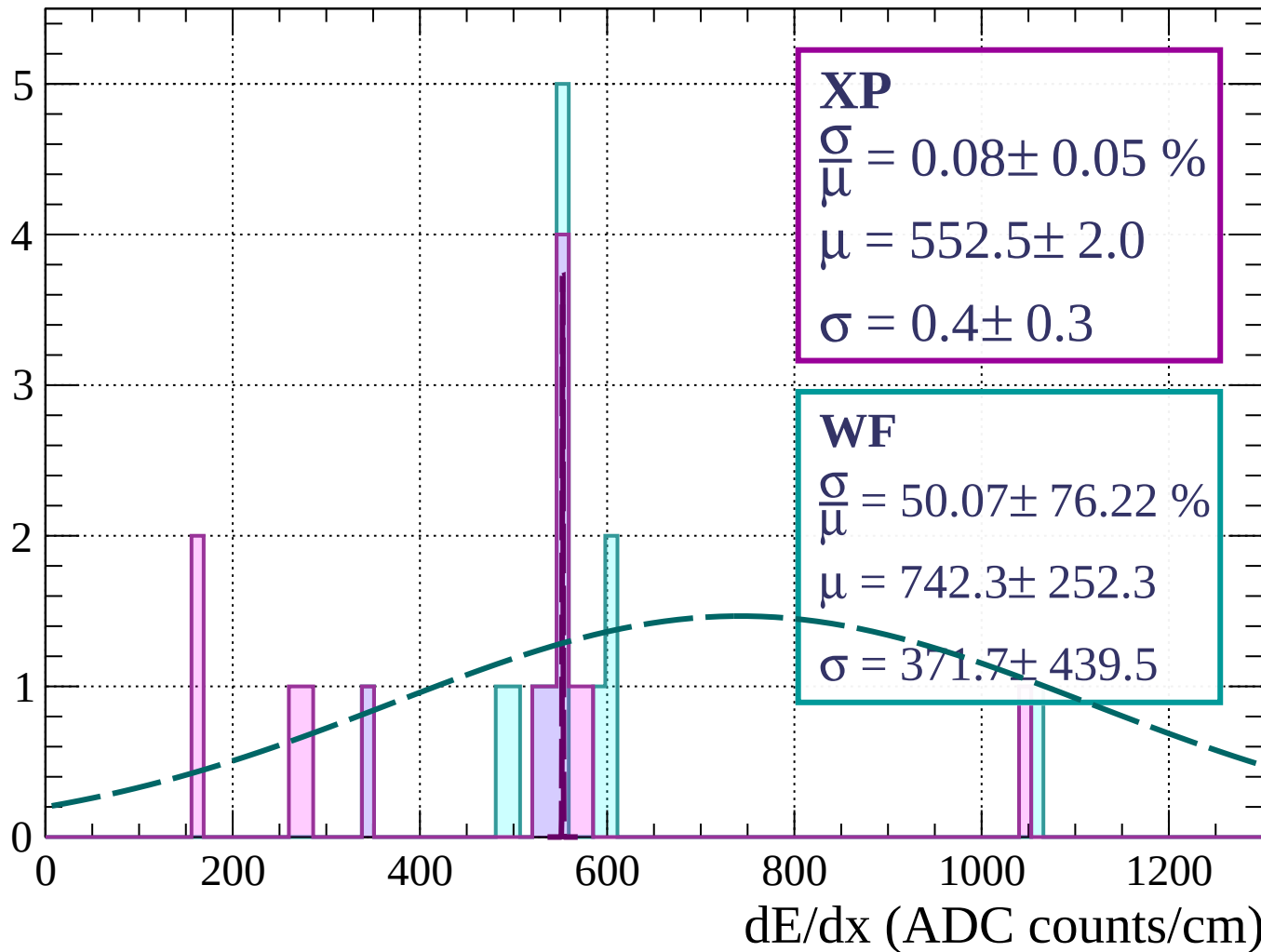






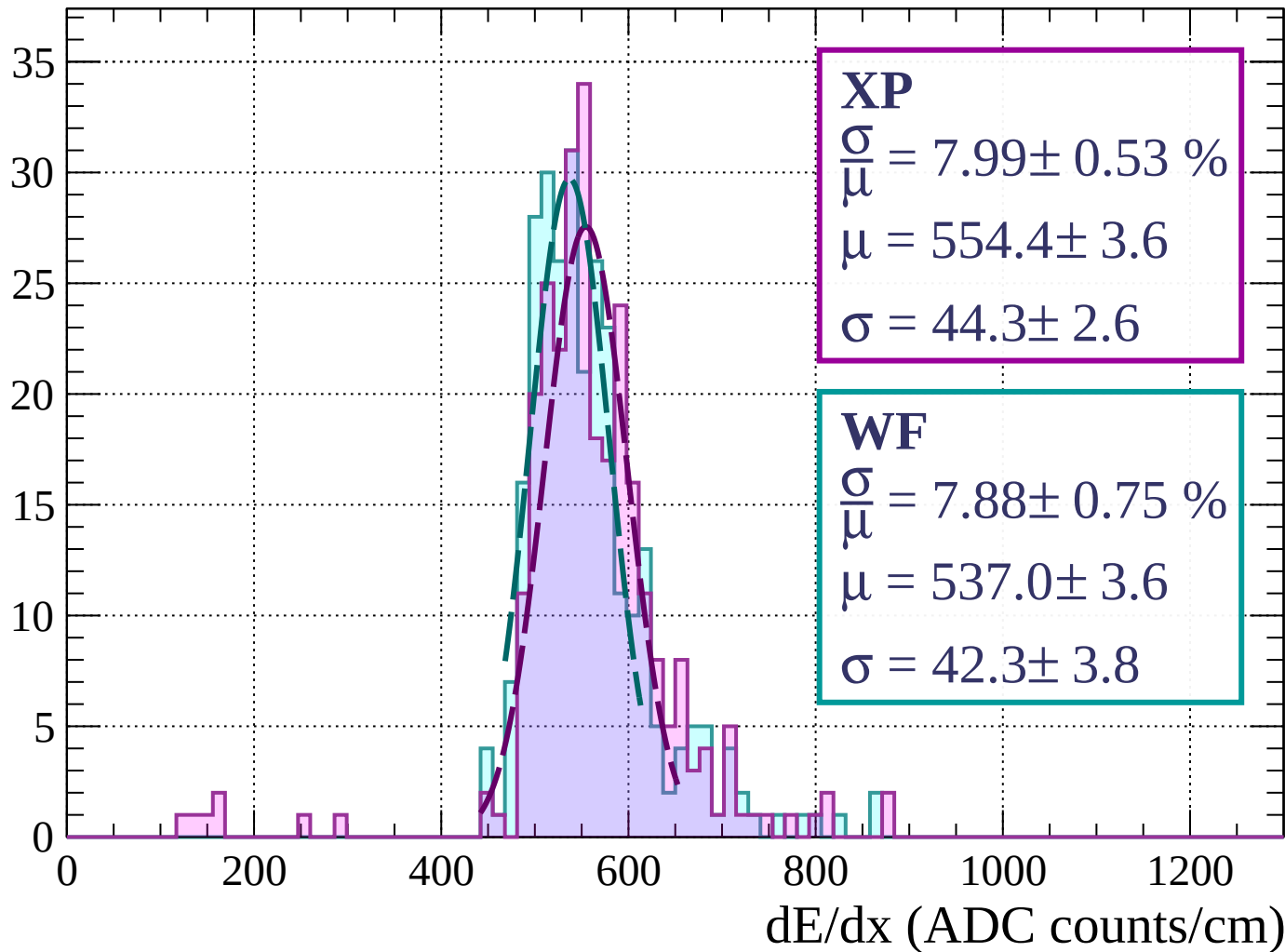
Energy loss | $0 < p < 80$

Count



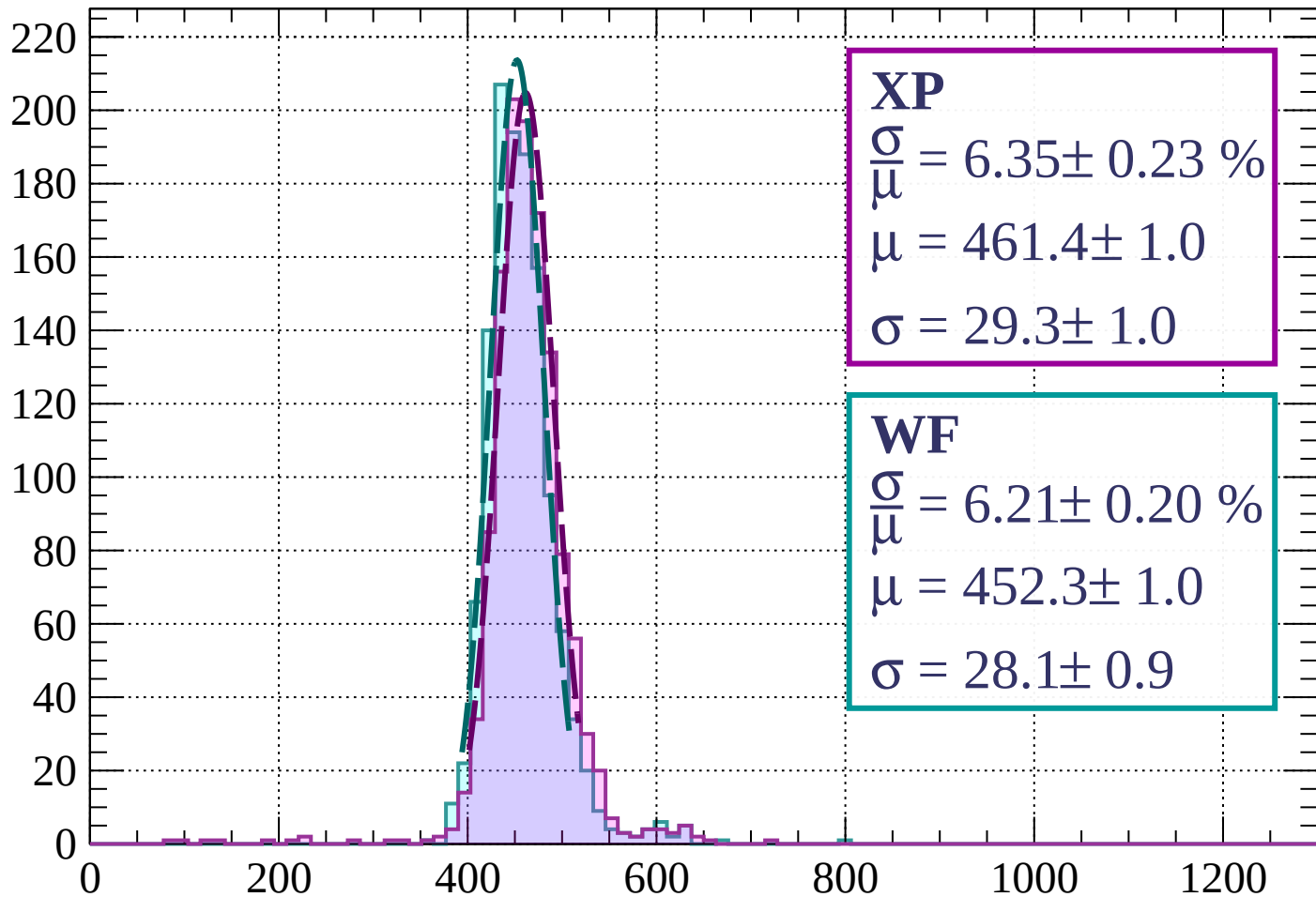
Energy loss | $80 < p < 160$

Count



Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 6.35 \pm 0.23 \%$$

$$\mu = 461.4 \pm 1.0$$

$$\sigma = 29.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.21 \pm 0.20 \%$$

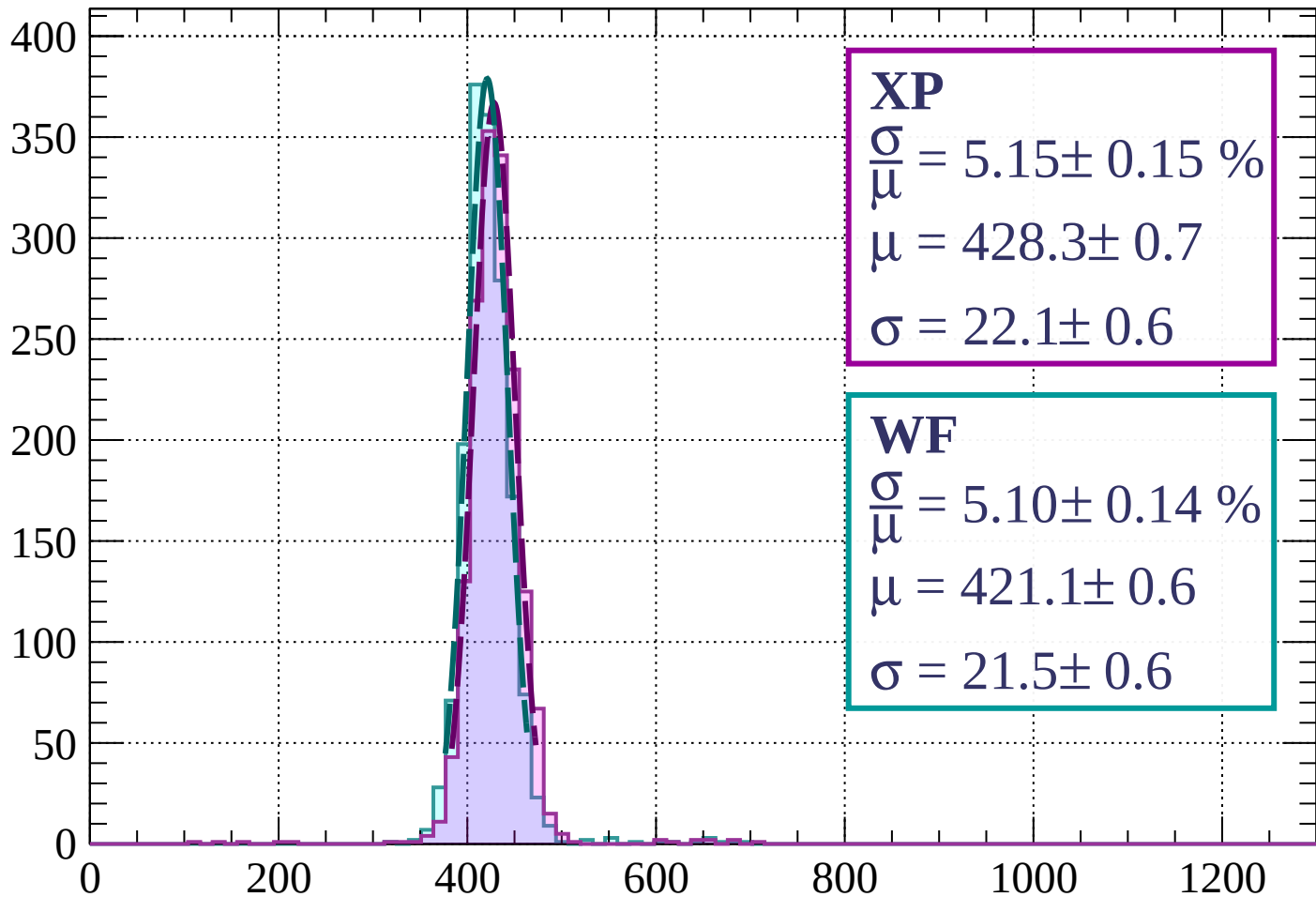
$$\mu = 452.3 \pm 1.0$$

$$\sigma = 28.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 5.15 \pm 0.15 \%$$

$$\mu = 428.3 \pm 0.7$$

$$\sigma = 22.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.10 \pm 0.14 \%$$

$$\mu = 421.1 \pm 0.6$$

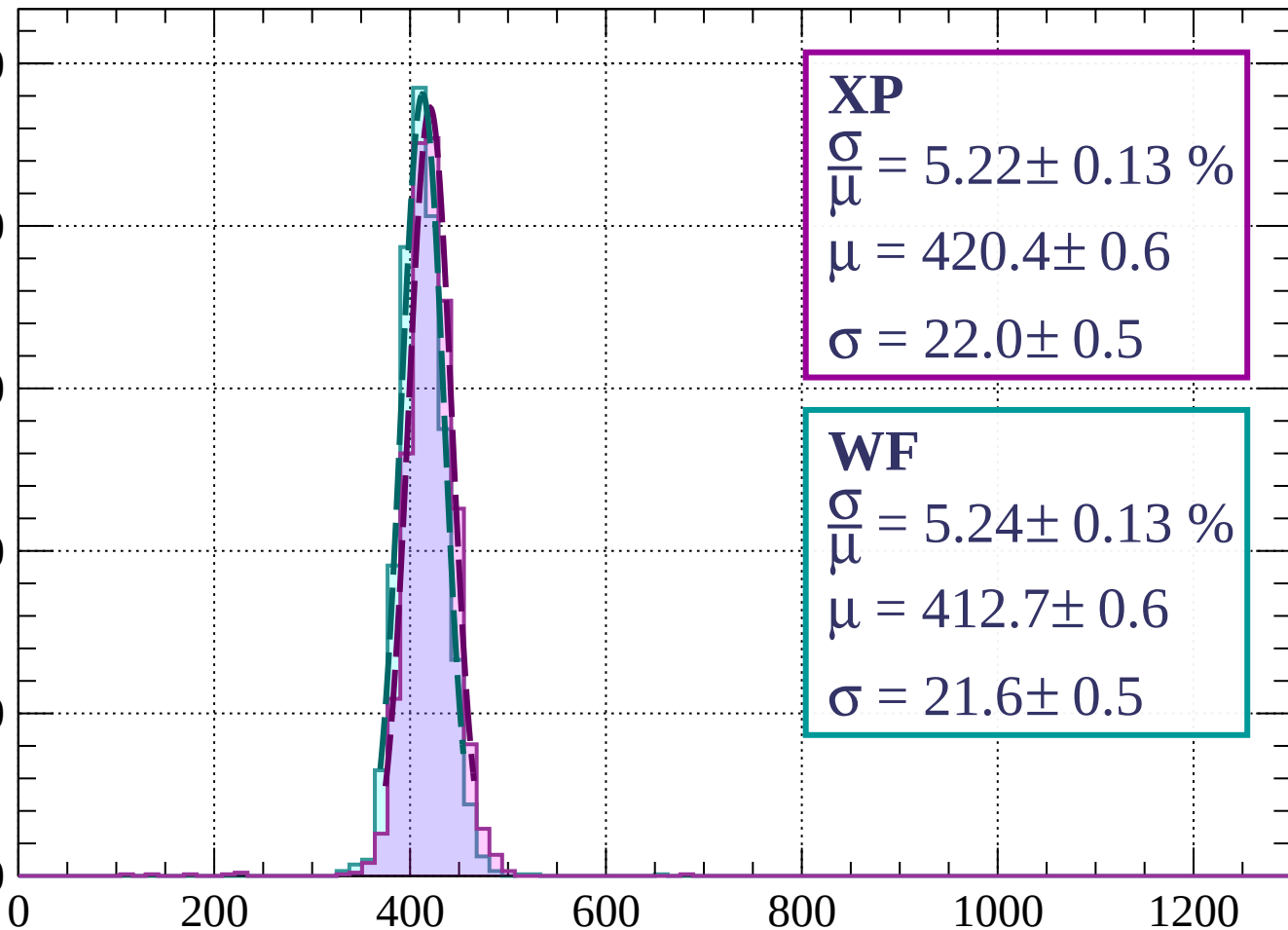
$$\sigma = 21.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 5.22 \pm 0.13 \%$$

$$\mu = 420.4 \pm 0.6$$

$$\sigma = 22.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.24 \pm 0.13 \%$$

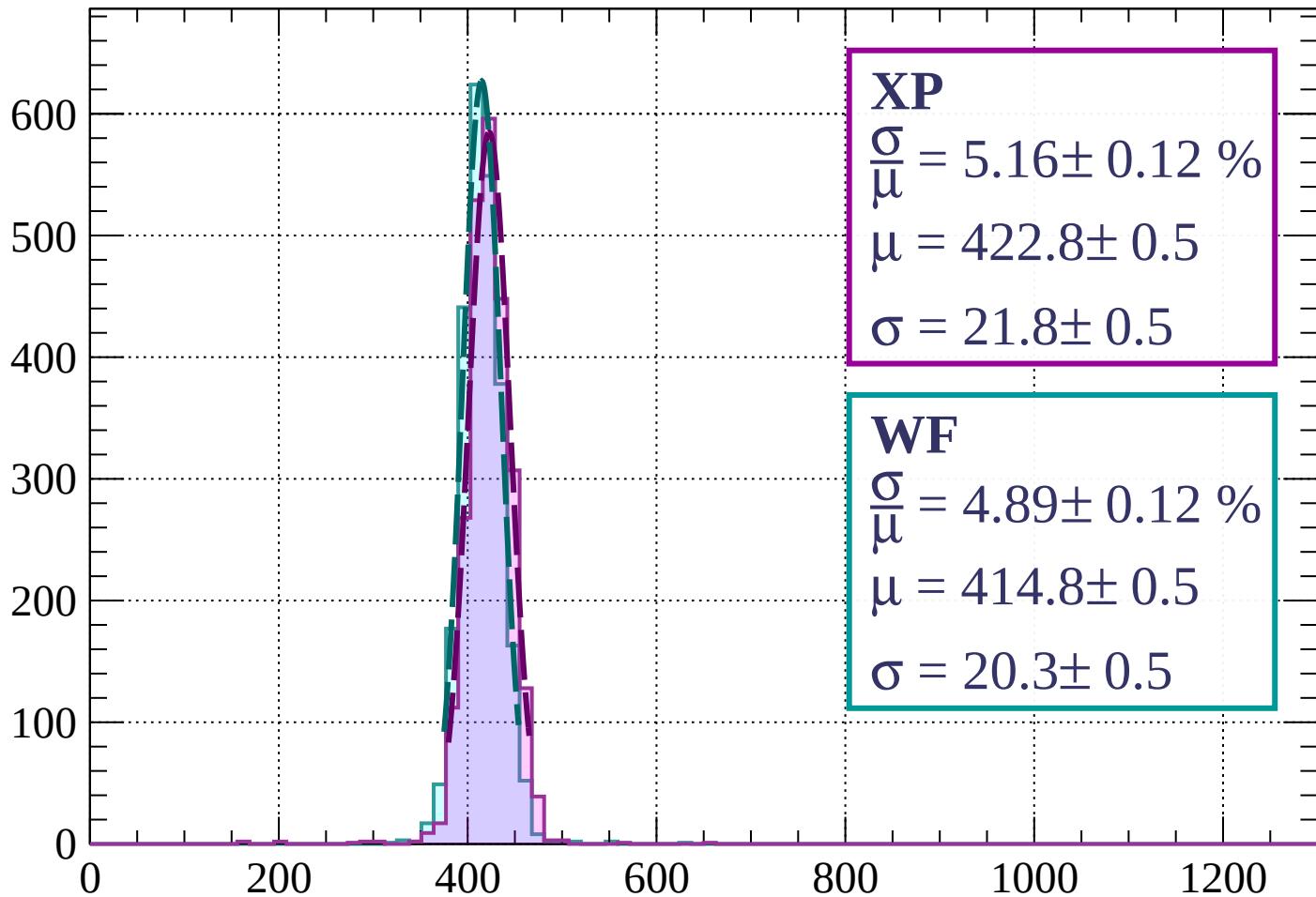
$$\mu = 412.7 \pm 0.6$$

$$\sigma = 21.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 5.16 \pm 0.12 \%$$

$$\mu = 422.8 \pm 0.5$$

$$\sigma = 21.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.89 \pm 0.12 \%$$

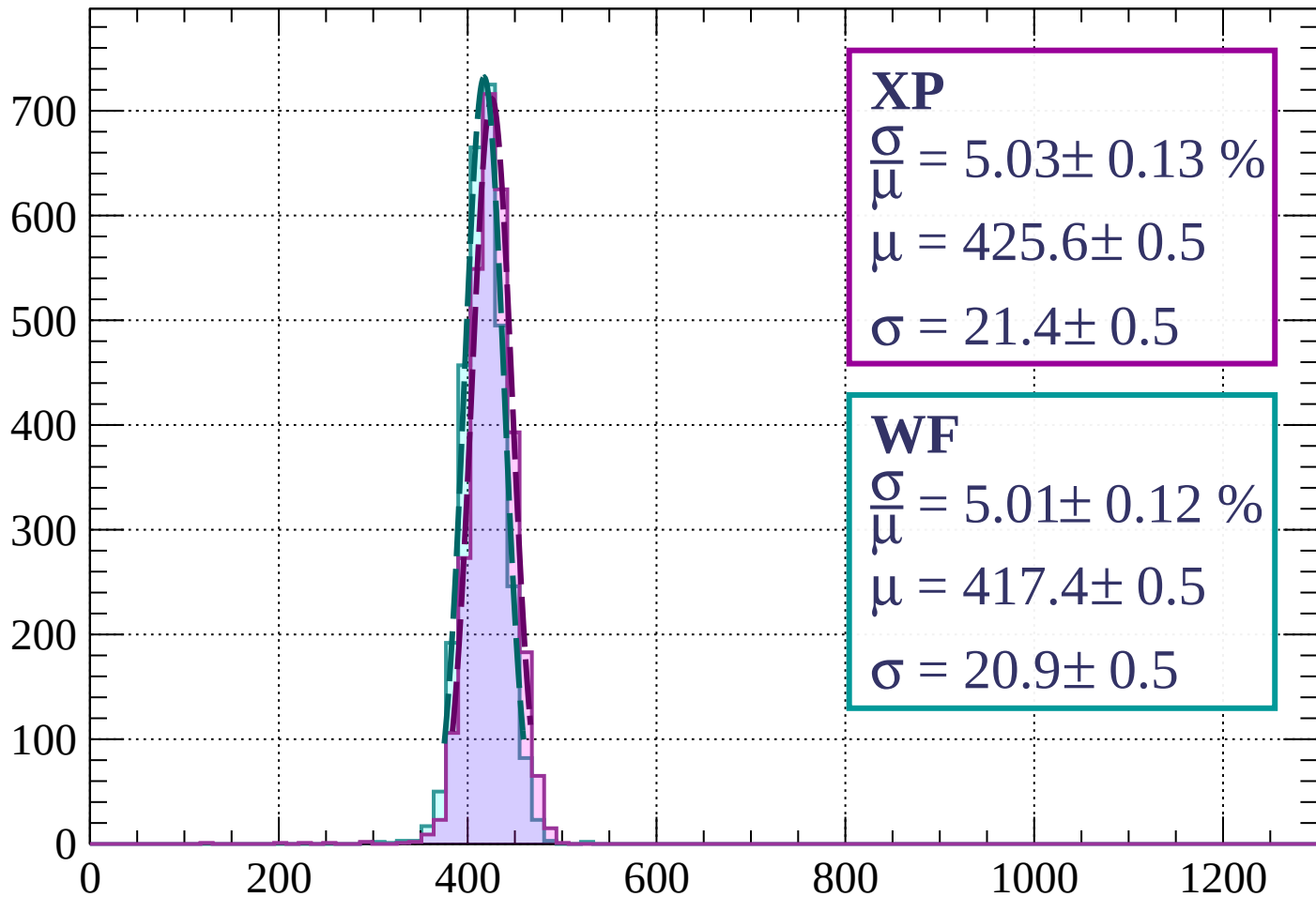
$$\mu = 414.8 \pm 0.5$$

$$\sigma = 20.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 5.03 \pm 0.13 \%$$

$$\mu = 425.6 \pm 0.5$$

$$\sigma = 21.4 \pm 0.5$$

WF

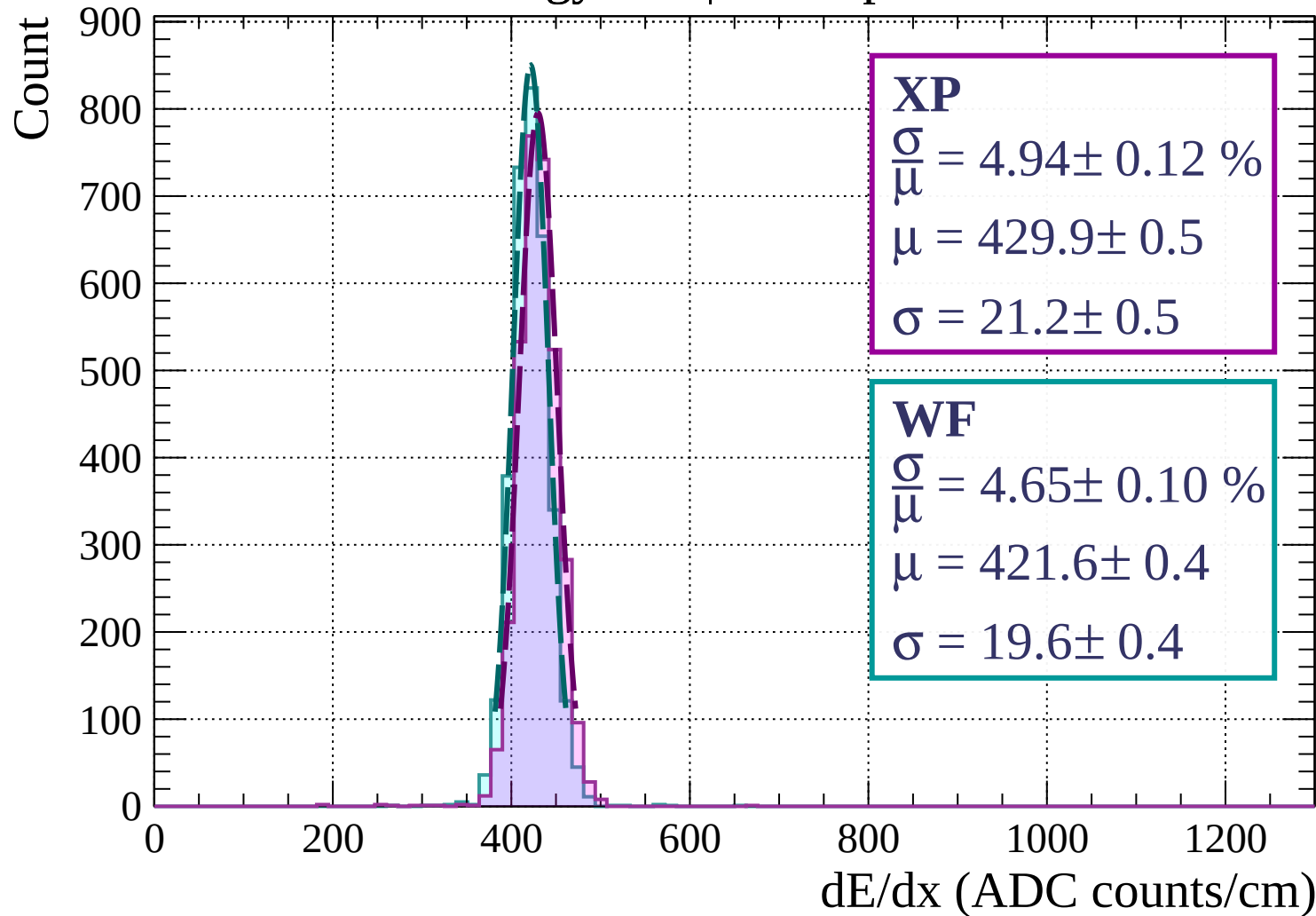
$$\frac{\sigma}{\mu} = 5.01 \pm 0.12 \%$$

$$\mu = 417.4 \pm 0.5$$

$$\sigma = 20.9 \pm 0.5$$

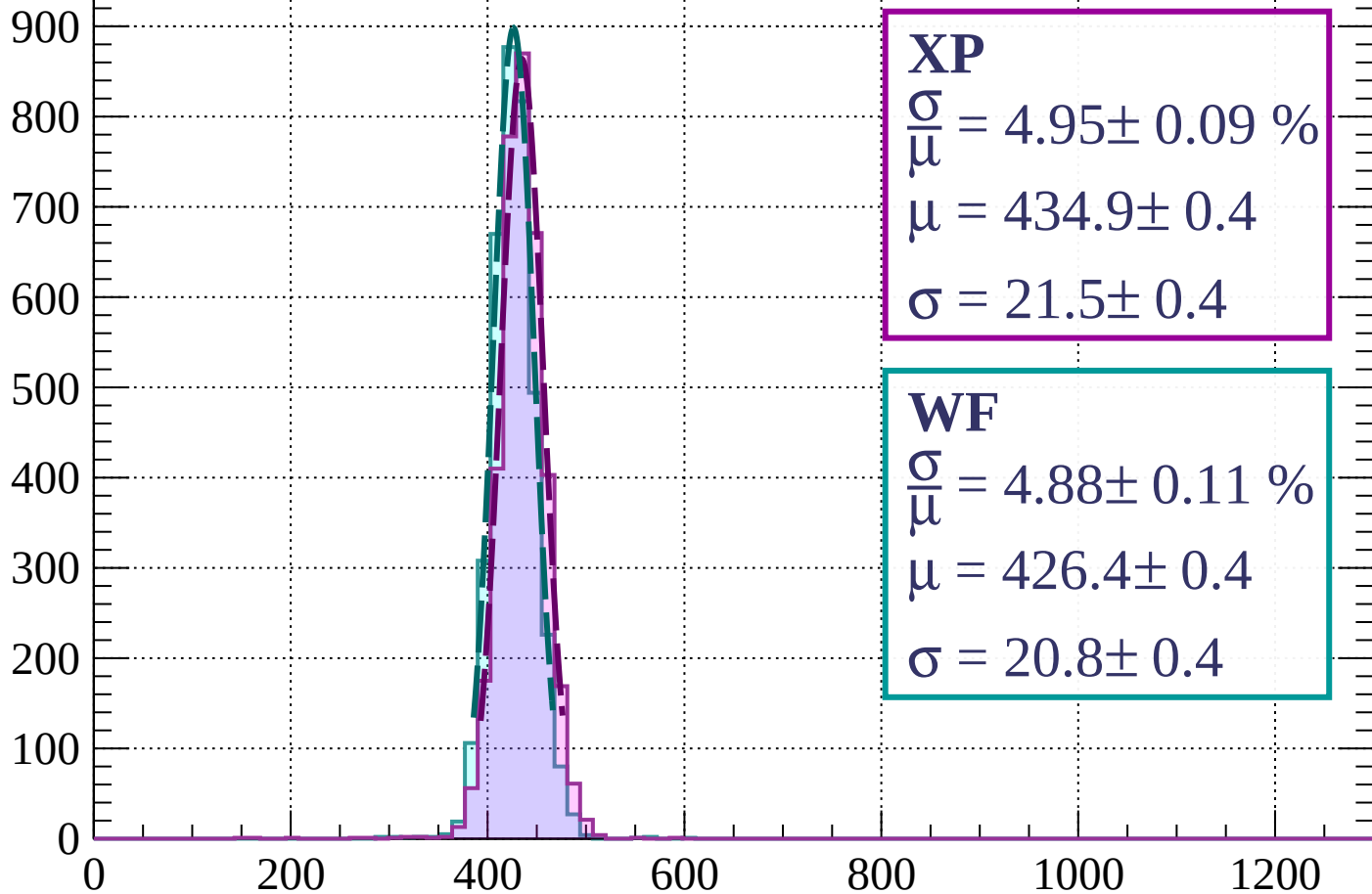
dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 4.95 \pm 0.09 \%$$

$$\mu = 434.9 \pm 0.4$$

$$\sigma = 21.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.88 \pm 0.11 \%$$

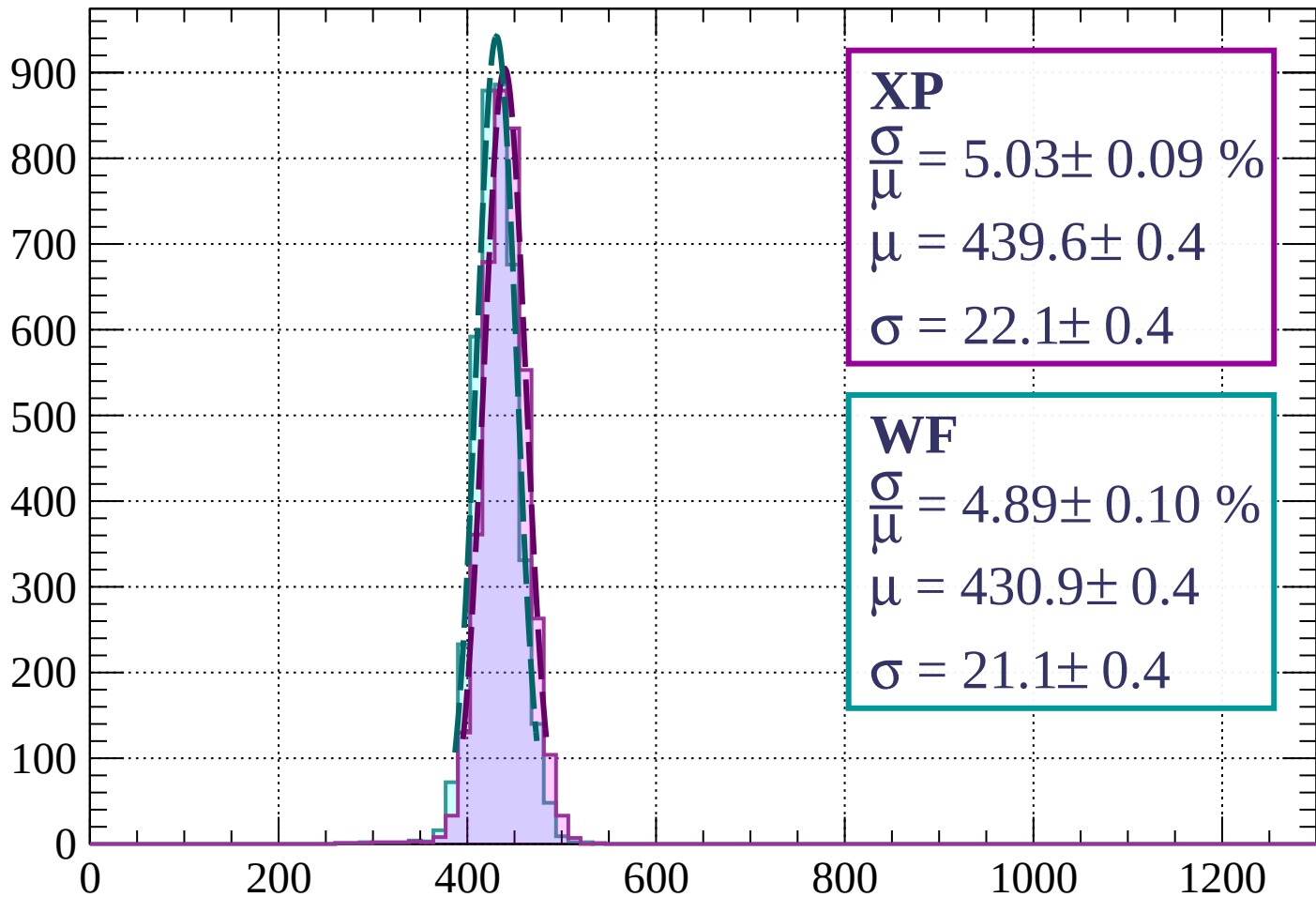
$$\mu = 426.4 \pm 0.4$$

$$\sigma = 20.8 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 5.03 \pm 0.09 \%$$

$$\mu = 439.6 \pm 0.4$$

$$\sigma = 22.1 \pm 0.4$$

WF

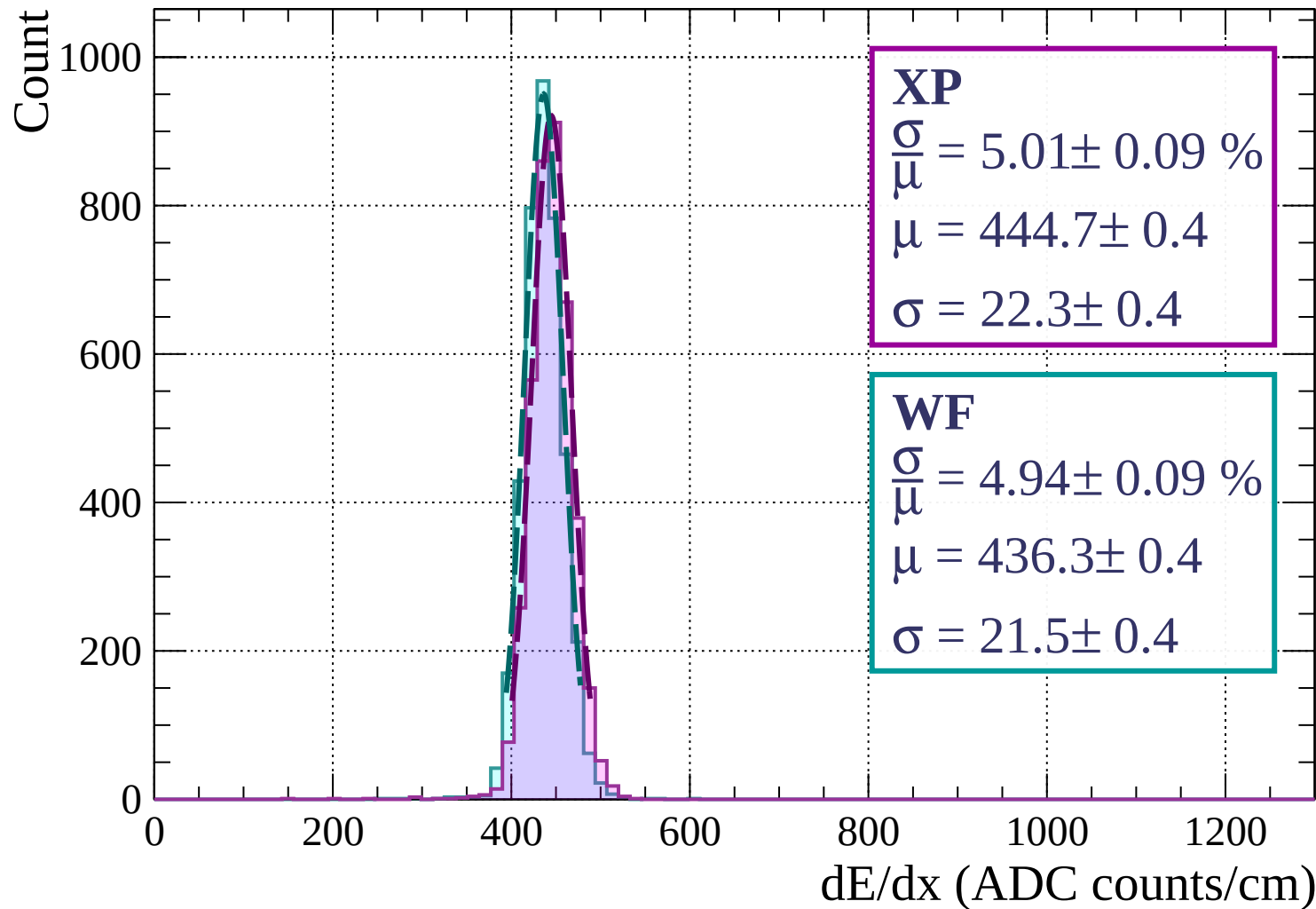
$$\frac{\sigma}{\mu} = 4.89 \pm 0.10 \%$$

$$\mu = 430.9 \pm 0.4$$

$$\sigma = 21.1 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$

Count

1000

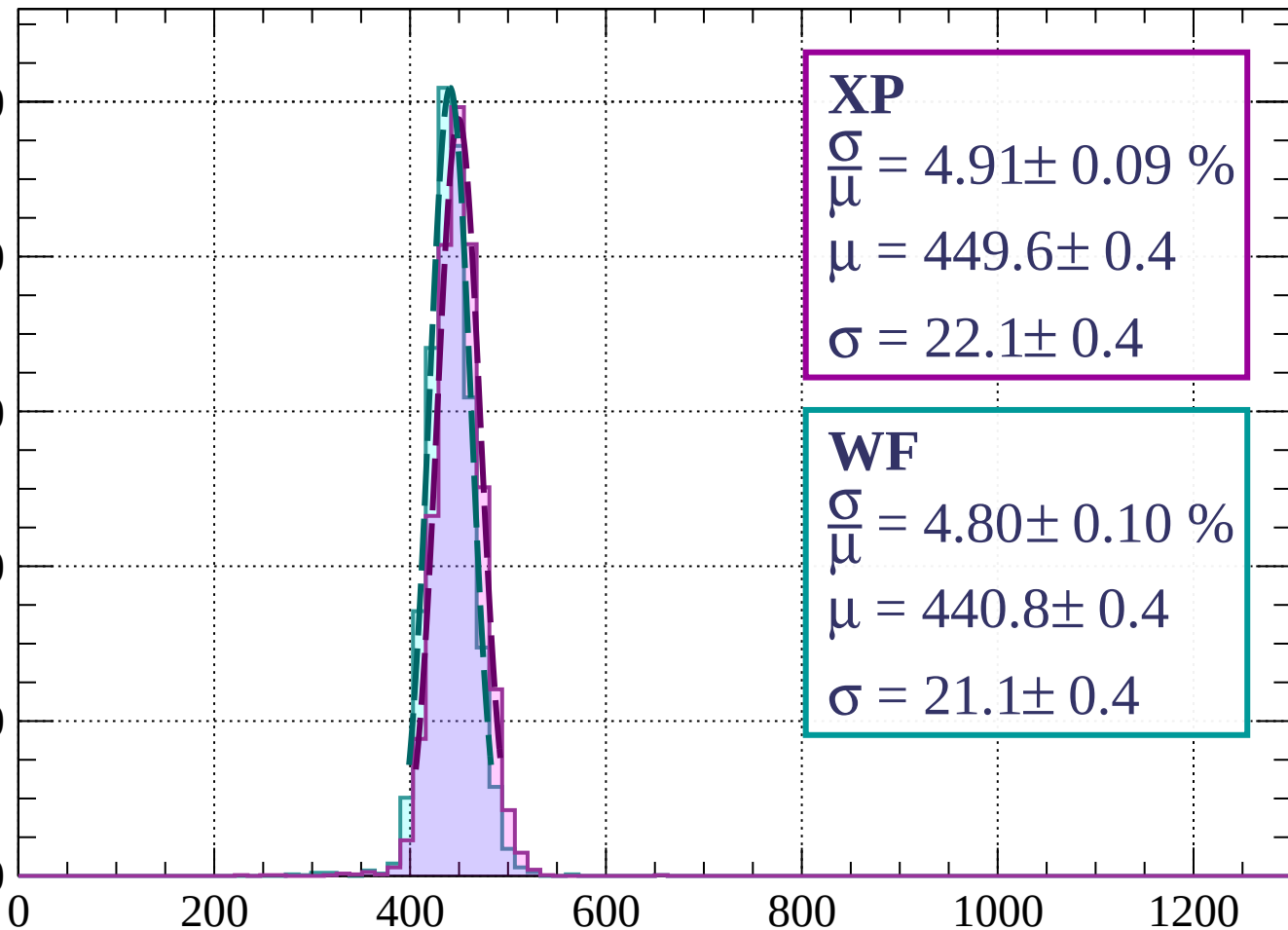
800

600

400

200

0



dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count

1000

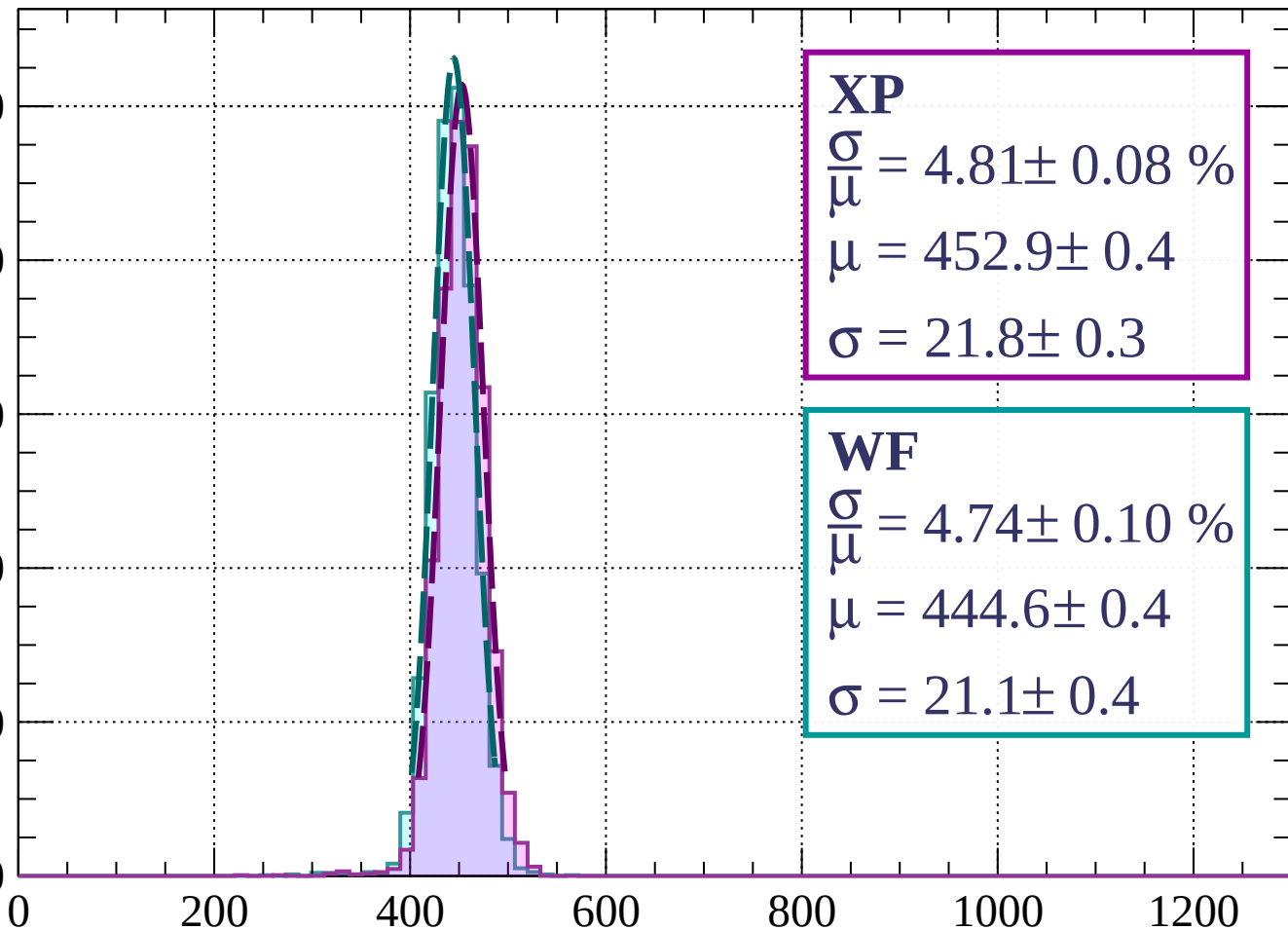
800

600

400

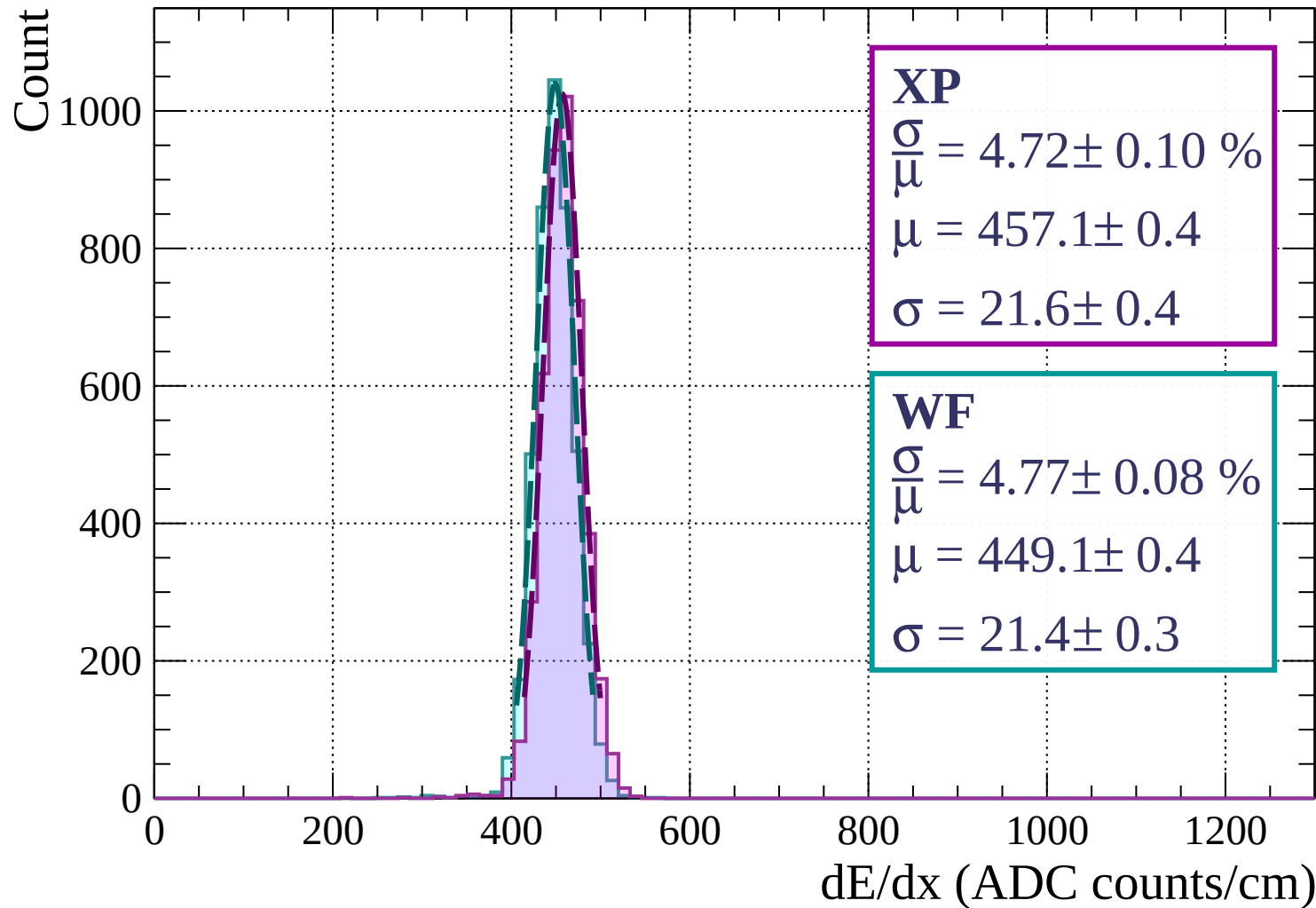
200

0

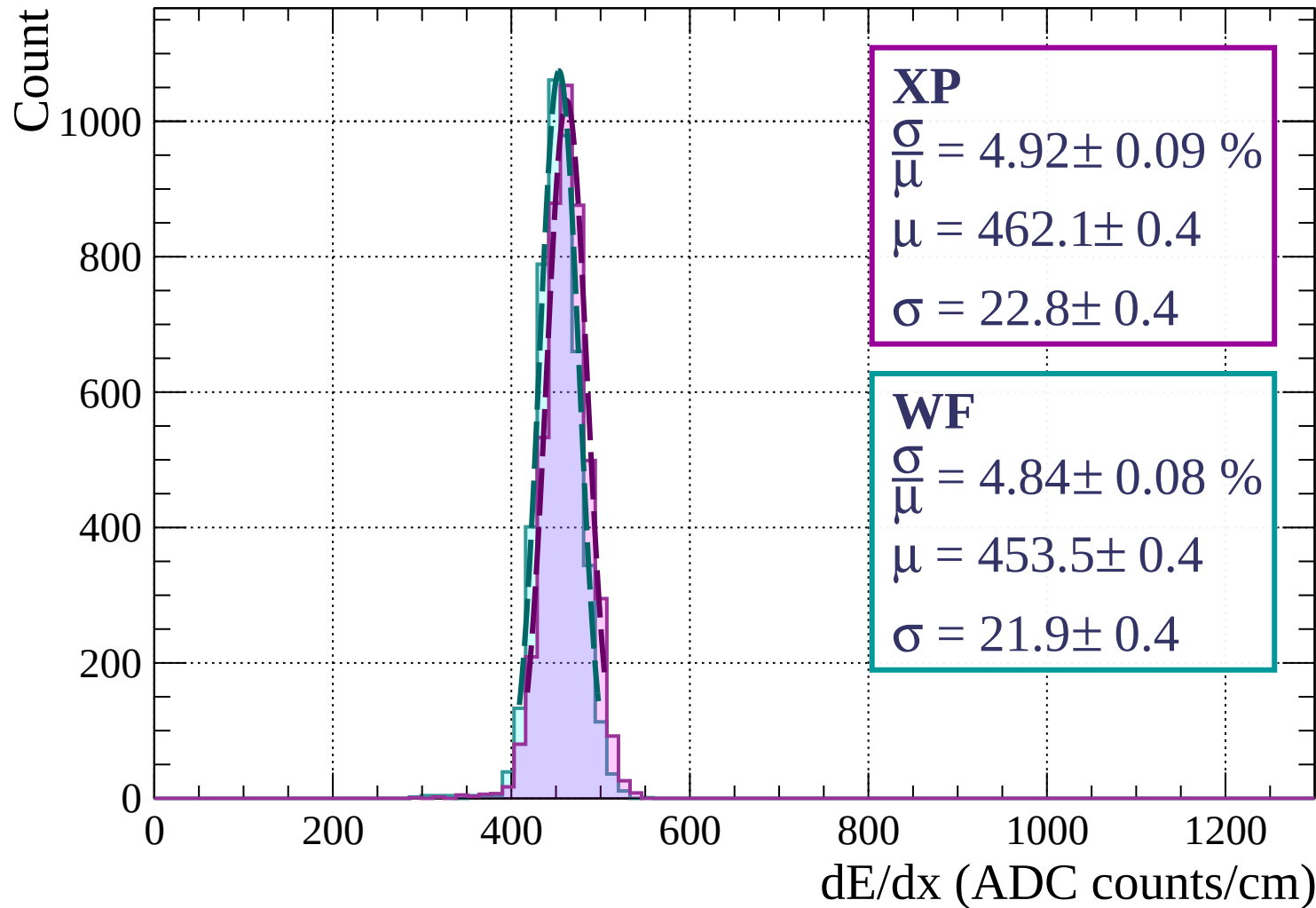


dE/dx (ADC counts/cm)

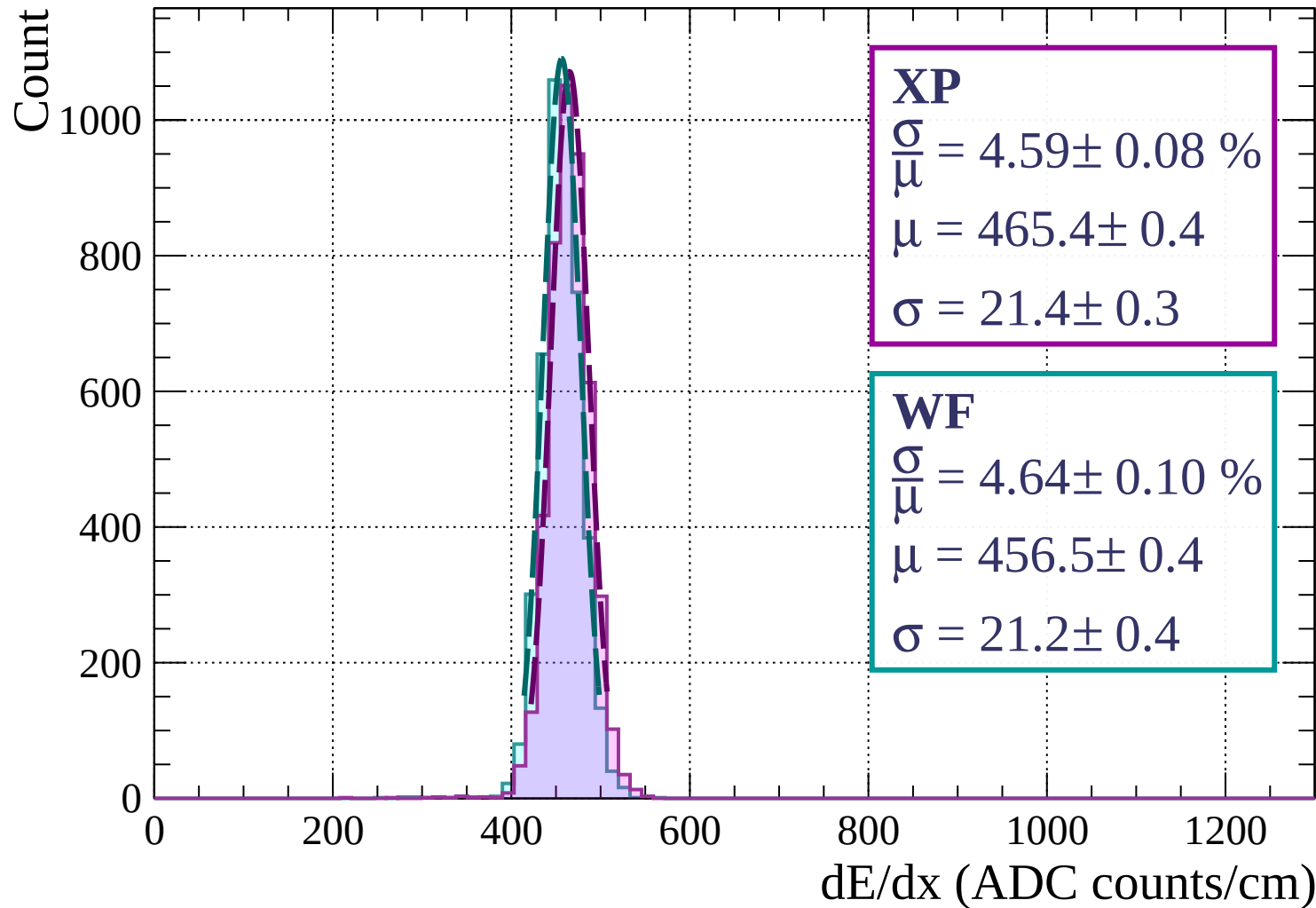
Energy loss | $1040 < p < 1120$



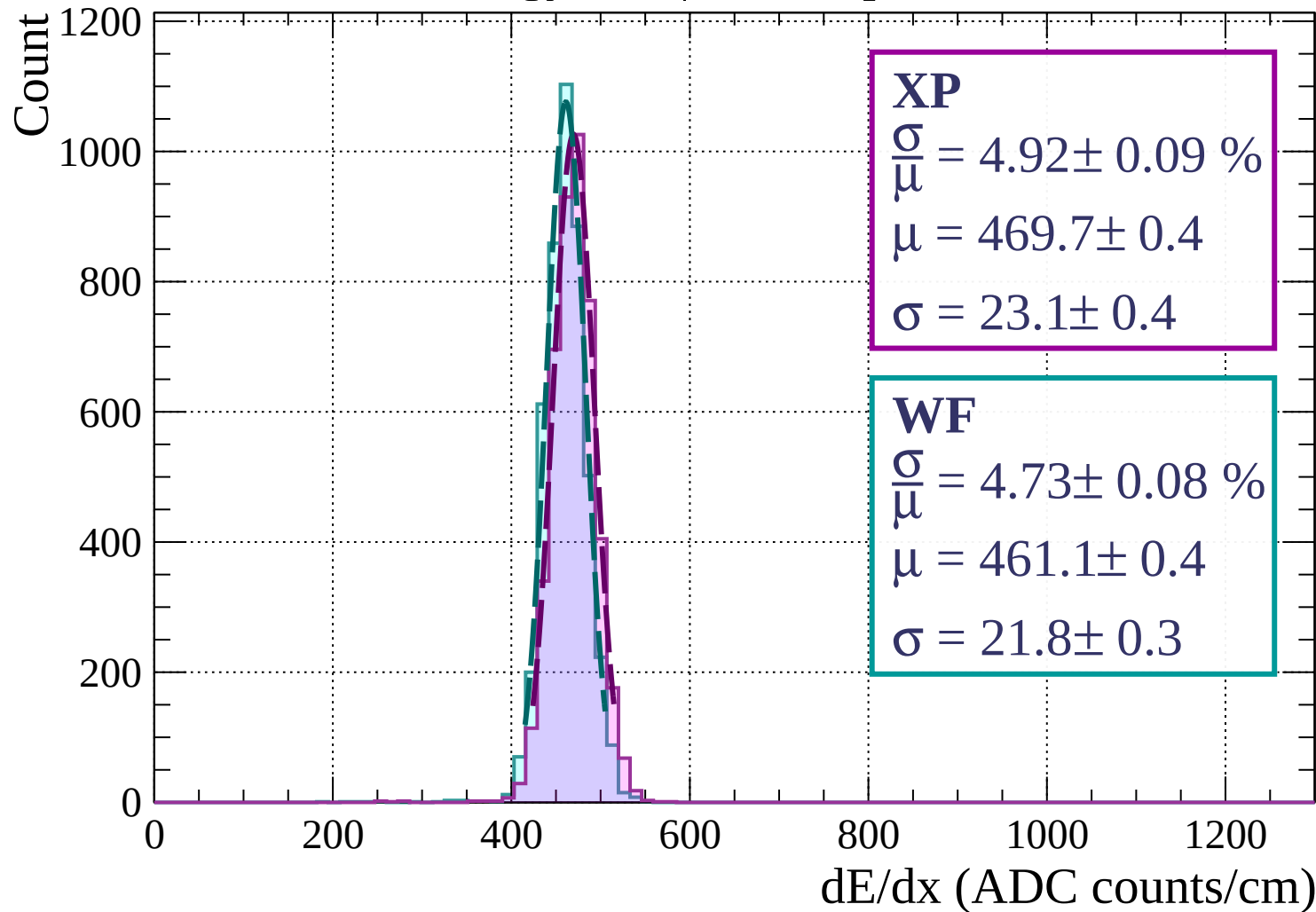
Energy loss | $1120 < p < 1200$



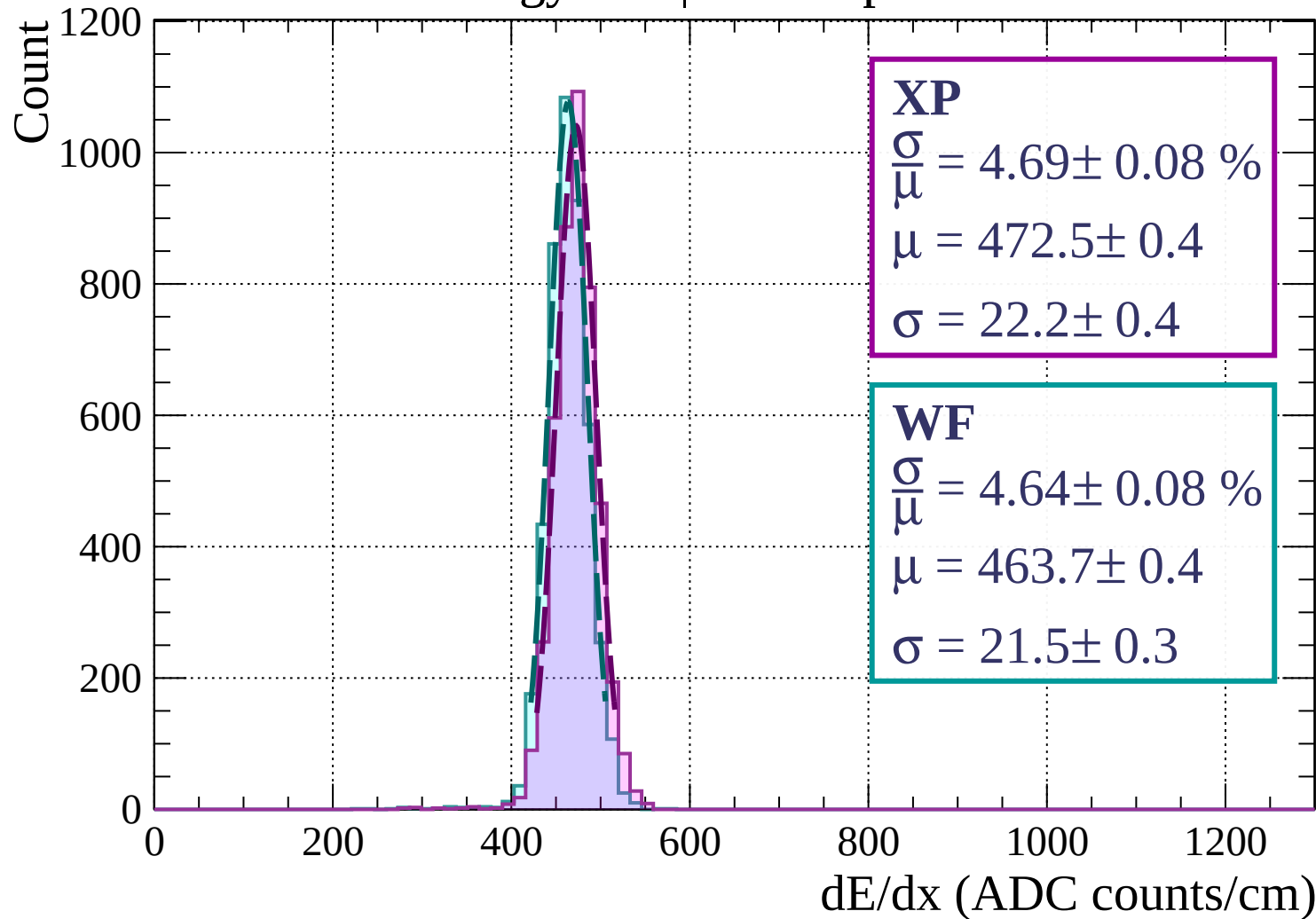
Energy loss | $1200 < p < 1280$



Energy loss | $1280 < p < 1360$



Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

Count

1000

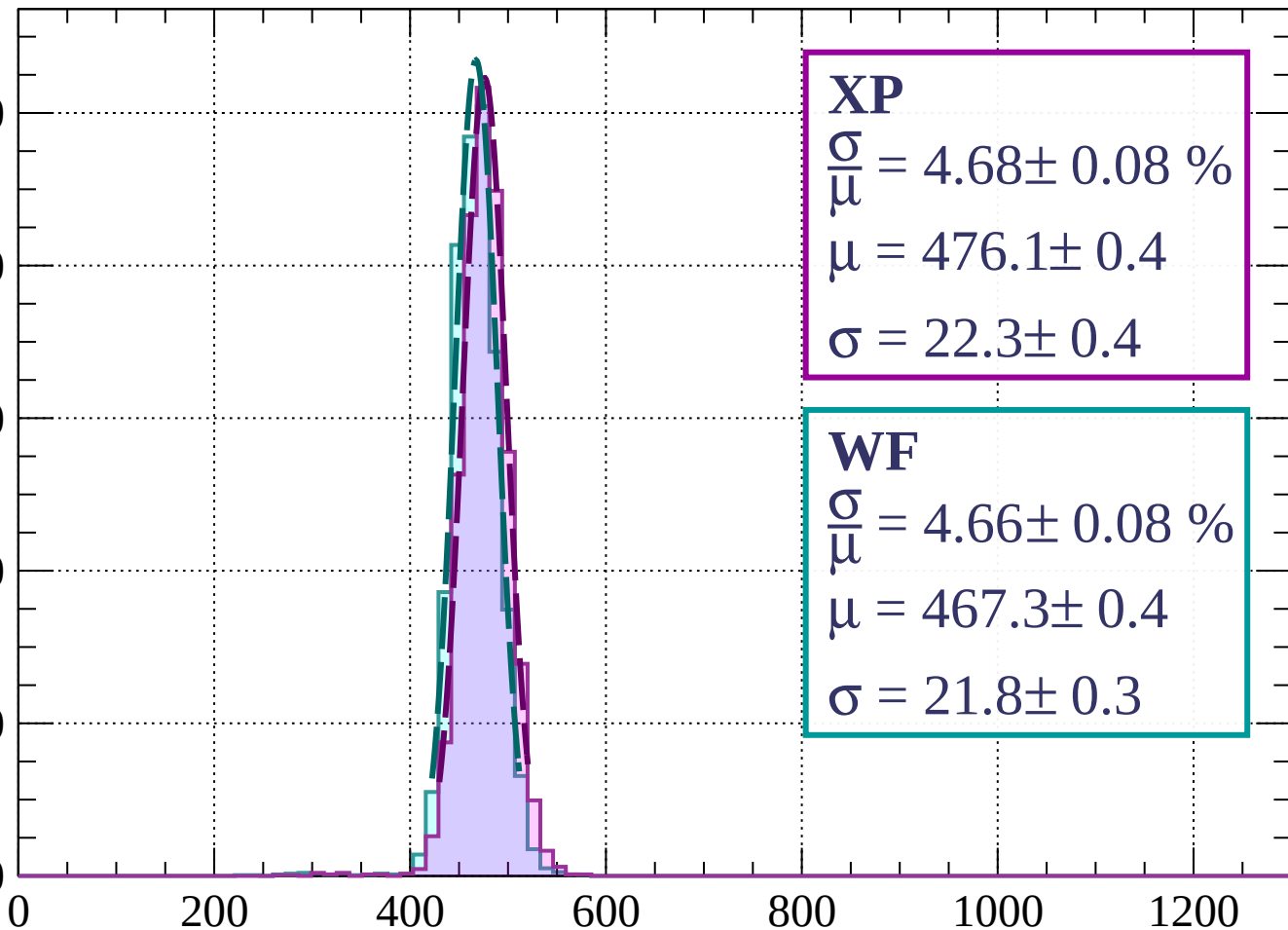
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 4.68 \pm 0.08 \%$$

$$\mu = 476.1 \pm 0.4$$

$$\sigma = 22.3 \pm 0.4$$

WF

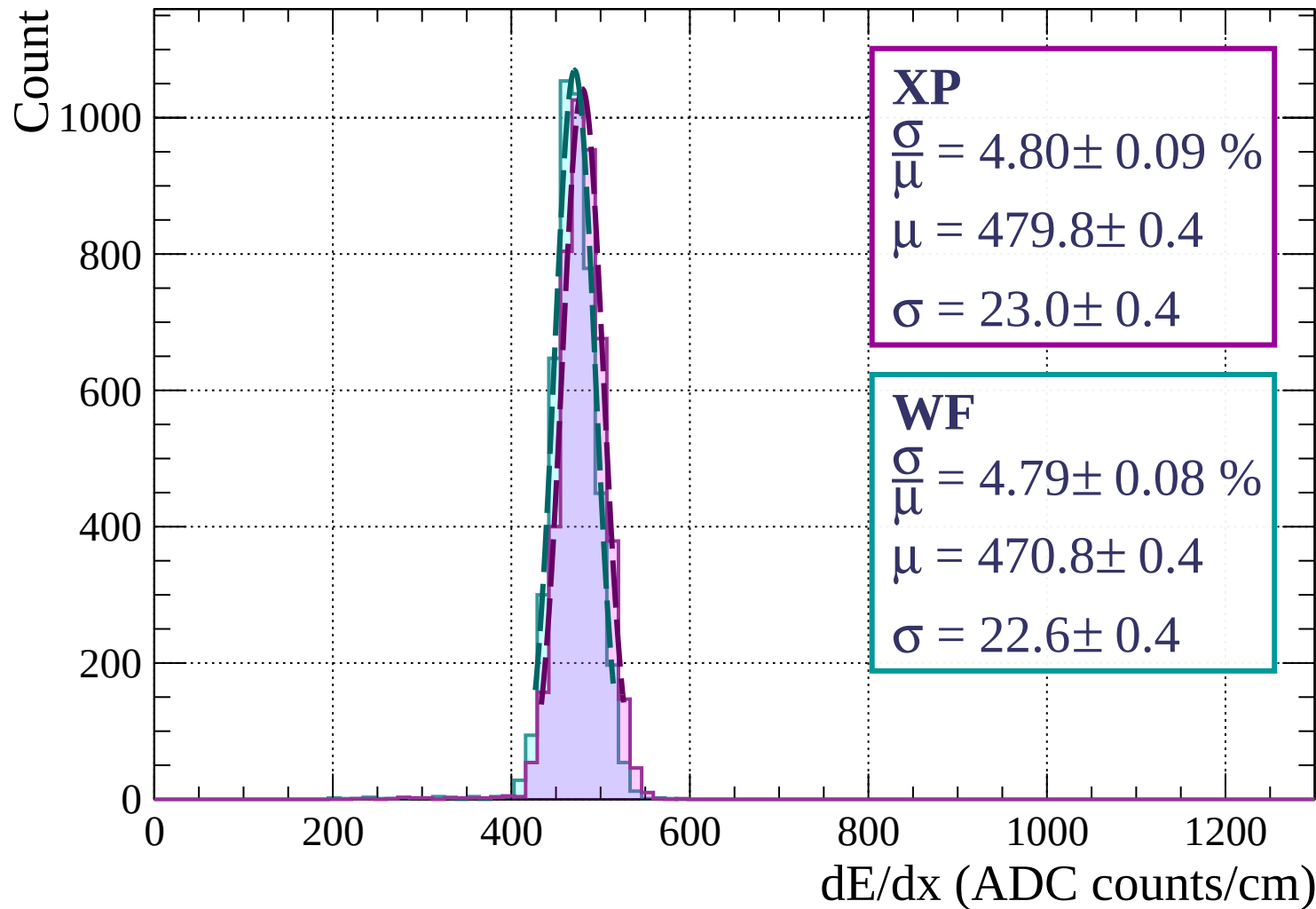
$$\frac{\sigma}{\mu} = 4.66 \pm 0.08 \%$$

$$\mu = 467.3 \pm 0.4$$

$$\sigma = 21.8 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$



Energy loss | $1600 < p < 1680$

Count

1000

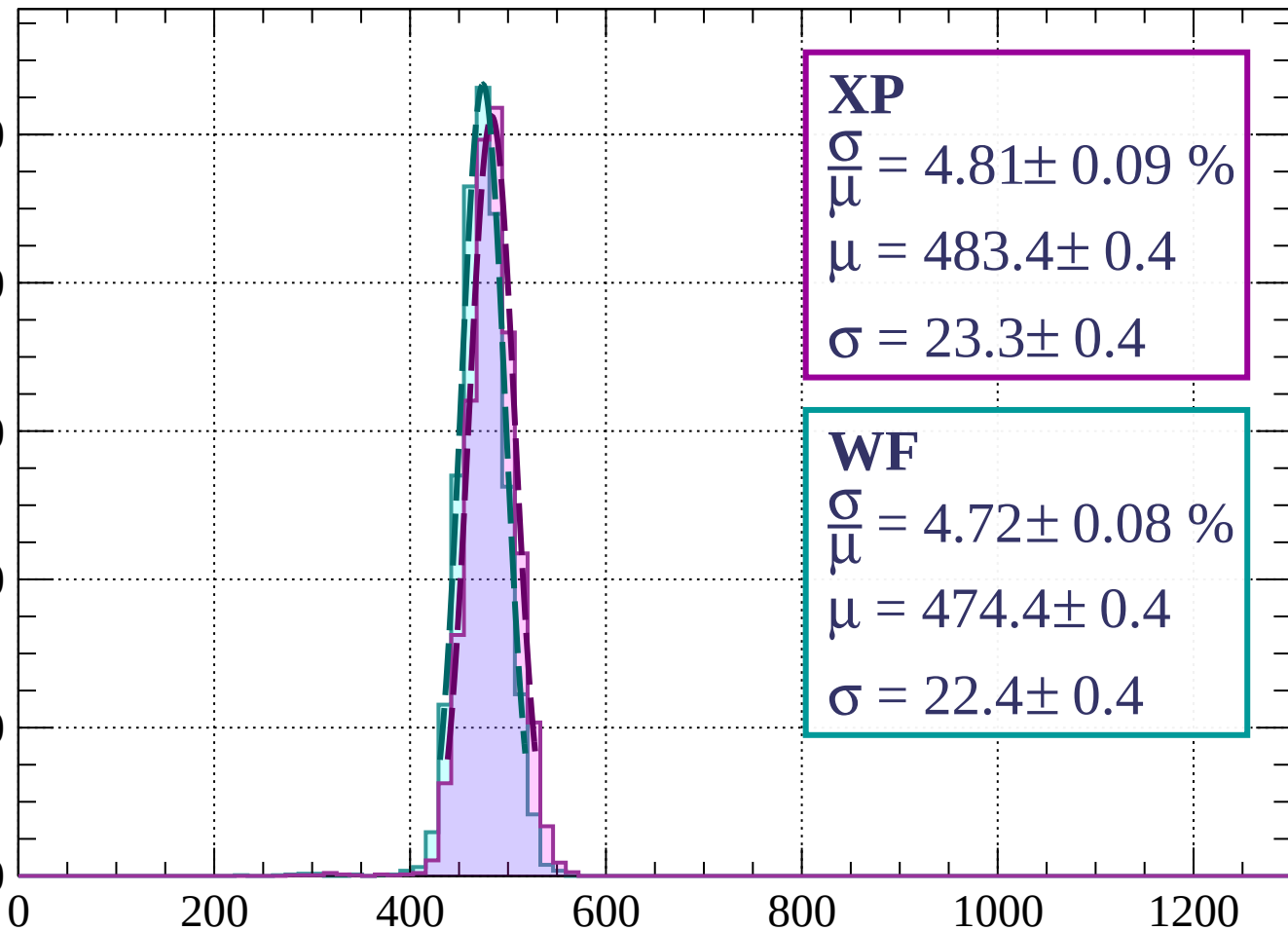
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.09 \%$$

$$\mu = 483.4 \pm 0.4$$

$$\sigma = 23.3 \pm 0.4$$

WF

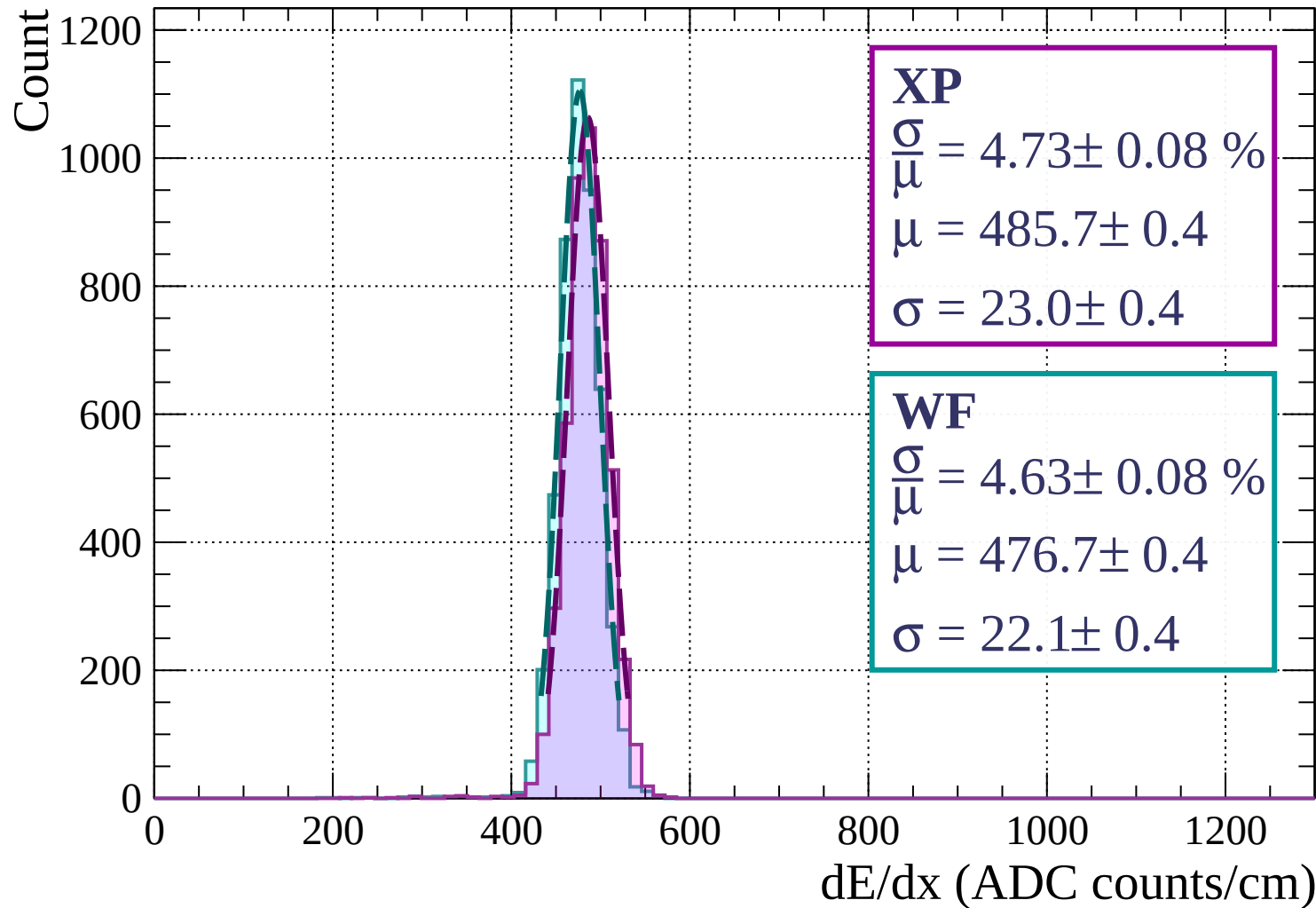
$$\frac{\sigma}{\mu} = 4.72 \pm 0.08 \%$$

$$\mu = 474.4 \pm 0.4$$

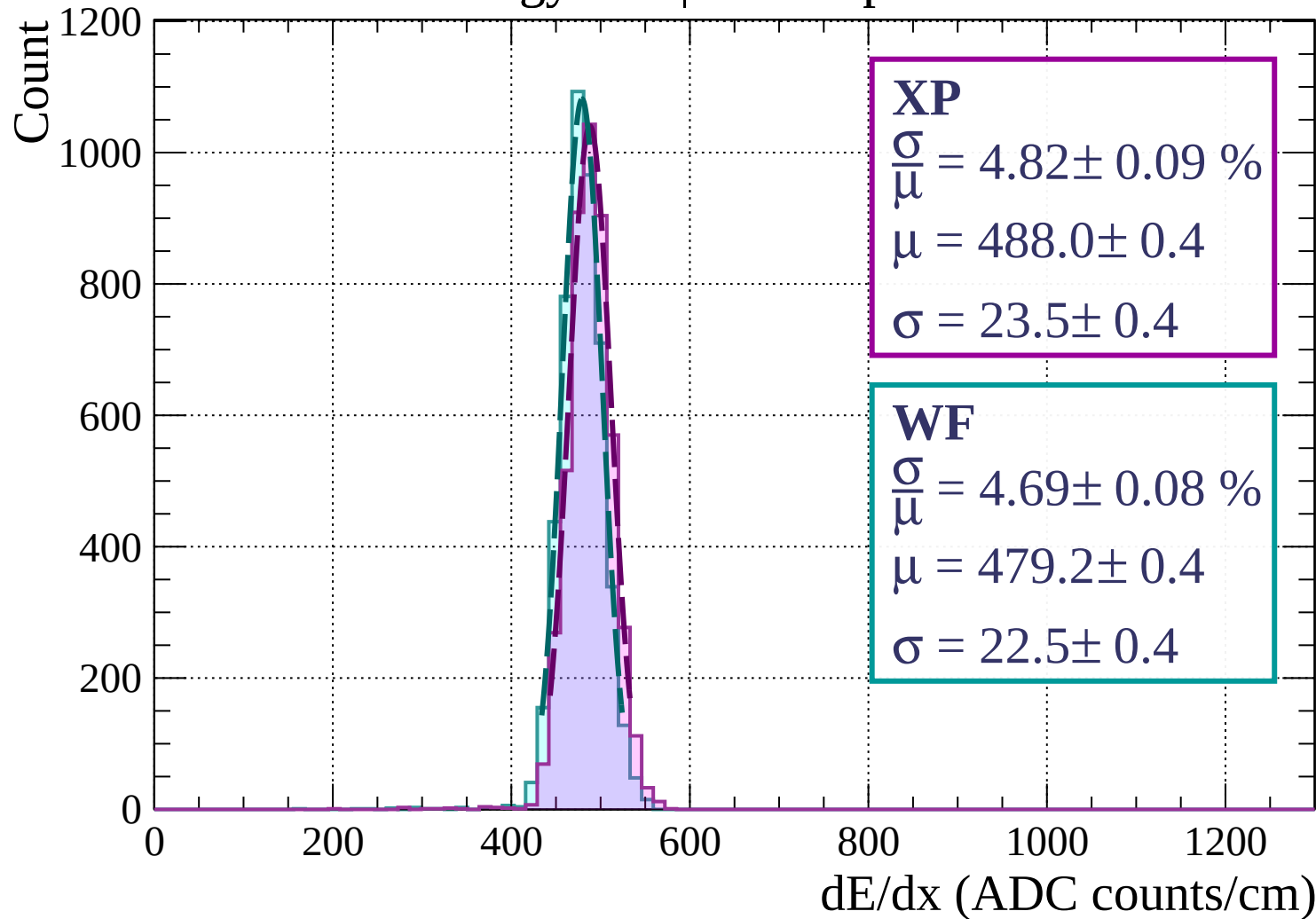
$$\sigma = 22.4 \pm 0.4$$

dE/dx (ADC counts/cm)

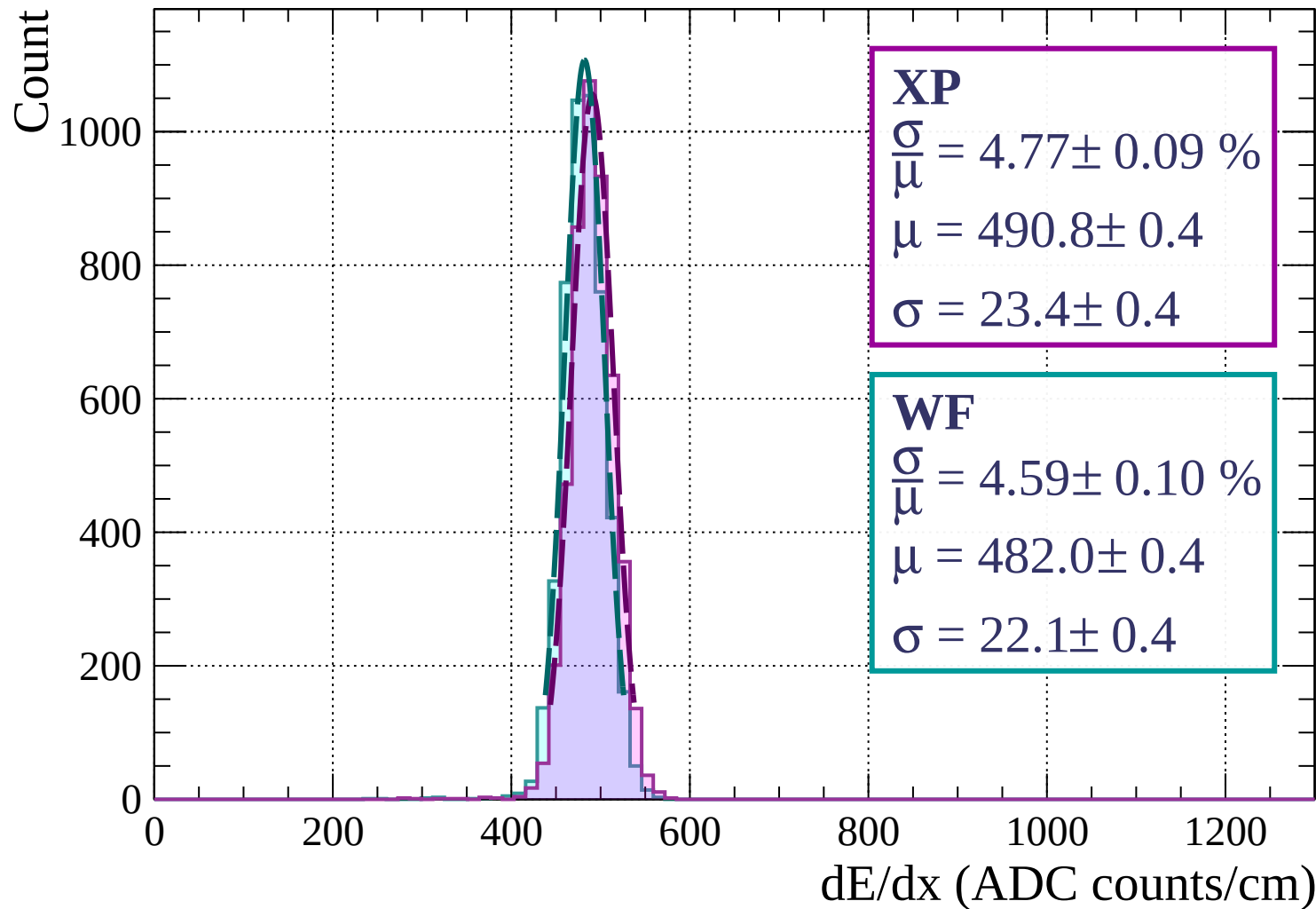
Energy loss | $1680 < p < 1760$



Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$

Count

1000

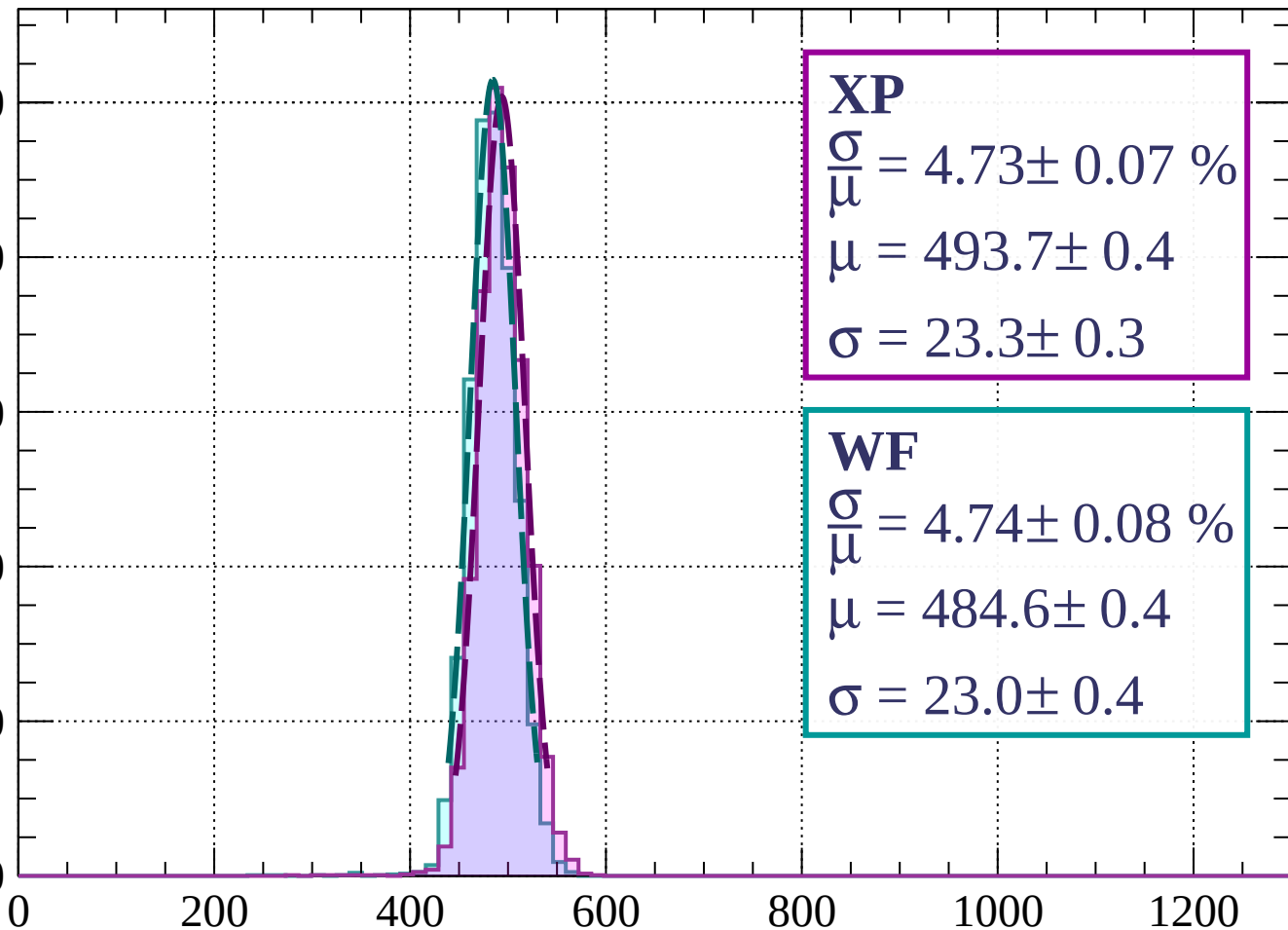
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 4.73 \pm 0.07 \%$$

$$\mu = 493.7 \pm 0.4$$

$$\sigma = 23.3 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 4.74 \pm 0.08 \%$$

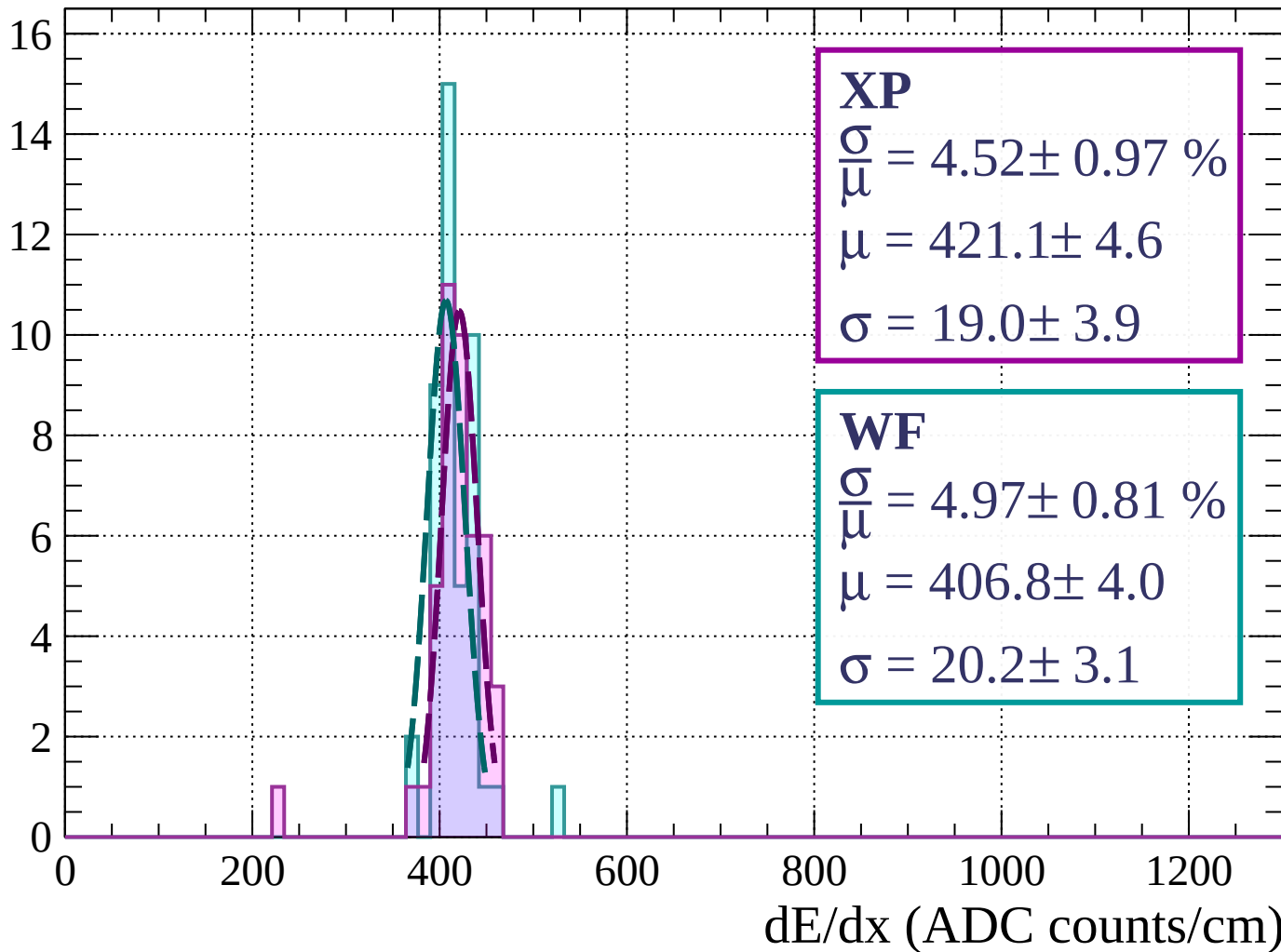
$$\mu = 484.6 \pm 0.4$$

$$\sigma = 23.0 \pm 0.4$$

dE/dx (ADC counts/cm)

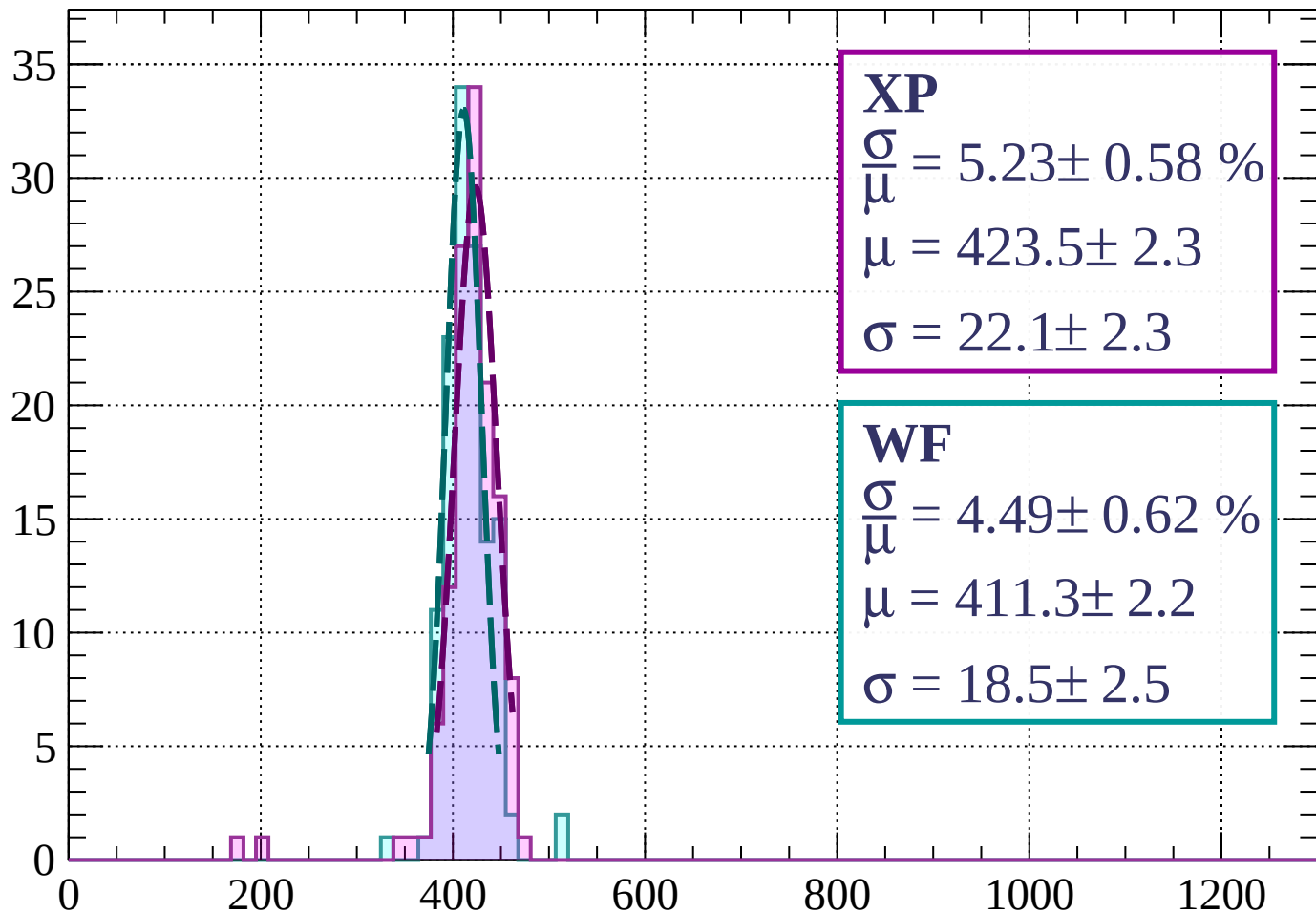
Energy loss | $0 < \phi < 3$

Count



Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.58 \%$$

$$\mu = 423.5 \pm 2.3$$

$$\sigma = 22.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 4.49 \pm 0.62 \%$$

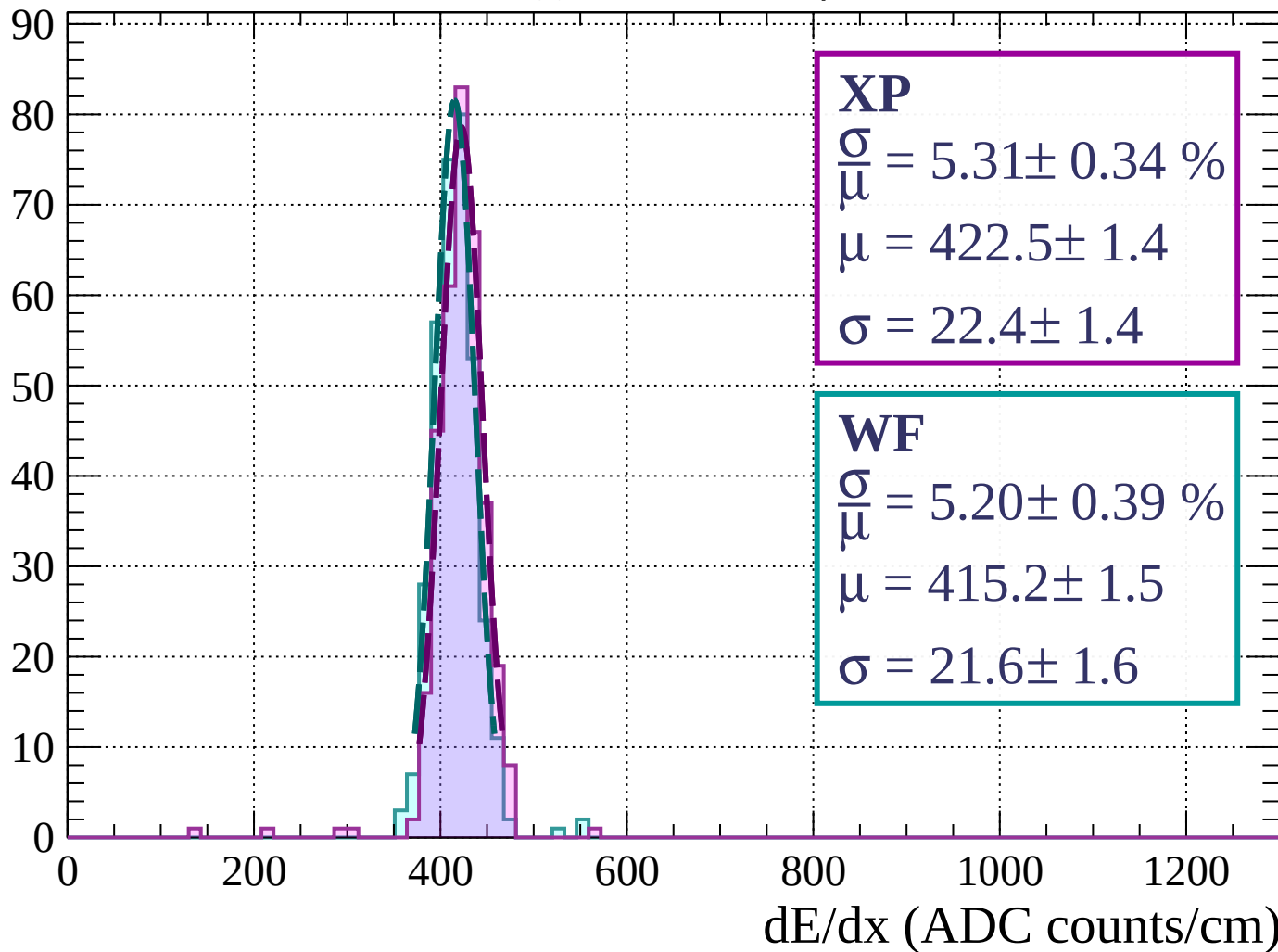
$$\mu = 411.3 \pm 2.2$$

$$\sigma = 18.5 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$

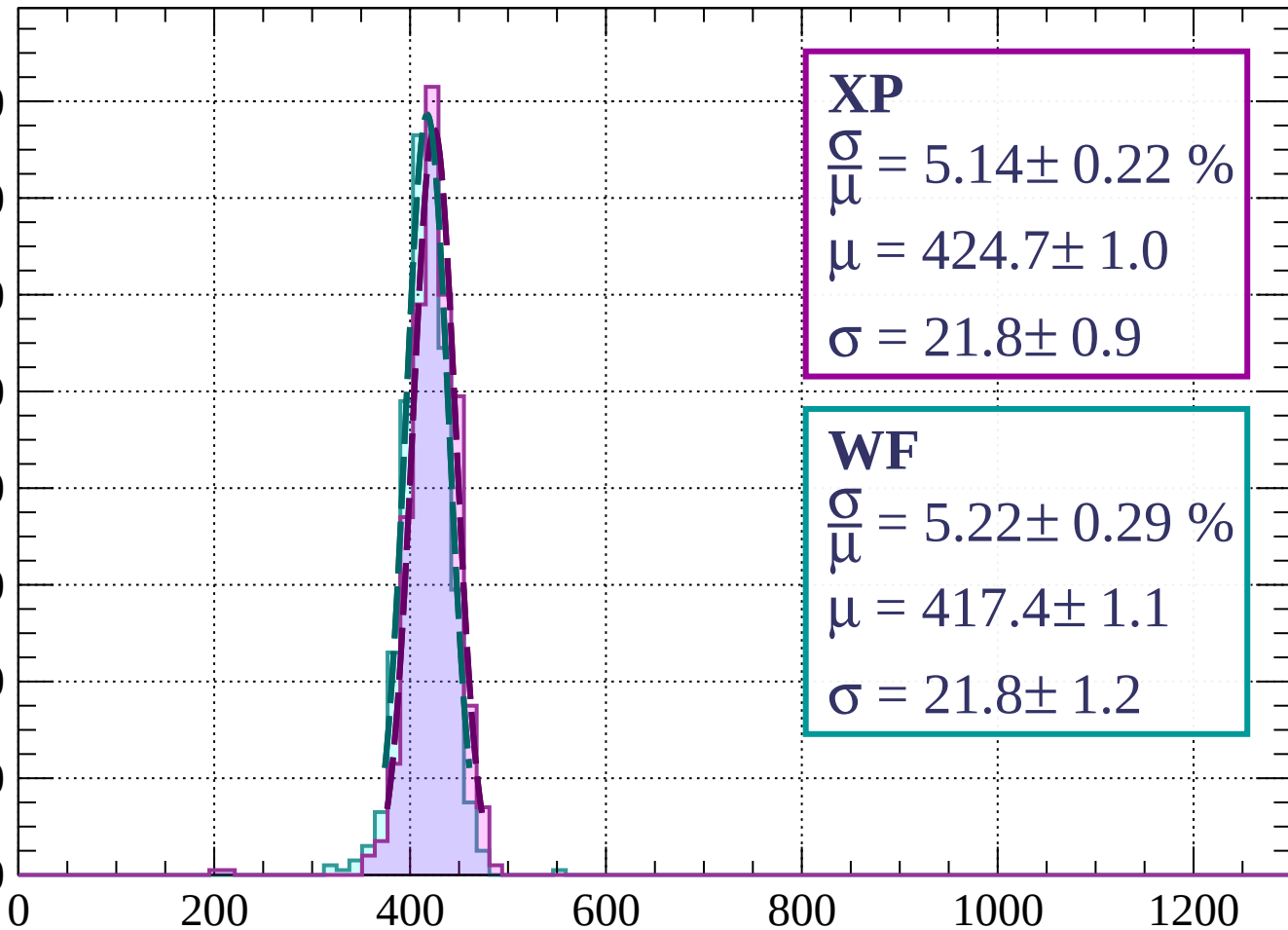
Count



Energy loss | $9 < \phi < 12$

Count

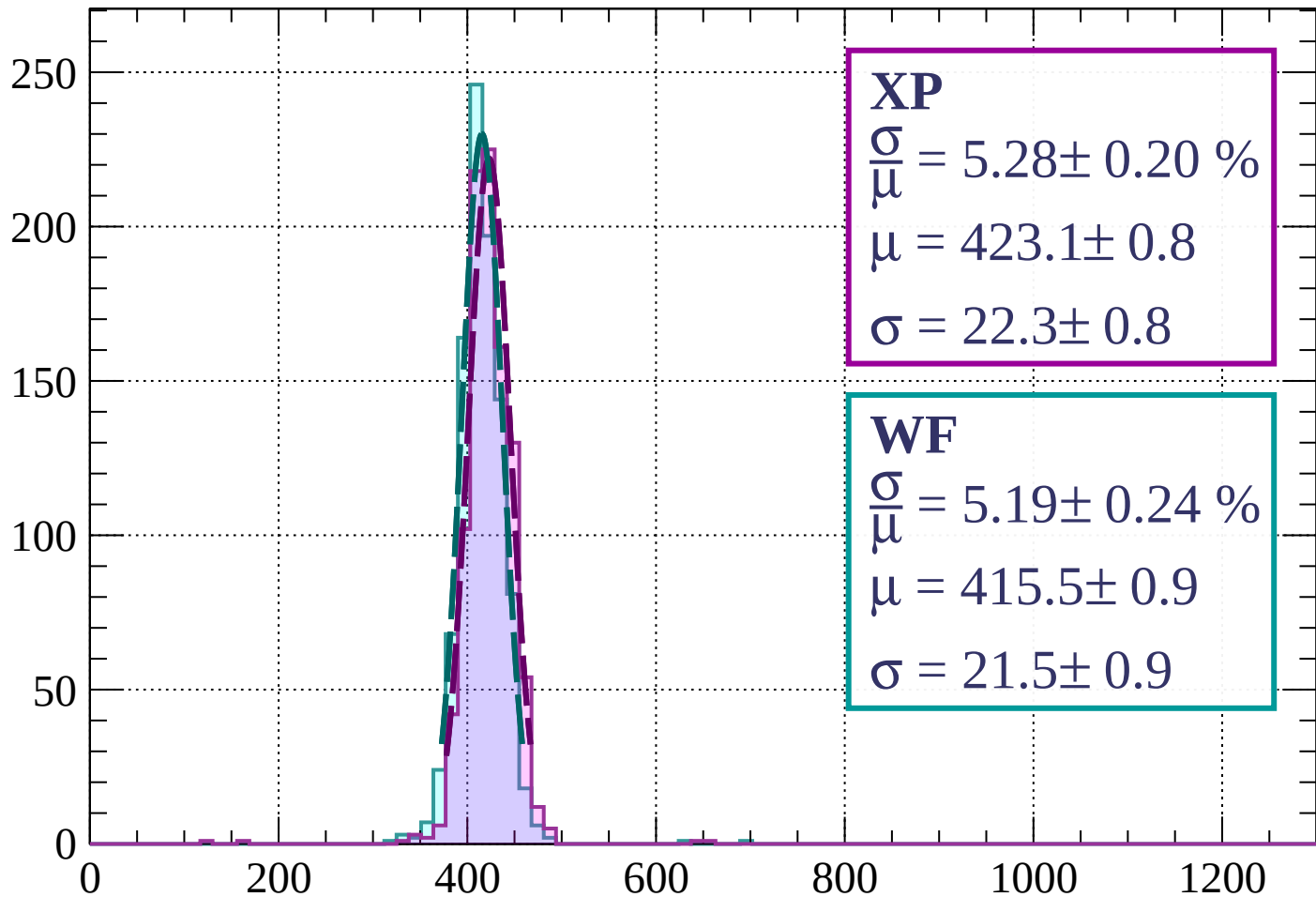
160
140
120
100
80
60
40
20
0



dE/dx (ADC counts/cm)

Energy loss | $12 < \phi < 15$

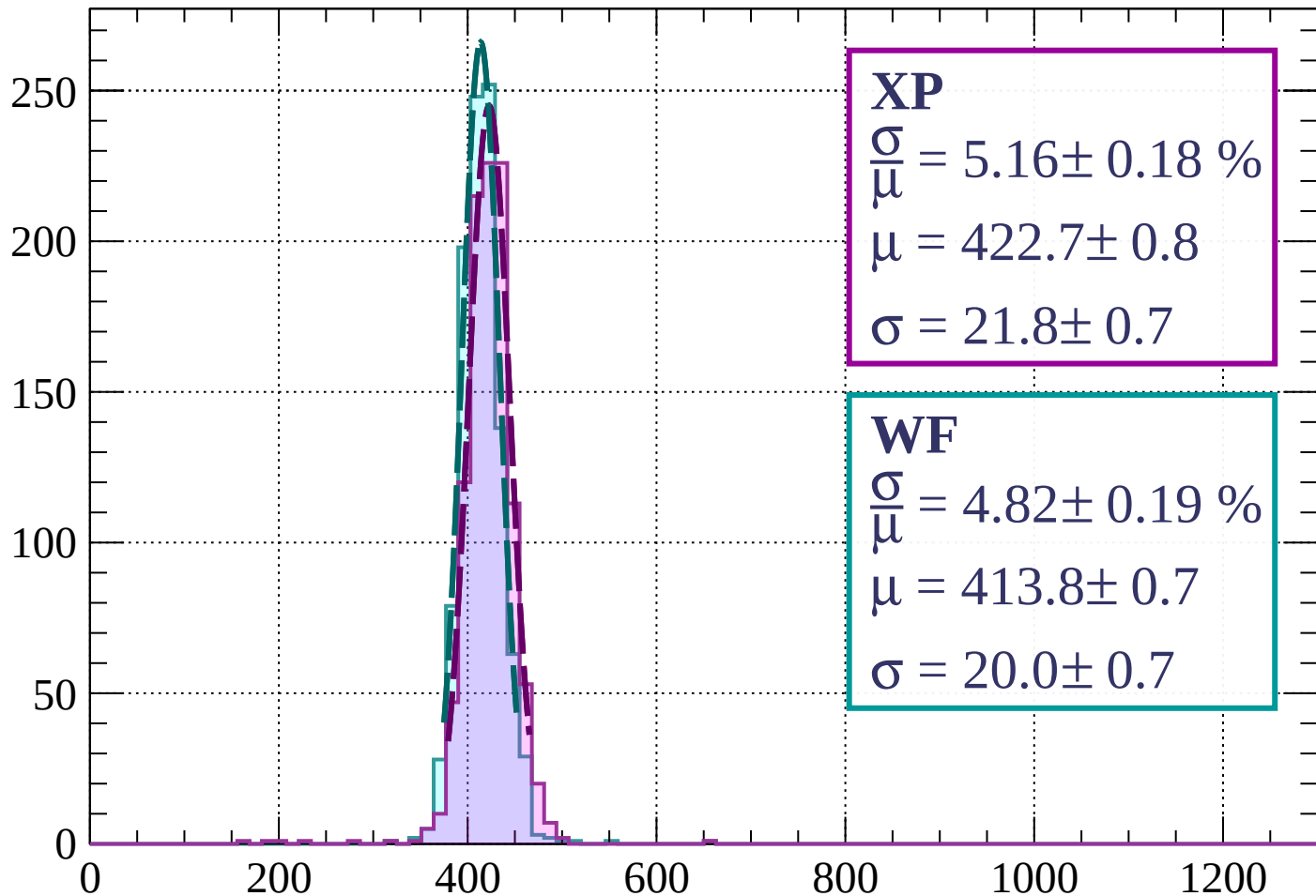
Count



dE/dx (ADC counts/cm)

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 5.16 \pm 0.18 \%$$

$$\mu = 422.7 \pm 0.8$$

$$\sigma = 21.8 \pm 0.7$$

WF

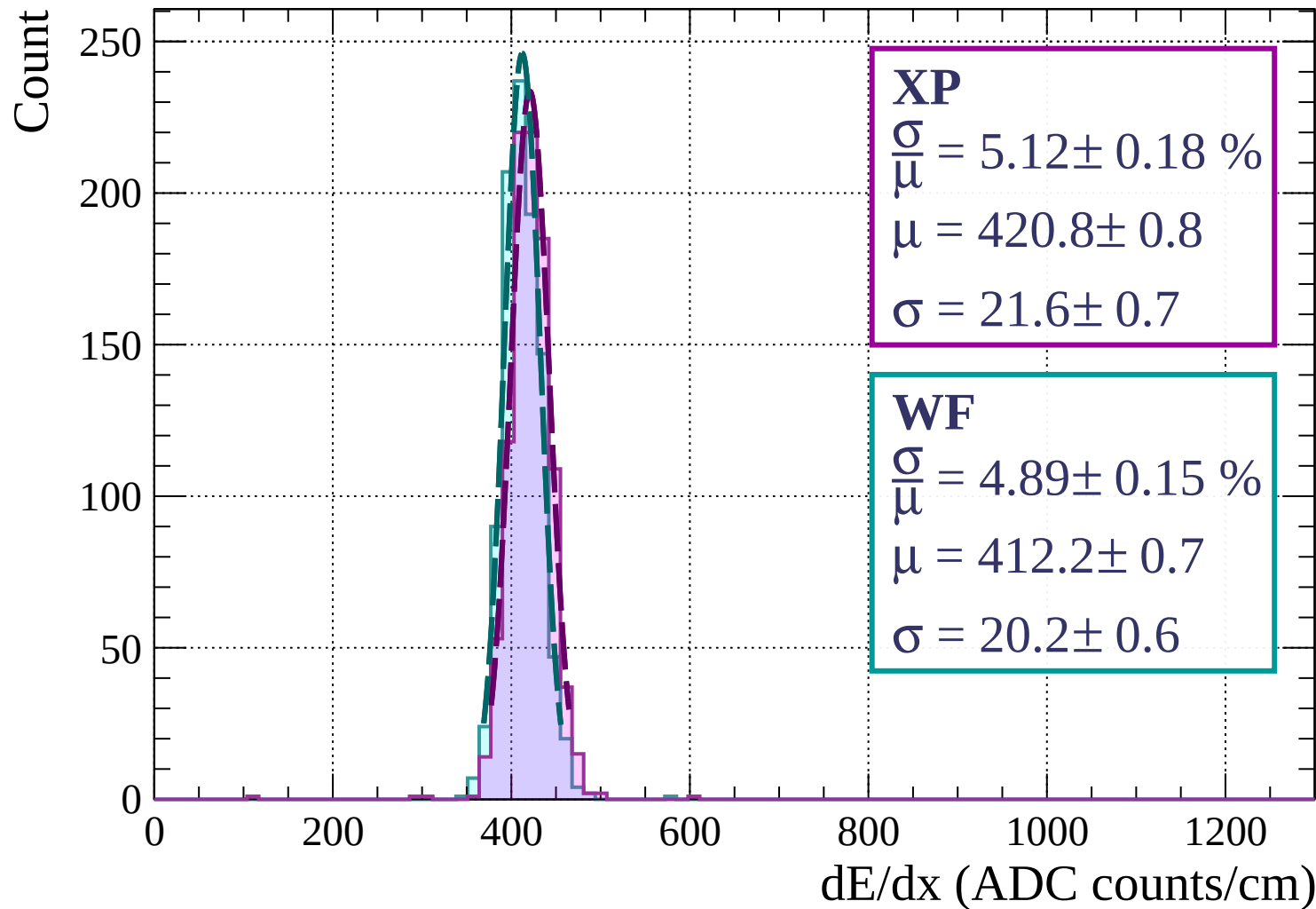
$$\frac{\sigma}{\mu} = 4.82 \pm 0.19 \%$$

$$\mu = 413.8 \pm 0.7$$

$$\sigma = 20.0 \pm 0.7$$

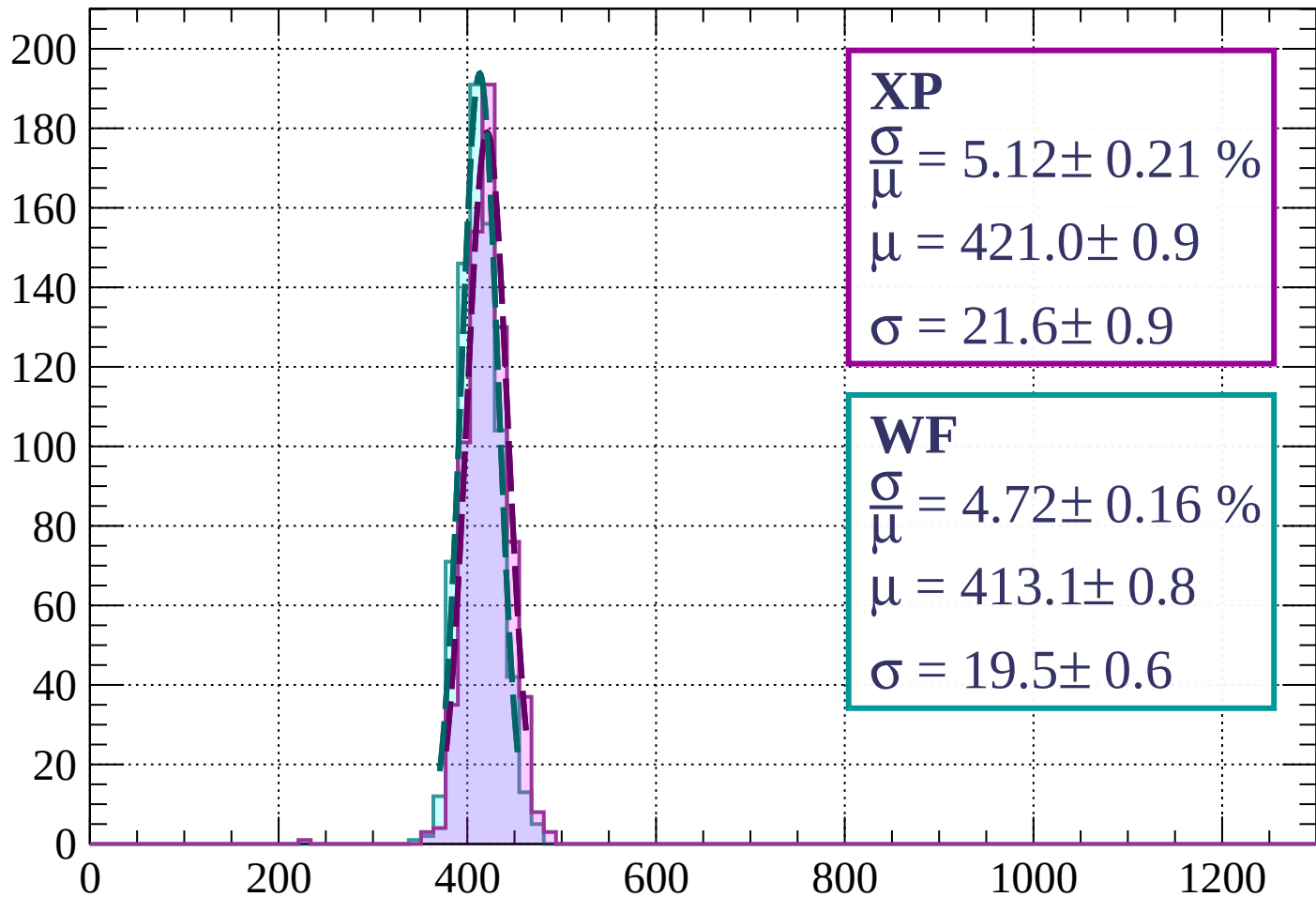
dE/dx (ADC counts/cm)

Energy loss | $18 < \phi < 21$



Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.12 \pm 0.21 \%$$

$$\mu = 421.0 \pm 0.9$$

$$\sigma = 21.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.72 \pm 0.16 \%$$

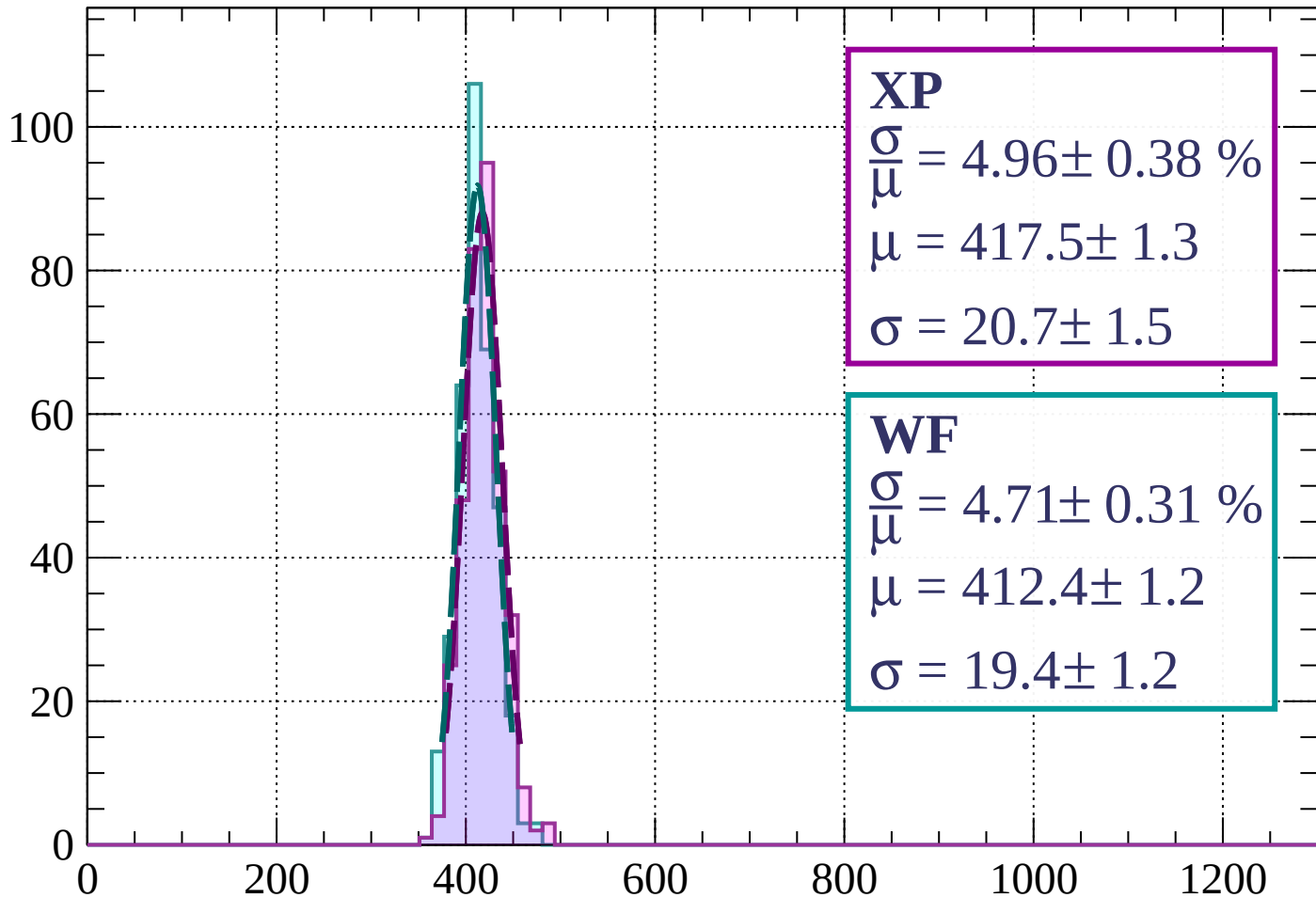
$$\mu = 413.1 \pm 0.8$$

$$\sigma = 19.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.38 \%$$

$$\mu = 417.5 \pm 1.3$$

$$\sigma = 20.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 4.71 \pm 0.31 \%$$

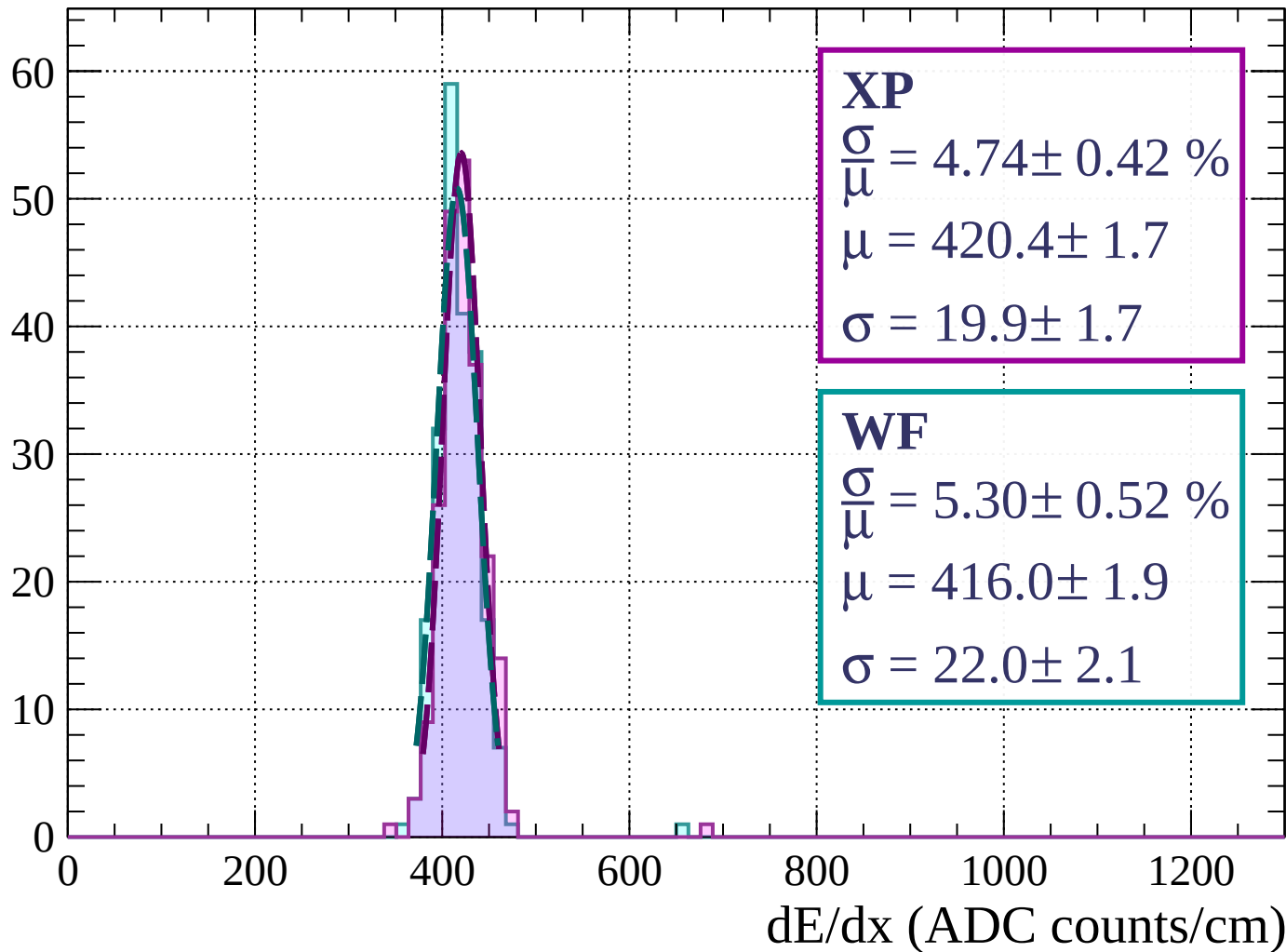
$$\mu = 412.4 \pm 1.2$$

$$\sigma = 19.4 \pm 1.2$$

dE/dx (ADC counts/cm)

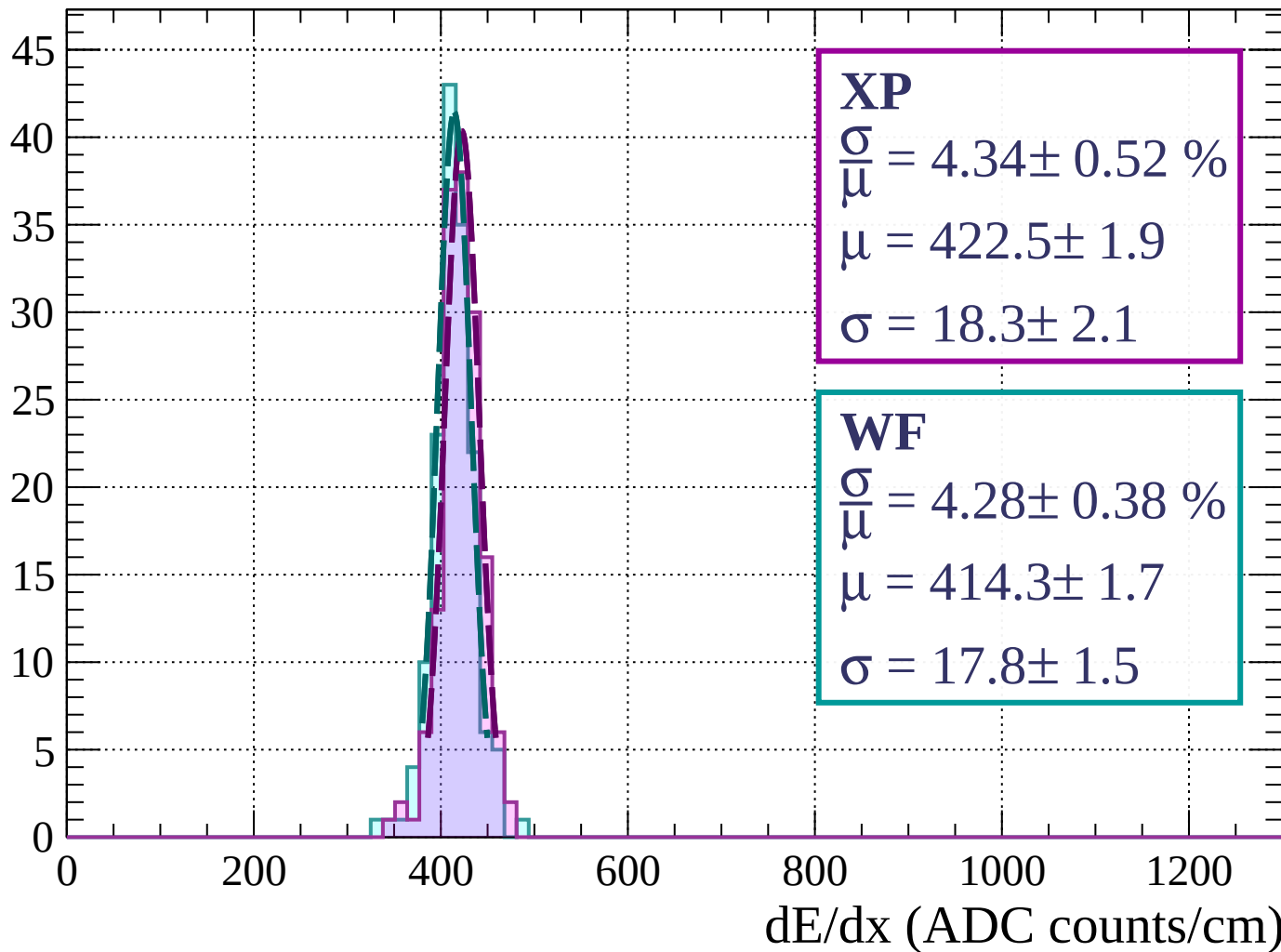
Energy loss | $27 < \phi < 30$

Count



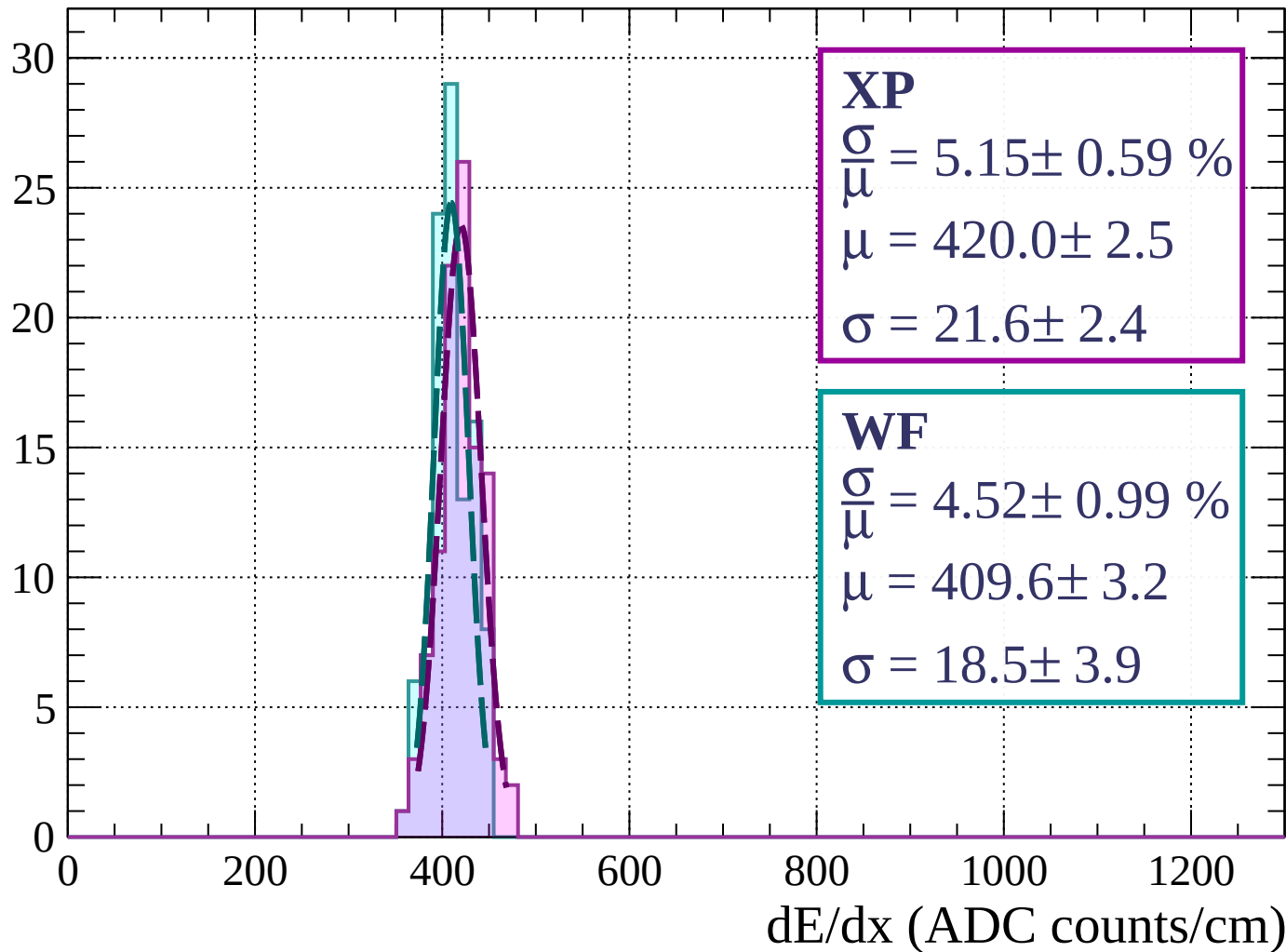
Energy loss | $30 < \phi < 33$

Count



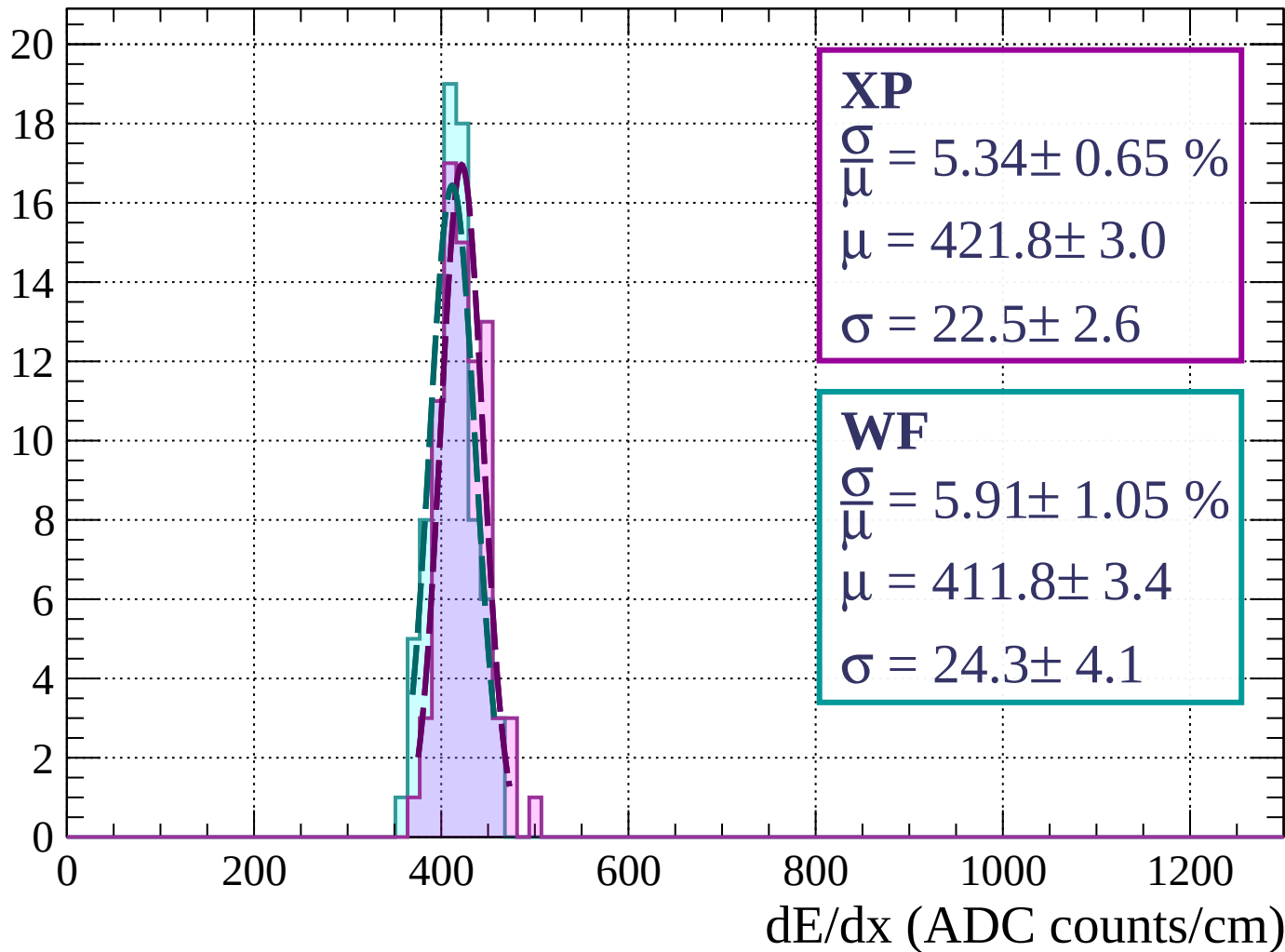
Energy loss | $33 < \phi < 36$

Count



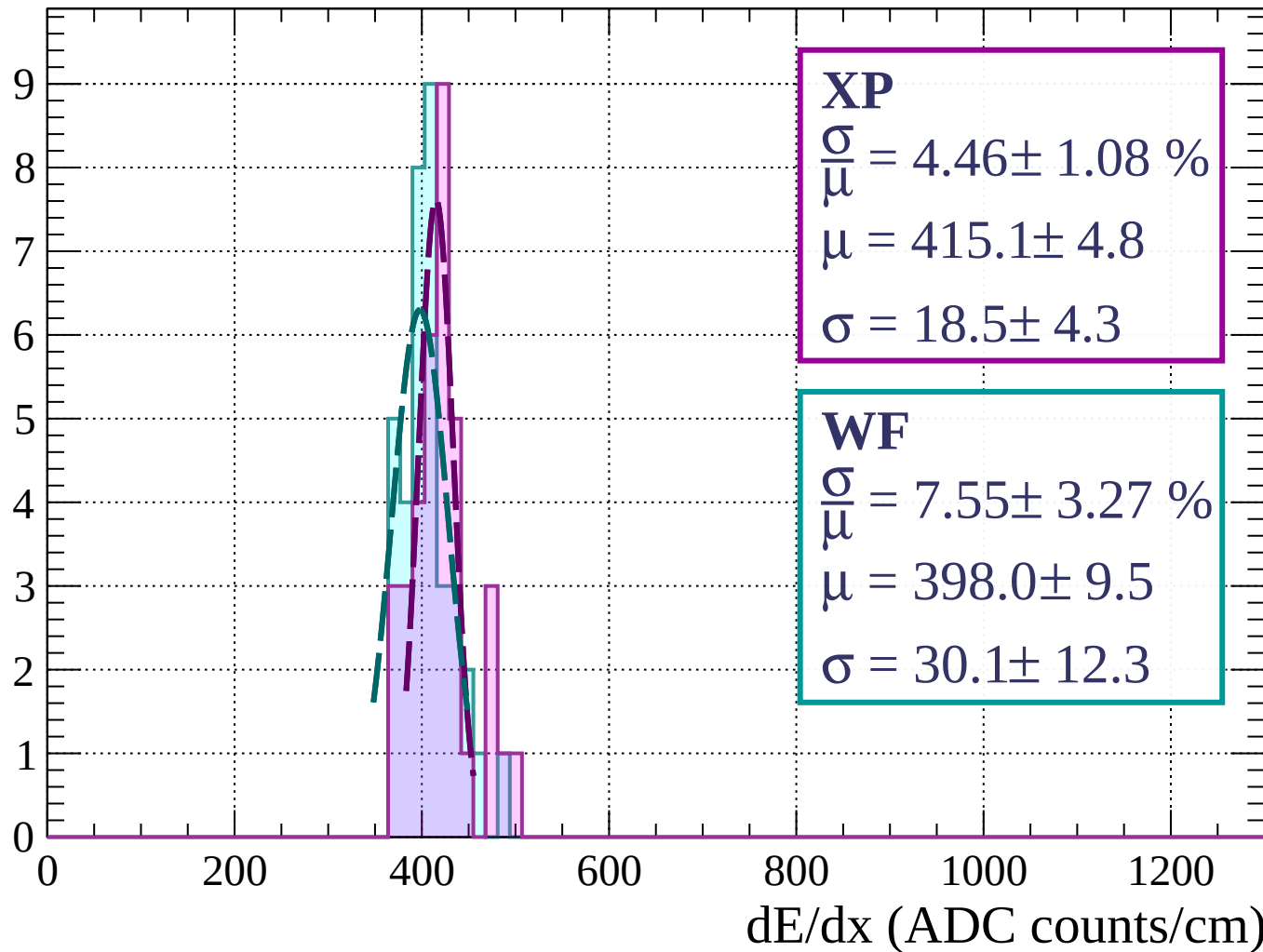
Energy loss | $36 < \phi < 39$

Count



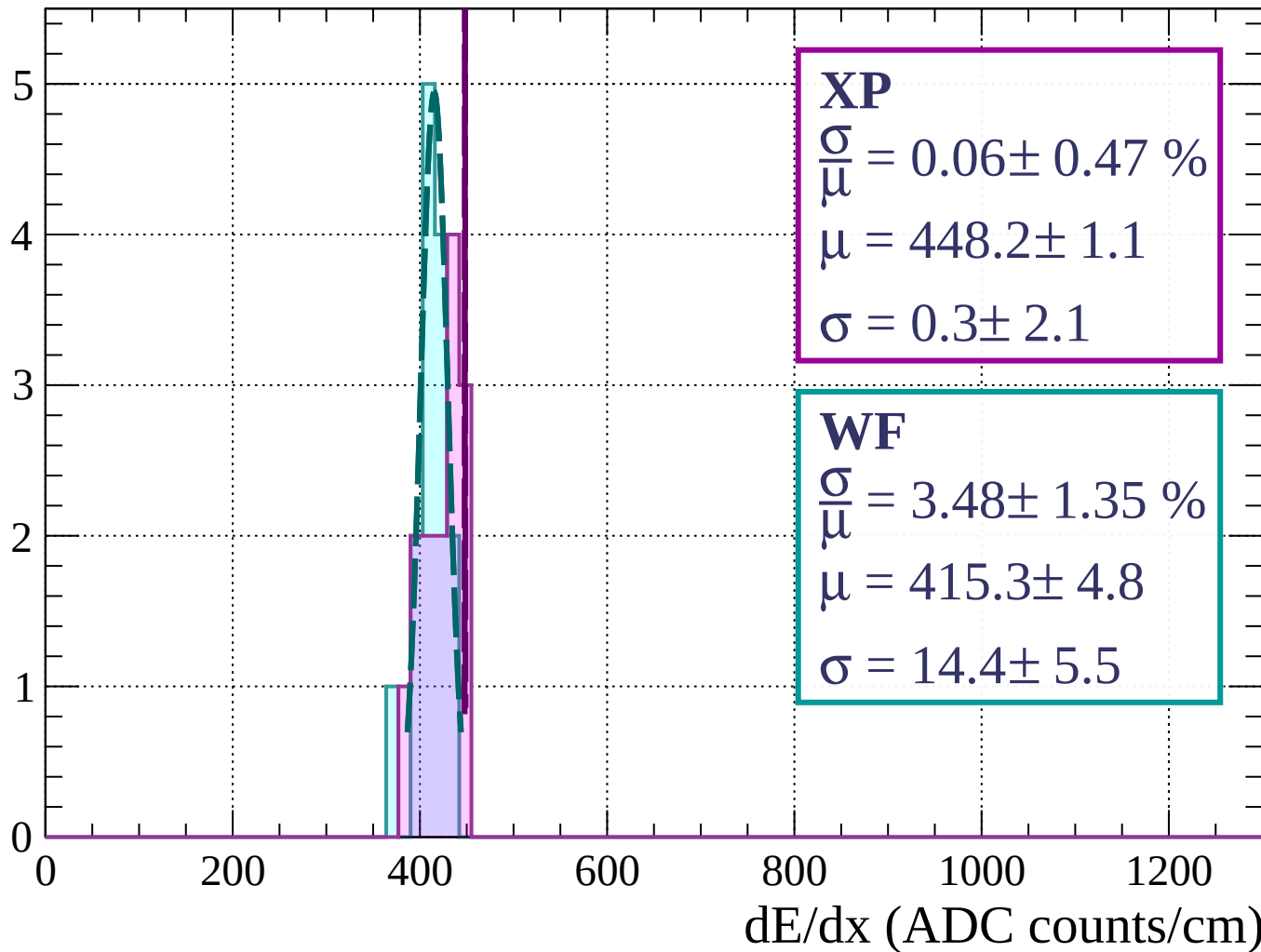
Energy loss | $39 < \phi < 42$

Count



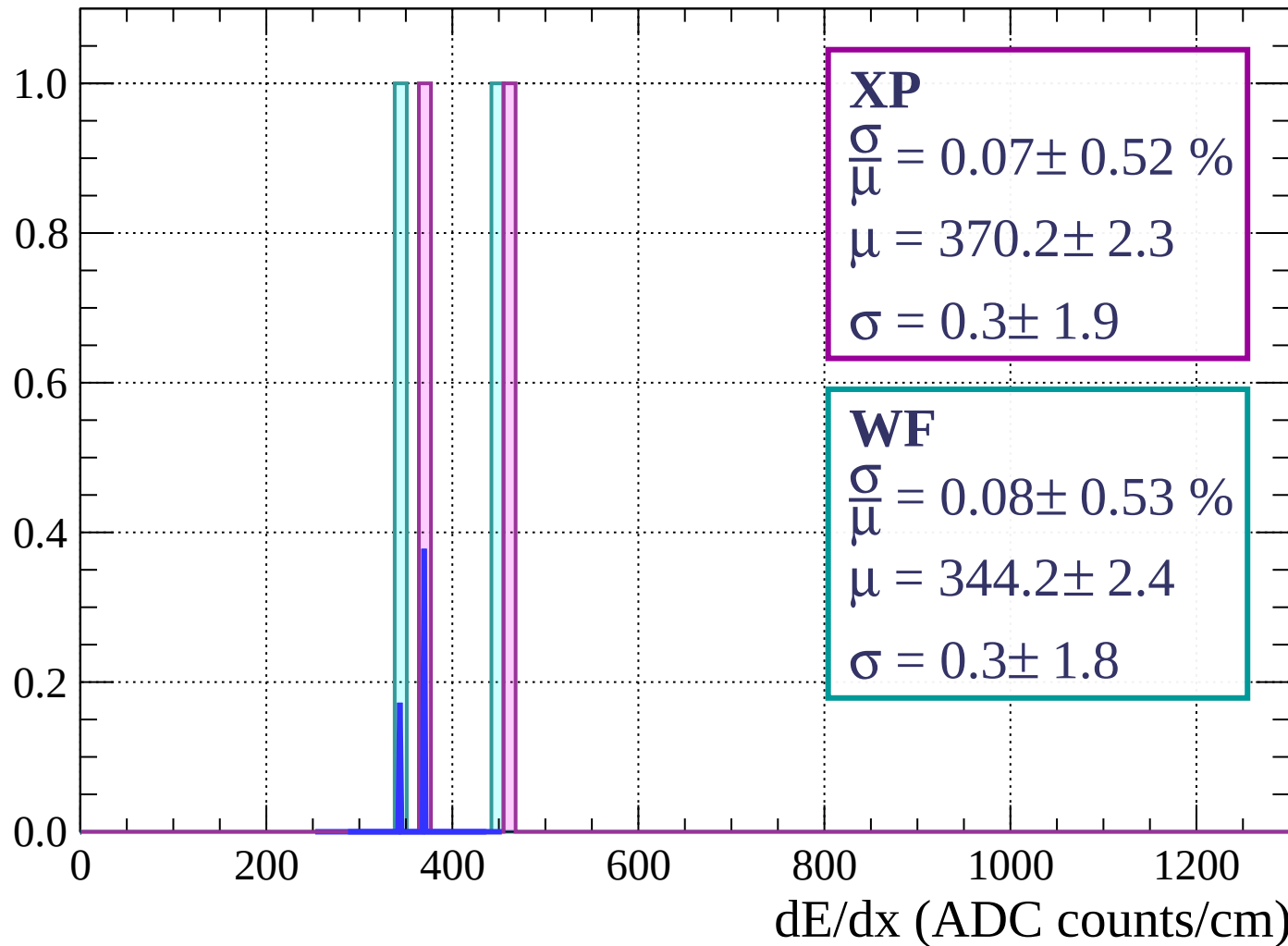
Energy loss | $42 < \phi < 45$

Count



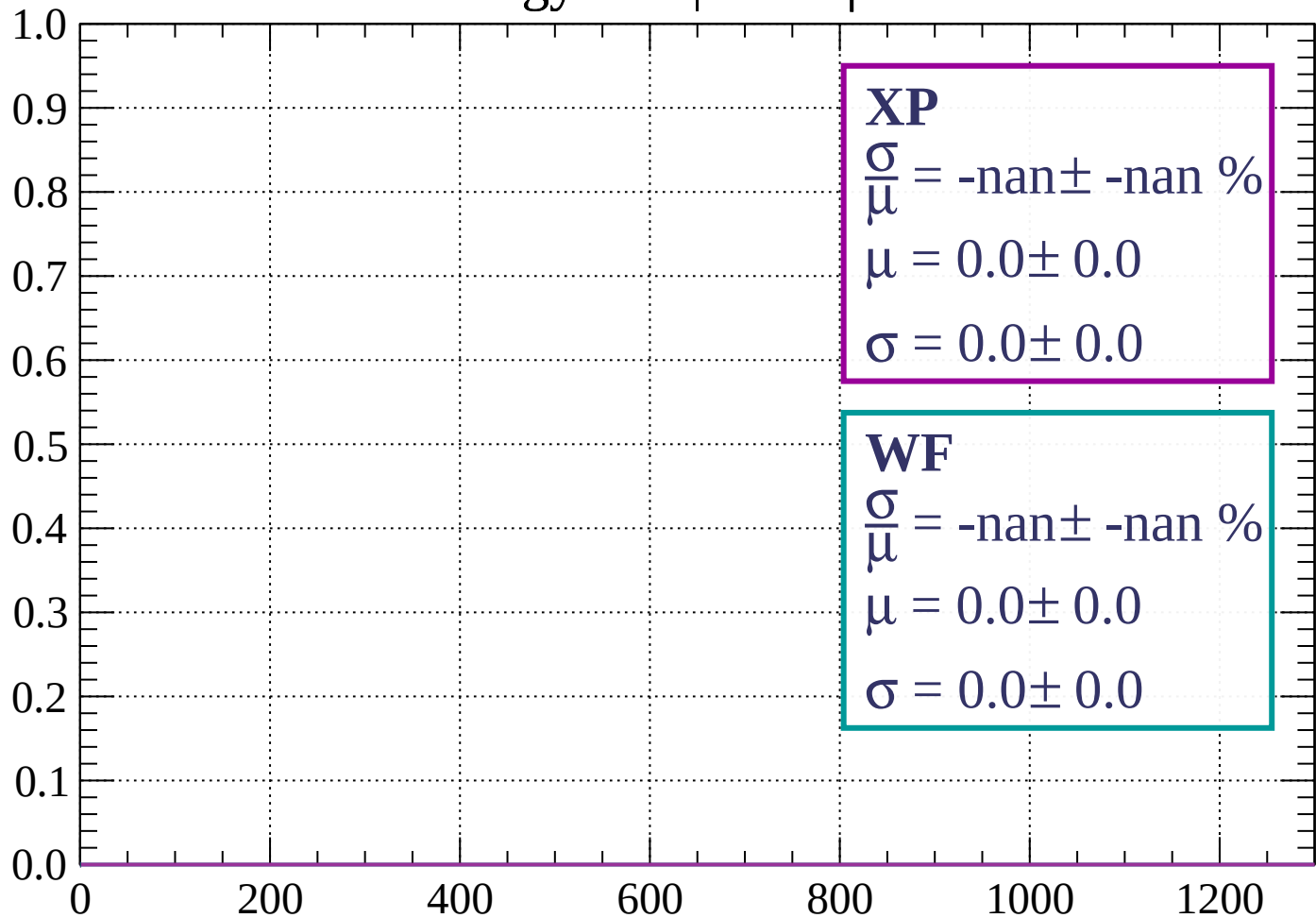
Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

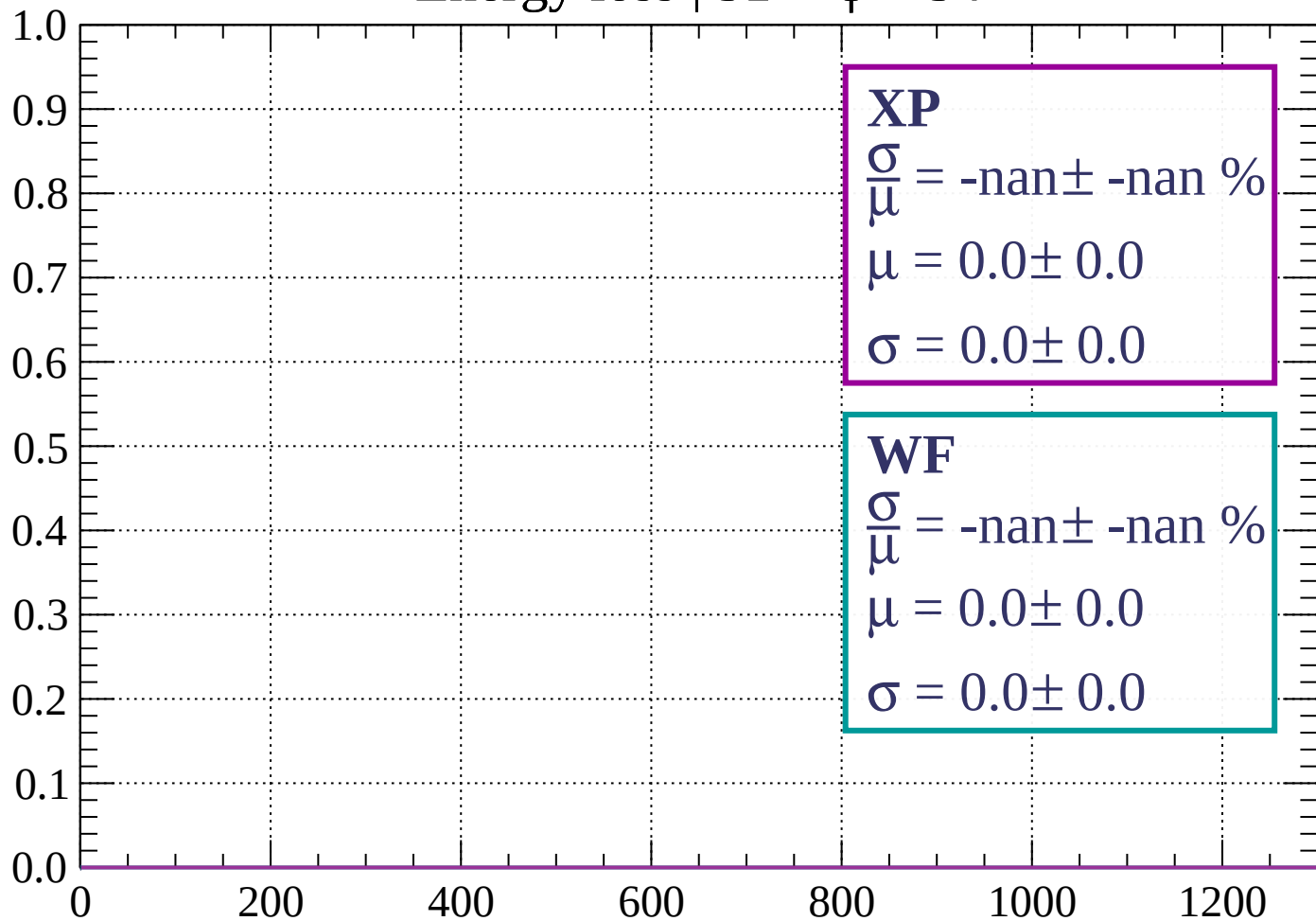
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

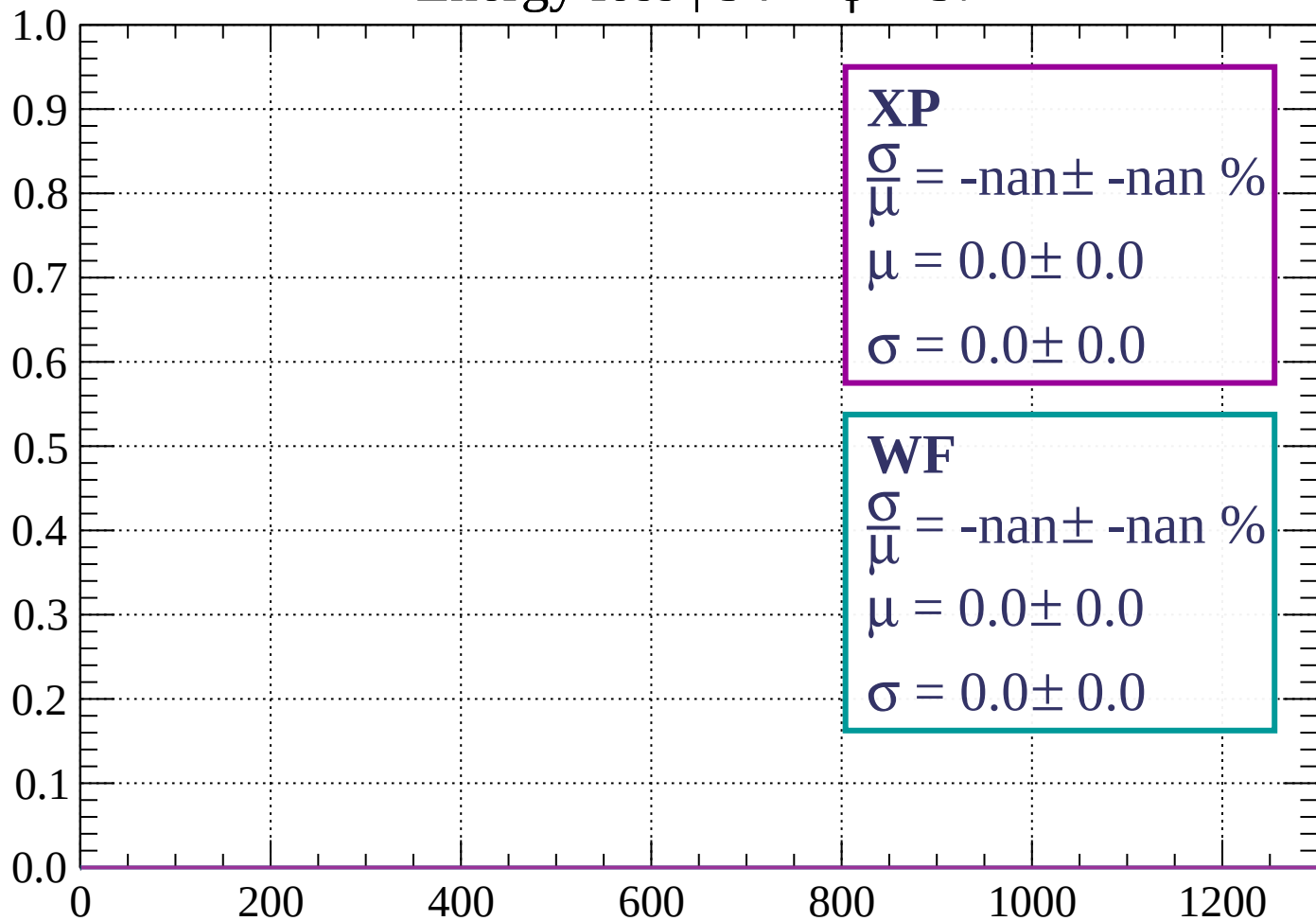
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

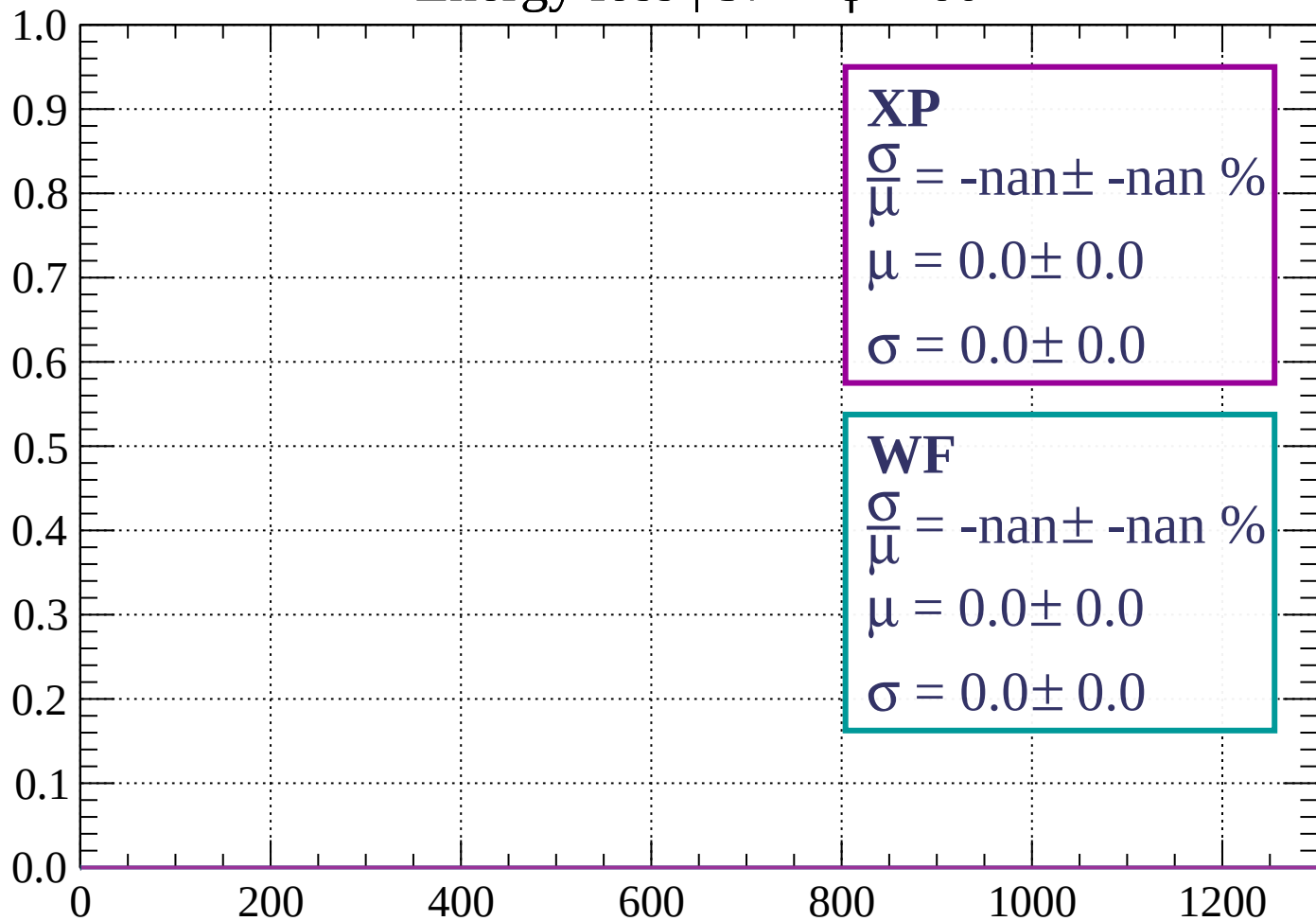
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

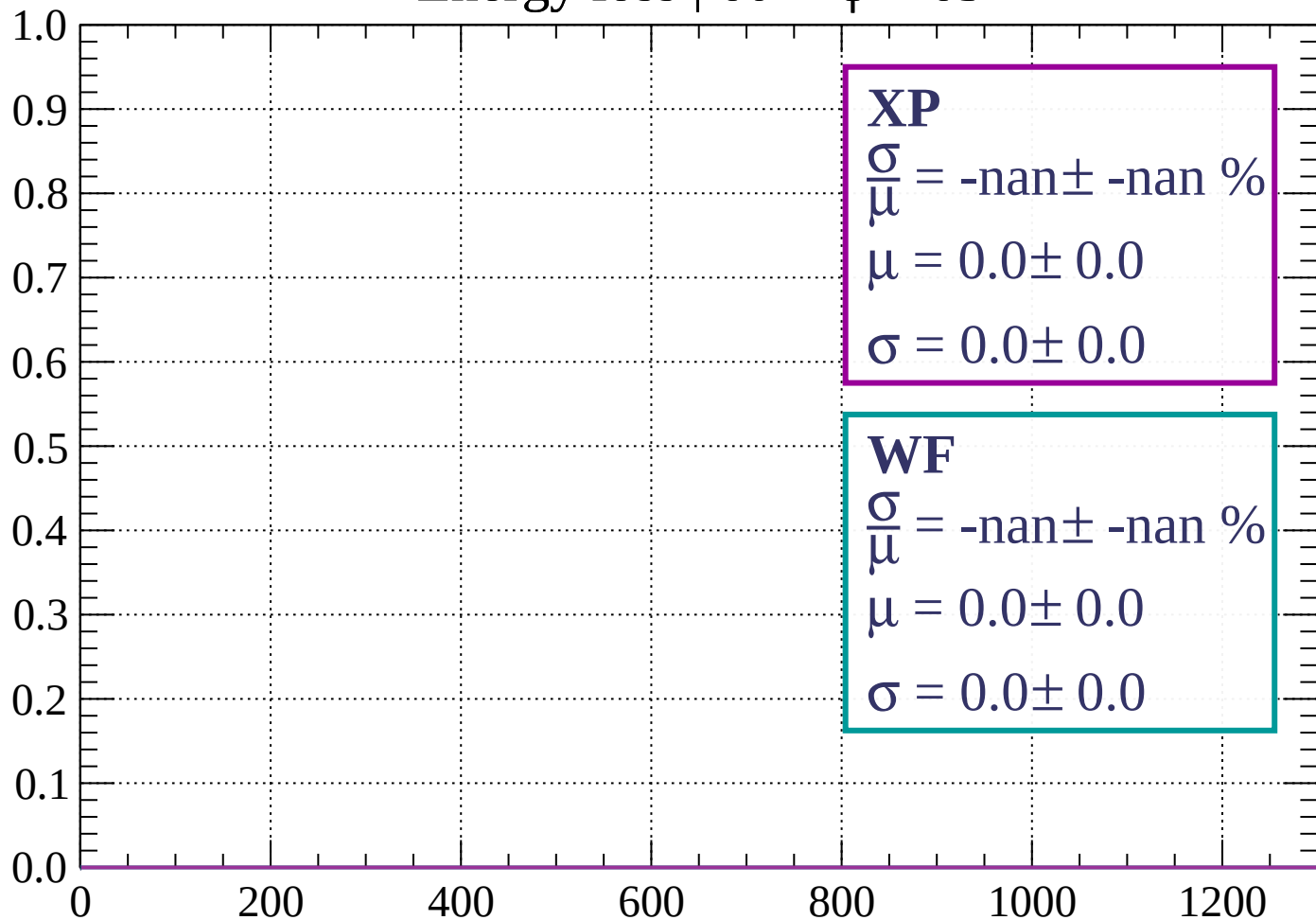
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

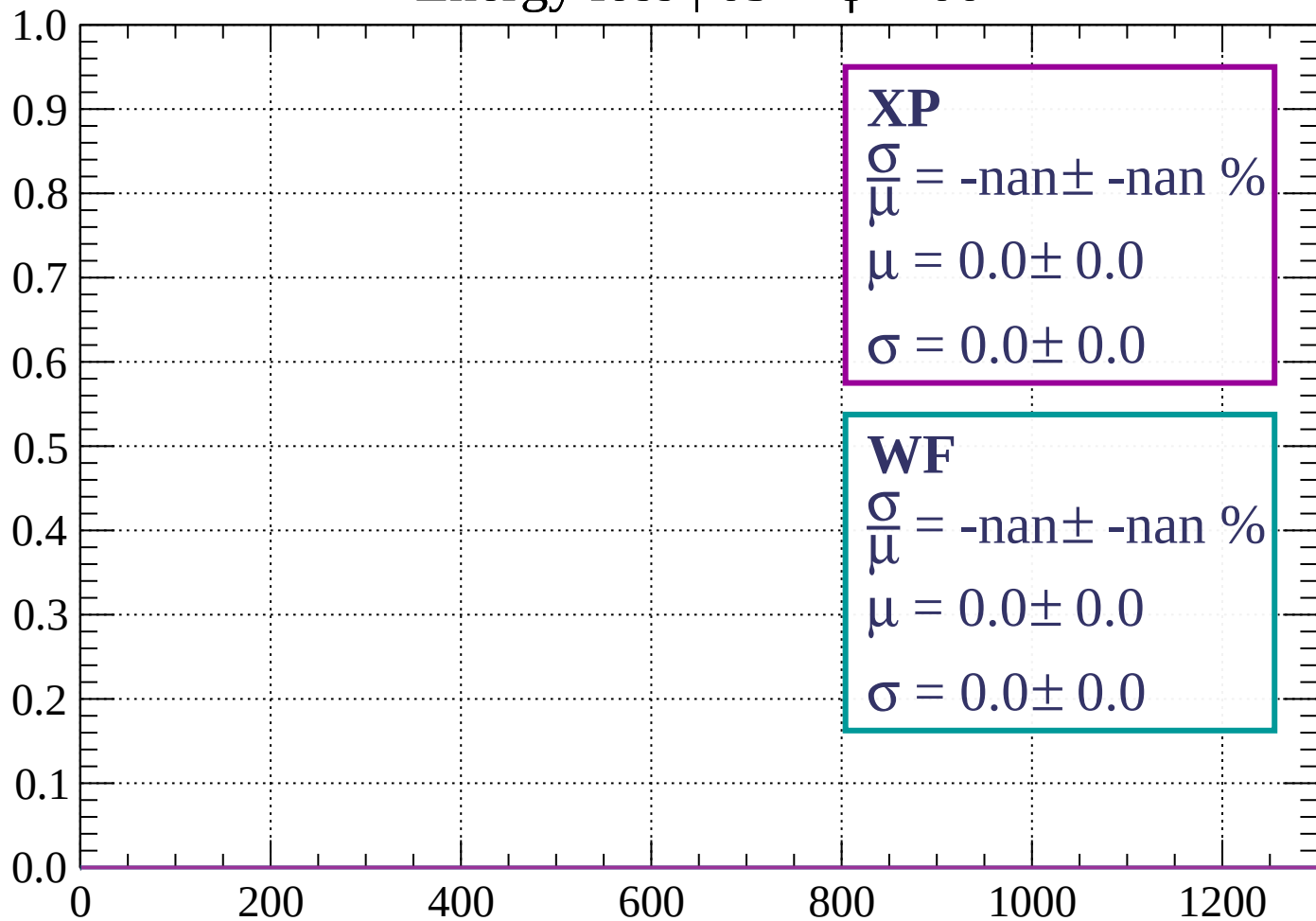
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

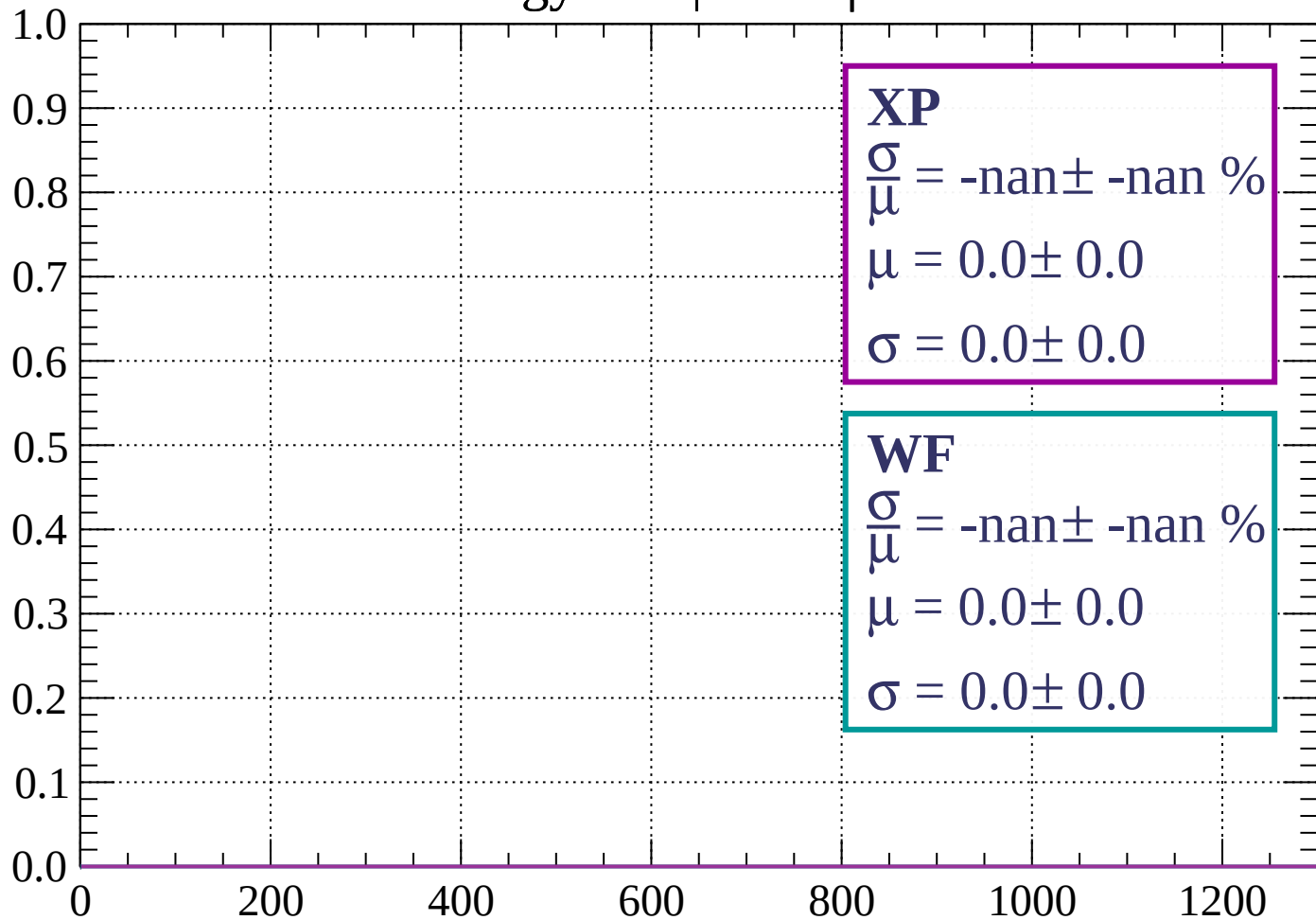
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

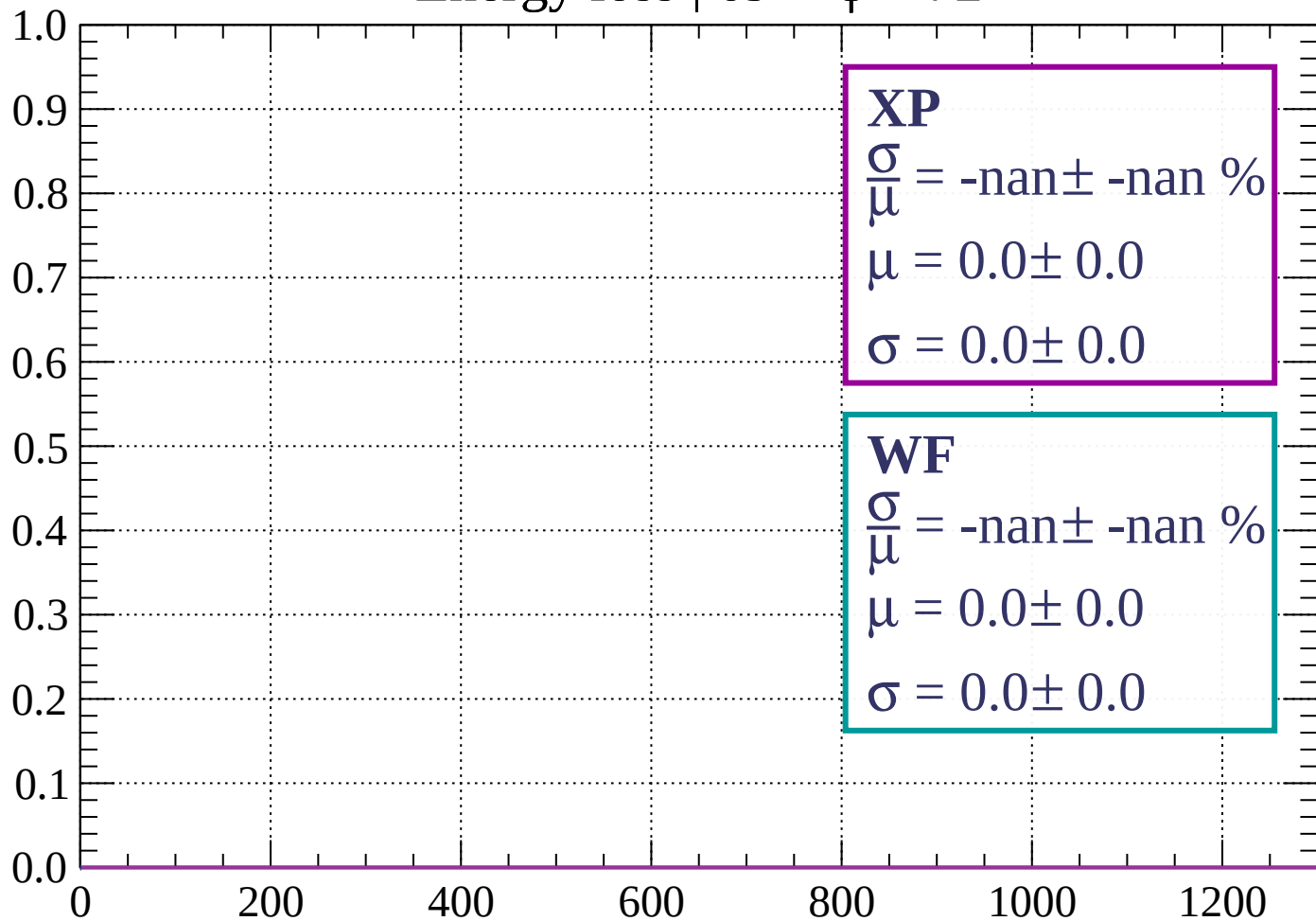
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

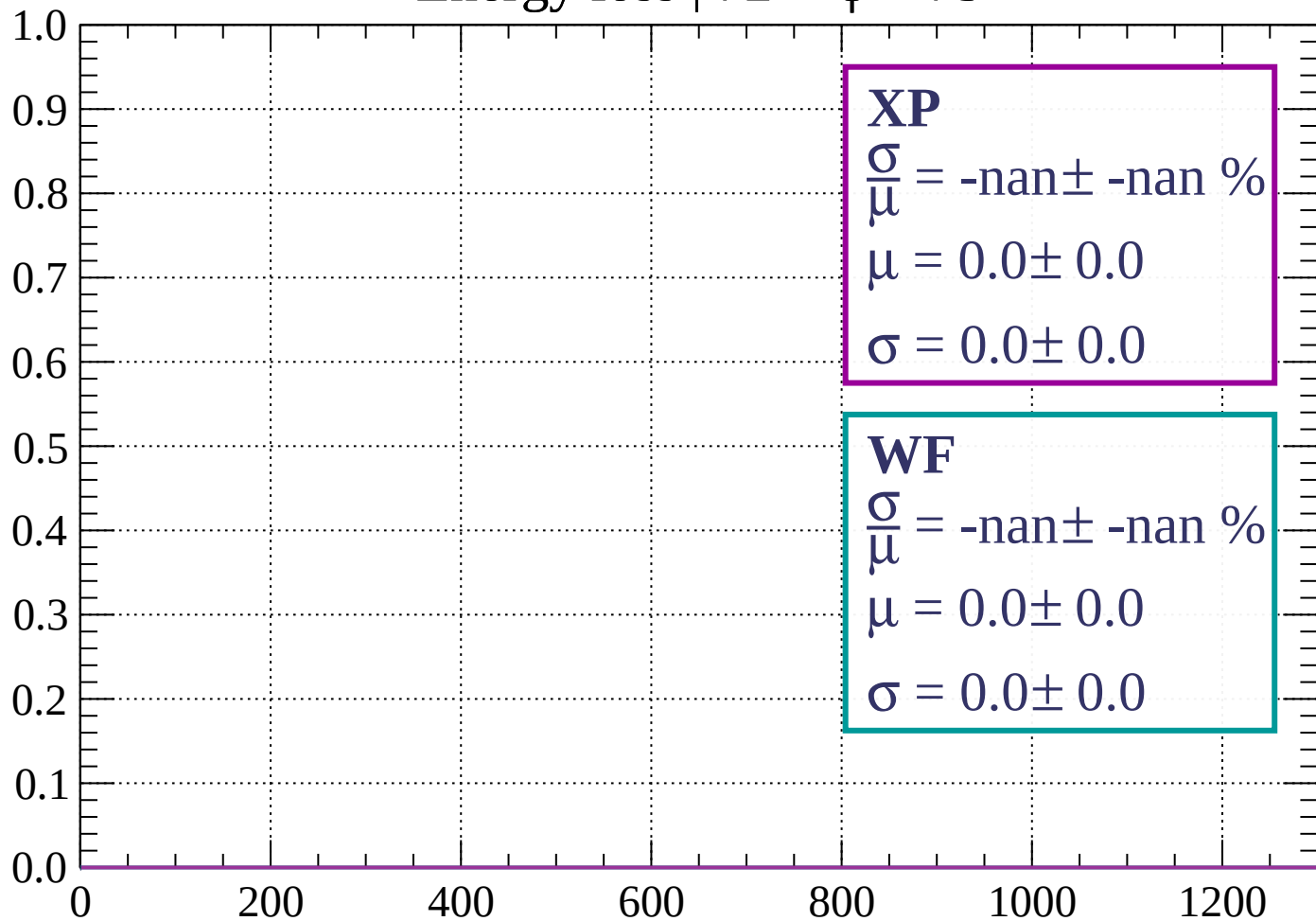
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

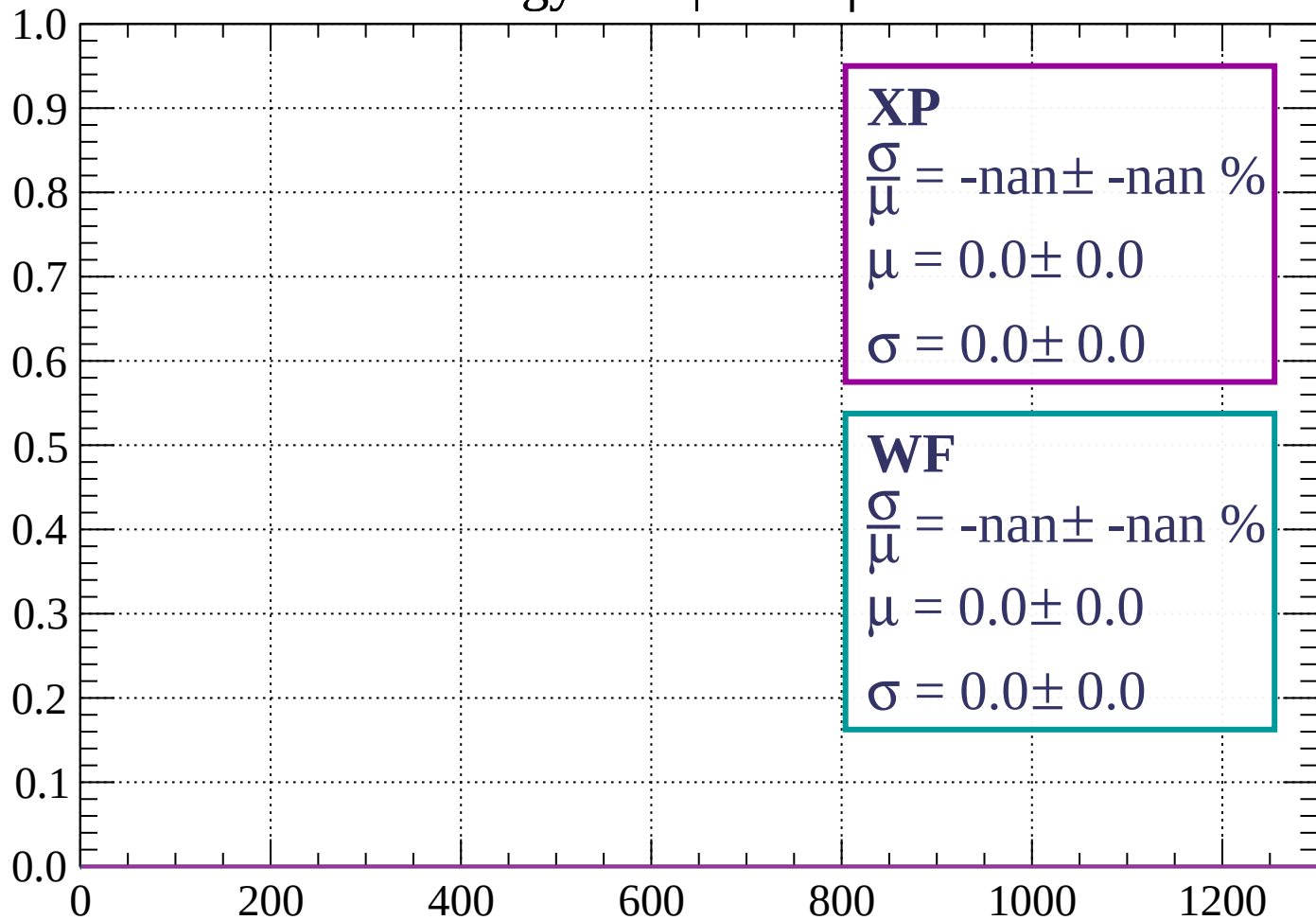
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

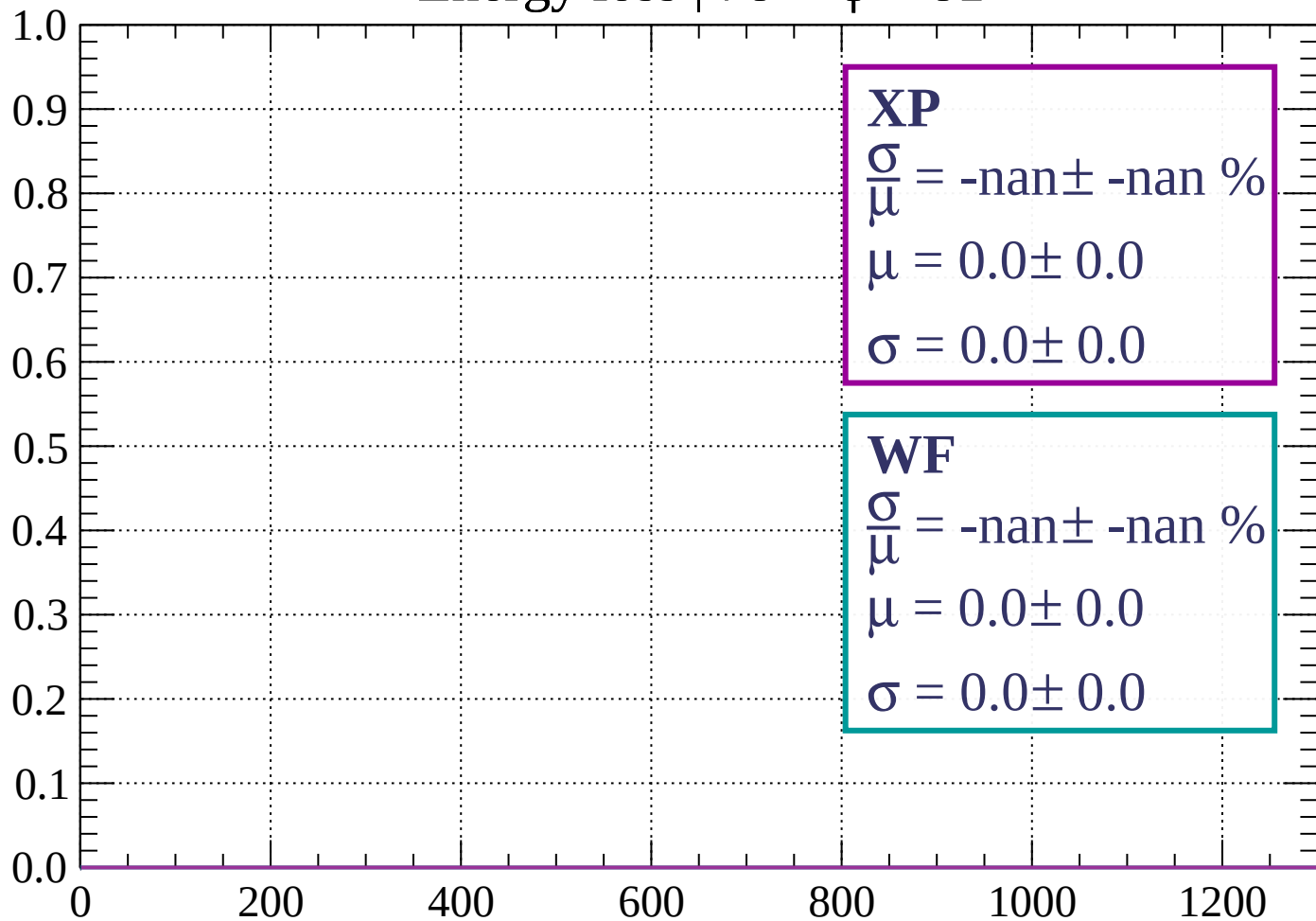
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

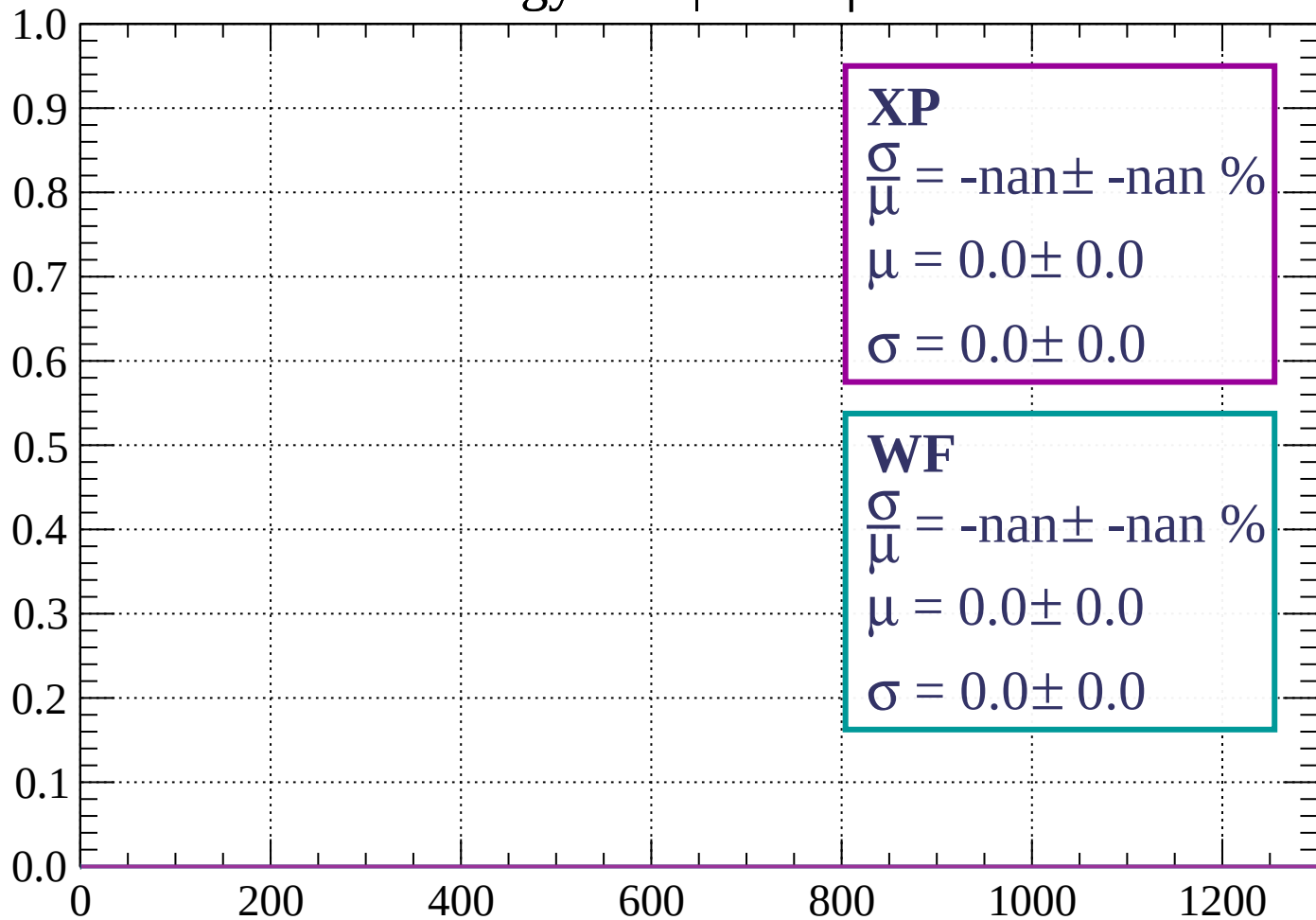
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \phi < 84$

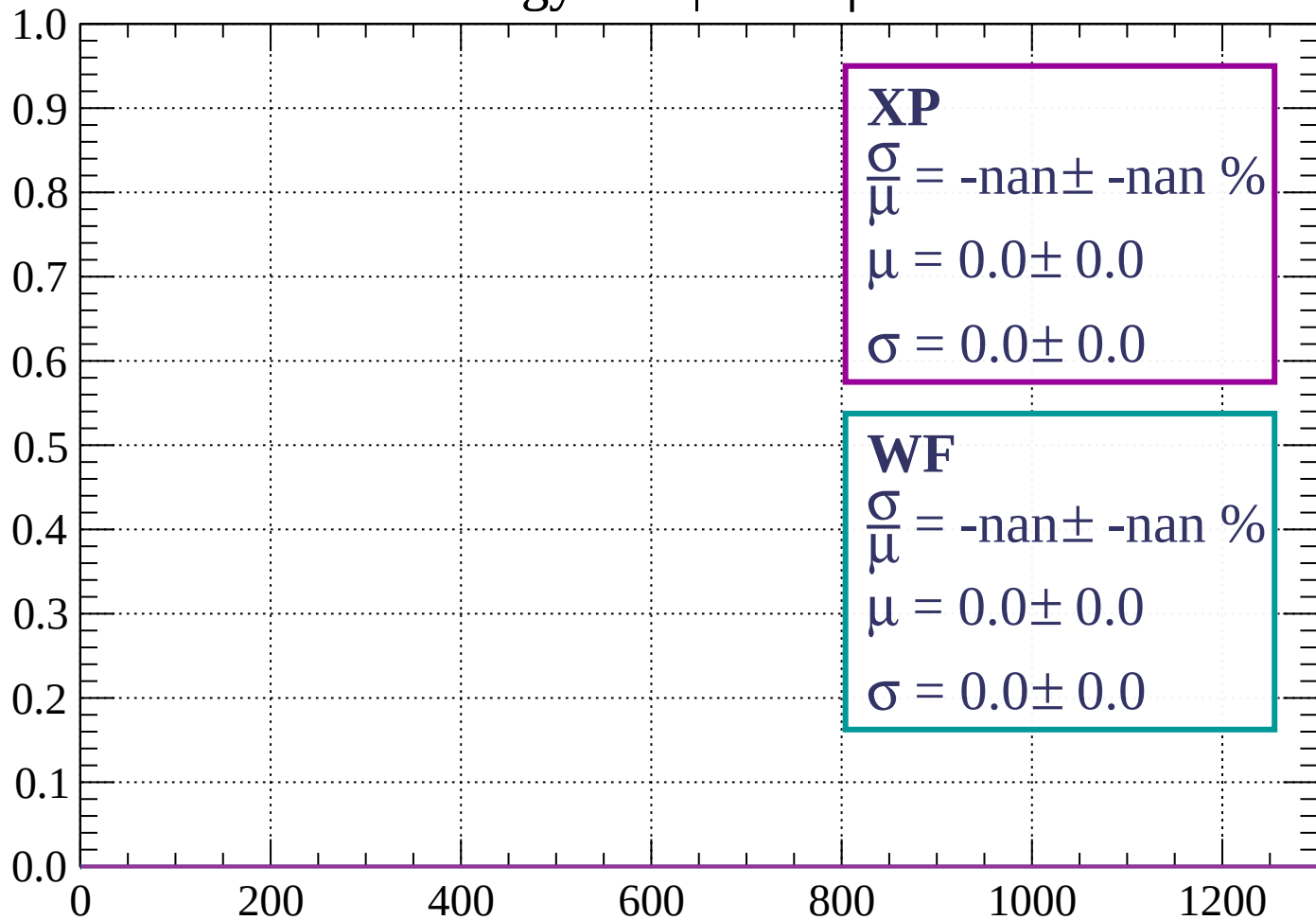
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \phi < 87$

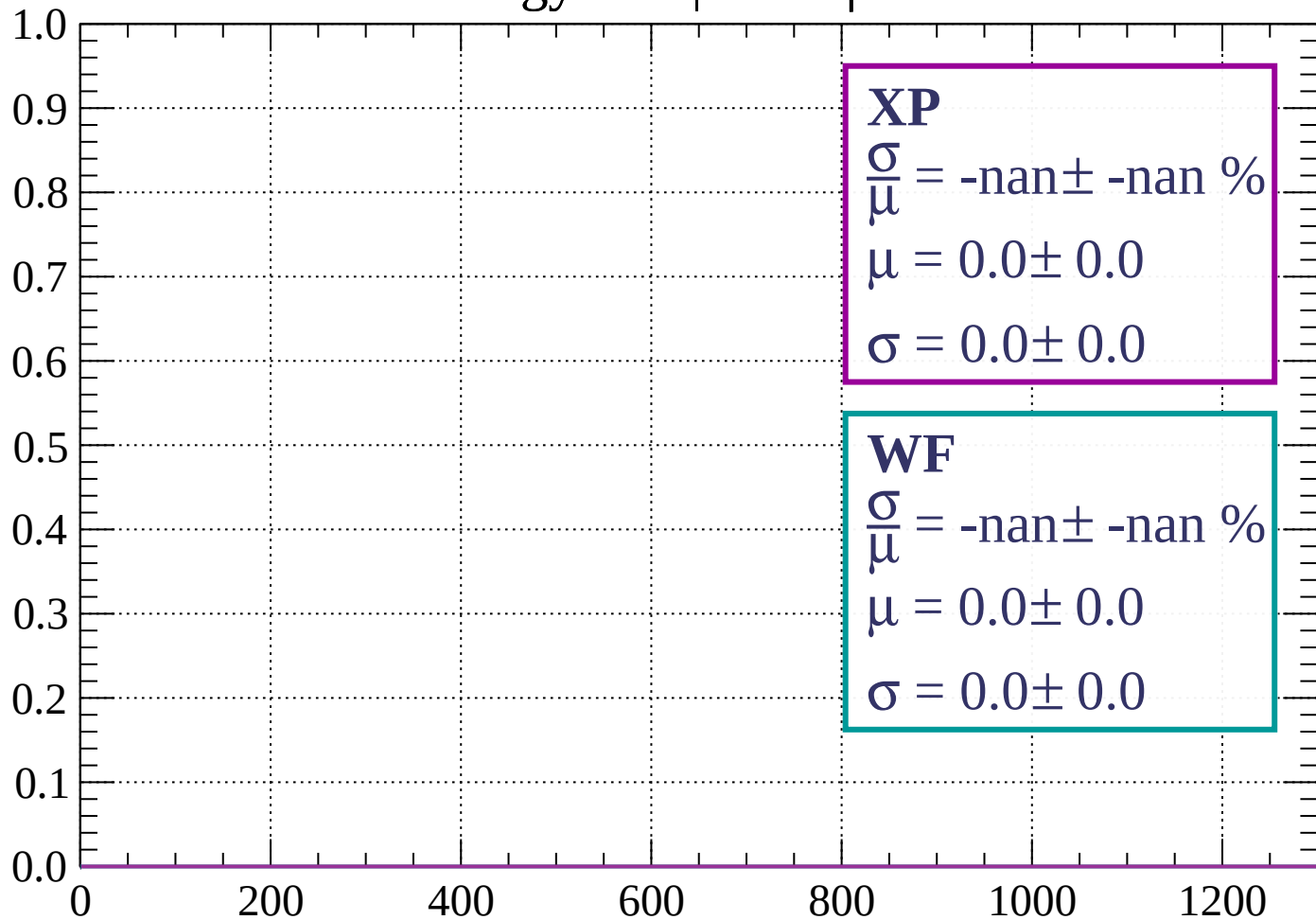
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \phi < 90$

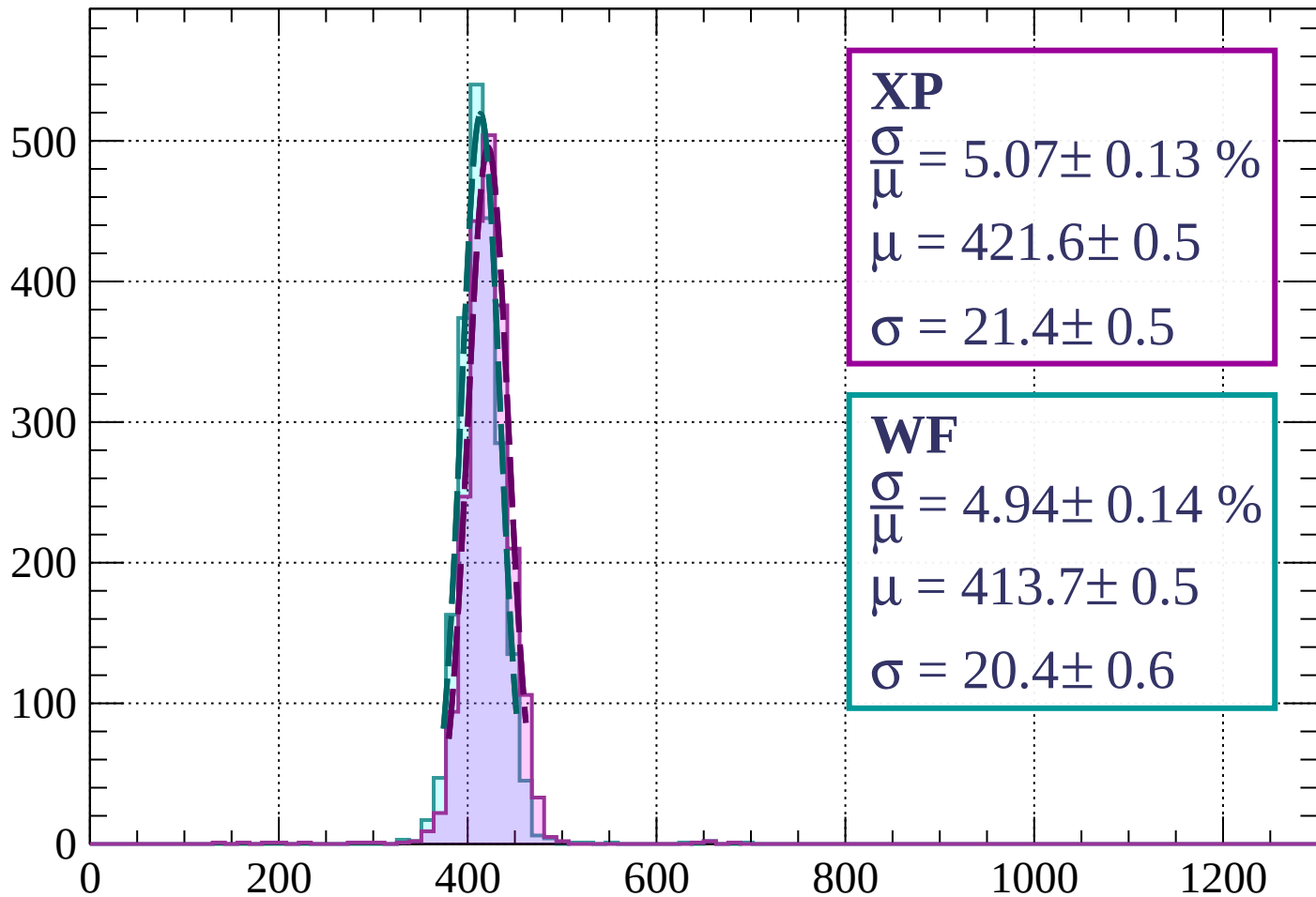
Count



dE/dx (ADC counts/cm)

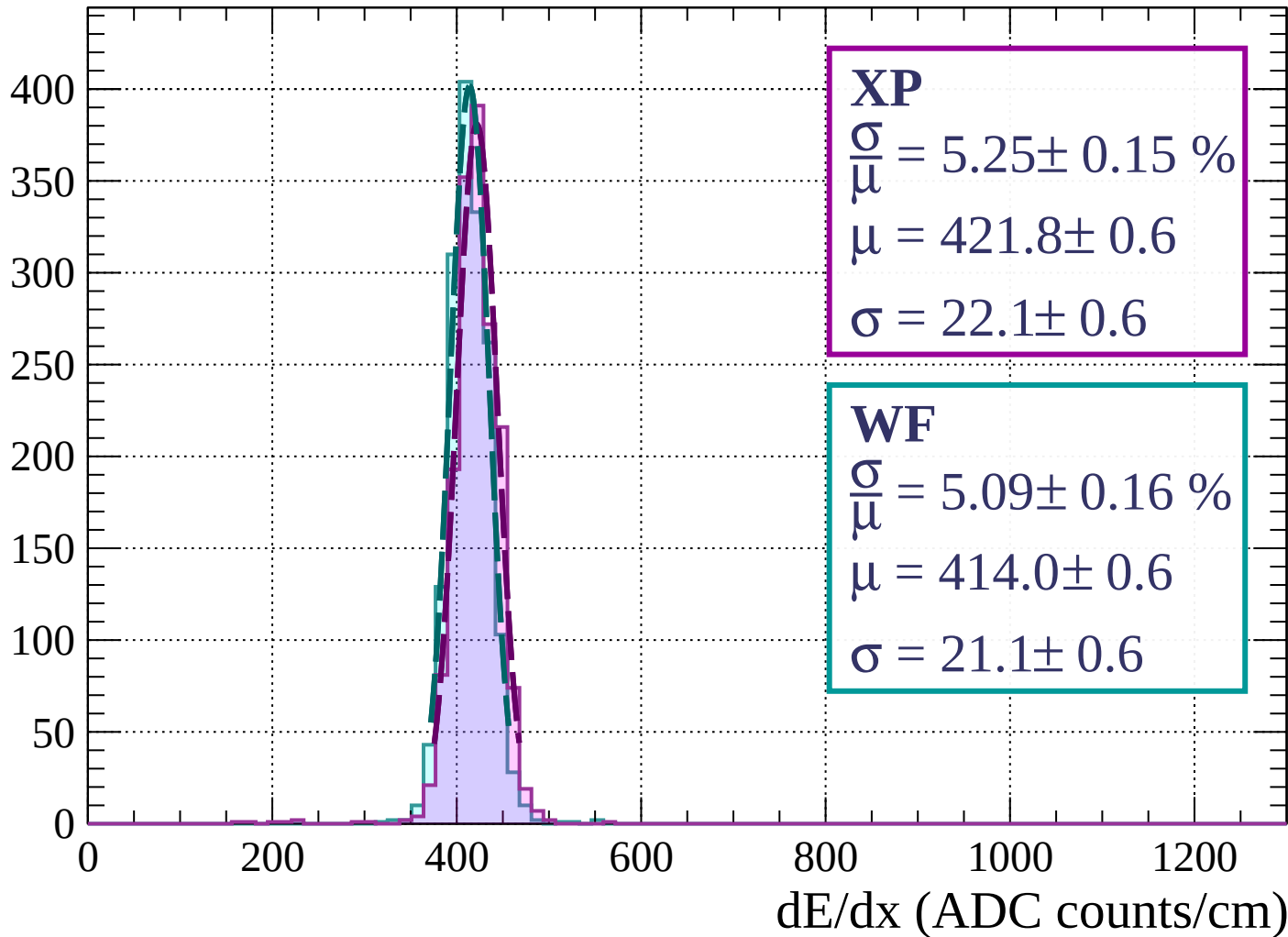
Energy loss | $0 < \theta < 3$

Count



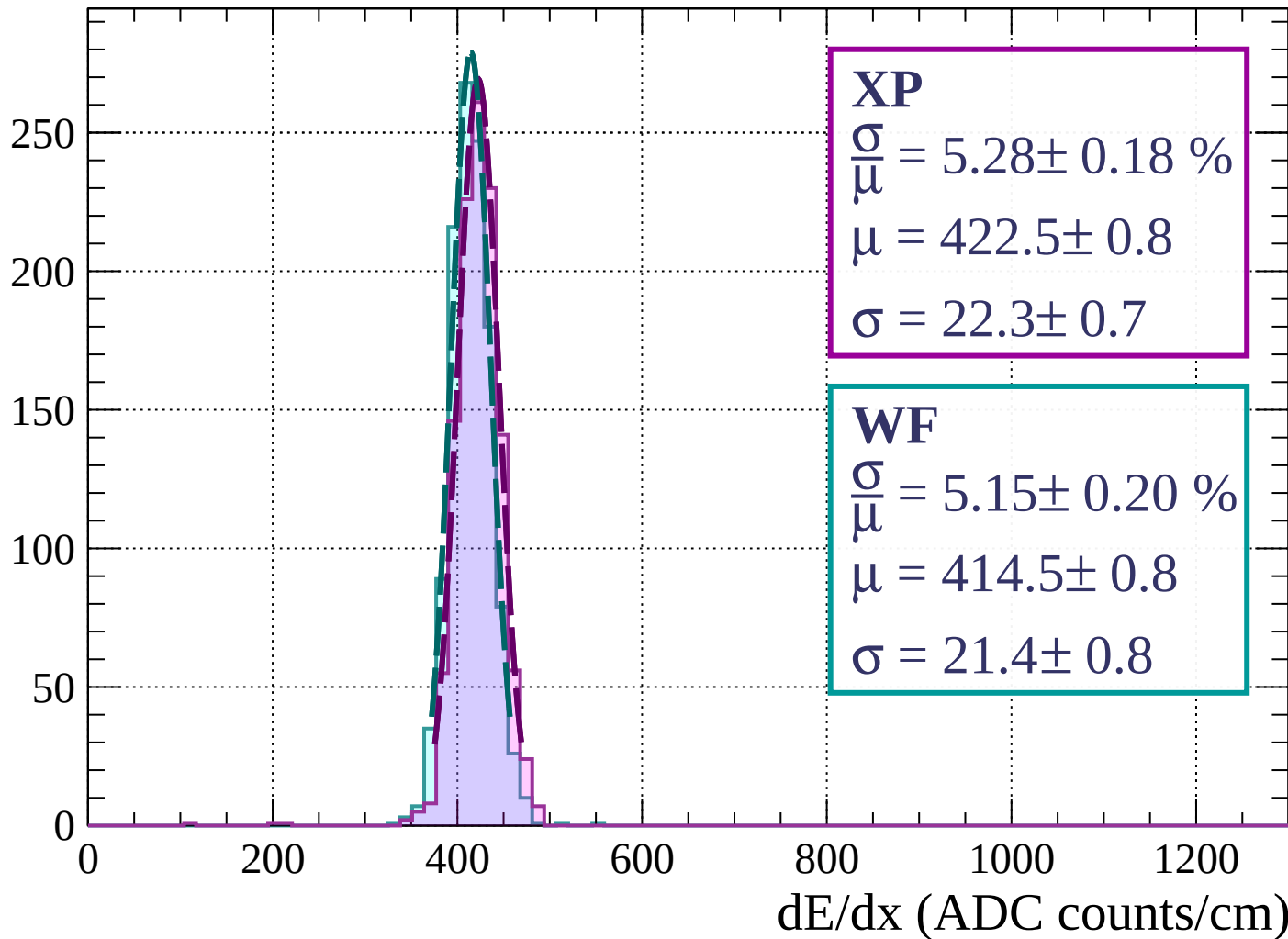
Energy loss | $3 < \theta < 6$

Count



Energy loss | $6 < \theta < 9$

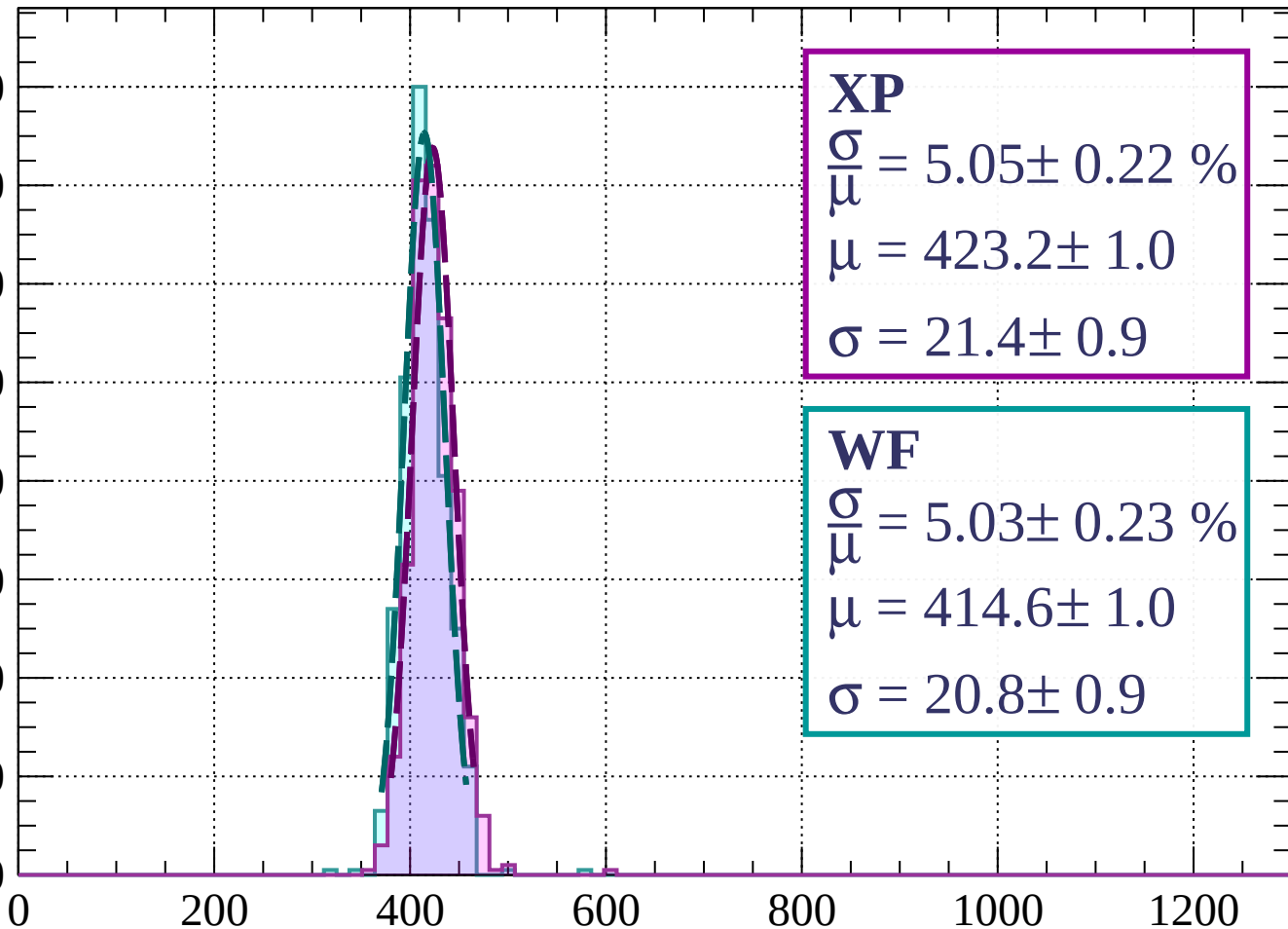
Count



Energy loss | $9 < \theta < 12$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 5.05 \pm 0.22 \%$$

$$\mu = 423.2 \pm 1.0$$

$$\sigma = 21.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.03 \pm 0.23 \%$$

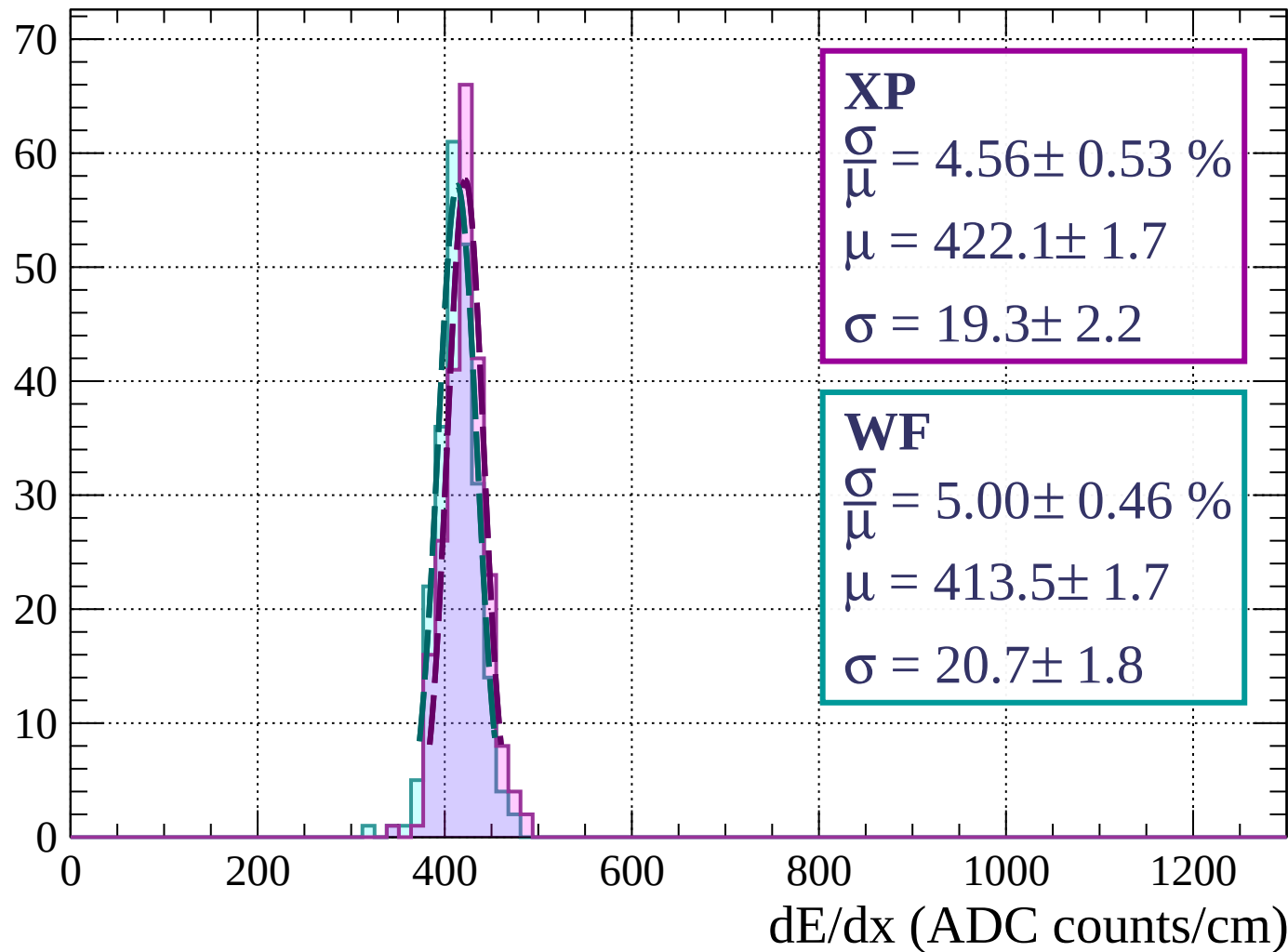
$$\mu = 414.6 \pm 1.0$$

$$\sigma = 20.8 \pm 0.9$$

dE/dx (ADC counts/cm)

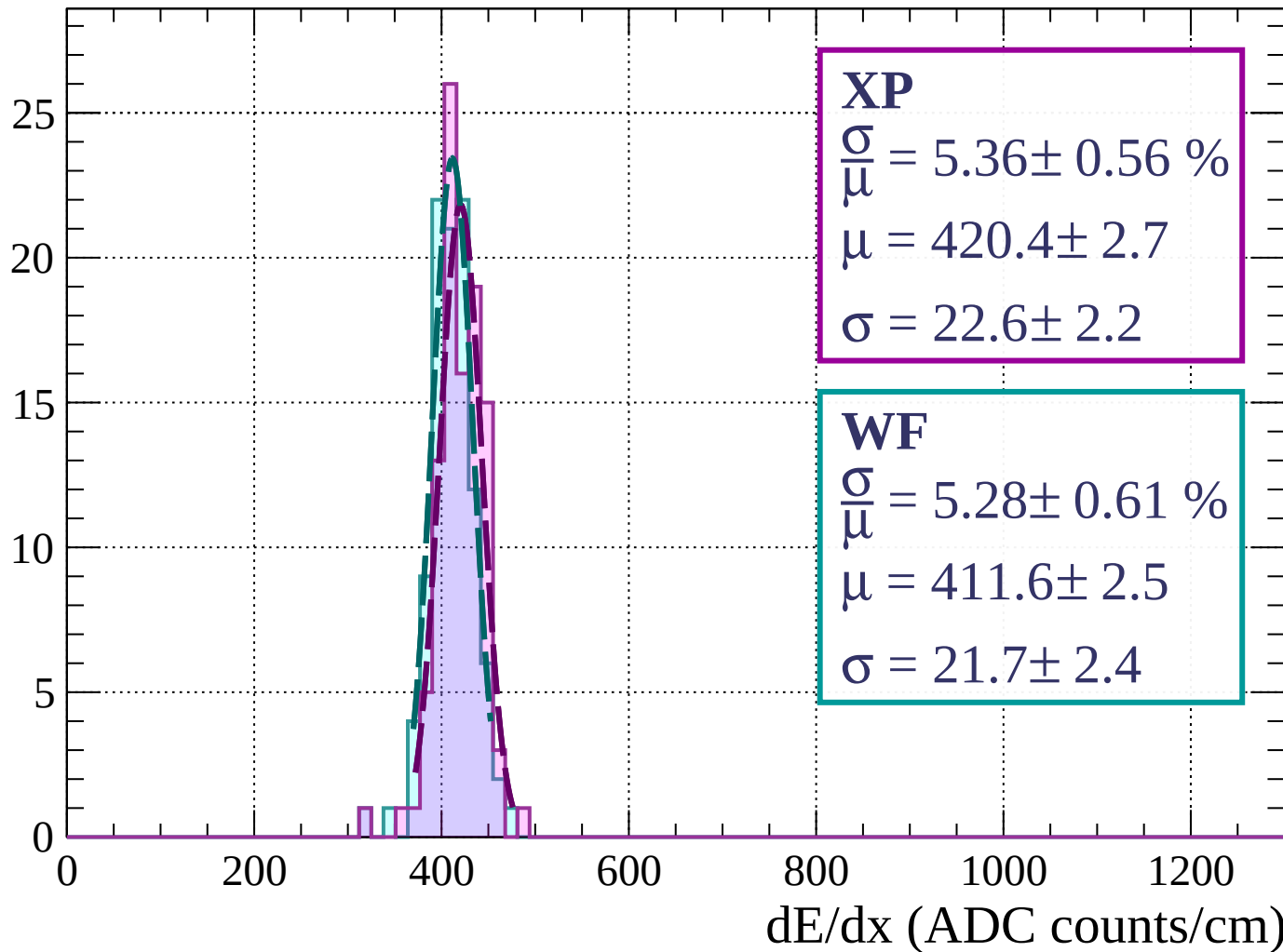
Energy loss | $12 < \theta < 15$

Count



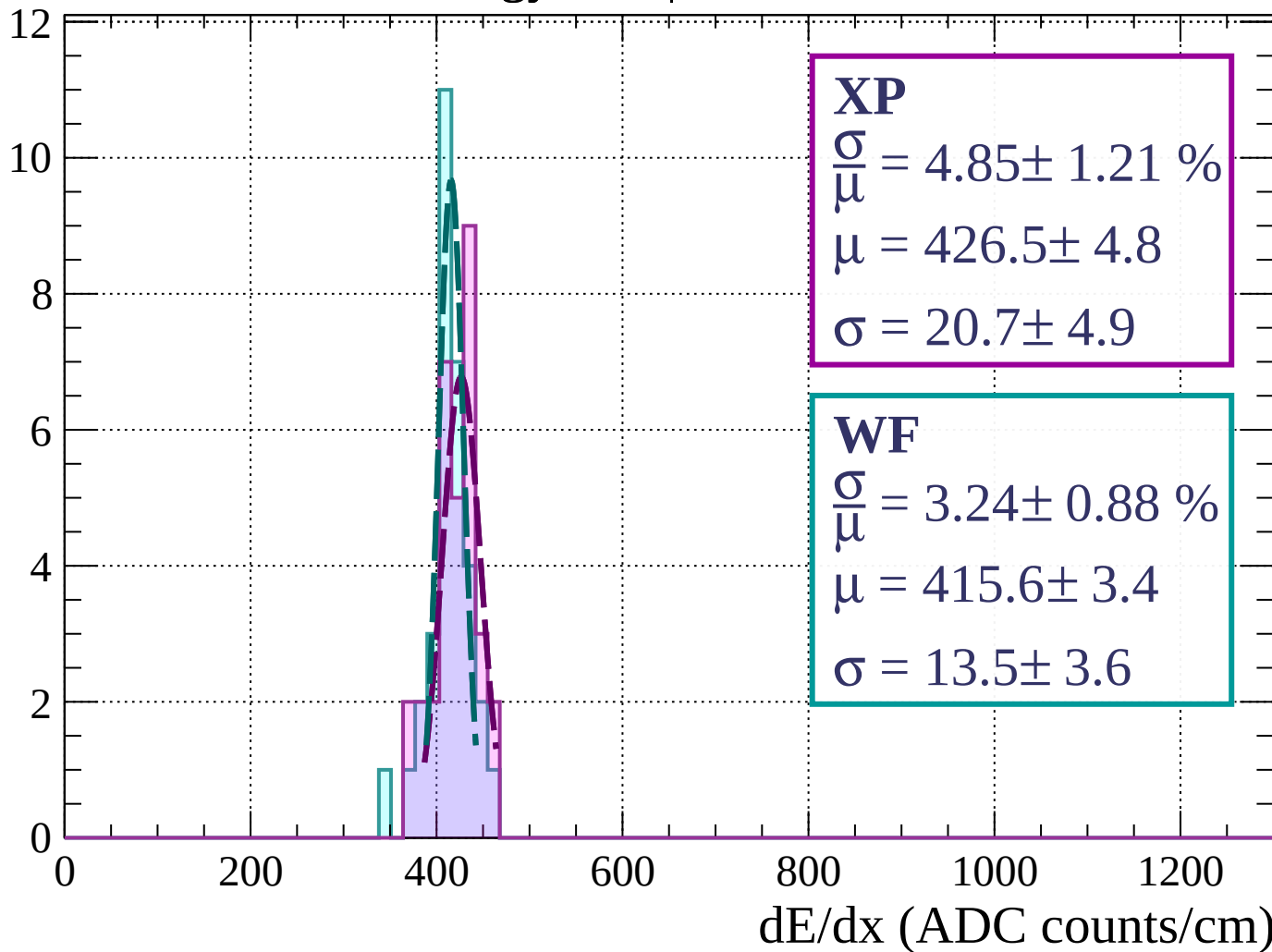
Energy loss | $15 < \theta < 18$

Count



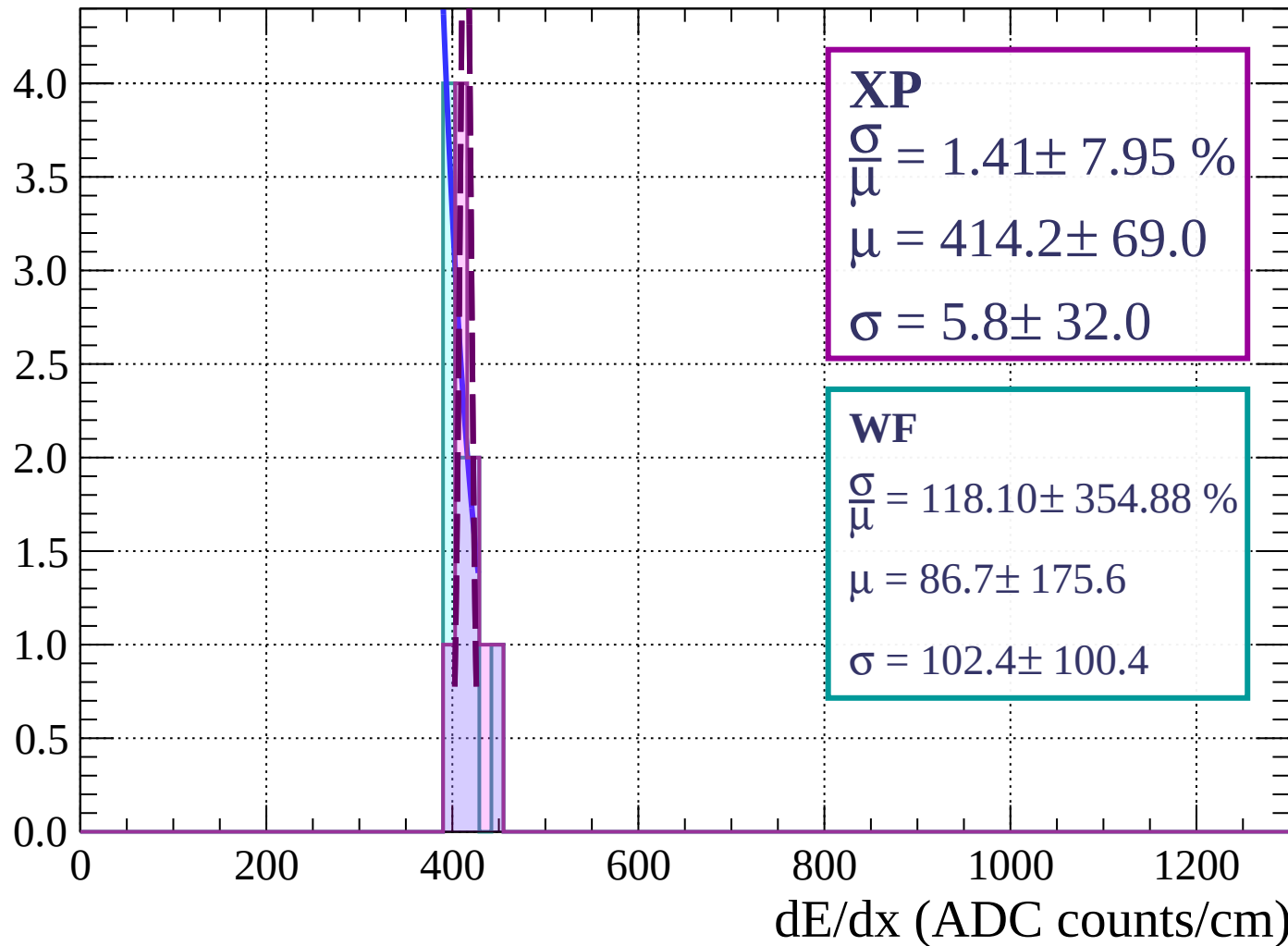
Energy loss | $18 < \theta < 21$

Count

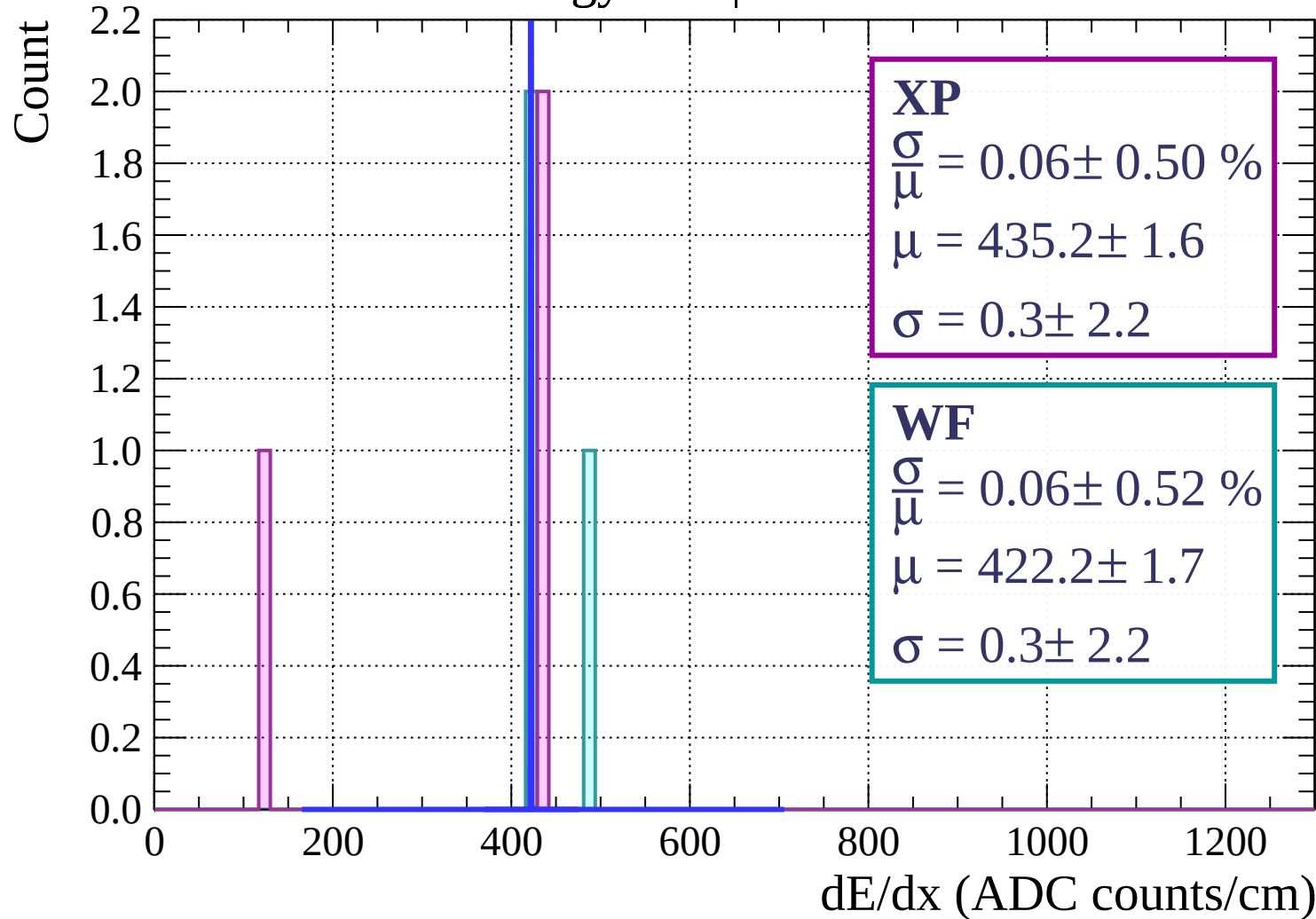


Energy loss | $21 < \theta < 24$

Count

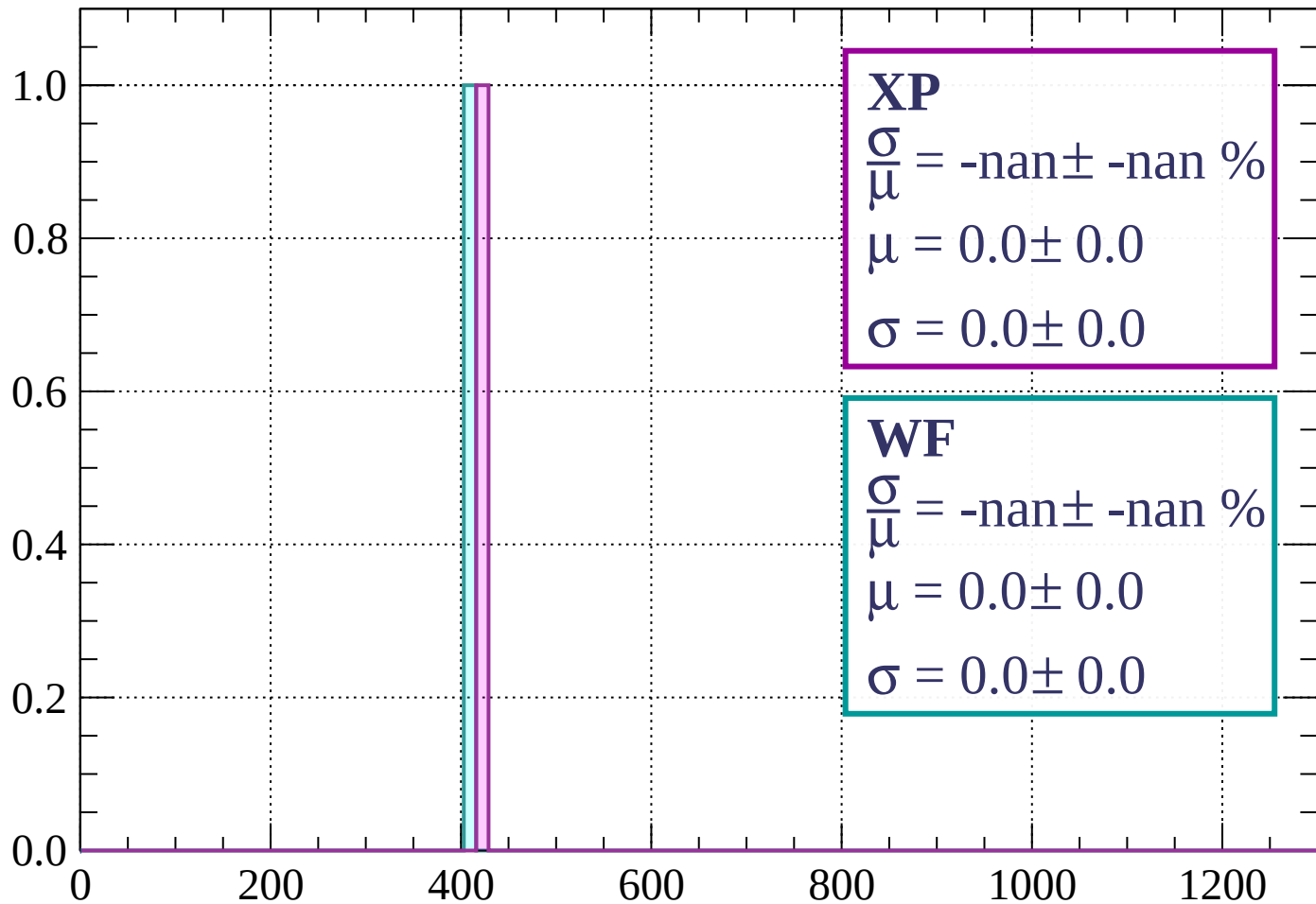


Energy loss | $24 < \theta < 27$



Energy loss | $27 < \theta < 30$

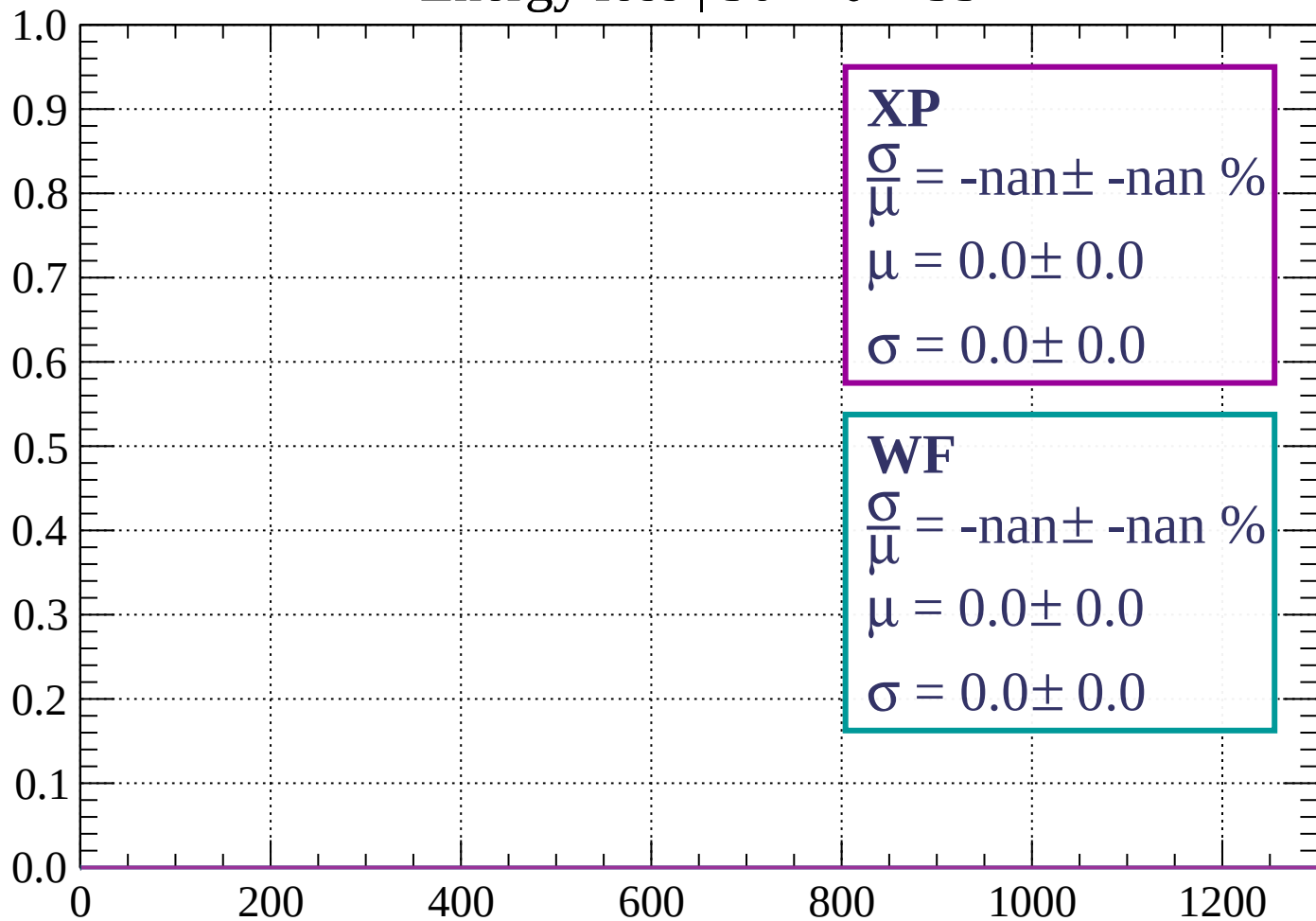
Count



dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 33$

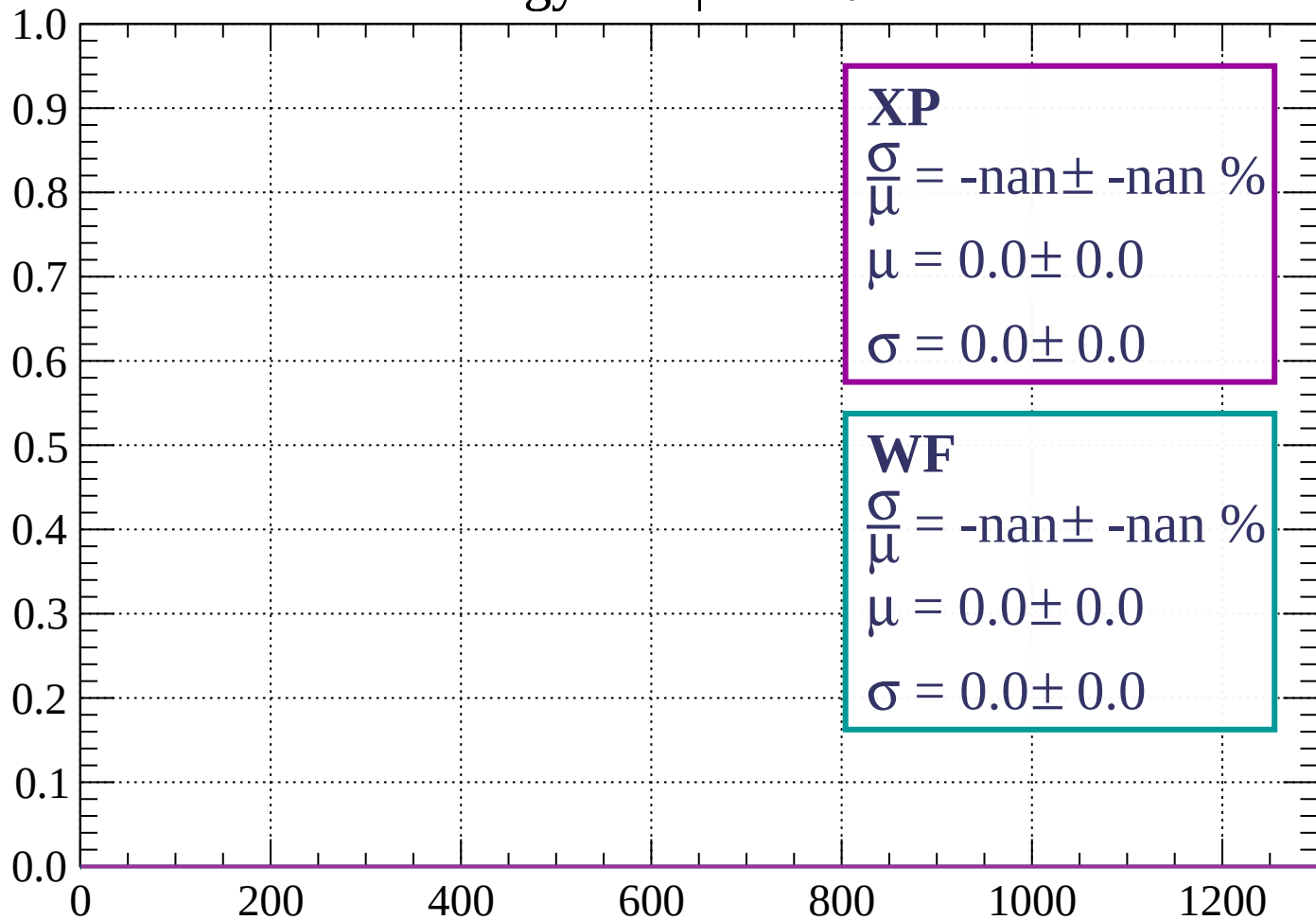
Count



dE/dx (ADC counts/cm)

Energy loss | $33 < \theta < 36$

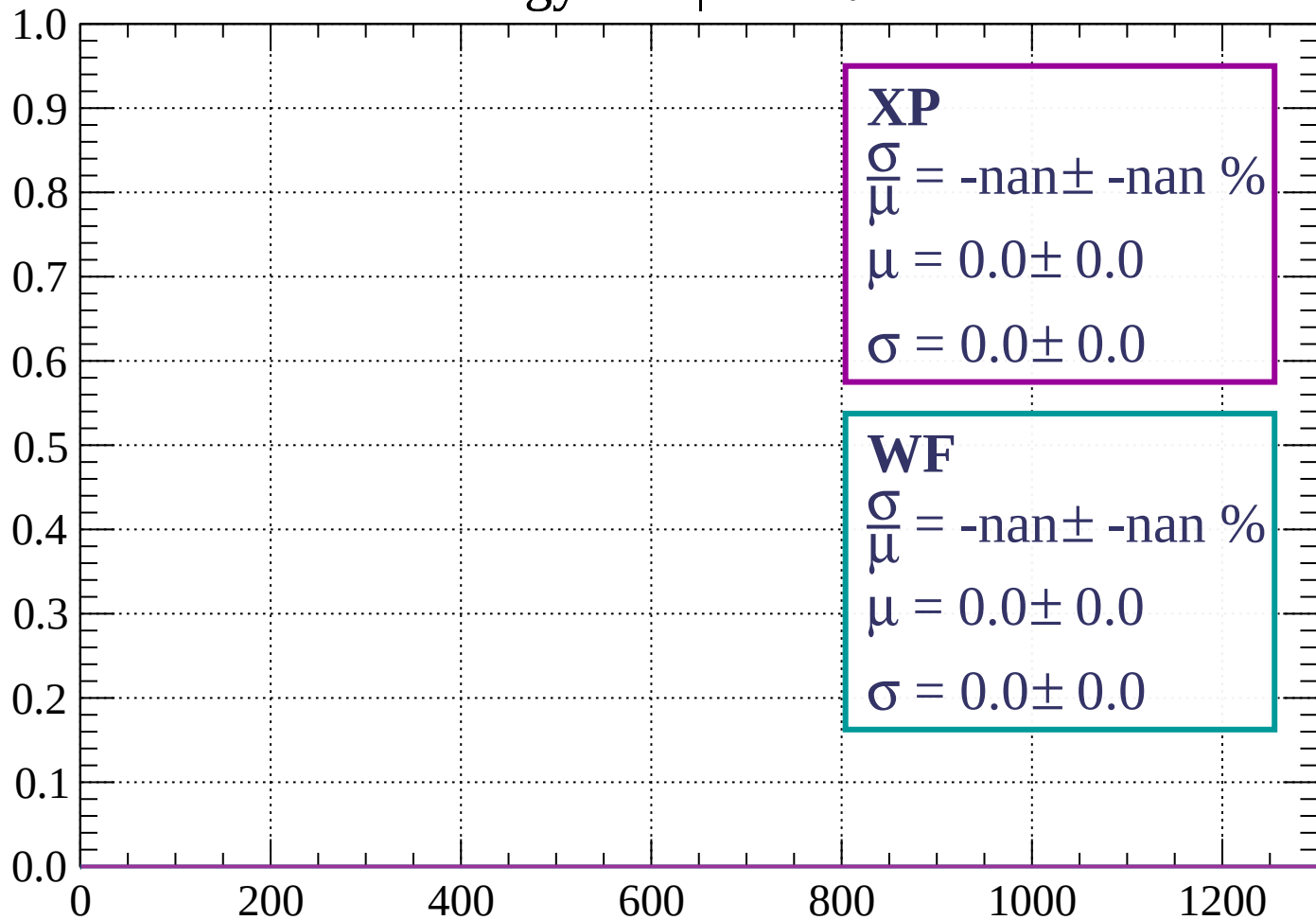
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 39$

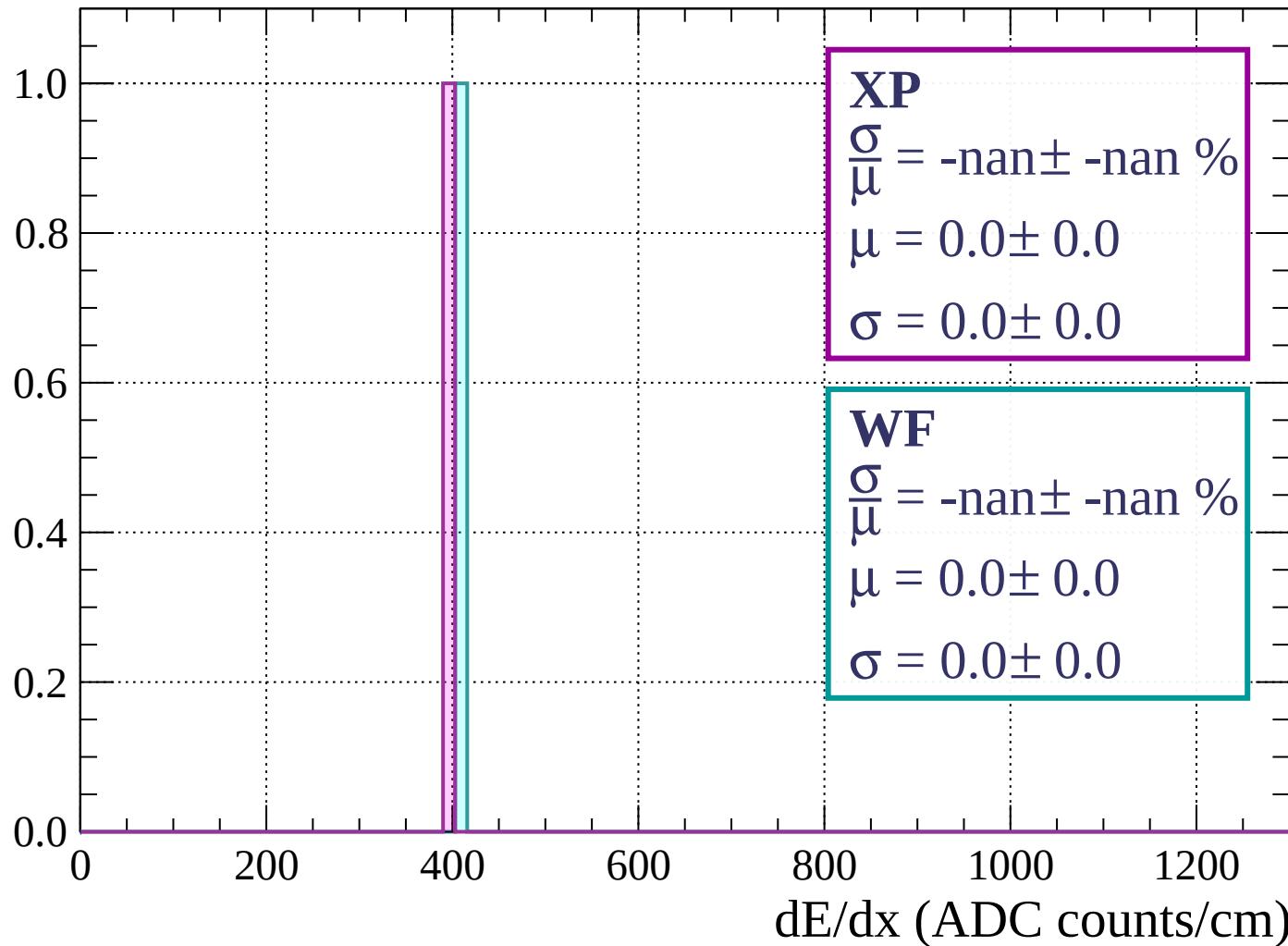
Count



dE/dx (ADC counts/cm)

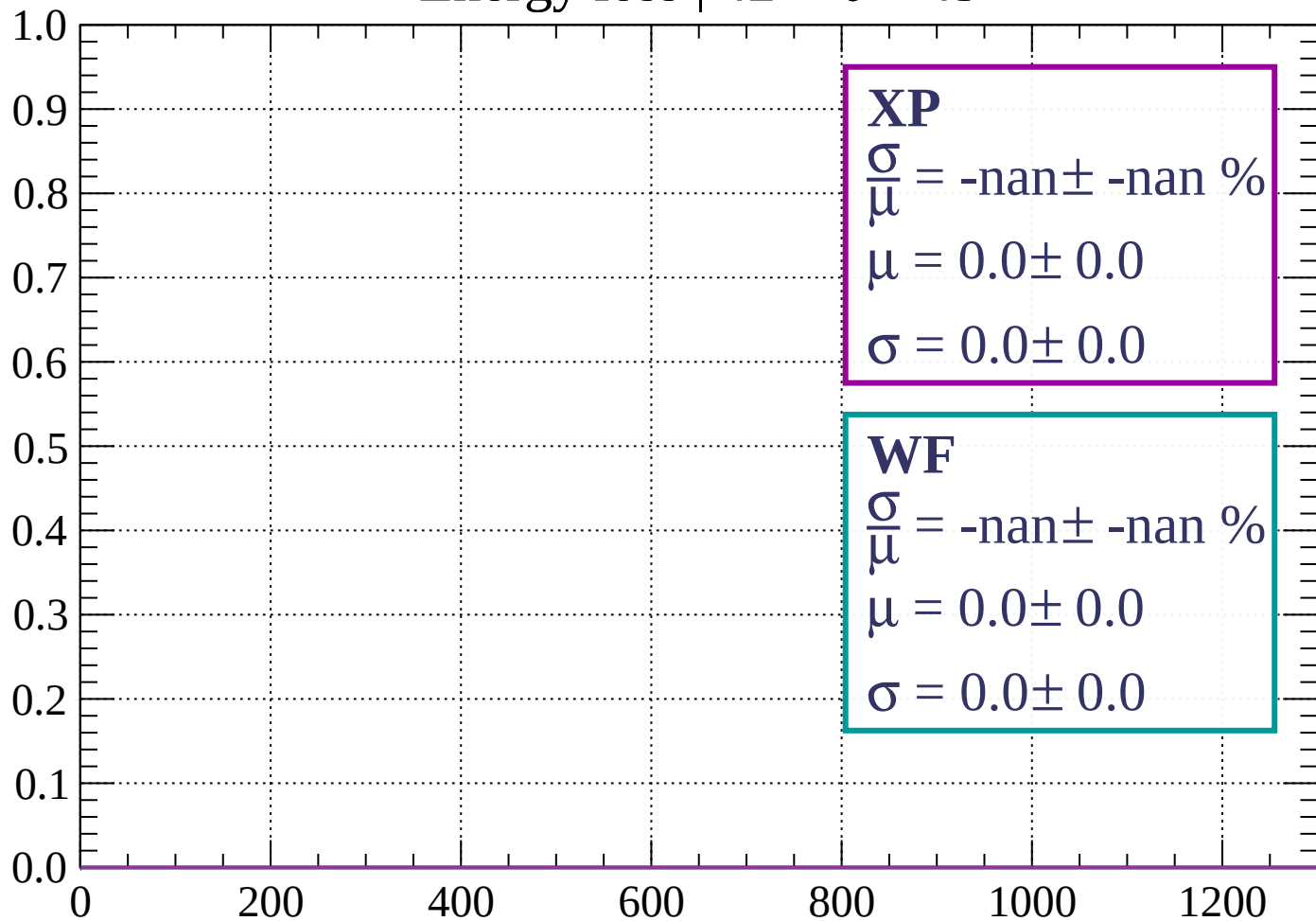
Energy loss | $39 < \theta < 42$

Count



Energy loss | $42 < \theta < 45$

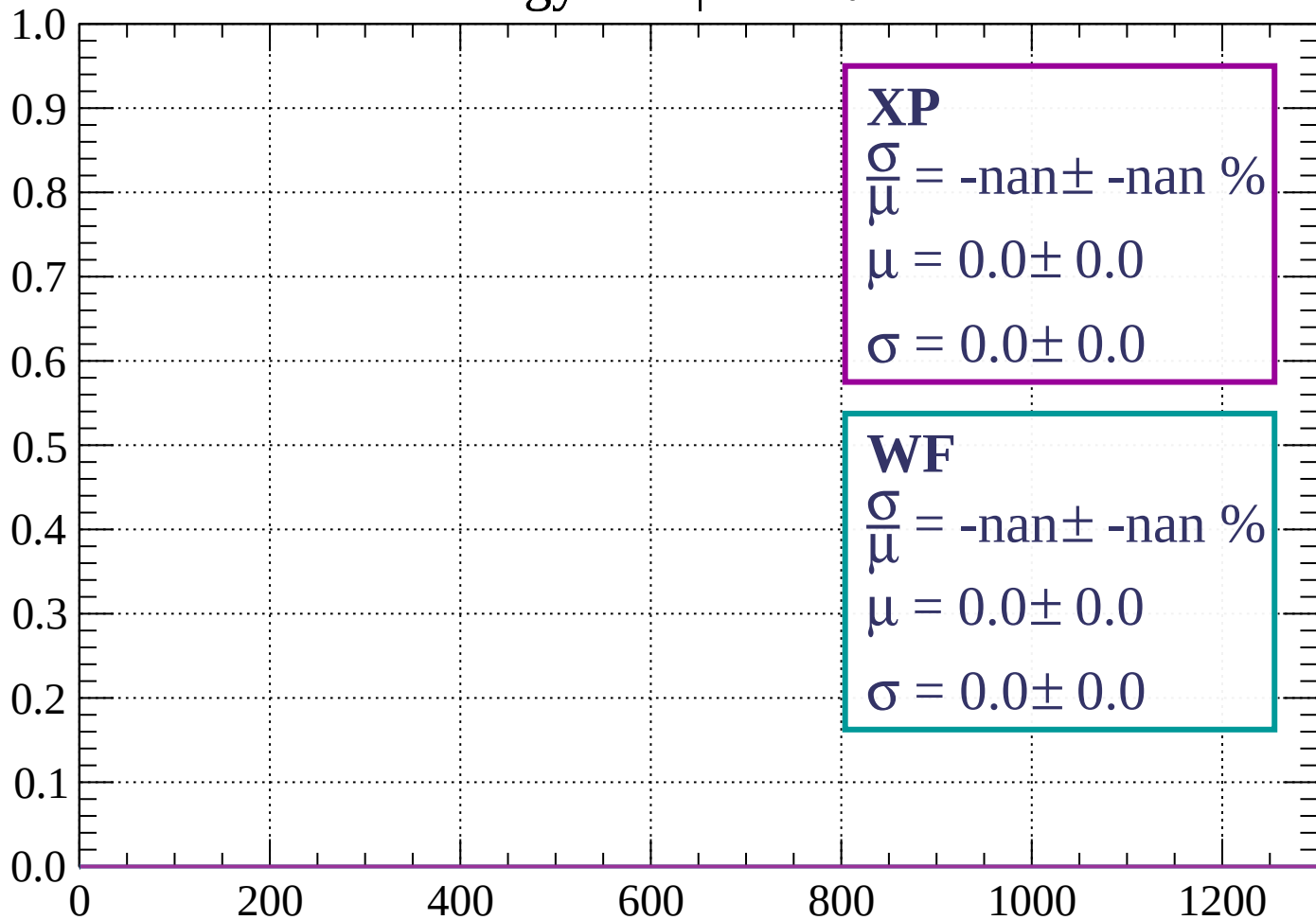
Count



dE/dx (ADC counts/cm)

Energy loss | $45 < \theta < 48$

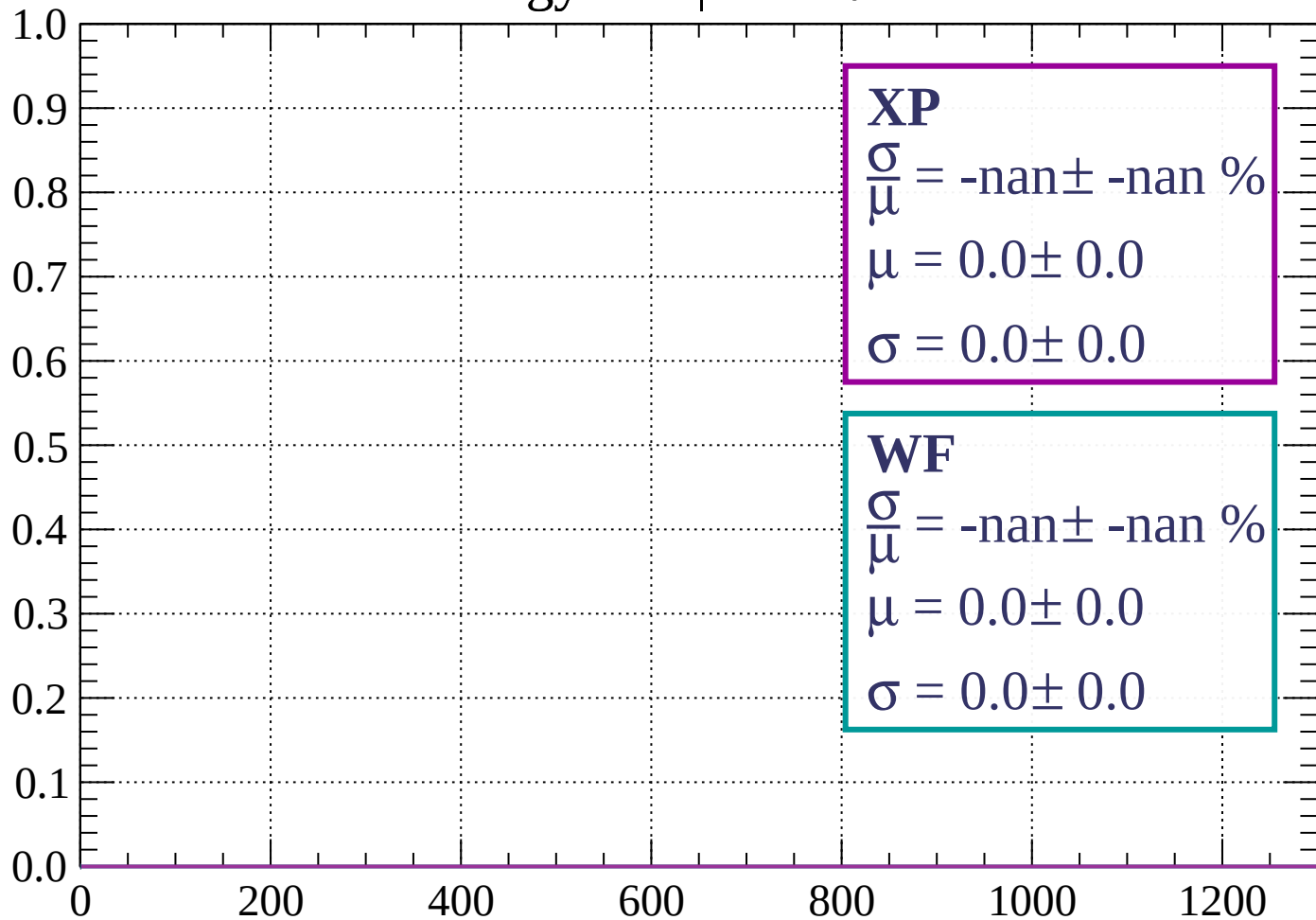
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 51$

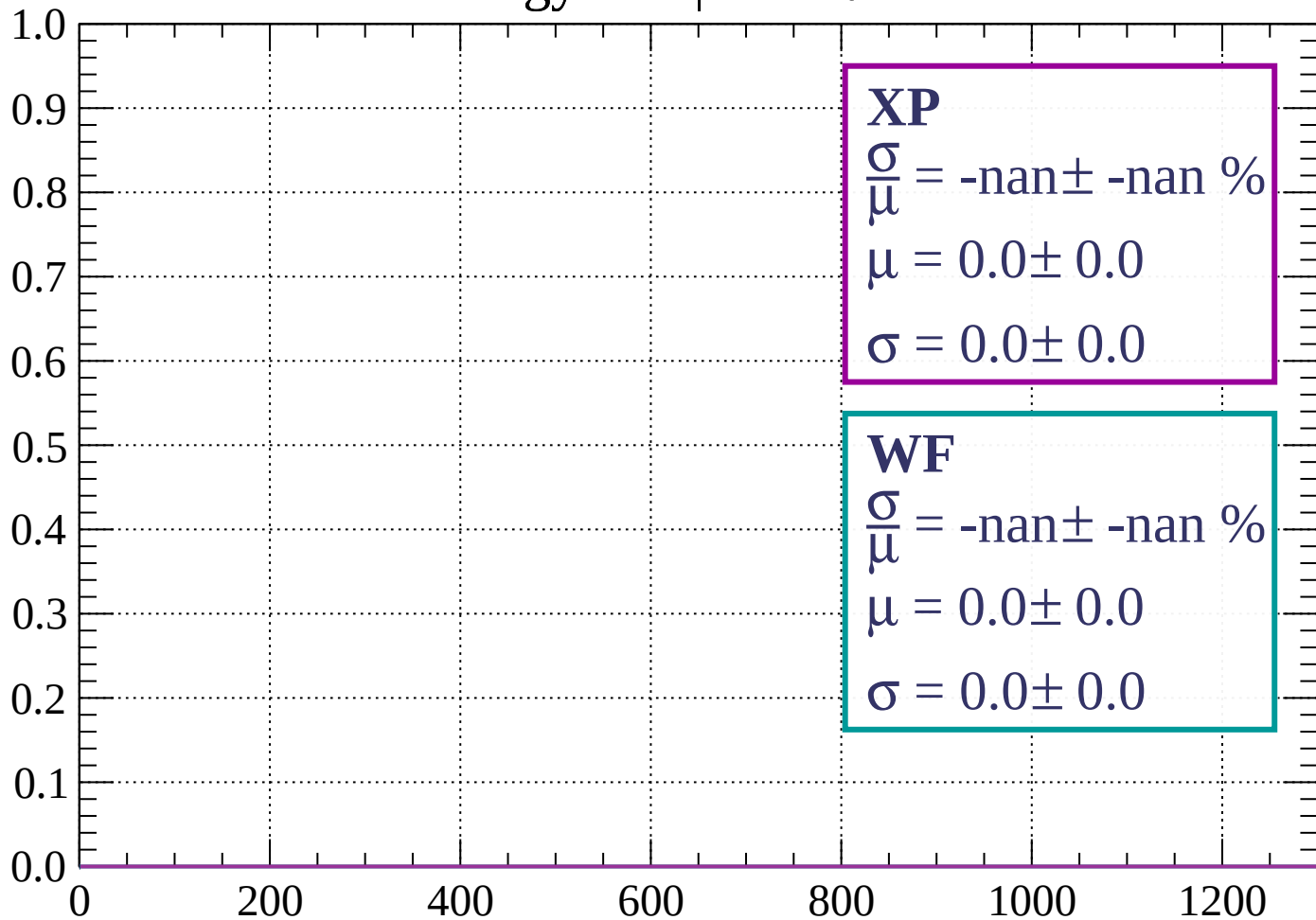
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \theta < 54$

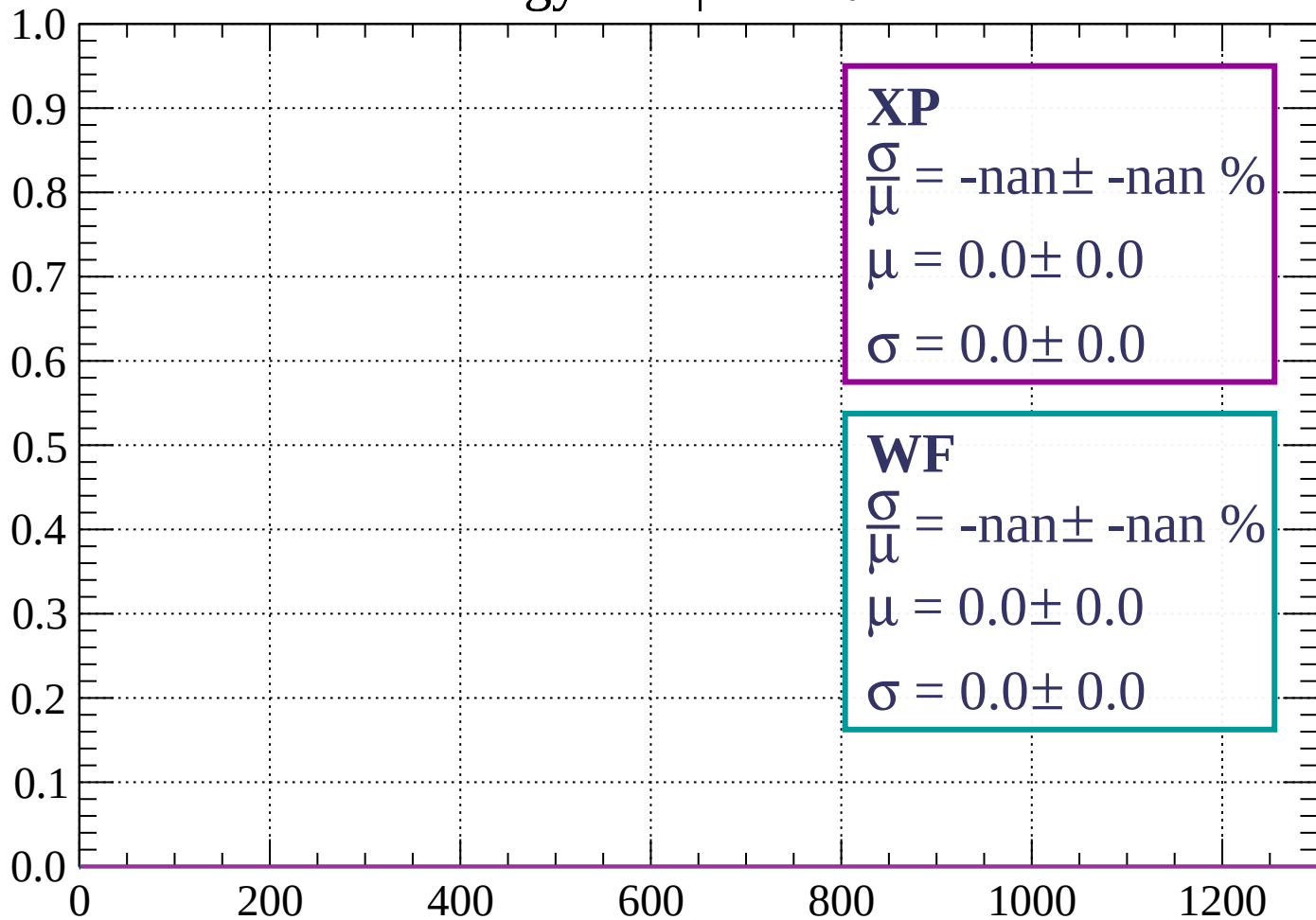
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 57$

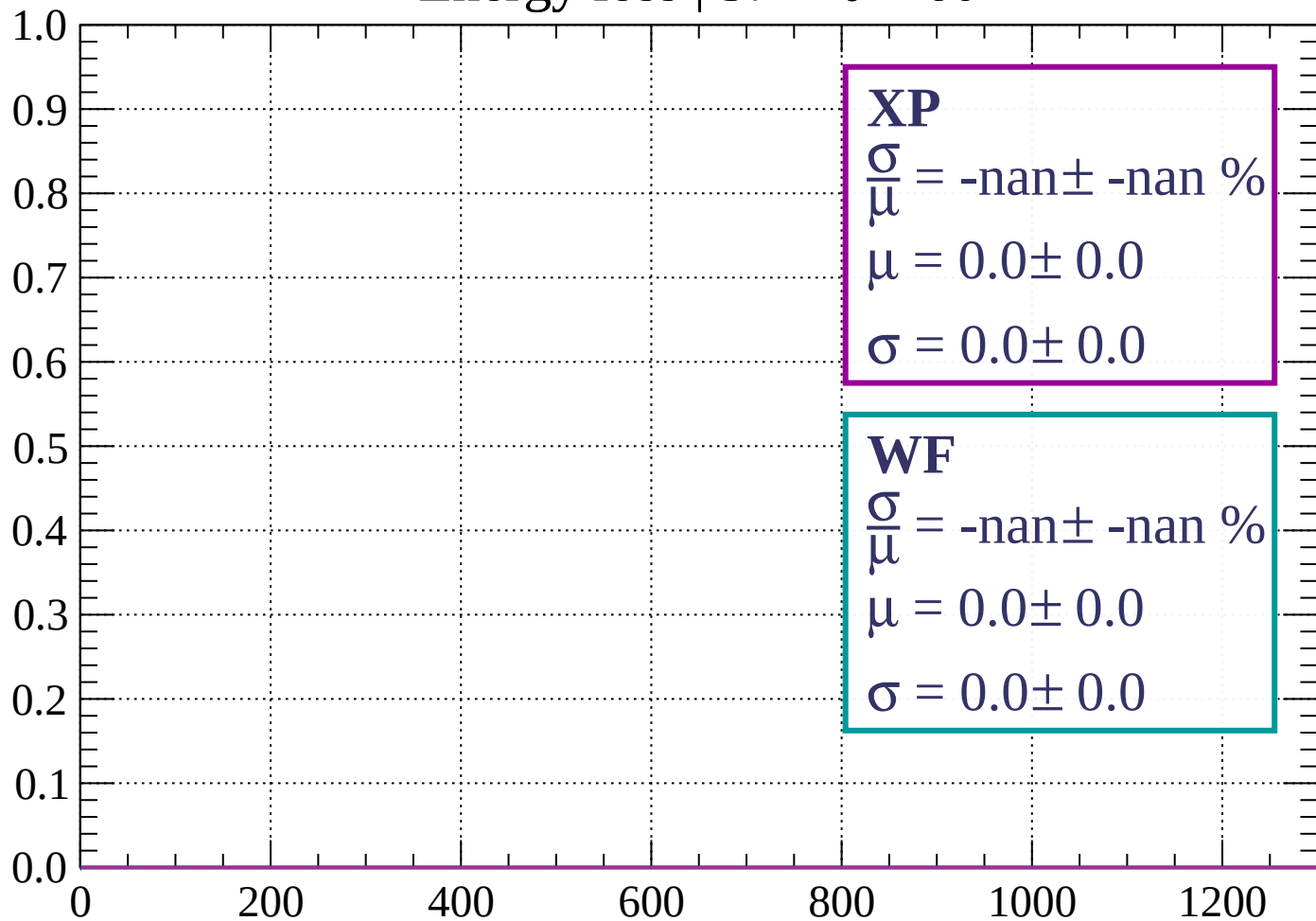
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \theta < 60$

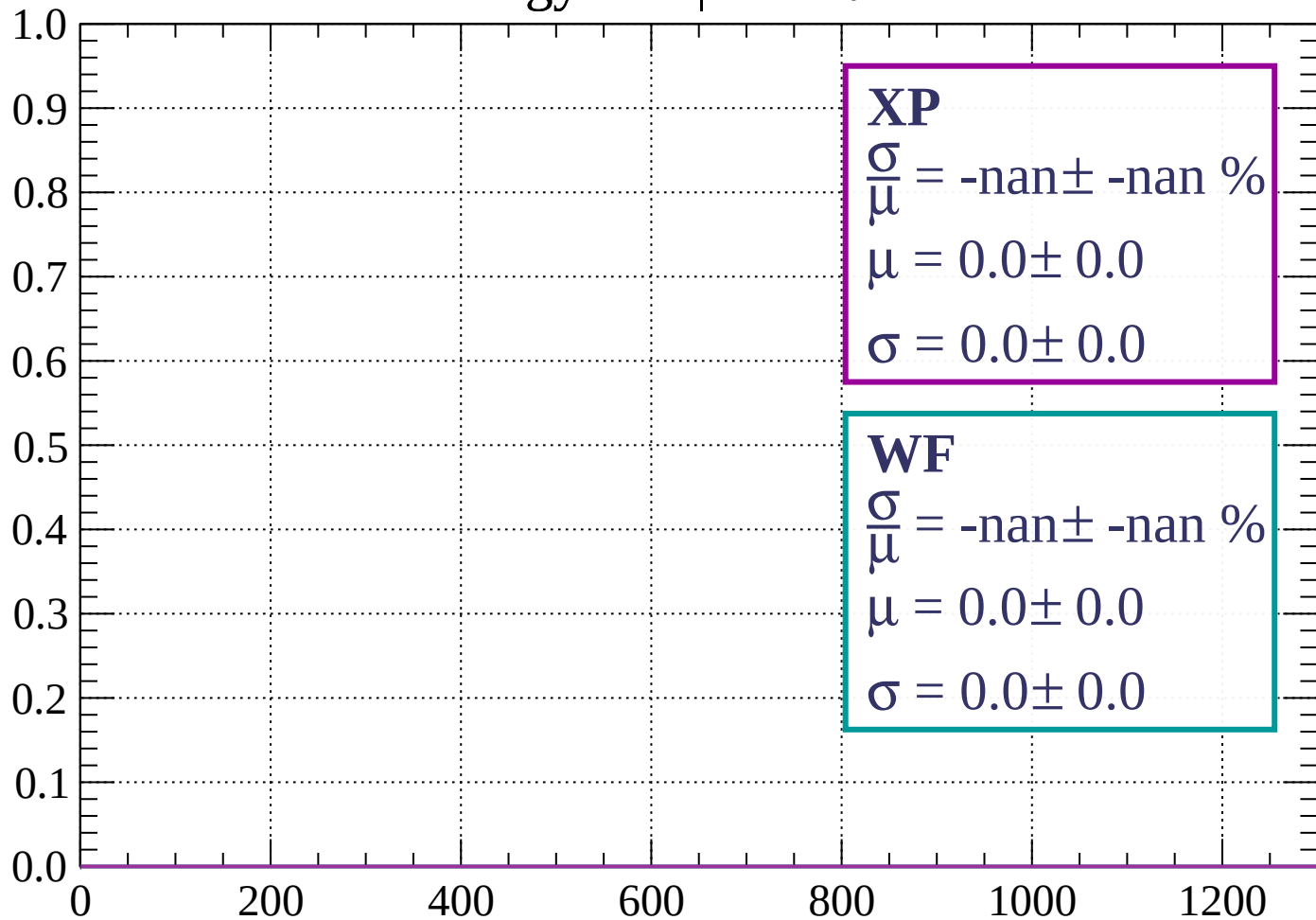
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

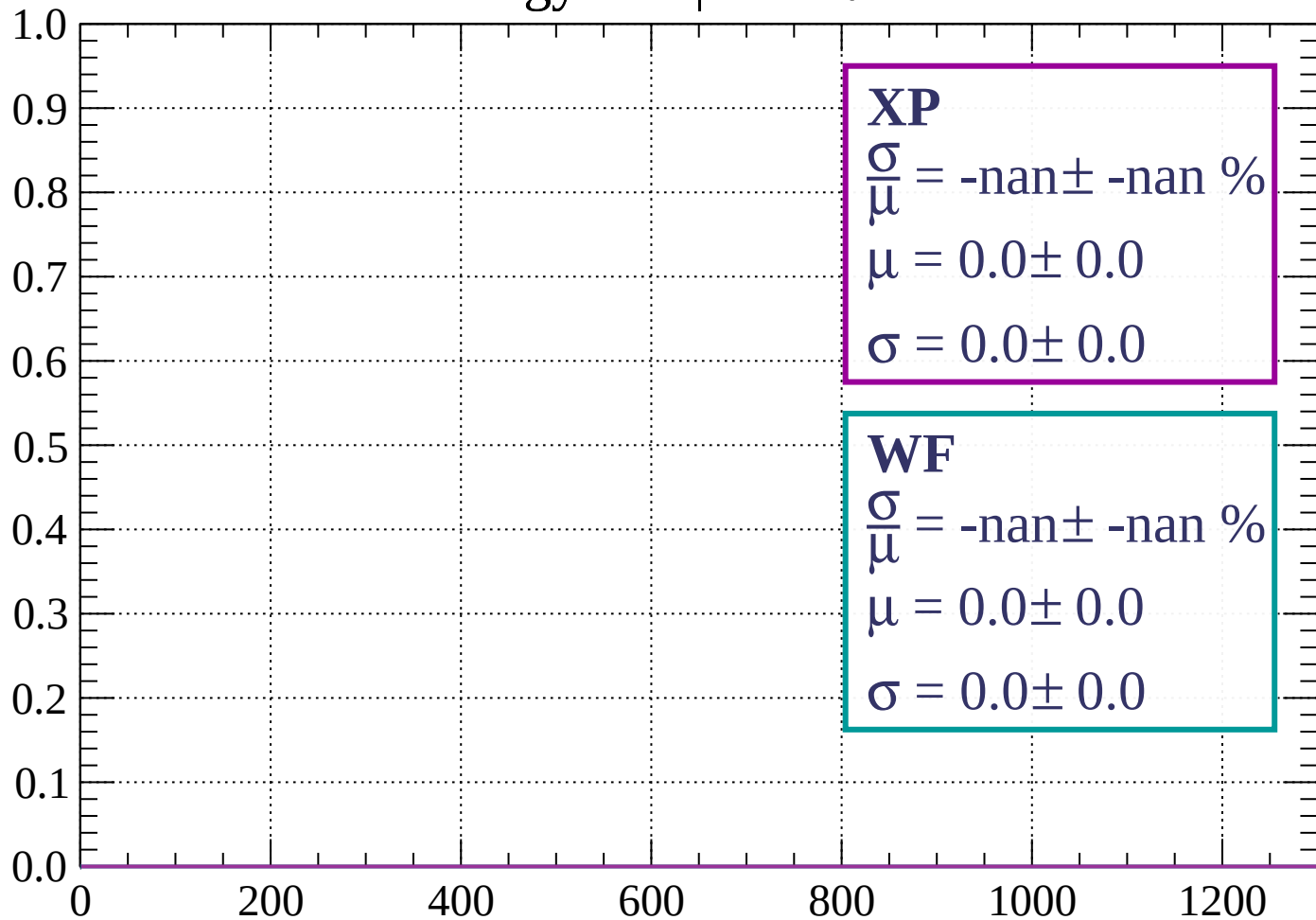
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

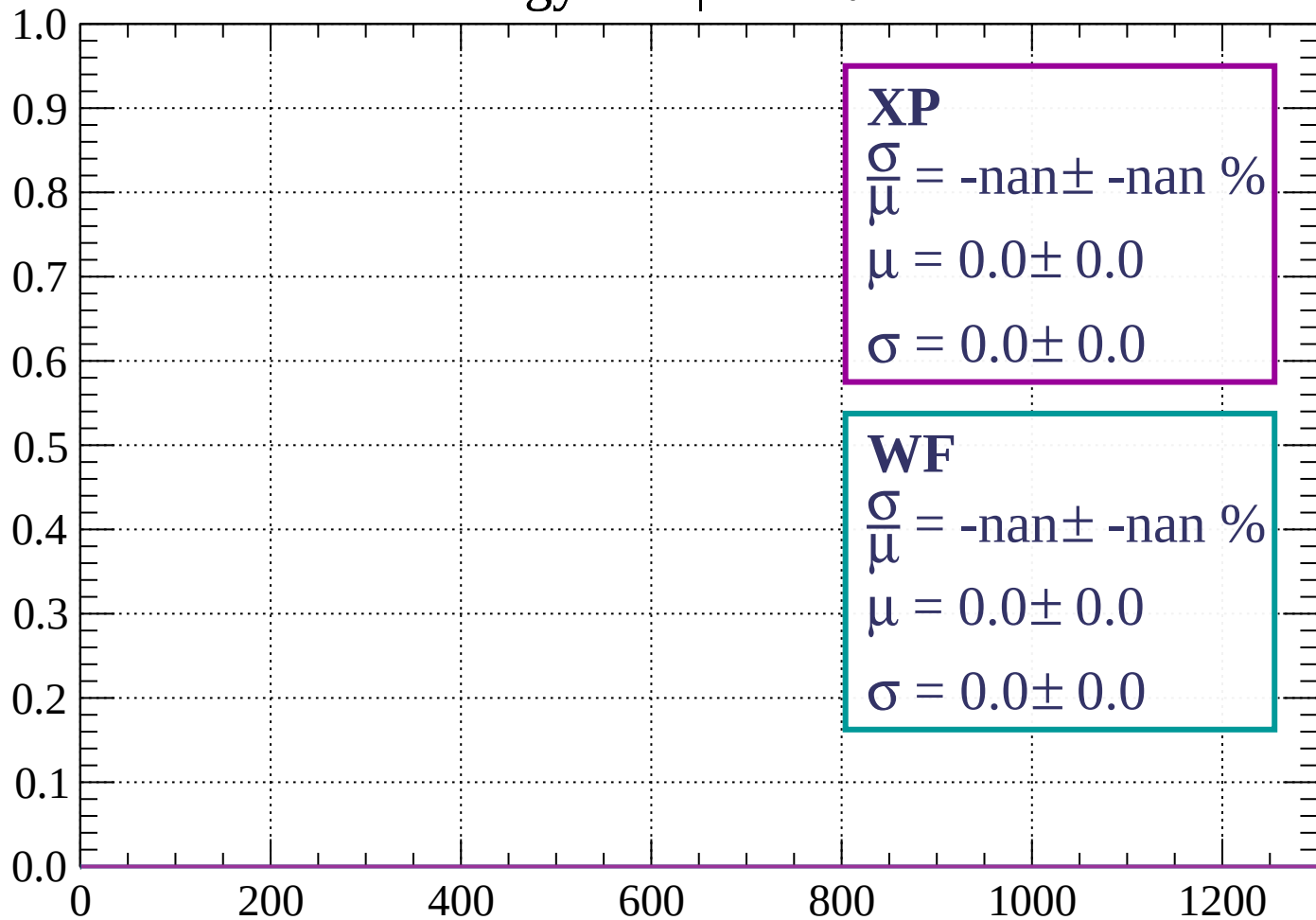
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

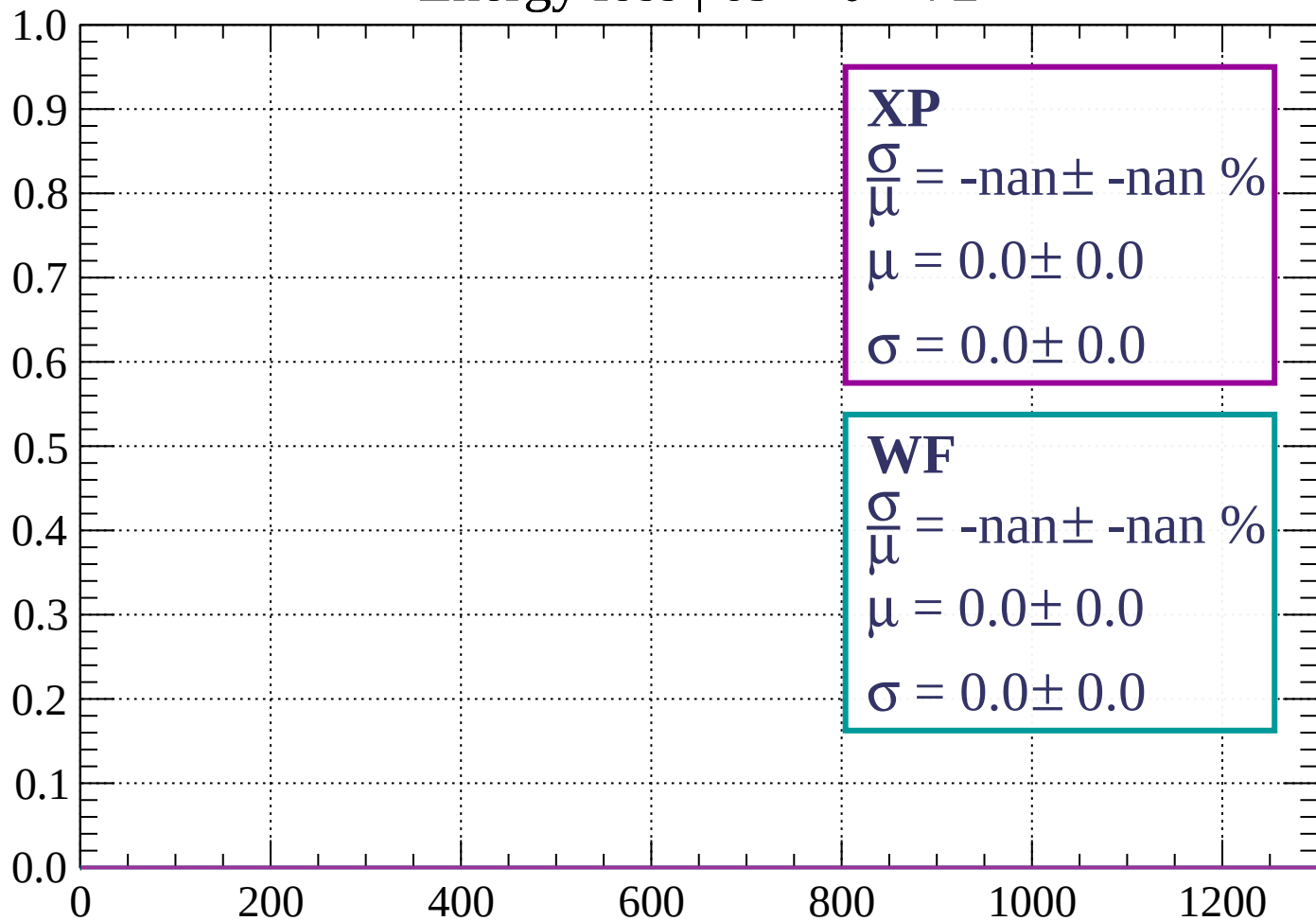
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

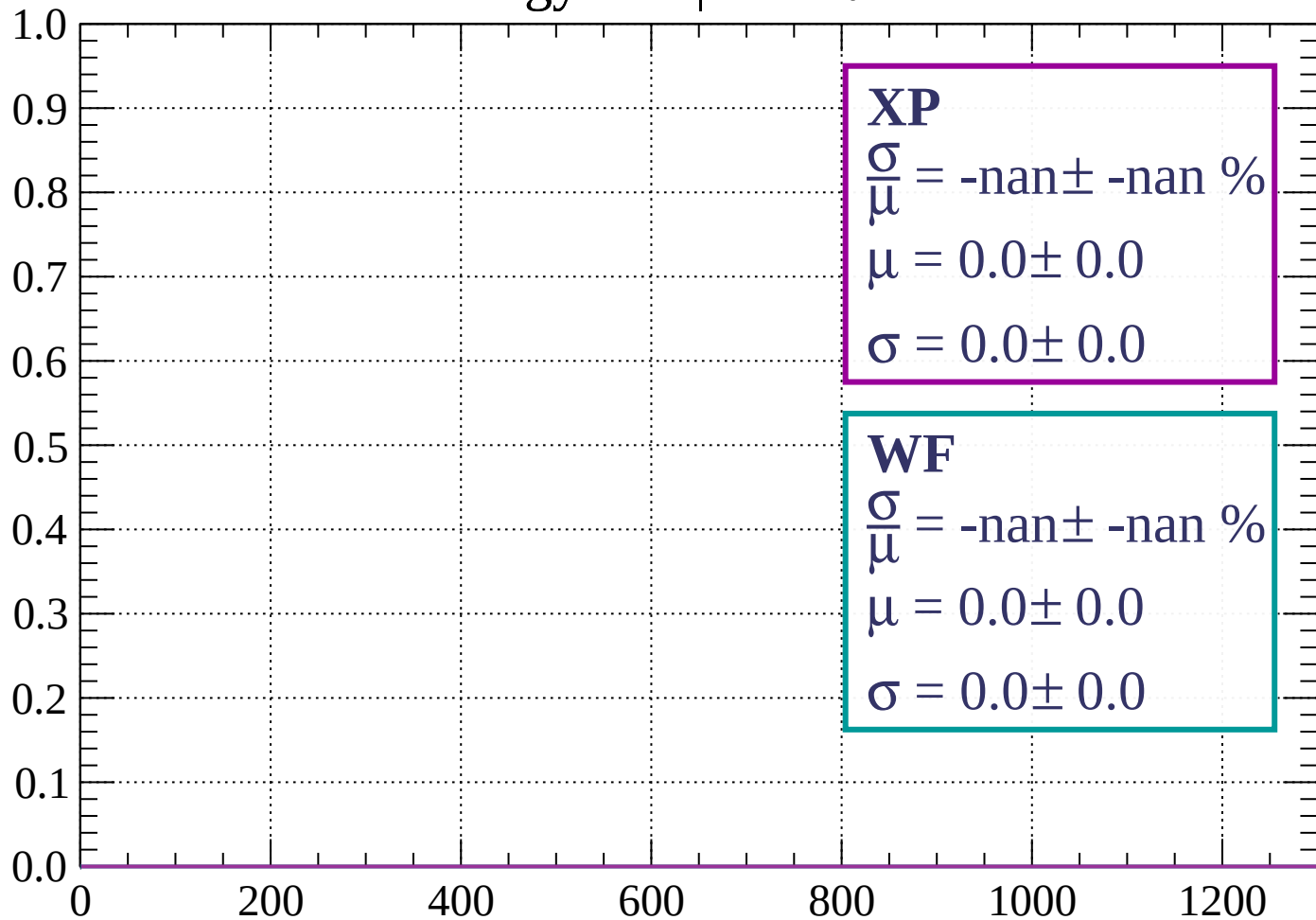
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

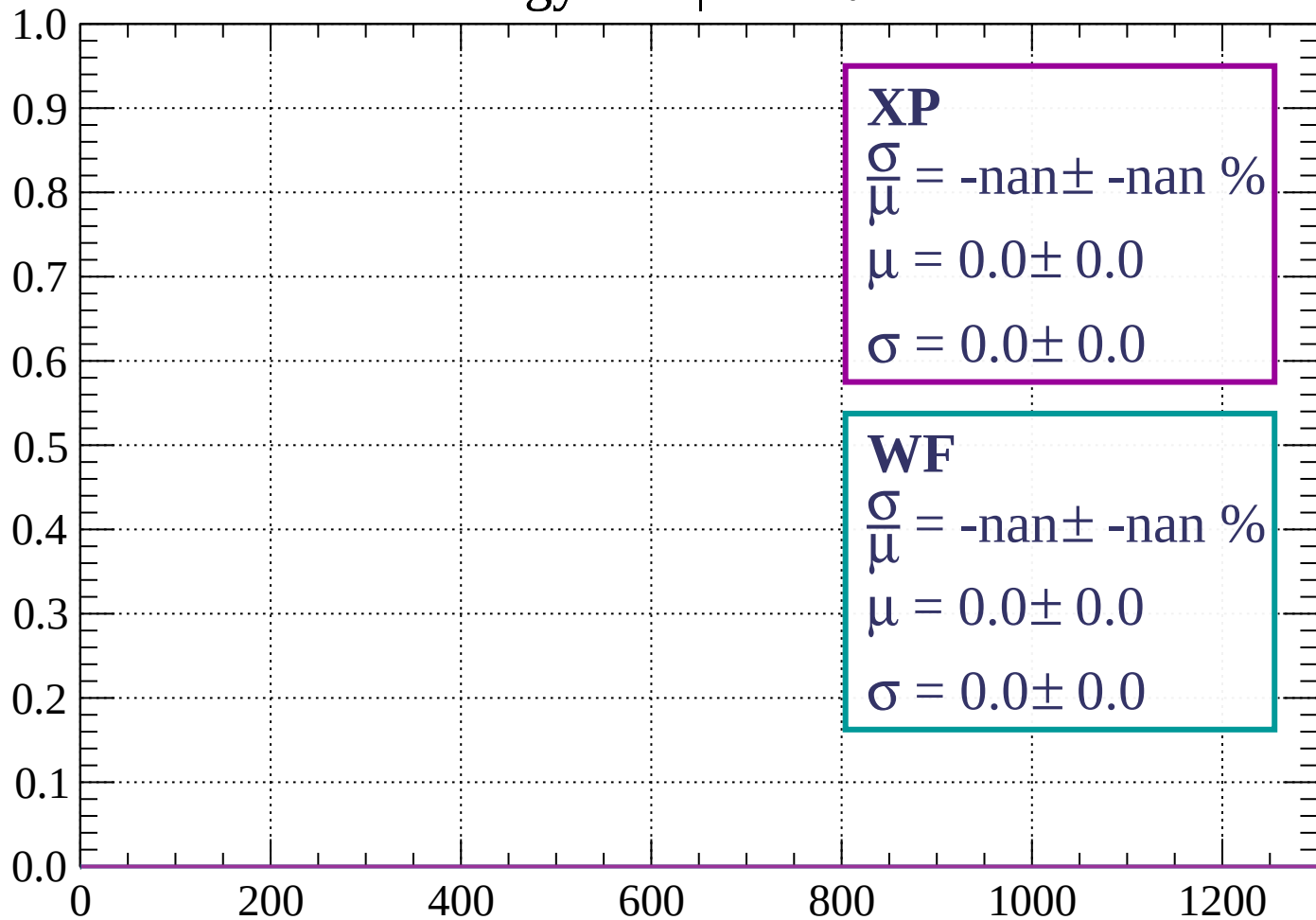
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

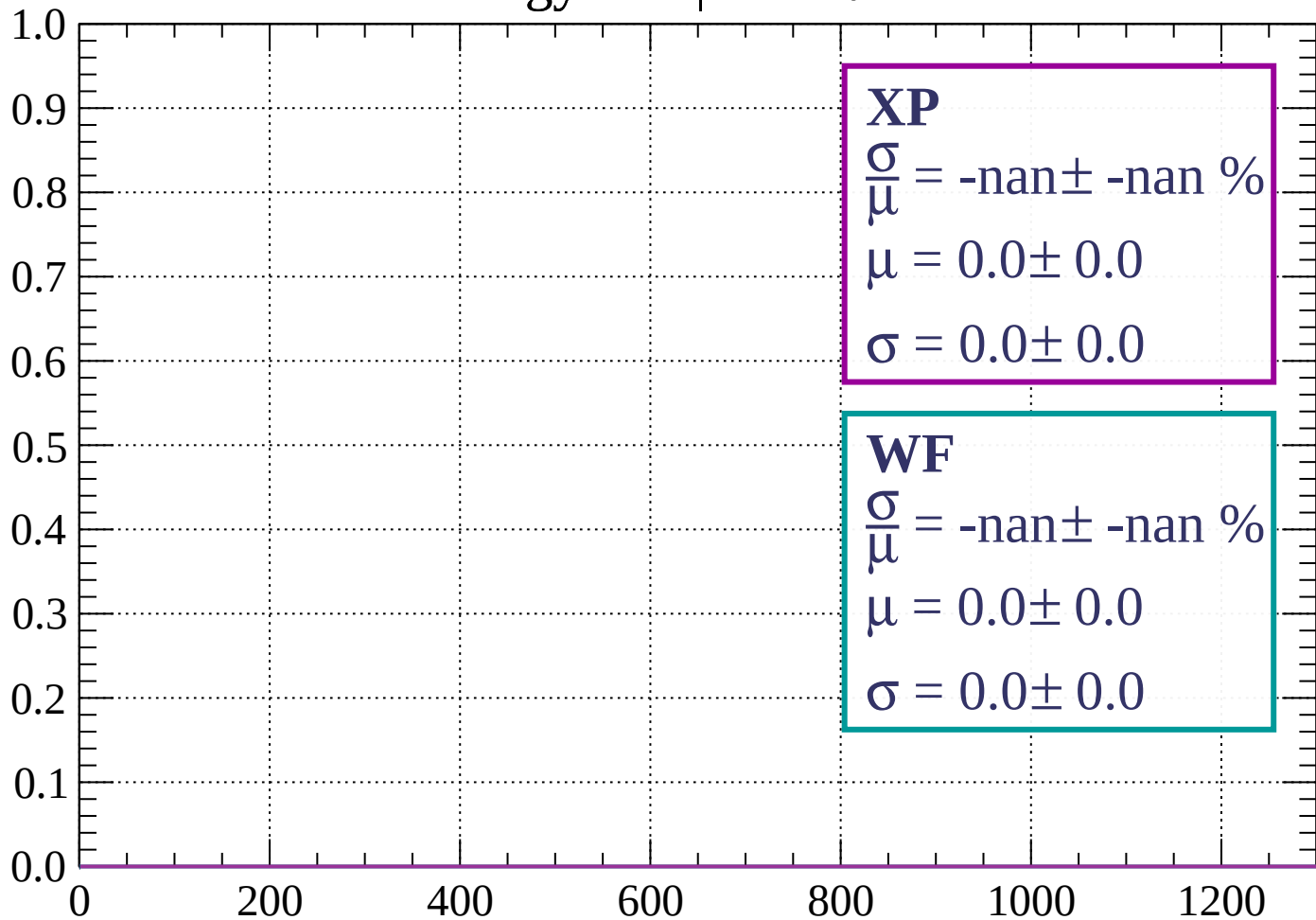
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

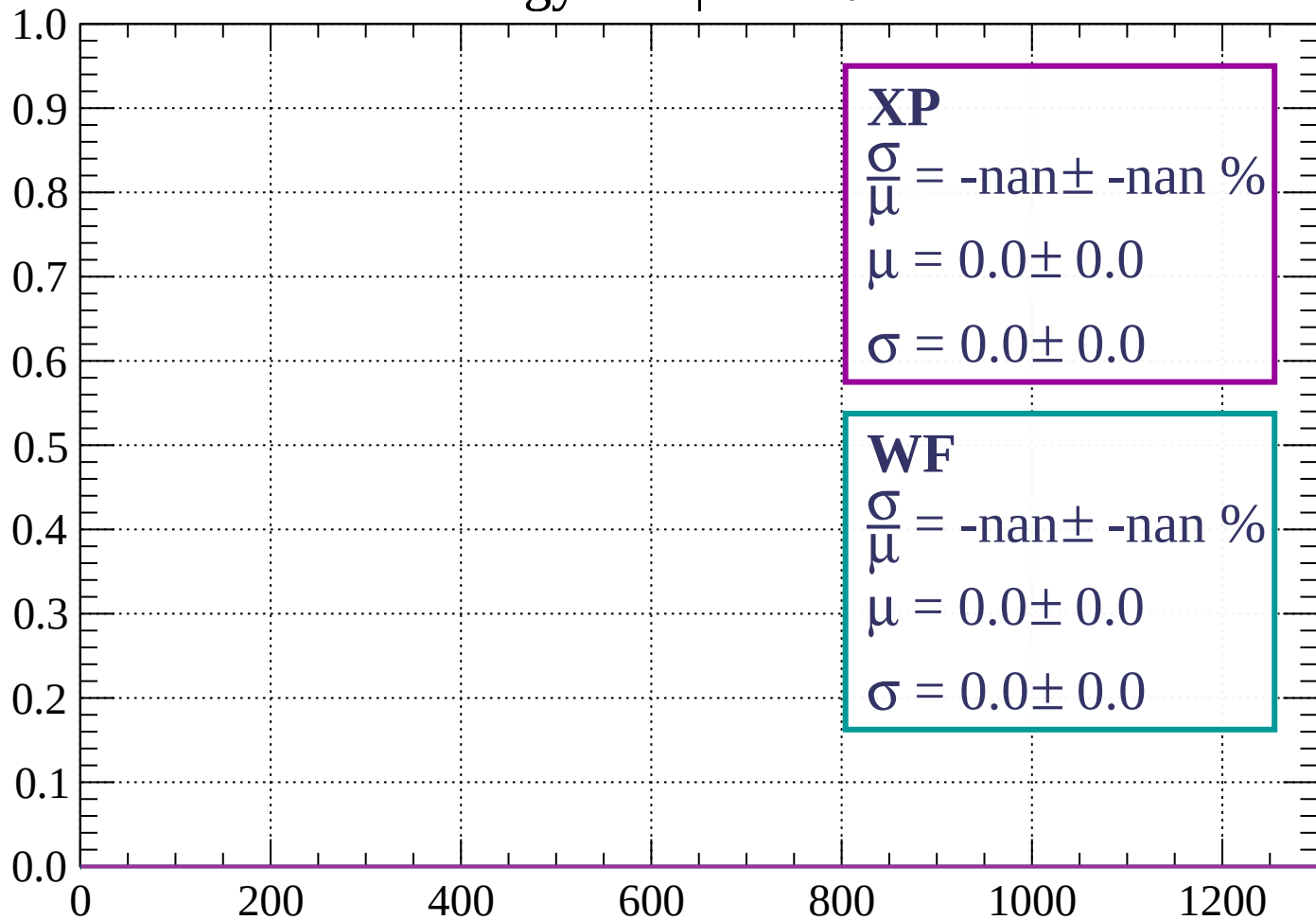
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

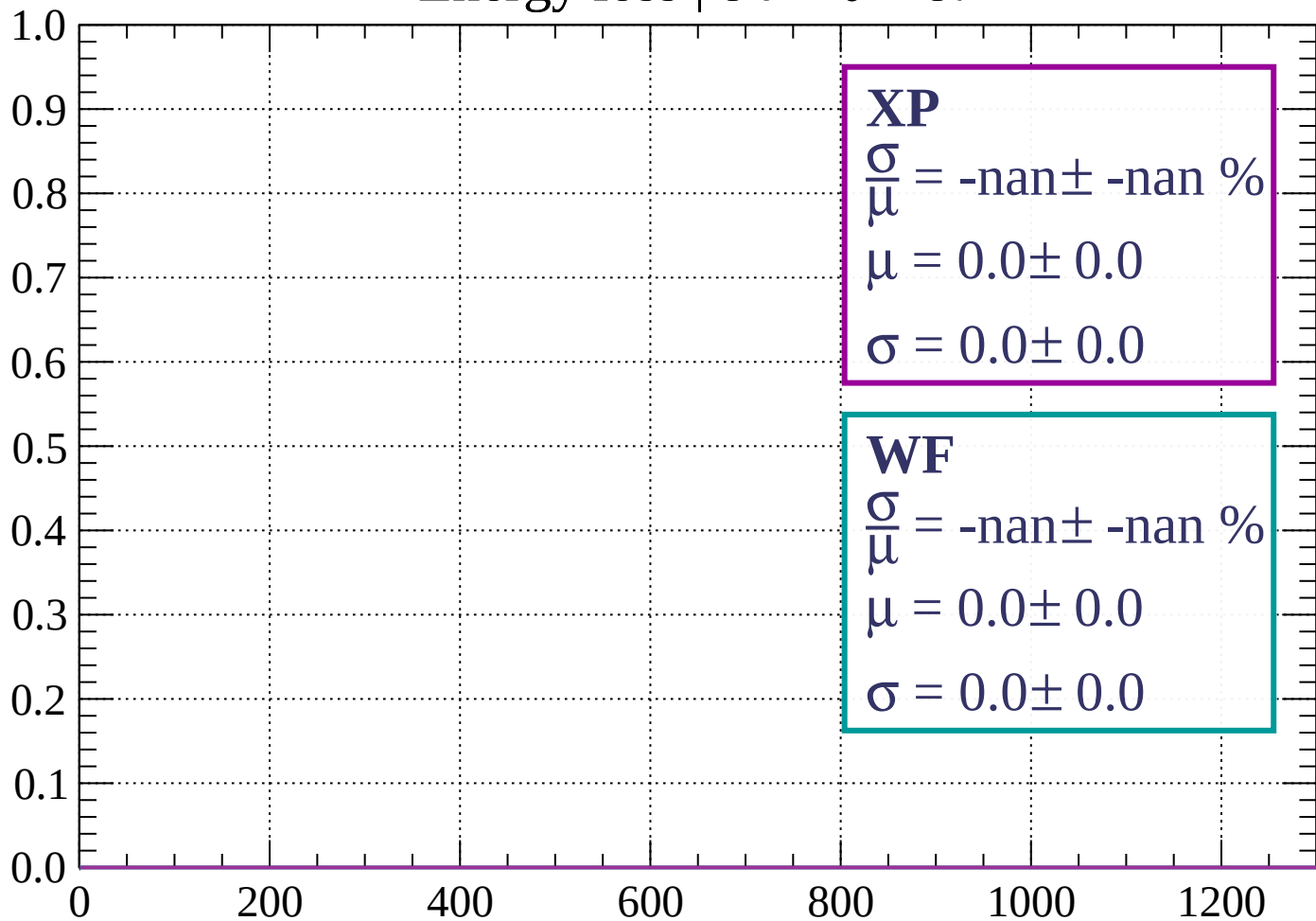
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

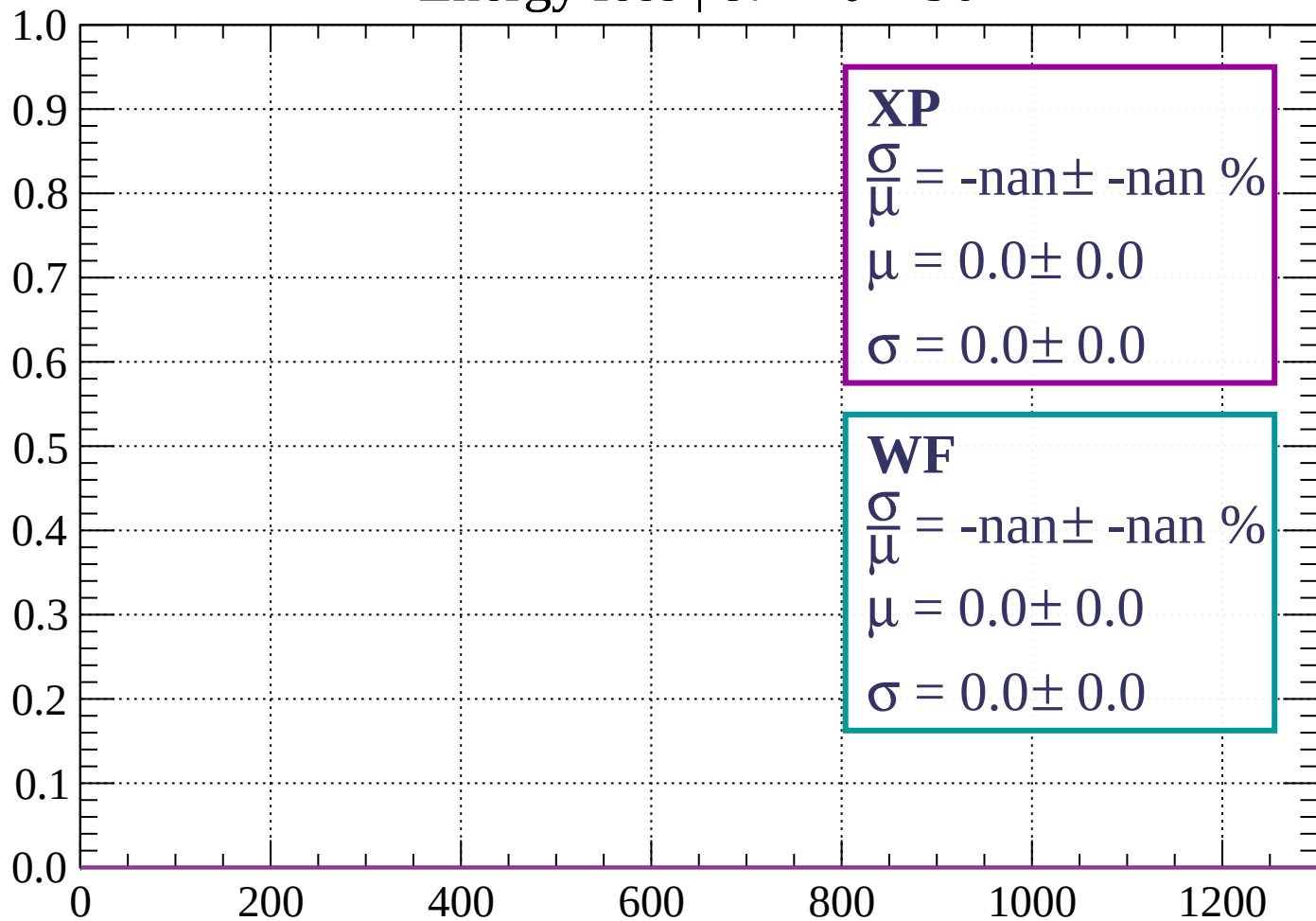
Count



dE/dx (ADC counts/cm)

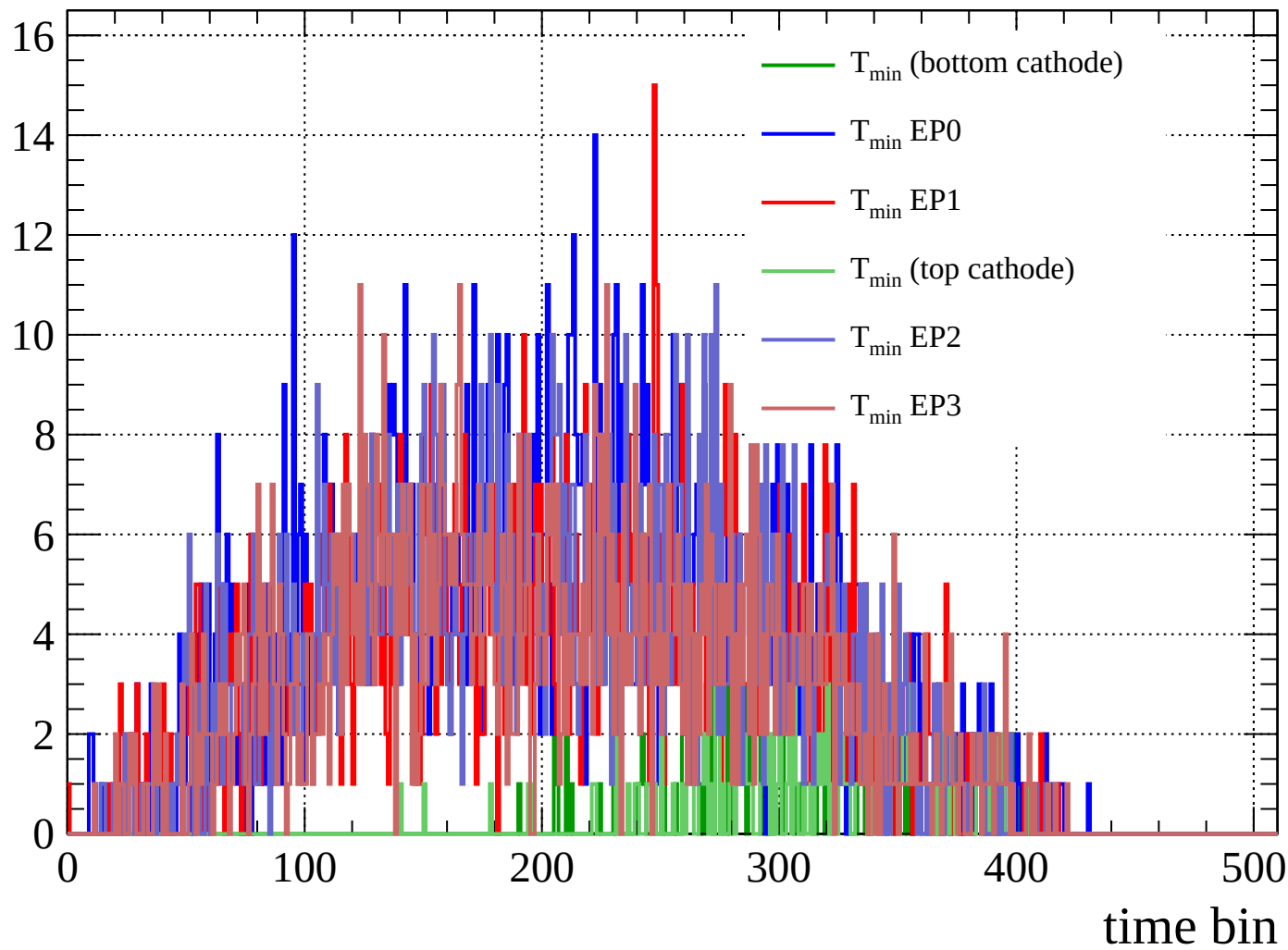
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

